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# MOLE CONCEPT

## SOME USEFUL CONVERSION FACTORS

$1 \text{ \AA} = 10^{-10} \text{ m}$ ,  $1 \text{ nm} = 10^{-9} \text{ m}$   
 $1 \text{ pm} = 10^{-12} \text{ m}$   
 $1 \text{ litre} = 10^{-3} \text{ m}^3 = 1 \text{ dm}^3$   
 $1 \text{ atm} = 760 \text{ mm or torr}$   
 $= 101325 \text{ Pa or Nm}^{-2}$   
 $1 \text{ bar} = 10^5 \text{ Nm}^{-2} = 10^5 \text{ Pa}$   
 $1 \text{ calorie} = 4.184 \text{ J}$   
 $1 \text{ electron volt (eV)} = 1.6022 \times 10^{-19} \text{ J}$   
 $(1 \text{ J} = 10^7 \text{ ergs})$   
 $(1 \text{ cal} > 1 \text{ J} > 1 \text{ erg} > 1 \text{ eV})$

## ATOMIC MASS OR MOLECULAR MASS

Mass of one atom or molecule in a.m.u.

$\text{C} \rightarrow 12 \text{ amu}$

$\text{H}_2\text{O} \rightarrow 18 \text{ amu}$

## ACTUAL MASS

mass of one atom or molecule in grams

$\text{C} \rightarrow 12 \times 1.6 \times 10^{-24} \text{ g}$

$\text{H}_2\text{O} \rightarrow 18 \times 1.6 \times 10^{-24} \text{ g}$

## RELATIVE ATOMIC MASS OR RELATIVE MOLECULAR MASS

Mass of one atom or molecule w.r.t.  $1/12^{\text{th}}$  of  $^{12}\text{C}$  atom

$\text{C} \rightarrow 12$

$\text{H}_2\text{O} \rightarrow 18$

It is unitless

## GRAMS ATOMIC MASS OR GRAM MOLECULAR MASS

Mass of one mole of atom or molecule

$\text{C} \rightarrow 12 \text{ g}$

$\text{H}_2\text{O} \rightarrow 18 \text{ g}$

It is also called molar mass

## DEFINITION OF MOLE

One mole is a collection of that many entities as there are number of atoms exactly in 12 gm of C-12 isotope. The number of atoms present in exactly 12 gm of C-12 isotope is called Avogadro's number [ $N_A = 6.022 \times 10^{23}$ ]

$$1 \text{ u} = 1 \text{ amu} = (1/12)^{\text{th}} \text{ of mass of 1 atom of } \text{C}^{12} = \frac{1 \text{ g}}{N_A} = 1.66 \times 10^{-24} \text{ g}$$

### For elements

- 1 g atom = 1 mole of atoms =  $N_A$  atoms
- g atomic mass (GAM) = mass of  $N_A$  atoms in g.
- Mole of atoms =  $\frac{\text{Mass (g)}}{\text{GAM or molar mass}}$

### For molecule

- 1 g molecule = 1 mole of molecule =  $N_A$  molecule
- g molecular mass (GMM) = mass of  $N_A$  molecule in g.
- Mole of molecule =  $\frac{\text{Mass (g)}}{\text{GMM or molar mass}}$

### For ionic compounds

- 1 g formula unit = 1 mole of formula unit =  $N_A$  formula unit.
- g formula mass (GFM) = mass of  $N_A$  formula unit in g.
- Mole of formula unit =  $\frac{\text{Mass (g)}}{\text{GFM or molar mass}}$

### 1 mole of substance

Contains  $6.022 \times 10^{23}$  particles

Weighs as much as molecular mass / atomic mass / ionic mass in grams

If it is a gas, one mole occupies a volume of 22.4 L at 1 atm & 273 K or 22.7 L at STP

### Average or mean molar mass

The average molar mass of the different substance present in the container  $M_{\text{avg}} = \frac{M_1 n_1 + M_2 n_2 + \dots}{n_1 + n_2 + \dots}$

Here  $M_1, M_2$  are molar mass of substances and  $n_1, n_2$  are mole of substances present in the container.

## DENSITIES

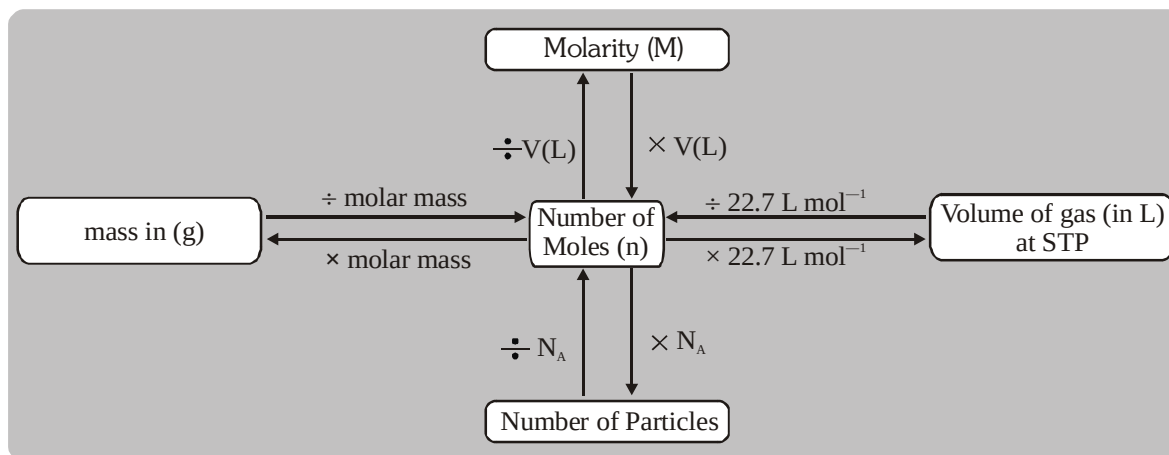
$$\text{Density} = \frac{\text{Mass}}{\text{volume}}$$

$$\text{Relative Density} = \frac{\text{Density of any substance}}{\text{Density of reference substance}}$$

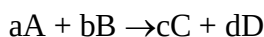
## VAPOUR DENSITY

Ratio of density of vapour to the density of hydrogen at similar pressure and temperature.

$$\text{Vapour density} = \frac{\text{Molar mass}}{2}$$



### STOICHIOMETRY BASED CONCEPT



- a,b,c,d, represents the ratios of moles, volumes [for gaseous] molecules in which the reactants react or products formed.
- a,b,c,d, does not represent the ratio of masses.
- The stoichiometric amount of components may be related as

$$\frac{\text{Moles of A reacted}}{a} = \frac{\text{Moles of B reacted}}{b} = \frac{\text{Moles of C reacted}}{c} = \frac{\text{Moles of D reacted}}{d}$$

- Amounts may also be related using POAC method. Moles of reactants and products may be related directly using conservation of suitable atoms.

### ☐ [Concept of limiting reagent]

If data of more than one reactant is given then first convert all the data into moles then divide the moles of reactants with their respective stoichiometric coefficient. The reactant having minimum ratio will be L.R. then find the moles of product formed or excess reagent left by comparing it with L.R. through stoichiometric concept.

❑ **PERCENTAGE YIELD :**

In general, when a reaction is carried out in the laboratory we do not obtain actually the theoretical amount of the product. The amount of the product that is actually obtained is called the actual yield. Knowing the actual yield and theoretical yield the percentage yield can be calculate as :

$$\% \text{ yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100$$

The percentage yield of any product is always equal to the percentage extent of that reaction.

□ **DEGREE OF DISSOCIATION, ( $\alpha$ ) :**

It represents the mole of substance dissociated per mole of the substance taken.

$$A \rightarrow n \text{ particles; } \alpha = \frac{M_o - M}{(n-1)M}$$

where,  $n$  = number of product particles per particle of reactant

$M_o$  = Molar mass of 'A'

$M$  = Molar mass of final mixture

Dissociation decreases the average molar mass of system while association increases it.

□ **PERCENTAGE PURITY :**

The percentage of a specified compound or element in an impure sample may be given as

$$\% \text{ purity} = \frac{\text{Actual mass of compound}}{\text{Total mass of sample}} \times 100$$

If impurity is unknown, it is always considered as inert (unreactive) material.

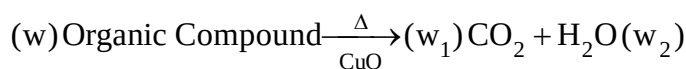
**PERCENTAGE DETERMINATION OF ELEMENTS IN COMPOUNDS**

- Mass % of an element in a compound

$$= \frac{\text{atomicity of an element} \times \text{atomic mass of an element}}{\text{molecular mass of compound}} \times 100$$

- Methods for organic compounds :

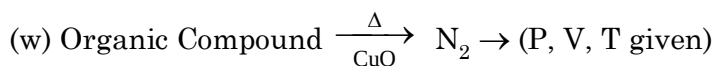
(a) **Liebig's method :** (For Carbon and hydrogen)



$$\% \text{ of C} = \frac{w_1}{44} \times \frac{12}{w} \times 100 ; \% \text{ of H} = \frac{w_2}{18} \times \frac{2}{w} \times 100$$

where  $w_1$  = wt. of  $\text{CO}_2$  produced,  $w_2$  = wt. of  $\text{H}_2\text{O}$  produced,  
 $w$  = wt. of organic compound taken

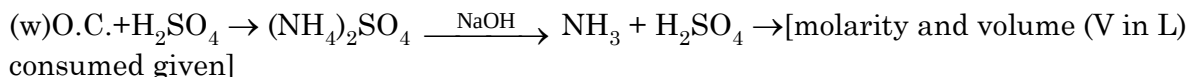
(b) **Duma's method :** (for nitrogen)



use  $PV = nRT$  to calculate moles of  $\text{N}_2$ ,  $n$ .

$$\therefore \% \text{ of N} = \frac{n \times 28}{w} \times 100$$

(c) **Kjeldahl's method :** (for nitrogen)



$$\Rightarrow \% \text{ of N} = \frac{MV \times 2 \times 14}{w} \times 100$$

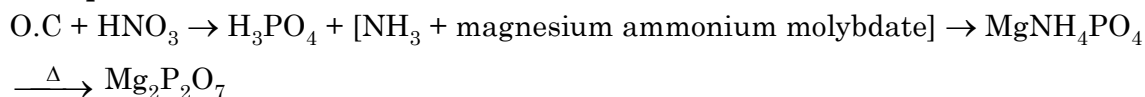
where  $M$  = molarity of  $\text{H}_2\text{SO}_4$ . Some N containing compounds do not give the above set of reaction as in Kjeldahl's method.

(d) **Sulphur :**

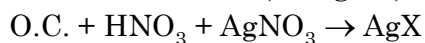


$$\Rightarrow \% \text{ of S} = \frac{w_1}{233} \times \frac{32}{w} \times 100\%$$

where  $w_1$  = wt. of  $\text{BaSO}_4$ ,  $w$  = wt. of organic compound

(e) **Phosphorus :**

$$\% \text{ of P} = \frac{w_1}{222} \times \frac{2 \times 31}{w} \times 100$$

(f) **Carius method : (Halogens)**

If X is Cl then colour = white

If X is Br then colour = dull yellow

If X is I then colour = bright yellow

Flourine can't be estimated by this

$$\% \text{ of X} = \frac{w_1}{(\text{Mol. wt. of AgX})} \times \frac{1 \times (\text{At. wt. of X})}{w} \times 100$$

## EMPIRICAL AND MOLECULAR FORMULA

- Empirical formula :** Formula depicting constituent atoms in their simplest ratio.
- Molecular formula :** Formula depicting actual number of atoms in one molecule of the compound.
- The molecular formula is generally an integral multiple of the empirical formula.  
i.e. molecular formula = empirical formula  $\times$  n

$$\text{where } n = \frac{\text{molecular formula mass}}{\text{empirical formula mass}}$$

## EXPERIMENTAL METHODS TO DETERMINE ATOMIC &amp; MOLECULAR MASSES

- For determination of atomic mass :**

**Dulong's & Petit's law :**

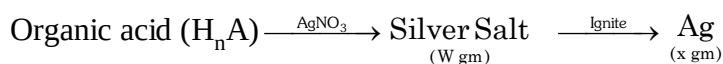
**Atomic weight of metal  $\times$  specific heat capacity ( $\text{cal/gm}^\circ\text{C}$ )  $\approx 6.4$ .**

It should be remembered that this law is an empirical observation and this gives an approximate value of atomic weight. This law gives better result for heavier solid elements, at high temperature conditions.

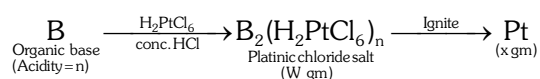
- Experimental methods for molecular mass determination.**

(a) **Victor Meyer's Method :**

Victor -Mayer's method is used to determine molecular weight of volatile compound.

(b) **Silver Salt Method :**

$$\text{Molar mass of acid} = \frac{108 \times nW}{x} - n \times 108 + n \times 1 = n \left( \frac{108W}{x} - 107 \right) \text{gmol}^{-1}$$

(c) **Chloroplatinate Salt Method :**

$$\text{Molar mass of base} = \frac{1}{2} (\text{Molar mass of salt} - n \times \text{Molar mass of H}_2\text{PtCl}_6)$$

$$= \frac{1}{2} \left( \frac{W \times 195 \times n}{x} - n \times 410 \right) = \frac{n}{2} \left( \frac{w \times 195}{x} - 410 \right) \text{gmol}^{-1}$$

CONCENTRATION TERMS

| Concentration Type     | Mathematical Formula                                                                                                                                                                               | Concept                                                                            |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Percentage by mass     | $\% \left( \frac{w}{w} \right) = \frac{\text{Mass of solute} \times 100}{\text{Mass of solution}}$                                                                                                 | Mass of solute (in gm) present in 100 gm of solution.                              |
| Volume percentage      | $\% \left( \frac{v}{v} \right) = \frac{\text{Volume of solute} \times 100}{\text{Volume of solution}}$                                                                                             | Volume of solute (in cm <sup>3</sup> ) present in 100 cm <sup>3</sup> of solution. |
| Mass-volume percentage | $\% \left( \frac{w}{v} \right) = \frac{\text{Mass of solute} \times 100}{\text{Volume of solution}}$                                                                                               | Mass of solute (in gm) present in 100 cm <sup>3</sup> of solution.                 |
| Parts per million      | $\text{ppm} = \frac{\text{Mass of solute} \times 10^6}{\text{Mass of solution}}$                                                                                                                   | Parts by mass of solute per million parts by mass of the solution                  |
| Mole fraction          | $X_A = \frac{\text{Mole of A}}{\text{Mole of A} + \text{Mole of B} + \text{Mole of C} + \dots}$<br>$X_B = \frac{\text{Mole of B}}{\text{Mole of A} + \text{Mole of B} + \text{Mole of C} + \dots}$ | Ratio of number of moles of one component to the total number of moles.            |
| Molarity               | $M = \frac{\text{Mole of solute}}{\text{Volume of solution (in L)}}$                                                                                                                               | Moles of solute in one liter of solution.                                          |
| Molality               | $m = \frac{\text{Mole of solute}}{\text{Mass of solvent (Kg)}}$                                                                                                                                    | Moles of solute in one kg of solvent                                               |

□ **MIXING OF SOLUTIONS :**

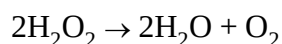
It is based on law of conservation of moles.

(i) **Two solutions having same solute :** Final molarity =  $\frac{\text{Total moles}}{\text{Total volume}} = \frac{M_1 V_1 + M_2 V_2}{V_1 + V_2}$

(ii) **Dilution Effect :** Final molarity,  $M_2 = \frac{M_1 V_1}{V_1 + V_2}$

□ **VOLUME STRENGTH OF H<sub>2</sub>O<sub>2</sub> SOLUTION :**

Labelled as 'volume H<sub>2</sub>O<sub>2</sub>' means volume of O<sub>2</sub> (in litre) at 1 bar & 273 K that can be obtained from 1 litre of such a sample when it decomposes according to



• Volume Strength of H<sub>2</sub>O<sub>2</sub> solution = 11.35 × molarity

### PERCENTAGE LABELLING OF OLEUM :

Labelled as '% oleum' means maximum amount of  $\text{H}_2\text{SO}_4$  that can be obtained from 100 gm of such oleum (mixture of  $\text{H}_2\text{SO}_4$  and  $\text{SO}_3$ ) by adding sufficient water. For example, 109 % oleum sample means, with the addition of sufficient water to 100 gm oleum sample 109 gm  $\text{H}_2\text{SO}_4$  is obtained.

% labelling of oleum sample =  $(100 + x)\%$

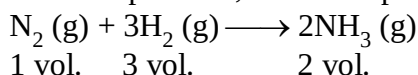
$x$  = mass of  $\text{H}_2\text{O}$  required for the complete conversion of  $\text{SO}_3$  in  $\text{H}_2\text{SO}_4$

$$\bullet \quad \% \text{ of free } \text{SO}_3 \text{ in oleum} = \left( \frac{40}{9} \times x \right) \%$$

### EUDIOMETRY

Some basic assumptions related with calculations are:

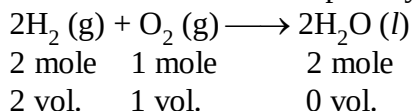
1. Gay-Lussac's law of volume combination holds good. According to this law, the volumes of gaseous reactants reacted and the volumes of gaseous products formed, all measured at the same temperature and pressure, bear a simple ratio.



Problem may be solved directly in terms of volume, in place of mole.

The stoichiometric coefficients of a balanced chemical reactions gives the ratio of volumes in which gaseous substances are reacting and products are formed at same temperature and pressure.

2. The volumes of solids or liquids is considered to be negligible in comparison to the volume of gas. It is due to the fact that the volume occupied by any substance in gaseous state is even more than thousand times the volume occupied by the same substance in solid or liquid states.



3. Air is considered as a mixture of oxygen and nitrogen gases only. It is due to the fact that about 99% volume of air is composed of oxygen and nitrogen gases only.
4. Nitrogen gas is considered as a non-reactive gas.
5. The total volume of non-reacting gaseous mixture is equal to sum of partial volumes of the component gases (**Amagat's law**).

$$V = V_1 + V_2 + \dots$$

Partial volume of gas in a non-reacting gaseous mixture is its volume when the entire pressure of the mixture is supposed to be exerted only by that gas.

6. The volume of gases produced is often given by certain solvent which absorb contain gases.

| Solvent                            | Gases absorb                                  |
|------------------------------------|-----------------------------------------------|
| KOH                                | $\text{CO}_2$ , $\text{SO}_2$ , $\text{Cl}_2$ |
| Ammonical $\text{Cu}_2\text{Cl}_2$ | CO                                            |
| Turpentine oil                     | $\text{O}_3$                                  |
| Alkaline pyrogallol                | $\text{O}_2$                                  |
| water                              | $\text{NH}_3$ , HCl                           |
| $\text{CuSO}_4/\text{CaCl}_2$      | $\text{H}_2\text{O}$                          |



## THERMODYNAMICS

### DEFINITION

Deals with interaction of one body with another in terms of energy.

**System** : Part of universe under investigation

**Surrounding** : Rest part of universe except system.

**Boundary** : Divide system & surrounding

### SYSTEM

| Open                                                                                                         | Closed                   | isolated                                       |
|--------------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------|
| Energy and matter can exchange                                                                               | Only energy can exchange | Neither energy nor matter                      |
| State function                                                                                               |                          | Path function                                  |
| Properties which depends only on initial & final state of system & not on process or path.<br>e.g. U, H etc. |                          | Depends on path or process.<br>e.g. work, heat |

### THERMODYNAMIC PROPERTIES

| Extensive                                                                | Intensive                                                                   |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Properties which are dependent of matter (size & mass) present in system | Properties which are independent of matter (size & mass) present in system. |

| Extensive Properties                                                                                                            | Intensive Properties                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Volume<br>Number of moles<br>Mass<br>Free Energy (G)<br>Entropy (S)<br>Enthalpy (H)<br>Internal energy (E & U)<br>Heat capacity | Molar volume<br>Density<br>Refractive index<br>Surface tension<br>Viscosity<br>Free energy per mole<br>specific heat<br>Pressure<br>Temperature<br>Boiling point, freezing point etc |

## PROCESSES

|                                          |                                         |                                        |                                                  |                                                           |
|------------------------------------------|-----------------------------------------|----------------------------------------|--------------------------------------------------|-----------------------------------------------------------|
| <b>Isothermal</b><br>$T = \text{const.}$ | <b>Isochoric</b><br>$V = \text{const.}$ | <b>Isobaric</b><br>$P = \text{const.}$ | <b>Adiabatic</b><br>No heat exchange<br>$dq = 0$ | <b>Cyclic</b><br>Initial & final state of system are same |
|------------------------------------------|-----------------------------------------|----------------------------------------|--------------------------------------------------|-----------------------------------------------------------|

| Reversible process                                                                                                                                                                                 | Irreversible process                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Slow process</li> <li>• At any time system and surrounding are in equilibrium.</li> <li>• <math>P_{\text{sys}} = P_{\text{surr}} \pm dP</math></li> </ul> | <ul style="list-style-type: none"> <li>• Fast process</li> <li>• No equilibrium between system and surrounding</li> <li>• <math>P_{\text{sys}} = P_{\text{surr}} \pm \Delta P</math></li> </ul> |

## HEAT (q)

Energy exchange due to temperature difference :

$$q = C\Delta T, \quad q = nC_m\Delta T, \quad q = ms\Delta T$$

$C$  = heat capacity

$C_m$  = molar heat capacity

$s$  = specific heat capacity

$m$  = Amount of substance

General values of  $C_v$  &  $C_p$  for an ideal gas can be taken as follows.

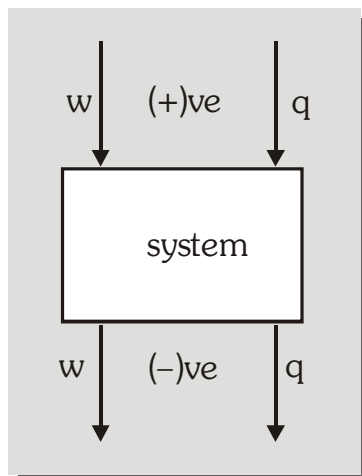
| Atomicity |            | $n_{tr}$ | $n_{rot}$ | $n_{vib}$ | $C_v$          |                 | $C_p$          |                 | $\gamma$      |                 |
|-----------|------------|----------|-----------|-----------|----------------|-----------------|----------------|-----------------|---------------|-----------------|
|           |            |          |           |           |                | Incl. Vib       | Excl. Vib      | Incl. Vib       | Excl. Vib     | Incl. Vib       |
| Mono      |            | 3        | 0         | 0         | $\frac{3}{2}R$ | $\frac{3}{2}R$  | $\frac{5}{2}R$ | $\frac{5}{2}R$  | $\frac{5}{3}$ | $\frac{5}{3}$   |
| Di        |            | 3        | 2         | 1         | $\frac{5}{2}R$ | $\frac{7}{2}R$  | $\frac{7}{2}R$ | $\frac{9}{2}R$  | $\frac{7}{5}$ | $\frac{9}{7}$   |
| Tri       | Linear     | 3        | 2         | 4         | $\frac{5}{2}R$ | $\frac{13}{2}R$ | $\frac{7}{2}R$ | $\frac{15}{2}R$ | $\frac{7}{5}$ | $\frac{15}{13}$ |
|           | Non Linear | 3        | 3         | 3         | $3R$           | $6R$            | $4R$           | $7R$            | $\frac{4}{3}$ | $\frac{7}{6}$   |

$$* \quad \gamma_{\text{mix}} = \frac{n_1 C_{p_1} + n C_{p_2} \dots}{n_1 C_{v_1} + n C_{v_2} \dots}$$

### WORK (W)

| Reversible                                                    | Irreversible                                    |
|---------------------------------------------------------------|-------------------------------------------------|
| $W_{\text{rev}} = - \int_{V_1}^{V_2} P_{\text{ext}} \cdot dV$ | $W_{\text{irr}} = - P_{\text{ext}} (V_2 - V_1)$ |

### SIGN CONVENTION



### INTERNAL ENERGY (E & U)

Every system having some quantity of matter is associated with a definite amount of energy, called internal energy.

$$U = U_{\text{Kinetics}} + U_{\text{Potential}} + U_{\text{Electronic}} + U_{\text{nuclear}} + \dots$$

U is a state function & is an extensive property.

$$\Delta U = U_{\text{final}} - U_{\text{initial}}$$

For a given closed system

$$U = f(T, V)$$

$$dU = \left( \frac{\partial U}{\partial T} \right)_V dT + \left( \frac{\partial U}{\partial V} \right)_T dV$$

### FIRST LAW OF THERMODYNAMICS (FLOT)

Law of conservation of energy

$$\Delta U = q + W$$

### ENTHALPY

Chemical reactions are generally carried out at constant pressure (atmospheric pressure) so it has been found useful to define a new state function **Enthalpy (H)** as.

$$H = U + PV$$

$$\Delta H = \Delta U + \Delta(PV)$$

at constant pressure  $\Delta H = \Delta U + P \Delta V$

combining with first law.  $\Delta H = q_p$

Hence transfer of heat at constant volume brings about a change in the internal energy of the system whereas that at constant pressure brings about a change in the enthalpy of the system.

\* For a given closed system  $H = f(P, T)$   $dH = \left(\frac{\partial H}{\partial T}\right)_P dT + \left(\frac{\partial H}{\partial P}\right)_T dP$

### RELATIONSHIP BETWEEN $\Delta H$ & $\Delta U$

The difference between  $\Delta H$  &  $\Delta U$  becomes significant only when gases are involved (insignificant in solids and liquids)  $\Delta H = \Delta U + \Delta(PV)$

If substance is not undergoing chemical reaction or phase change.  $\Delta H = \Delta U + nR\Delta T$

In case of chemical reaction  $\Delta H = \Delta U + (\Delta n_g)RT$

### WORK DONE IN VARIOUS PROCESS

| Isochoric                                | Isobaric                         | Free expansion                                       |
|------------------------------------------|----------------------------------|------------------------------------------------------|
| $W = 0$<br>$\Delta U = q = nC_V\Delta T$ | $W = -P_{\text{ext}}(V_2 - V_1)$ | $P_{\text{ext}} = 0$<br>$W = 0, \Delta U = 0, q = 0$ |

### ISOTHERMAL

$dT = 0$ ;  $\Delta U = 0$  (for ideal gas);  $q = -W$

| Reversible Isothermal                                                                                   | Irreversible Isothermal                                                                  |
|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| $W_{\text{rev, iso}} = nRT \ln\left(\frac{V_2}{V_1}\right)$<br>$= -nRT \ln\left(\frac{P_1}{P_2}\right)$ | $W_{\text{irr, iso}} = -P_{\text{ext}} \left[ \frac{nRT}{P_2} - \frac{nRT}{P_1} \right]$ |

### ADIABATIC :

$$q = 0 \Rightarrow \Delta U = W = nC_V\Delta T \Rightarrow W = \frac{P_2V_2 - P_1V_1}{\gamma - 1}$$

$$\gamma = \frac{C_P}{C_V} \Rightarrow C_P = \text{molar heat capacity at constant } P.$$

$$C_P - C_V = R \Rightarrow C_V = \text{molar heat capacity at constant } V.$$

#### For Reversible adiabatic :

$$PV^\gamma = \text{constant (Ideal gas)}$$

$$TV^{\gamma-1} = \text{constant (Ideal gas)}$$

#### For irreversible adiabatic process :

$$\Delta U = -P_{\text{ext}}(V_2 - V_1)$$

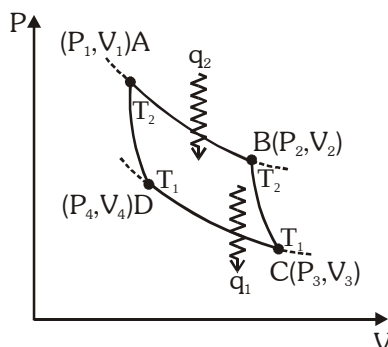
$$nC_{V,m}(T_2 - T_1) = -P_{\text{ext}} \cdot nR \left( \frac{T_2}{P_2} - \frac{T_1}{P_1} \right)$$

| Process                        | Expression for w                                                                                         | Expression for q                                                                                                          | $\Delta U$                | $\Delta H$                | Work on PV-graph |
|--------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------|------------------|
| Reversible isothermal          | $w = -nRT \ln \frac{V_2}{V_1}$<br>$= -nRT \ln \frac{P_1}{P_2}$                                           | $q = nRT \ln \left( \frac{V_2}{V_1} \right)$<br>$q = nRT \ln \left( \frac{P_1}{P_2} \right)$                              | 0 process                 | 0                         |                  |
| Irreversible isothermal        | $w = -P_{\text{ext}}(V_2 - V_1)$<br>$= -P_{\text{ext}} \left( \frac{nRT}{P_2} - \frac{nRT}{P_1} \right)$ | $q = P_{\text{ext}}(V_2 - V_1)$                                                                                           | 0                         | 0                         |                  |
| Isobaric process               | $w = -P_{\text{ext}}(V_2 - V_1)$<br>$= -nR\Delta T$                                                      | $q = \Delta H = nC_p\Delta T$                                                                                             | $\Delta U = nC_v\Delta T$ | $\Delta H = nC_p\Delta T$ |                  |
| Isochoric process              | $w = 0$                                                                                                  | $q = \Delta U = nC_v\Delta T$                                                                                             | $\Delta U = nC_v\Delta T$ | $\Delta H = nC_p\Delta T$ |                  |
| Reversible adiabatic process   | $w = nC_v(T_2 - T_1)$<br>$= \frac{P_2V_2 - P_1V_1}{\gamma - 1}$                                          | $q = 0$<br>$PV^\gamma = \text{constant}$<br>$TV^{\gamma-1} = \text{constant}$<br>$TP^{1-\gamma/\gamma} = \text{constant}$ | $\Delta U = nC_v\Delta T$ | $\Delta H = nC_p\Delta T$ |                  |
| Irreversible adiabatic process | $w = nC_v(T_2 - T_1)$<br>$= \frac{P_2V_2 - P_1V_1}{\gamma - 1}$                                          | $q = 0$<br>$nC_v(T_2 - T_1) = -P_{\text{ext}} \left( \frac{nRT_2}{P_2} - \frac{nRT_1}{P_1} \right)$                       | $\Delta U = nC_v\Delta T$ | $\Delta H = nC_p\Delta T$ |                  |
| Polytropic                     | $w = \frac{P_2V_2 - P_1V_1}{n - 1}$<br>$w = \frac{R(T_2 - T_1)}{(n - 1)}$                                | $q = \int_{T_1}^{T_2} C_v dT$<br>$w = \frac{R(T_2 - T_1)}{(n - 1)}$                                                       | $\Delta U = nC_v\Delta T$ | $\Delta H = nC_p\Delta T$ |                  |
| Cyclic Process                 | Area enclosed in PV-diagram<br>For clockwise it is -ive<br>it is +ive                                    | $q = -w$                                                                                                                  | 0                         | 0                         |                  |

## SPONTANEOUS PROCESS :

A process which takes place on its own (without any external assistance). The driving force of a spontaneous process is large or finite.

## CARNOT CYCLE



AB – Isothermal reversible expansion  $q_2 = -w_{AB} = nRT_2 \ln \frac{V_2}{V_1}$

BC – adiabatic reversible expansion  $w_{BC} = nC_V(T_1 - T_2)$

CD – Isothermal reversible compression  $q_1 = -w_{CD} = nRT_1 \ln \left( \frac{V_4}{V_3} \right)$

DA – adiabatic reversible compression  $w_{DA} = nC_V(T_2 - T_1)$

carnot efficiency  $\eta = \frac{-w_{Total}}{q_2} = \frac{q_1 + q_2}{q_2} = \frac{T_2 - T_1}{T_2}$

$\frac{q_1}{T_1} + \frac{q_2}{T_2} = 0$  for rev. cycle

$\oint \frac{q_{rev}}{T} = \oint dS = 0$

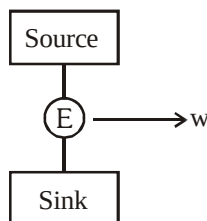
Entropy (denoted by S) is state function

$\Delta S = \int \frac{q_{rev}}{T}$

The direction of a spontaneous process and that it eventually reaches equilibrium, can be understood on the basis of entropy concept introduced through the second law of thermodynamics.

## □ STATEMENTS OF SECOND LAW OF THERMODYNAMICS

- (i) No cyclic engine is possible which take heat from one single source and in a cycle completely convert it into work without producing any change in surrounding.



- (ii) In an irreversible process entropy of universe increases but it remains constant in a reversible process.

$$\Delta S_{\text{svt}} + \Delta S_{\text{sur}} = 0 \quad \text{for rev. process}$$

$$\Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0 \quad \text{for irrev. process}$$

$$\Delta S_{\text{syt}} + \Delta S_{\text{surr}} \geq 0 \quad (\text{In general})$$

## PHYSICAL SIGNIFICANCE OF ENTROPY

One can think entropy as a measure of the degree of randomness or disorder in a system. The greater the disorder in a system, the higher is the entropy.

- (i)** The entropies of substance follow the order,

$$S(g) > S(l) > S(s)$$

- (ii) If more no. of gaseous moles are present on product side,  $\Delta_r S$  will be +ive (since gas is more disordered than solid or liquid).

- (iii) Entropy rises with increasing mass, other thing being same e.g. atomicity in gas phase.

**e.g.**  $F_2(g)$   $S^\circ = 203 \text{ J/K-mole}$

$$\text{Cl}_2(\text{g}) \quad S^\circ = 223 \text{ J/K-mole}$$

$$\text{Br}_2(\text{g}) \quad S^\circ = 245 \text{ J/K-mole}$$

- (iv)** Entropy increases with chemical complexity

For  $\text{CuSO}_4 \cdot n\text{H}_2\text{O}$

n = 0            n = 1            n = 3            n = 5

$$S^\circ = 113 \qquad 150 \qquad 225 \qquad 305$$

## □ CALCULATION OF ENTROPY CHANGE FOR AN IDEAL GAS

## General Expression

$$\Delta S = nC_V \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1}$$

$$= nC_p \ln \frac{T_2}{T_1} + nR \ln \frac{P_1}{P_2}$$

- \* Reversible & irreversible isothermal expansion or contraction of an ideal gas

$$\Delta S = nR \ln \frac{V_2}{V_1}$$

- \* Isobaric heating or cooling :

$$\Delta S = nC_p \ln \left( \frac{T_2}{T_1} \right)$$

- \* Isochoric heating or cooling :

$$\Delta S = nC_v \ln \left( \frac{T_2}{T_1} \right)$$

- \* Adiabatic process :

$$\Delta S = nC_v \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1} \text{ for irreversible process}$$

$\Delta S = 0$  for reversible adiabatic compression and expansion.

## □ ENTROPY CALCULATION

| Process                 | $\Delta S_{\text{Sys.}}$                                                     | $\Delta S_{\text{Surr.}}$                                                                                               |
|-------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Isothermal reversible   | $\Delta S_{\text{Sys.}} = nR \ln \frac{V_2}{V_1}$                            | $\Delta S_{\text{Surr.}} = -\Delta S_{\text{Sys.}}$                                                                     |
| Isothermal irreversible | $\Delta S_{\text{Sys.}} = nR \ln \frac{V_2}{V_1}$                            | $\Delta S_{\text{Surr.}} = \frac{-q_{\text{Sys}}}{T} = \frac{W_{\text{Sys}}}{T} = \frac{-P_{\text{ext}}(V_2 - V_1)}{T}$ |
| Adiabatic reversible    | $\Delta S_{\text{Sys.}} = 0$                                                 | $\Delta S_{\text{Surr.}} = 0$                                                                                           |
| Adiabatic irreversible  | $\Delta S_{\text{Sys.}} = nC_p \ln \frac{T_2}{T_1} + nR \ln \frac{P_1}{P_2}$ | $\Delta S_{\text{Surr.}} = 0$                                                                                           |
| Isochoric reversible    | $\Delta S_{\text{Sys.}} = nC_v \ln \frac{T_2}{T_1}$                          | $\Delta S_{\text{Surr.}} = -\Delta S_{\text{Sys.}}$                                                                     |
| Isochoric irreversible  | $\Delta S_{\text{Sys.}} = nC_v \ln \frac{T_2}{T_1}$                          | $\Delta S_{\text{Surr.}} = \frac{-q_{\text{sys}}}{T_{\text{surr}}} = \frac{-nC_v \Delta T}{T_{\text{surr}}}$            |



### ❑ THIRD LAW OF THERMODYNAMICS

“At absolute zero, the entropy of a perfectly crystalline substance is zero”, which means that at absolute zero every crystalline solid is in a state of perfect order and its entropy should be zero. By virtue of the third law, the absolute value of entropy (unlike absolute value of enthalpy) for any pure substance can be calculated at room temperature.

$$S_T - S_{0K} = \int_0^T \frac{q_{rev}}{T}$$

Since  $S_{0K} = 0$

$$S_T = \int_0^T \frac{q_{rev}}{T}$$

Absolute entropies of various substances have been tabulated and these values are used to calculate entropy changes for the reactions by the formula;

$$\Delta S_r = \sum S (\text{products}) - \sum S (\text{reactants})$$

### ❑ VARIATION OF $\Delta S_r$ WITH TEMPERATURE & PRESSURE :

$$(\Delta S_r)_{T_2} - (\Delta S_r)_{T_1} = (\Delta C_p)_r \ln \frac{T_2}{T_1}$$

$$(\Delta S_r)_{p_2} - (\Delta S_r)_{p_1} = \Delta n_g R \ln \frac{p_1}{p_2}$$

Similarly

$$(\Delta H_r)_{T_2} - (\Delta H_r)_{T_1} = (\Delta C_p)_r (T_2 - T_1) \quad \{\text{Krichoff's equation}\}$$

$$(\Delta U_r)_{T_2} - (\Delta U_r)_{T_1} = (\Delta C_v)_r (T_2 - T_1)$$

### ❑ GIBBS FREE ENERGY (G) AND SPONTANEITY :

A new thermodynamic state function G, the Gibbs free energy is defined as :

$$G = H - TS$$

at constant temperature and pressure

$$\Delta G = \Delta H - T \Delta S$$

|                           |                                         |
|---------------------------|-----------------------------------------|
| If $(\Delta G)_{T,P} < 0$ | Process is irreversible (spontaneous)   |
| $(\Delta G)_{T,P} = 0$    | Process is reversible                   |
| $(\Delta G)_{T,P} > 0$    | process is impossible (non spontaneous) |

The use of Gibbs free energy has the advantage that it refers to the system only (and not surroundings).

To summarise, the spontaneity of a chemical reaction is decided by two factors taken together

- The enthalpy factor
- The entropy factor

The equation  $\Delta G = \Delta H - T \Delta S$  takes both the factors into consideration.

| $(\Delta H_r)_{T,P}$ | $(\Delta S_r)_{T,P}$ | $(\Delta G_r)$                                                                | Remarks                        |
|----------------------|----------------------|-------------------------------------------------------------------------------|--------------------------------|
| - ve                 | + ve                 | Always -ve                                                                    | Reaction is spontaneous        |
| + ve                 | - ve                 | Always +ve                                                                    | Reaction non spontaneous       |
| + ve                 | + ve                 | At low temperature, $\Delta G = +ve$<br>At high temperature, $\Delta G = -ve$ | Non spontaneous<br>Spontaneous |
| - ve                 | - ve                 | At low temperature, - ve                                                      | Spontaneous                    |
| - ve                 | - ve                 | At high temperature, + ve                                                     | Non spontaneous                |

❑ **VARIATION OF GIBB'S FUNCTION (G) WITH TEMPERATURE AND PRESSURE :**

$$G = H - TS$$

$$= U + PV - TS$$

$$dG = dU + P dV - T dS + V dP - S dT$$

$$dG = V dP - S dT$$

\* At constant temperature

$$\Delta G = V dP$$

$$\text{or } \left( \frac{\partial G}{\partial P} \right)_T = V$$

\* At constant pressure

$$\Delta G = -S dT$$

$$\left( \frac{\partial G}{\partial T} \right)_P = -S$$

**Relationship between  $\Delta G$  &  $W_{\text{non-PV}}$**

$$dU = q + W_{\text{PV}} + W_{\text{non-PV}}$$

for reversible process at constant T & P

$$dU + p dV - T dS = W_{\text{non-PV}}$$

$$dH - T dS = W_{\text{non-PV}}$$

$$(dG_{\text{system}})_{T,P} = W_{\text{non-PV}}$$

$$(dG_{\text{system}})_{T,P} = (W_{\text{non-PV}})_{\text{system}}$$

Non-PV work done by the system = decrease in gibbs free energy

❑ **SOME FACTS TO BE REMEMBERED :**

(a) Standard condition

\* For gases/solid / liquid

$$P = 1 \text{ bar}$$

\* For ion / substance in solution

$$\text{Concentration} = 1M$$

$$(b) \Delta G_r = (\Delta G_f)_{\text{product}} - (\Delta G_f)_{\text{reactant}}$$

$$\Delta H_r = (\Delta H_f)_{\text{product}} - (\Delta H_f)_{\text{reactant}}$$

$$\Delta S_r = (\Delta S_f)_{\text{product}} - (\Delta S_f)_{\text{reactant}}$$

(All above equation will be derived in thermochemistry)

Relationship between ,  $\Delta G^\circ$  and equilibrium constant

$$\Delta G = \Delta G^\circ + RT \ln Q$$

At equilibrium  $\Delta G = 0$

$$\Delta G^\circ = -RT \ln K_{eq}$$

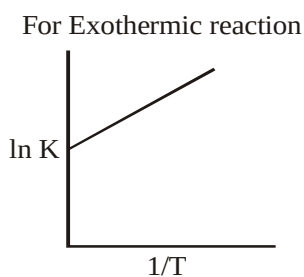
$$\Delta H^\circ - T\Delta S^\circ = -RT \ln K_{eq}$$

$$\ln K_{eq} = \frac{-\Delta_r H^\circ}{RT} + \frac{\Delta_r S^\circ}{R}$$

$$\ln K_1 = \frac{\Delta_r S^\circ}{R} - \frac{\Delta_r H^\circ}{RT_1}$$

$$\ln K_2 = \frac{\Delta_r S^\circ}{R} - \frac{\Delta_r H^\circ}{RT_2}$$

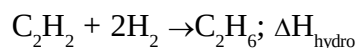
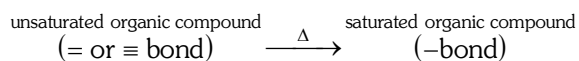
$$\ln \left( \frac{K_2}{K_1} \right) = \frac{\Delta H^\circ}{R} \left( \frac{1}{T_1} - \frac{1}{T_2} \right)$$





**ENTHALPY OF HYDROGENATION ( $\Delta H_{\text{hydro}}$ )**  
(Always exothermic)

Enthalpy change during the complete hydrogenation of one mole unsaturated organic compound into its saturated compound.



**NOTE :**

If in a reaction heat of reactant & products are given then heat of that reaction can be measured as follows:

(a) For heat of combustion & for bond enthalpy

$$\Delta_r H = \sum (\Delta H_C)_{\text{reactant}} - \sum (\Delta H_C)_{\text{product}}$$

(b) For heat of formation

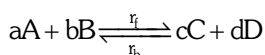
$$\Delta_r H = \sum (\Delta H_f)_{\text{product}} - \sum (\Delta H_f)_{\text{reactant}}$$

**IMPORTANT NOTES**

## CHEMICAL EQUILIBRIUM

- Equilibrium represents the state of a process in which the measurable properties like :- temperature, pressure, color, concentration of the system do not show any change with the passage of time.
- Equilibrium is a dynamic process, chemical equilibrium can be approached from both sides.
- The state of equilibrium is not affected by the presence of catalyst. It only helps to attain the equilibrium state in less or more time.
- Equilibrium can be attained both in homogeneous & heterogeneous system.

Consider a reversible reaction,



## AT EQUILIBRIUM STATE

Rate of forward reaction ( $r_f$ )  
= rate of backward reaction ( $r_b$ )

So, at equilibrium,

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b} = \frac{K_f}{K_b} \quad \text{In terms of active mass}$$

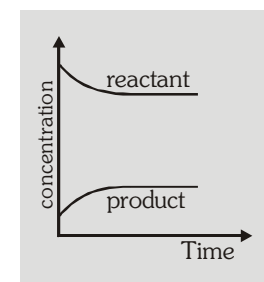
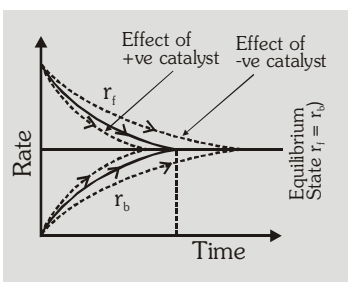
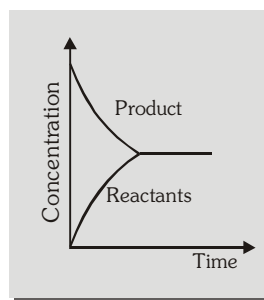
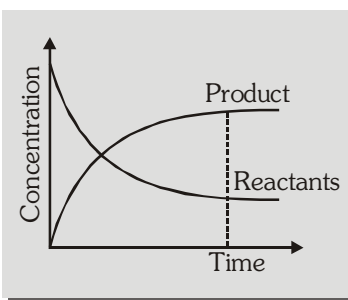
$$K_p = \frac{[P_C]^c [P_D]^d}{[P_A]^a [P_B]^b} \quad \text{In terms of partial pressure}$$

$$K_x = \frac{[X_C]^c [X_D]^d}{[X_A]^a [X_B]^b} \quad \text{In terms of mole fraction}$$

• Partial pressure of solid is taken as unity & in calculation of partial pressure of solids, their number of moles are not considered.

- $K_p = K_c(RT)^{\Delta n_g}$  then  $K_p = K_c$   
when  $\Delta n_g = 0$  then  $K_p = K_c$   
when  $\Delta n_g > 0$  then  $K_p > K_c$   
when  $\Delta n_g < 0$  then  $K_p < K_c$
- While determining  $\Delta n_g$  take only gaseous species.
- The active mass of solid & pure liquid is a constant quantity (unity) because it is an intensive property.

## GRAPHS



- Unit of Equilibrium constant:**

$$K_c = (\text{mol L}^{-1})^{\Delta n_g}; K_p = (\text{atm})^{\Delta n_g}$$

- Application of  $K_c$  or  $K_p$**

- More is the value of  $K_p$  or  $K_c$  more is the extent of reaction.
- Stability of reactant increases when value of  $K$  decreases
- Stability of Product increases when value of  $K$  increases.

### CHARACTERISTICS OF EQUILIBRIUM CONSTANT

Equilibrium constant depends upon temperature & way of writing the reaction

#### (i) Temperature :

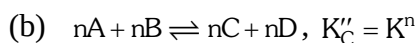
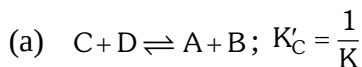
Let  $K_1$  &  $K_2$  be equilibrium constant at  $T_1$  &  $T_2$  then

$$\log\left(\frac{K_2}{K_1}\right) = \frac{\Delta H^\circ}{2.303R} \left[ \frac{1}{T_1} - \frac{1}{T_2} \right]$$

(Van't Hoff equation)

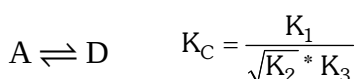
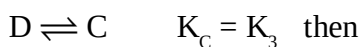
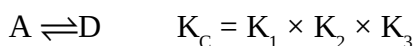
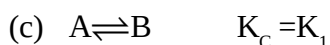
#### (ii) Way of writing the reaction :

For  $A + B \rightleftharpoons C + D$   $K_c = K$  then



when  $n=2$ , then  $K''_c = K^2$

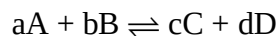
$n = \frac{1}{3}$  then  $K''_c = K^{1/3}$



#### • Predicting the direction of reaction :

Reaction Quotient (Q) is expressed in the same way as for equilibrium constant, except that the concentrations may not necessarily be at equilibrium.

In general for the reversible reaction :



$$Q = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

$$Q = \frac{[P_c]^c [P_D]^d}{[P_A]^a [P_B]^b} \text{ (in terms of pressure)}$$

If  $Q = K_{eq}$  then system is in equilibrium

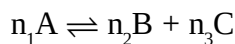
If  $Q > K_{eq}$  then system proceed in backward direction to attain equilibrium.

If  $Q < K_{eq}$  then system proceed in forward direction to attain equilibrium.

#### • Degree of dissociation ( $\alpha$ )

$$\frac{\text{No. of moles of reactant dissociated}}{\text{No. of mole of reactant present initially}}$$

#### • Degree of Dissociation from Vapour pressure



$$\alpha = \frac{n_1}{\Delta n} \left( \frac{D_T - D_0}{D_0} \right); \Delta n = (n_2 + n_3) - (n_1)$$

$$D_T = \frac{\text{theoretical vapour density}}{\text{Molecular weight}} = \frac{\text{Molecular weight}}{2}$$

$D_0$  = Observed vapour density

## PHYSICAL EQUILIBRIUM

## Physical reaction :

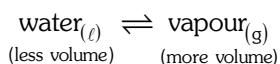
Those reactions in which change in only & only physical states of substances takes place without any chemical change.

## (i) Ice-water system (melting of ice) :



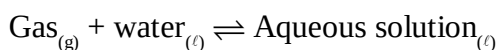
It is an endothermic process & there is decrease in volume. Thus, the favourable conditions for melting of ice are high temperature, & High-pressure.

## (ii) Water -Water vapour system (vapourisation of water) :



It is an endothermic process & there is increase in volume. Thus, the favourable conditions for vaporisation of water are high temperature, & low-pressure.

## (iii) Solubility of gases in liquids :



When a gas dissolve in liquid, there is decrease in volume. Thus, increase in pressure will favour the dissolution of a gas in liquid.

## LE-CHATELIER'S

## PRINCIPLE

If a system at equilibrium is subjected to a change of any one of the factors such as concentration, pressure or temperature then the equilibrium is shifted in such a way as to nullify the effect of change.

Le-Chatelier's principle is applicable for both chemical and physical equilibrium.

## CHEMICAL EQUILIBRIUM

| S. No. | Effect due to change in |                                                                | $\Delta n_g = 0$<br>$A \rightleftharpoons B$ | $\Delta n_g > 0$<br>$A \rightleftharpoons 2B$    | $\Delta n_g < 0$<br>$2A \rightleftharpoons B$    |
|--------|-------------------------|----------------------------------------------------------------|----------------------------------------------|--------------------------------------------------|--------------------------------------------------|
| a)     | Concentration           | (i) $\uparrow [A]$<br>(ii) $\downarrow [A]$                    | Forward direction<br>Backward direction      | Forward direction<br>Backward direction          | Forward direction<br>Backward direction          |
| b)     | Pressure                | (i) $\uparrow$ in pressure<br>(ii) $\downarrow$ in pressure    | Unchanged<br>Unchanged                       | Backward direction<br>Forward direction          | Forward direction<br>Backward direction          |
| c)     | Temperature             | (i) $\uparrow$ in Endothermic<br>(ii) $\uparrow$ in Exothermic | Forward direction<br>Backward direction      | Forward direction<br>Backward direction          | Forward direction<br>Backward direction          |
| d)     | Dissociation            | (i) $\uparrow$ in pressure<br>(ii) $\uparrow$ in volume        | Unchanged<br>Unchanged                       | Dissociation Decreases<br>Dissociation Increases | Dissociation Increases<br>Dissociation Decreases |
| e)     | Mixing of inert gas     | (i) at constant P<br>(ii) at constant V                        | Unchanged<br>Unchanged                       | Dissociation Increases<br>Unchanged              | Dissociation Decreases<br>Unchanged              |



**IONIC EQUILIBRIUM**
**CLASSIFICATION OF SUBSTANCES**

On the basis of their dissociation nature :

- (i) **Strong electrolytes** : Substances which are largely dissociated and are strong electrolytes. e.g. HCl, H<sub>2</sub>SO<sub>4</sub>, HNO<sub>3</sub> etc.
- (ii) **Weak electrolytes** Substances which dissociate only to a small extent in aqueous solution. e.g. HCN, H<sub>3</sub>BO<sub>3</sub> etc.
- (iii) **Non electrolytes** Which do not dissociate.

**pH CALCULATUION**
**Case (i) A weak acid in water**

- (a) if  $\alpha = \sqrt{\frac{K_a}{C}}$  is  $< 0.1$ , then  $[H^+] \approx \sqrt{K_a C}$ .
- (b) **General Expression** :  $[H^+] = 0.5(-K_a + \sqrt{K_a^2 + 4K_a C})$   
 Similarly for a weak base, substitute  $[OH^-]$  and  $K_b$  instead of  $[H^+]$  and  $K_a$  respectively in these expressions.

**Case (ii) A weak acid and a strong acid** : Due to strong acid degree of dissociation of weak acid decreases.

**Case (iii) Two (or more) weak acids**

The accurate treatment yields a cubic equation. Assuming that acids dissociate to a negligible extent [ i.e.  $C - x \approx C$  ]  $[H^+] = (K_1 C_1 + K_2 C_2 + \dots + K_w)^{1/2}$

**Case (iv) When dissociation of water becomes significant:**

Dissociation of water contributes significantly to  $[H^+]$  or  $[OH^-]$  only when for

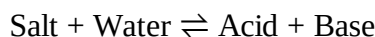
- (i) **Strong acids (or bases)** :  
 $10^{-8}M < C < 10^{-6}M$ . Neglecting ionisation of water at  $10^{-6}M$  causes 1% error (approvable). Below  $10^{-8}M$ , contribution of acid (or base) can be neglected and pH can be taken to be practically 7.
- (ii) **Weak acids (or bases)** : When  $K_a C < 10^{-12}$ , then consider dissociation of water as well.

**Case (v) pH of solution involving a polyprotic acid or base** depend upon  $K_1$ ,  $K_2$ . Successive dissociation can be neglected.

$$K_1 = \frac{x^2}{c - x}$$

## ❑ SALT HYDROLYSIS

The phenomenon of interaction of cations and anions of a salt with  $H_2O$  in order to produce acidic nature or alkaline nature is known as salt hydrolysis.



The process of salt hydrolysis is actually the reverse process of neutralization.

(i) **Salts of strong acids and strong bases** do not undergo hydrolysis.

(ii) **Salts of a strong acids and weak bases** give an acidic solution.

e.g.  $NH_4Cl$  when dissolved, it dissociates to give  $NH_4^+$  ions.



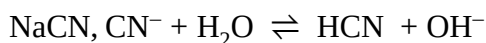
**Important !** In general :  $K_a$  (of an acid).  $K_b$  (of its conjugate base) =  $K_w$

If the degree of hydrolysis ( $h$ ) is small ( $\ll 1$ ),  $h = \sqrt{K_h / C}$ .

$$pH = \frac{1}{2} (pk_w - pk_b - \log C)$$

$$\text{Otherwise } h = \frac{-K_h + \sqrt{K_h^2 + 4K_h C}}{2C}, \quad [H^+] = Ch$$

(iii) **Salts of strong base and weak acid** give a basic solution ( $pH > 7$ ) when dissolved in water, e.g.



$$[OH^-] = Ch, \quad h = \sqrt{K_h / C}$$

$$pH = \frac{1}{2} (pk_w + pk_a + \log C)$$

(iv) **Salts of weak base and weak acid**

Assuming degree of hydrolysis to be same for the both the ions,

$$K_h = K_w / (K_a \cdot K_b), \quad [H^+] = [K_a K_w / K_b]^{1/2}$$

$$pH = \frac{1}{2} (pk_w + pk_a - pk_b)$$

(v) **Amphiprotic salts**

e.g.  $NaHCO_3$

$$pH = \frac{1}{2} (pk_1 + pk_2)$$

**Note:** Exact treatment of case (iv) & (v) is difficult to solve. So use this assumption in general cases.

Also, degree of anion or cation will be much higher in the case of a salt of weak acid and weak base. This is because each of them gets hydrolysed, producing  $H^+$  and  $OH^-$  ions. These ions combine to form water and the hydrolysis equilibrium is shifted in the forward direction.

## ❑ BUFFER SOLUTIONS

A solution whose pH does not change significantly on addition of a small amount of acid or alkali.

### Type of Buffers

#### 1. Simple buffers

(i) A salt of weak acid and weak base in water e.g.  $CH_3COONH_4$ ,  $NH_4CN$

(ii) Proteins and amino acids

## 2. Mixed buffers

These are of two types :

### (i) Acidic buffer mixtures ; A weak acid with its conjugate base :

$\text{NaHCO}_3 + \text{H}_2\text{CO}_3$  ( $\text{H}_2\text{CO}_3$  is weak acid and  $\text{HCO}_3^-$  is its conjugate base ) ;

$\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$  ;  $\text{NaH}_2\text{PO}_4 + \text{H}_3\text{PO}_4$ .

The pH of the buffer solution of this category not necessarily lie in between 0 to 7. It may be in the range of 7 to 14 depending upon the dissociation constant of acid.

#### Henderson's Equation

$\text{pH} = \text{pK}_a + \log \{ [\text{salt}] / [\text{acid}] \}$  for weak acid with its conjugate base.

### (ii) Basic buffer mixtures ; A weak base with its conjugate Acid :

$\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$  (  $\text{NH}_4\text{OH}$  is weak base and  $\text{NH}_4^+$  is its conjugate acid.)

#### Henderson's Equation

$\text{pOH} = \text{pK}_b + \log \{ [\text{salt}] / [\text{base}] \}$  for weak base with its conjugate acid.

**Important :** For good buffer capacity, [salt] : [acid] ratio should be as close to one as possible. In such a case,  $\text{pH} = \text{pK}_a$ . (This also is the case at midpoint of titration)

Buffer capacity = (no. of moles of acid (or base) added to 1L) / (change in pH)

## □ INDICATORS

Indicator is a substance which indicates the point of equivalence in a titration by undergoing a change in its colour. They are weak acids or weak bases.

**Theory of Indicators.** The ionized and unionized forms of indicators have different colours. If 90 % or more of a particular form (ionised or unionised) is present, then its colour can be distinctly seen. In general, for an indicator which is weak acid,  $\text{HIn} \rightleftharpoons \text{H}^+ + \text{In}^-$ , the ratio of ionized to unionized form can be determined from

$$\text{pH} = \text{pK}_a + \log \frac{[\text{In}^-]}{[\text{HIn}]}$$

So, for detectable colour change,  $\text{pH} = \text{pK}_a \pm 1$

This roughly gives the range of indicators. Ranges for some popular indicators are

**Table 1 : Indicators**

| Indicators       | pH range | Colour      |              |
|------------------|----------|-------------|--------------|
|                  |          | acid medium | basic medium |
| Methyl Orange    | 3.1-4.4  | pink        | yellow       |
| Methyl red       | 4.2-6.3  | red         | yellow       |
| Litmus           | 5.5-7.5  | red         | blue         |
| Phenol red       | 6.8-8.4  | yellow      | red          |
| Phenolphthaleine | 8.3-10   | colourless  | pink         |
| Thymol blue      | 1.2-2.8  | red         | yellow       |

**Equivalence point.** The point at which exactly equivalent amounts of acid and base have been mixed.



THE ATLAS

Ionic Equilibrium

The concept of ionic equilibria as equilibria involving ions in solution

- (i) Arrhenius theory of electrolytes.
- (ii) Ostwards dilution law for weak electrolyte  $K_a = \left(\frac{\alpha^2}{1-\alpha}\right)\left(\frac{1}{V}\right)$
- (iii) Acid & Bases
  - (a) Arrhenius  $H^+/OH^-$  theory.
  - (b) Bronsted lowery - protonic concept.
  - (c) Lewis concept - electronic concept of acids and bases.

Some basic concept

- pH scale :  $pH = -\log[H^+]$ .
- (i) Water as amphiprotic solvent.
- (ii) Autoionization of water :  $K_w = [H^+][OH^-]$ .
- (iii)  $K_a[H_2O] = K_w/[H_2O]$
- (iv) Change in pH of neutral  $H_2O$  with temperature.

Homogenons Ionic equilibria

- (i) Acid/ base equilibrium
  - (a) Strong acid  $[H^+] = c/2 + \sqrt{\frac{c^2}{4} + K_w}$ ;  $c = \text{conc. of (acid)}$
  - (b) pH due to polyprotic weak acids
  - (c) Weak monobasic acid  $[H^+] = \sqrt{K_a \cdot c}$  (if  $\alpha < 0.1$ )
  - (d) Mixture of S.A. & W.A.
  - (e) Mixture of two W.A.  $[H^+] = \sqrt{K_1 c_1 + K_2 c_2}$
  - (f) Buffer solutions :  $pH = pK_a + \log\left(\frac{\text{salt}}{\text{acid}}\right)$   
 $pOH = pK_b + \log\left(\frac{\text{salt}}{\text{base}}\right)$
  - (g) Salt hydrolysis – (W.A. & S.B)  $pH = \frac{1}{2}(pK_w + pK_a + \log c)$   
 (W.B & S.A.)  $pH = \frac{1}{2}(pK_w - pK_b - \log c)$   
 (W.A. & W.B.)  $pH = \frac{1}{2}(pK_w + pK_a - pK_b)$

Heterogenons equilibria

- Solubility of sparingly soluble salt's  $(AB, AB_2, A_x B_y) K_{sp} = (S^{x+y}) X^x Y^y$
- (i) Effect of pH on solubility.
- (ii) Simultaneous solubility.

Application of both heterogenous and homogenous equilibrium

- (i) Extent of hydrolysis in buffer solution.
- (ii) Change in solubility due to complex formation.
- (iii) Solubility and hydrolysis.

## REDOX

### OXIDISING AND REDUCING AGENT

#### ◆ Oxidising agent or Oxidant :

Oxidising agents are those compound which can oxidise others and reduced itself during the chemical reaction. Those reagents whose O.N. decrease or which gain electrons in a redox reaction are termed as oxidants

e.g.  $\text{KMnO}_4$ ,  $\text{K}_2\text{Cr}_2\text{O}_7$ ,  $\text{HNO}_3$ , conc.  $\text{H}_2\text{SO}_4$  etc, are powerful oxidising agents.

#### ◆ Reducing agent or Reductant :

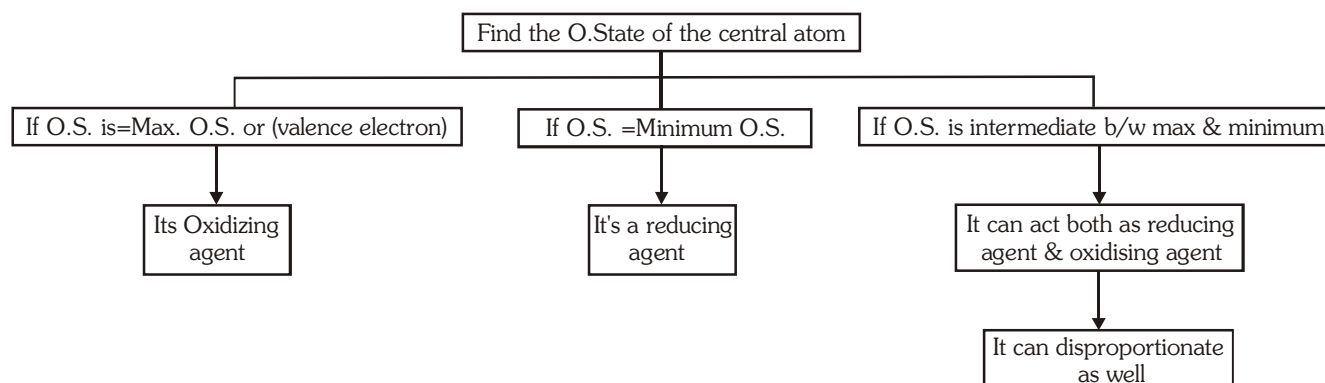
Reducing agents are those compound which can reduce others and oxidise itself during the chemical reaction. Those reagents whose O.N. increase or which loses electrons in a redox reaction are termed as reductants.

e.g.  $\text{KI}$ ,  $\text{Na}_2\text{S}_2\text{O}_3$  are powerful reducing agents.

**Note :** There are some compounds also which can work both oxidising agent and reducing agent.

e.g.  $\text{H}_2\text{O}_2$ ,  $\text{NO}_2^-$

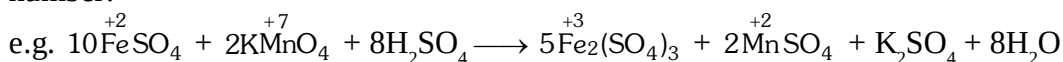
### HOW TO IDENTIFY WHETHER A PARTICULAR SUBSTANCE IS AN OXIDISING OR REDUCING AGENT



### REDOX REACTION

A reaction in which oxidation and reduction simultaneously take place.

In all redox reactions the total increase in oxidation number must equal the total decrease in oxidation number.



#### Equivalent weight (E) :

$$\text{Eq. wt (E)} = \frac{\text{Molecular weight}}{\text{valency factor (v.f)}} = \frac{\text{Mol. wt.}}{n - \text{factor}}$$

$$\text{no of Equivalents} = \frac{\text{mass of a sample}}{\text{eq. wt. of that species}}$$

- ◆ Equivalent mass is a pure number when expressed in gram, it is called gram equivalent mass.
- ◆ The equivalent mass of substance may have different values under different conditions.

(c) Eq. wt. of oxidising / reducing agents in redox reaction :

The equivalent weight of an oxidising agent is that weight which accepts one mole electron in a chemical reaction.

- (a) Equivalent wt. of an oxidant (get reduced)

$$= \frac{\text{Mol.wt.}}{\text{No. of electrons gained by one mole}}$$

## □ n-FACTOR IN VARIOUS CASES

### In Non Redox Change

- ◆ **n-factor for element** : Valency of the element
- ◆ **For acids** : Acids will be treated as species which furnish  $H^+$  ions when dissolved in a solvent.

The n-factor of an acid is the no. of acidic  $H^+$  ions that a molecule of the acid would give when dissolved in a solvent (Basicity).

For example, for HCl ( $n = 1$ ), HNO<sub>3</sub> ( $n = 1$ ), H<sub>2</sub>SO<sub>4</sub> ( $n = 2$ ), H<sub>3</sub>PO<sub>4</sub> ( $n = 3$ ) and H<sub>3</sub>PO<sub>3</sub> ( $n = 2$ )

- ◆ **For bases :** Bases will be treated as species which furnish  $\text{OH}^-$  ions when dissolved in a solvent . The n-factor of a base is the no. of  $\text{OH}^-$  ions that a molecule of the base would give when dissolved in a solvent (**Acidity**).

For example, NaOH ( $n = 1$ ), Ba(OH)<sub>2</sub> ( $n = 2$ ), Al(OH)<sub>3</sub> ( $n = 3$ ), etc.

- ◆ **For salts :** A salt reacting such that no atom of the salt undergoes any change in oxidation state.

For example,  $2\text{AgNO}_3 + \text{MgCl}_2 \rightarrow \text{Mg}(\text{NO}_3)_2 + 2\text{AgCl}$

In this reaction, it can be seen that the oxidation state of Ag, N, O, Mg and Cl remains the same even in the product. The n-factor for such a salt is the total **charge on cation or anion**.

## In Redox Change

For oxidizing agent or reducing agent n-factor is the **change in oxidation number per mole of the substance**.

**☐ SOME OXIDIZING AGENTS/REDUCING AGENTS WITH EQ. WT.**

| Species                             | Changed to                             | Reaction                                                                                               | Electrons<br>exchanged<br>or change in<br>O.N. | Eq. wt.           |
|-------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------|-------------------|
| $\text{MnO}_4^-$ (O.A.)             | $\text{Mn}^{+2}$<br>in acidic medium   | $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \longrightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$      | 5                                              | $E = \frac{M}{5}$ |
| $\text{MnO}_4^-$ (O.A.)             | $\text{MnO}_2$<br>in neutral medium    | $\text{MnO}_4^- + 3\text{e}^- + 2\text{H}_2\text{O} \longrightarrow \text{MnO}_2 + 4\text{OH}^-$       | 3                                              | $E = \frac{M}{3}$ |
| $\text{MnO}_4^-$ (O.A.)             | $\text{MnO}_4^{2-}$<br>in basic medium | $\text{MnO}_4^- + \text{e}^- \longrightarrow \text{MnO}_4^{2-}$                                        | 1                                              | $E = \frac{M}{1}$ |
| $\text{Cr}_2\text{O}_7^{2-}$ (O.A.) | $\text{Cr}^{3+}$<br>in acidic medium   | $\text{CrO}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$ | 6                                              | $E = \frac{M}{6}$ |
| $\text{MnO}_2$ (O.A.)               | $\text{Mn}^{2+}$<br>in acidic medium   | $\text{MnO}_2 + 4\text{H}^+ + 2\text{e}^- \longrightarrow \text{Mn}^{2+} + 2\text{H}_2\text{O}$        | 2                                              | $E = \frac{M}{2}$ |

|                                                   |                             |                                                                                                                                           |   |                        |
|---------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------|
| $\text{Cl}_2$ (O.A.)<br>in bleaching powder       | $\text{Cl}^-$               | $\text{Cl}_2 + 2\text{e}^- \longrightarrow 2\text{Cl}^-$                                                                                  | 2 | $E = \frac{M}{2}$      |
| $\text{CuSO}_4$ (O.A.)<br>in iodometric titration | $\text{Cu}^+$               | $\text{Cu}^{2+} + \text{e}^- \longrightarrow \text{Cu}^+$                                                                                 | 1 | $E = \frac{M}{1}$      |
| $\text{S}_2\text{O}_3^{2-}$ (R.A.)                | $\text{S}_4\text{O}_6^{2-}$ | $2\text{S}_2\text{O}_3^{2-} \longrightarrow \text{S}_4\text{O}_6^{2-} + 2\text{e}^-$                                                      | 2 | $E = \frac{2M}{2} = M$ |
| (for two molecules)                               |                             |                                                                                                                                           |   |                        |
| $\text{H}_2\text{O}_2$ (O.A.)                     | $\text{H}_2\text{O}$        | $\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \longrightarrow 2\text{H}_2\text{O}$                                                    | 2 | $E = \frac{M}{2}$      |
| $\text{H}_2\text{O}_2$ (R.A.)                     | $\text{O}_2$                | $\text{H}_2\text{O}_2 \longrightarrow \text{O}_2 + 2\text{H}^+ + 2\text{e}^-$ (O.N. of oxygen in $\text{H}_2\text{O}_2$ is $-1$ per atom) | 2 | $E = \frac{M}{2}$      |
| $\text{Fe}^{2+}$ (R.A.)                           | $\text{Fe}^{3+}$            | $\text{Fe}^{2+} \longrightarrow \text{Fe}^{3+} + \text{e}^-$                                                                              | 1 | $E = \frac{M}{1}$      |

## □ NORMALITY

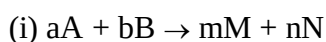
Normality of solution is defined as the number of equivalent of solute present in one litre (1000 mL) solutions. Let a solution is prepared by dissolving W g of solute of eq. wt. E in V mL water.

- ◆ No. of equivalent of solute =  $\frac{W}{E}$
- ◆ V mL of solution have  $\frac{W}{E}$  equivalent of solute
- ◆ 1000 mL solution have  $\frac{W \times 1000}{E \times V_{\text{mL}}}$
- ◆ **Normality (N) =  $\frac{W \times 1000}{E \times V_{\text{mL}}}$**
- ◆ **Normality (N) = Molarity  $\times$  Valence factor**  
 Normality (N) = molarity  $\times$  Valence factor (n)  
**or**  $N \times V$  (in mL) =  $M \times V$  (in mL)  $\times$  n  
**or** milli equivalents = millimoles  $\times$  n

## □ LAW OF EQUIVALENCE

The law states that one equivalent of an element combine with one equivalent of the other, and in a chemical reaction equivalent and mill equivalent of reactants react in equal amount to give same no. of equivalent or milli equivalents of products separately.

**According :**



$$m.\text{eq of A} = m.\text{eq of B} = m.\text{eq of M} = m.\text{eq of N}$$

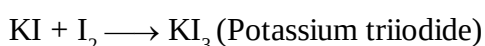


$$m.\text{eq of M}_x\text{N}_y = m.\text{eq of M} = m.\text{eq of N}$$

**Solved Examples :**

## Iodimetric Titration

These are the titrations in which free iodine is used as it is difficult to prepare the solution of iodine (volatile and less soluble in water), it is dissolved in KI solution :





This solution is first standardized before using with the standard solution of substance such as sulphite, thiosulphate, arsenite etc, are estimated.

In iodimetric and iodometric titration, starch solution is used as an indicator. Starch solution gives blue or violet colour with free iodine. At the end point, the blue or violet colour disappears when iodine is completely changed to iodide.

□ **SOME IODOMETRIC TITRATIONS (TITRATING SOLUTIONS IS  $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$ )**

| Estimation of    | Reaction                                                                                                                                                                                                                                                                                                                                                                                                  | Relation between O.A. and R.A.                                                                                                                                        |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\text{I}_2$     | $\text{I}_2 + 2\text{Na}_2\text{S}_2\text{O}_3 \longrightarrow 2\text{NaI} + \text{Na}_2\text{S}_4\text{O}_6$ <p style="text-align: center;">or</p> $\text{I}_2 + 2\text{S}_2\text{O}_3^{2-} \longrightarrow 2\text{I}^- + \text{S}_4\text{O}_6^{2-}$                                                                                                                                                     | $\text{I}_2 = 2\text{I} \equiv 2\text{Na}_2\text{S}_2\text{O}_3$ <p>Eq. wt. of <math>\text{Na}_2\text{S}_2\text{O}_3 = \text{M}/1</math></p>                          |
| $\text{CuSO}_4$  | $2\text{CuSO}_4 + 4\text{KI} \longrightarrow 2\text{Cu}_2\text{I}_2 + 2\text{K}_2\text{SO}_4 + \text{I}_2$ $\text{Cu}^{2+} + 4\text{I}^- \longrightarrow \text{Cu}_2\text{I}_2 + \text{I}_2$ <p style="text-align: center;">(White ppt.)</p>                                                                                                                                                              | $2\text{CuSO}_4 \equiv \text{I}_2 \equiv 2\text{I} \equiv 2\text{Na}_2\text{S}_2\text{O}_3$ <p>Eq. wt. of <math>\text{CuSO}_4 = \text{M}/1</math></p>                 |
| $\text{CaOCl}_2$ | $\text{CaOCl}_2 + \text{H}_2\text{O} \longrightarrow \text{Ca(OH)}_2 + \text{Cl}_2$ $\text{Cl}_2 + 2\text{KI} \longrightarrow 2\text{KCl} + \text{I}_2$ $\text{CaOCl}_2 \equiv \text{Cl}_2 \equiv \text{I}_2 \equiv 2\text{I} \equiv 2\text{Na}_2\text{S}_2\text{O}_3$ $\text{Cl}_2 + 2\text{I}^- \longrightarrow 2\text{Cl}^- + \text{I}_2$                                                              | <p>Eq. wt. of <math>\text{CaOCl}_2 = \text{M}/2</math></p>                                                                                                            |
| $\text{MnO}_2$   | $\text{MnO}_2 + 4\text{HCl (conc)} \xrightarrow{\Delta} \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$ $\text{Cl}_2 + 2\text{KI} \longrightarrow 2\text{KCl} + \text{I}_2$ <p style="text-align: center;">or</p> $\text{MnO}_2 + 4\text{H}^+ + 2\text{Cl}^- \longrightarrow \text{Mn}^{2+} + 2\text{H}_2\text{O} + \text{Cl}_2$ $\text{Cl}_2 + 2\text{I}^- \longrightarrow \text{I}_2 + 2\text{Cl}^-$ | $\text{MnO}_2 \equiv \text{Cl}_2 \equiv \text{I}_2 \equiv 2\text{I} \equiv 2\text{Na}_2\text{S}_2\text{O}_3$ <p>Eq. wt. of <math>\text{MnO}_2 = \text{M}/2</math></p> |
| $\text{IO}_3^-$  | $\text{IO}_3^- + 5\text{I}^- + 6\text{H}^+ \longrightarrow 3\text{I}_2 + 3\text{H}_2\text{O}$                                                                                                                                                                                                                                                                                                             | $\text{IO}_3^- \equiv 3\text{I}_2 \equiv 6\text{I} \equiv 6\text{Na}_2\text{S}_2\text{O}_3$ <p>Eq. wt. of <math>\text{IO}_3^- = \text{M}/6</math></p>                 |

**ELECTROCHEMISTRY**
**ELECTROCHEMICAL CELLS**

An electrochemical cell consists of two electrodes (metallic conductors) in contact with an electrolyte (an ionic conductor).

An electrode and its electrolyte comprise an **Electrode Compartment**.

Electrochemical Cells can be classified as:

- (i) **Electrolytic Cells** in which a non-spontaneous reaction is driven by an external source of current.
- (ii) **Galvanic Cells** which produce electricity as a result of a spontaneous cell reaction

Note: In a **galvanic cell**, cathode is positive with respect to anode.

In a **electrolytic cell**, anode is made positive with respect to cathode.

**REPRESENTATION OF A CELL ( IUPAC CONVENTIONS ):**

Let us illustrate the convention taking the example of Daniel cell.

- (i) Anodic half cell is written on left and cathodic half cell on right hand side.  

$$\text{Zn(s)} \mid \text{ZnSO}_4(\text{sol}) \parallel \text{CuSO}_4(\text{sol}) \mid \text{Cu(s)}$$
- (ii) Two half cells are separated by double vertical lines: Double vertical lines indicate salt bridge or any type of porous partition.
- (iii) EMF (electromotive force) may be written on the right hand side of the cell.
- (iv) Single vertical lines indicate the phase separation between electrode and electrolyte solution.  

$$\text{Zn} \mid \text{Zn}^{2+} \parallel \text{Cu}^{2+} \mid \text{Cu}$$
- (v) Inert electrodes are represented in the bracket  

$$\text{Zn} \mid \text{ZnSO}_4 \parallel \text{H}^+ \mid \text{H}_2, \text{Pt}$$

**RELATIONSHIP BETWEEN  $\Delta G$  AND ELECTRODE POTENTIAL**

Let  $n$ , Faraday charge is taken out from a cell of e.m.f. ( $E$ ) then electrical work done by the cell may be calculated as,

$$\text{Work done} = \text{Charge} \times \text{Potential} = nFE$$

From thermodynamics we know that decrease in Gibbs free energy of a system is a measure of reversible or maximum obtainable work by the system if there is no work due to volume expansion

$$\therefore \Delta G = -nFE$$

$$\text{Under standard state} \quad \Delta G^0 = -nFE^0 \quad (i)$$

- (i) From thermodynamics we know,  $\Delta G$  = negative for spontaneous process. Thus from e(i) it is clear that the EMF should be +ve for a cell process to be feasible or spontaneous.

(ii) When  $\Delta G = \text{positive}$ ,  $E = \text{negative}$  and the cell process will be non spontaneous.

| Reactions        | $\Delta G$ | E   |
|------------------|------------|-----|
| Spontaneous      | (-)        | (+) |
| Non- spontaneous | (+)        | (-) |
| Equilibrium      | 0          | 0   |

Standard free energy change of a cell may be calculated by electrode potential data.

Substituting the value of  $E^0$  (i.e., standard reduction potential of cathode- standard reduction potential of anode) in e (i) we may get  $\Delta G^0$ .

## CONCEPT OF ELECTROMOTIVE FORCE (EMF) OF A CELL

Electron flows from anode to cathode in external circuit due to a pushing effect called or electromotive force (e.m.f.). EMF is called as *cell potential*. Unit of e.m.f. of cell is volt.

EMF of cell may be calculated as :

$$E_{\text{cell}} = \text{reduction potential of cathode} - \text{Reduction potential of anode}$$

Similarly, standard e.m.f. of the cell ( $E^\circ$ ) may be calculated as

$$E^\circ_{\text{cell}} = \text{Standard reduction potential of cathode} - \text{Standard}$$

## NERNST EQUATION

Walter Nernst derived a relation between cell potential and concentration or Reaction quotient.

$$\Delta G = \Delta G^\circ + RT \ln Q \quad \text{..(A)}$$

where  $\Delta G$  and  $\Delta G^\circ$  are free energy and standard free energy change; 'Q' is reaction quotient.

$$\therefore -\Delta G = nFE \quad \text{and} \quad -\Delta G^\circ = nFE^\circ$$

Thus from E (i), we get  $-nFE = -nFE^\circ + RT \ln Q$

At 25°C, above equation may be written as  $E = E^0 - \frac{0.0591}{n} \log Q$

Where 'n' represents number of moles of electrons involved in process.

$E, E^\circ$  are e.m.f. and standard e.m.f. of the cell respectively.

In general , for a redox cell reaction involving the transference of  $n$  electrons

$aA + bB \longrightarrow cC + dD$ , the EMF can be calculated as:

$$E_{\text{Cell}} = E^{\circ}_{\text{Cell}} - \frac{0.0591}{n} \log \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

**(ii) Determination of equilibrium constant :** We know, that

$$E = E^0 - \frac{0.0591}{n} \log Q \quad \text{..(i)}$$

At equilibrium, the cell potential is zero because cell reactions are balanced, i.e.  $E = 0$

∴ From E (i), we have

$$0 = E^0 - \frac{0.0591}{n} \log K_{eq} \quad \text{or} \quad K_{eq} = \text{anti log} \left[ \frac{nE^0}{0.0591} \right]$$

- (iii) **Heat of Reaction inside the cell:** Let  $n$  Faraday charge flows out of a cell of e.m.f.  $E$ , then

$$-\Delta G = nFE \quad (i)$$

Gibbs Helmholtz equation (from thermodynamics) may be given as,

$$\Delta G = \Delta H + T \left[ \frac{\partial \Delta G}{\partial T} \right]_p \quad (ii)$$

From Eqs. (i) and (ii), we have

$$-nFE = \Delta H + T \left[ \frac{\partial (-nFE)}{\partial T} \right]_p = \Delta H - nFT \left[ \frac{\partial E}{\partial T} \right]_p$$

$$\therefore \Delta H = -nFE + nFT \left[ \frac{\partial E}{\partial T} \right]_p$$

- (iv) **Entropy change inside the cell :** We know that  $G = H - TS$  or  $\Delta G = \Delta H - T\Delta S$  ... (i)

where  $\Delta G$  = Free energy change ;  $\Delta H$  = Enthalpy change and  $\Delta S$  = entropy change.

According to Gibbs Helmholtz equation,

$$\Delta G = \Delta H + T \left[ \frac{\partial \Delta G}{\partial T} \right]_p \quad \dots (ii)$$

From Eqs. (i) and (ii), we have

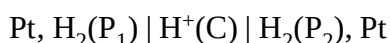
$$-T\Delta S = T \left[ \frac{\partial \Delta G}{\partial T} \right]_p \quad \text{or} \quad \Delta S = - \left[ \frac{\partial \Delta G}{\partial T} \right]_p$$

$$\text{or} \quad \Delta S = nF \left[ \frac{\partial E}{\partial T} \right]_p$$

## CONCENTRATION CELL

The cells in which electrical current is produced due to transport of a substance from higher to lower concentration. Concentration gradient may arise either in electrode material or in electrolyte. Thus there are two types of concentration cell.

- (i) **Electrode Gas concentration cell :**



Here, hydrogen gas is bubbled at two different partial pressures at electrode dipped in the solution of same electrolyte.

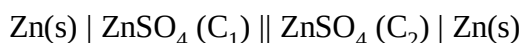
Cell process :  $1/2\text{H}_2(p_1) \rightarrow \text{H}^+(\text{c}) + \text{e}^-$  (Anode process)

$$\frac{H^+(c) + e^- \rightarrow 1/2 H_2(p_2)}{1/2 H_2(p_1) \rightleftharpoons 1/2 H_2(p_2)} \quad \therefore \quad E = -\frac{2.303RT}{F} \log \left[ \frac{p_2}{p_1} \right]^{1/2}$$

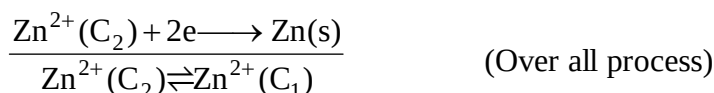
$$\text{or} \quad E = \left[ \frac{2.303RT}{2F} \right] \log \left[ \frac{p_2}{p_1} \right], \text{ At } 25^\circ\text{C}, E = \frac{0.059}{2F} \log \left[ \frac{p_1}{p_2} \right]$$

For spontaneity of such cell reaction,  $p_1 > p_2$

## (ii) Electrolyte concentration cells:



In such cells, concentration gradient arise in electrolyte solutions. Cell process may be given as,



$\therefore$  From Nernst equation, we have

$$E = 0 - \frac{2.303RT}{2F} \log \left[ \frac{C_1}{C_2} \right] \quad \text{or} \quad E = \frac{2.303RT}{2F} \log \left[ \frac{C_2}{C_1} \right]$$

For spontaneity of such cell reaction,  $C_2 > C_1$

## ELECTROLYSIS

The decomposition of electrolyte solution by passage of electric current, resulting into deposition of metals or liberation of gases at electrodes is known as electrolysis.

## ELECTROLYTIC CELL

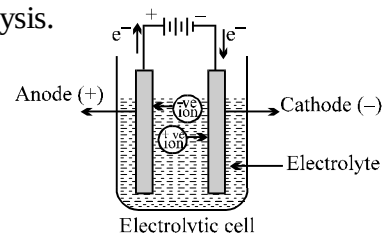
This cell converts electrical energy into chemical energy.

The entire assembly except that of the external battery is known as the electrolytic cell

## ELECTRODES

The metal strip at which positive current enters is called **anode**; anode is positively charged in electrolytic cell. On the other hand, the electrode at which current leaves is called **cathode**. Cathodes are negatively charged.

|         |          |                                                 |                               |
|---------|----------|-------------------------------------------------|-------------------------------|
| Anode   | Positive | Loss of electron<br>or oxidation<br>takes place | Positive<br>current<br>enters |
| Cathode | Negative | Gain of electron<br>or reduction<br>takes place | Current<br>leaves             |





**SPECIFIC CONDUCTANCE IS CONDUCTANCE OF 1 CM<sup>3</sup> OF AN ELECTROLYTE SOLUTION.**

In case of electrolytic solution, the specific conductance is defined as the conductance of a solution of definite concentration enclosed in a cell having two electrodes of unit area separated by 1 cm apart.

## 1. Equivalent Conductance

Equivalent conductance is the conductance of an electrolyte solution containing 1 gm equivalent of electrolyte. It is denoted by  $\wedge$ .

$$\wedge = K \times V$$

$$(\wedge = \text{ohm}^{-1} \text{ cm}^{-1} \times \text{cm}^3 = \text{ohm}^{-1} \text{ cm}^2)$$

Usually concern ration of electrolyte solution is expressed as C gm equivalent per litre.

Thus,  $V = \frac{1000}{C}$

{Volume having 1 gm equivalent electrolyte in the solution} Thus,  $\wedge = K \times \frac{1000}{C}$

## 2. Molar Conductance

Molar conductance may be defined as conductance of an electrolyte solution having 1 gm mole electrolyte in a litre. It is denoted by  $\wedge_m$ .

$$\wedge_m = K \quad V$$

Usually concentration of electrolyte solution is expressed as 'M' gm mole electrclyte per litre.

Thus,  $V = \frac{1000}{M}$

Hence,  $\wedge_m = K_x \frac{1000}{M}$

**Relation between  $\wedge$  and  $\wedge_m$  :**  $\wedge_m = n \times \wedge$

### Application of Kohlrausch's law :

(A) Determination of  $\wedge_m^0$  of a weak electrolyte:

In order to calculate  $\wedge_m^0$  of a weak electrolyte say  $\text{CH}_3\text{COOH}$ , we determine experimentally

 $\Lambda_m^0$  values of the following three strong electrolytes:

- A strong electrolyte containing same cation as in the test electrolyte, say HCl
- A strong electrolyte containing same anion as in the test electrolyte, say  $\text{CH}_3\text{COONa}$
- A strong electrolyte containing same anion of (a) and cation of (b) i.e. NaCl.

 $\Delta_m^0$  of  $\text{CH}_3\text{COOH}$  is then given as:

$$\Delta_m^0(\text{CH}_3\text{COOH}) = \Delta_m^0(\text{HCl}) + \Delta_m^0(\text{CH}_3\text{COONa}) - \Delta_m^0(\text{NaCl})$$

**Proof :**

$$\Lambda_m^0(\text{HCl}) = \lambda_{\text{H}^+}^0 + \lambda_{\text{Cl}^-}^0 \quad \dots\dots\dots\text{(i)}$$

$$\Lambda_m^0(\text{CH}_3\text{COONa}) = \lambda_{\text{CH}_3\text{COO}^-}^0 + \lambda_{\text{Na}^+}^0 \quad \dots\dots\dots\text{(ii)}$$

$$\Lambda_m^0(\text{NaCl}) = \lambda_{\text{Na}^+}^0 + \lambda_{\text{Cl}^-}^0 \quad \dots\dots\dots\text{(iii)}$$

Adding equation (i) and equation (ii) and subtracting (iii) from them:

$$\Lambda_m^0(\text{HCl}) + \Lambda_m^0(\text{CH}_3\text{COONa}) - \Lambda_m^0(\text{NaCl}) = \lambda_{(\text{H}^+)}^0 + \lambda_{(\text{CH}_3\text{COO}^-)}^0 = \Lambda_m^0(\text{CH}_3\text{COOH})$$

(B) Determination of degree of dissociation ( $\alpha$ ) :

$$\alpha = \frac{\text{No. of molecules ionised}}{\text{total number of molecules dissolved}} = \frac{\Lambda_m}{\Lambda_m^0}$$

(C) Determination of solubility of sparingly soluble salt

The specific conductivity of a saturated solution of the test electrolyte (sparingly soluble) made in conductivity water is determined by the method as described above. From this the specific conductivity of conductivity water is deducted. The molar conductance of the saturated solution is taken to be equal to  $\Lambda_m^0$  as the saturated solution of a sparingly soluble salt is extremely dilute. Hence from equation (D).

$$\Lambda_m^0 = \frac{1000\kappa}{C},$$

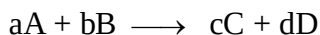
where C is the molarity of solution and hence the solubility.



# CHEMICAL KINETICS

➤ **Rate of reaction (ROR)** =  $\frac{\text{Rate of disappearance of reactant (appearance of products)}}{\text{Stoichiometric coefficient of reactant (products)}}$

➤ **For a reaction :**

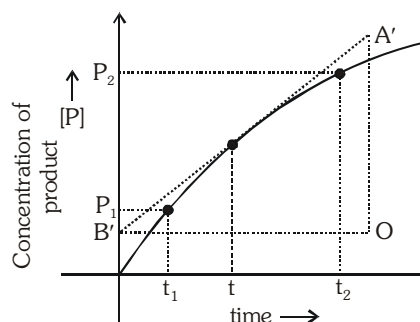
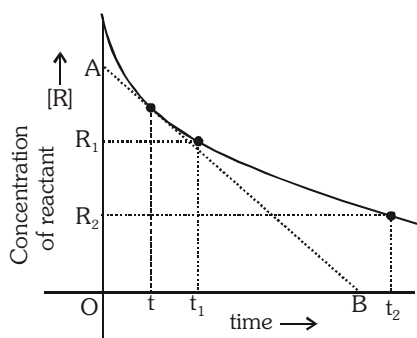


➤ **Instantaneous rate :**  $-\frac{1}{a}\left(\frac{d[A]}{dt}\right) = -\frac{1}{b}\left(\frac{d[B]}{dt}\right) = \frac{1}{c}\left(\frac{d[C]}{dt}\right) = \frac{1}{d}\left(\frac{d[D]}{dt}\right)$

Relationship between rate of reaction and rate of disappearance of reactant (rate of appearance of product).

◆ **Average rate :**  $-\frac{1}{a}\left(\frac{\Delta[A]}{\Delta t}\right) = -\frac{1}{b}\left(\frac{\Delta[B]}{\Delta t}\right) = \frac{1}{c}\left(\frac{\Delta[C]}{\Delta t}\right) = \frac{1}{d}\left(\frac{\Delta[D]}{d\Delta t}\right)$

⇒ Graphical method for determining rate :



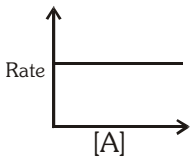
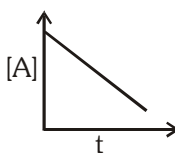
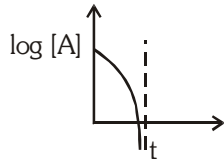
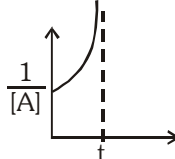
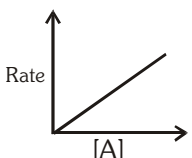
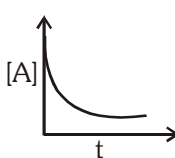
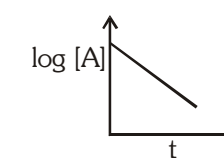
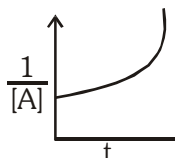
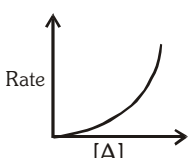
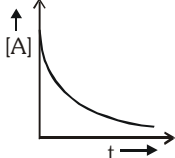
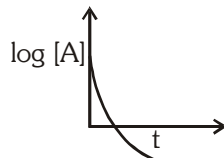
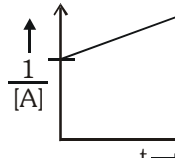
$$\text{Avg. Rate} = -\left(\frac{[\text{R}]_2 - [\text{R}]_1}{t_2 - t_1}\right) = \frac{([\text{P}]_2 - [\text{P}]_1)}{t_2 - t_1}$$

$$\text{Instantaneous rate} = -\left(\frac{\text{OA}}{\text{OB}}\right) = +\frac{\text{OA}'}{\text{OB}'} = \pm \text{slope of tangent}$$

➤ **Important kinetic expression for reaction of type  $A \longrightarrow B$  :**

| Order                         | Zero                                   | 1st                                   | 2nd                                    | nth                                                                                  |
|-------------------------------|----------------------------------------|---------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------|
| Differential rate law         | Rate = k                               | Rate= k[A]                            | Rate = k[A] <sup>2</sup>               | Rate = k[A] <sup>n</sup>                                                             |
| Integrated rate law           | [A <sub>0</sub> ]-[A]= kt              | kt = ln $\frac{[A]_0}{[A]}$           | kt = $\frac{1}{[A]} - \frac{1}{[A]_0}$ | kt = $\frac{1}{(n-1)} \left[ \frac{1}{[A]^{n-1}} - \frac{1}{[A]_0^{n-1}} \right]$    |
| Half life (t <sub>1/2</sub> ) | t <sub>1/2</sub> = $\frac{[A]_0}{2k}$  | t <sub>1/2</sub> = $\frac{\ln 2}{k}$  | t <sub>1/2</sub> = $\frac{1}{[A]_0 k}$ | t <sub>1/2</sub> = $\frac{1}{k(n-1)} \left[ \frac{2^{n-1} - 1}{[A]_0^{n-1}} \right]$ |
| (t <sub>3/4</sub> )           | t <sub>3/4</sub> =1.5 t <sub>1/2</sub> | t <sub>3/4</sub> = 2 t <sub>1/2</sub> | t <sub>3/4</sub> = 3 t <sub>1/2</sub>  | t <sub>3/4</sub> = (2 <sup>n-1</sup> + 1) t <sub>1/2</sub>                           |

## ➤ Graphs of various order

| Order        | Rate vs [A]                                                                       | [A] vs t                                                                          | log [A] vs t                                                                       | $\frac{1}{[A]}$ vs t                                                                |
|--------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Zero order   |  |  |  |  |
| First order  |  |  |  |  |
| Second order |  |  |  |  |

## Where

$[A]_0 \Rightarrow$  initial concentration

$[A] \Rightarrow$  concentration at time  $t$

$t_{1/2} \Rightarrow$  time taken for initial concentration of reactant to finish by 50%

$t_{3/4} \Rightarrow$  time taken for initial concentration of reactant to finish by 75%

## ➤ Monitoring Kinetics Experimentally :

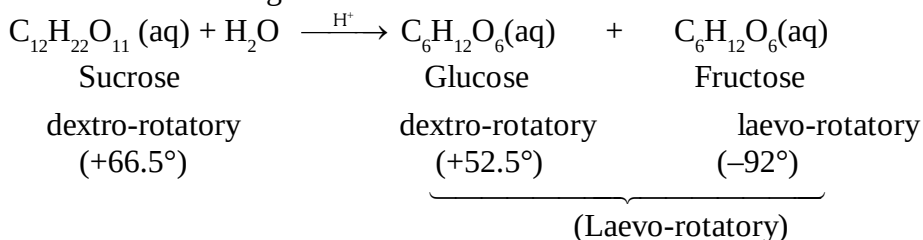
The kinetics of reaction can be followed (i.e. order, rate constant etc. can be established) by measuring a property which changes with time.

e.g. (i) Total pressure in a gaseous reaction.

$$k = \frac{1}{t} \ln \left( \frac{P_0}{P_0 - x} \right)$$

## ➤ Examples : (For Monitoring Kinetics Experimentally)

(i) Inversion of cane sugar :



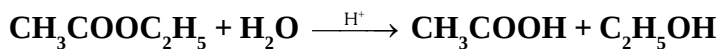
$$k = \frac{2.303}{t} \log \left( \frac{r_\infty - r_0}{r_\infty - r_t} \right)$$

$r_0$  = rotation at time,  $t = 0$

$r_t$  = rotation at time,  $t = t$

$r_\infty$  = rotation at time,  $t = \infty$

(ii) **Acidic hydrolysis of ethyl acetate :**



$$k = \frac{2.303}{t} \log \left( \frac{V_\infty - V_0}{V_\infty - V_t} \right)$$

$V_0$  = Volume of NaOH solution used at time,  $t = 0$

$V_t$  = Volume of NaOH solution used at time,  $t = t$

$V_\infty$  = Volume of NaOH solution used at time,  $t = \infty$

**Note :** Here NaOH acts as a reagent. Acetic acid is one of the product the amount of which can be found by titration against standard NaOH solution. But being an acid-catalysed reaction, the acid present originally as catalyst, also reacts with NaOH solution.

➤ **Important characteristics of first order reaction :**

- ◆  $t_{1/2}$  is independent of initial concentration.
- ◆ In equal time interval, reactions finishes by equal fraction.

|                |         |         |           |                      |
|----------------|---------|---------|-----------|----------------------|
|                | $t = 0$ | $t = t$ | $t = 2t$  | $t = 3t \dots$       |
| Reactant conc. | $a_0$   | $a_0 x$ | $a_0 x^2$ | $a_0 x^3 \dots\dots$ |

$x$  = fraction by which reaction complete in time 't'.

- ◆ Graph of  $\ln[A]$  vs  $t$  is straight line with slope =  $\frac{k}{2.303}$
- ◆ Graph of  $[A]$  vs  $t$  is exponentially decreasing.

➤ **Zero order :**

- $t_{1/2}$  of zero order is directly proportional to initial concentration.
- In equal time interval, reaction finishes by equal amount.

|         |           |            |                  |
|---------|-----------|------------|------------------|
| $t = 0$ | $t = t$   | $t = 2t$   | $t = 3t \dots$   |
| $C_0$   | $C_0 - x$ | $C_0 - 2x$ | $C_0 - 3x \dots$ |

- Graph of  $[A]$  vs  $t$  is straight line.

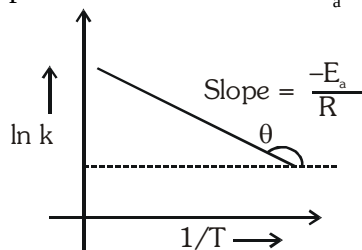
A zero order reaction finishes in  $t = \frac{[A]_0}{k}$

➤ **Temperature dependence :**

- Arrhenious equation :  $k = A.e^{-E_a/RT}$
- $E_a$  = minimum energy over and above the avg. energy of reactant which must be possessed by reacting molecule for collision to be succesful.
- $A$  = frequency factor - proportional to number of collisions per unit volume per second.
- $e^{-E_a/RT}$  = Fraction of collision in which energy is greater than  $E_a$ .
- $A$  and  $E_a$  are constant i.e. do not vary with temperature

$$\ln k = \ln A - \frac{E_a}{RT}$$

**Graph :** Graphical determination of  $E_a$ .

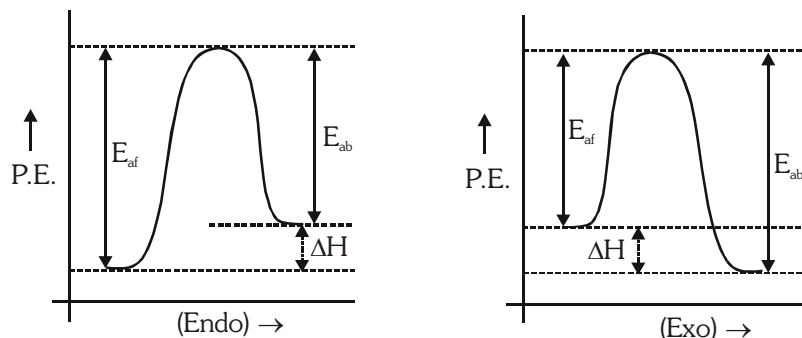


$$\text{Temperature coefficient} = \frac{k_{T+10}}{k_T}$$

By default  $T = 298 \text{ K}$

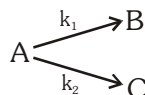
$$\text{Variation of rate constant with temperature} \Rightarrow \ln \frac{k_2}{k_1} = \frac{E_a}{R} \left[ \frac{1}{T_1} - \frac{1}{T_2} \right]$$

➤ **Endothermic and exothermic reactions :**



$$\Delta H = E_{af} - E_{ab}$$

➤ **Parallel reaction :**



(i) Rate =  $(k_1 + k_2) [A]$  - (differential rate law)

$$(ii) \frac{k_1}{k_2} = \frac{[B]}{[C]}$$

$$(iii) t_{1/2} = \frac{0.693}{k_1 + k_2}$$

$$(iv) \% \text{ of } B = \frac{k_1}{k_1 + k_2} \times 100 ; \% \text{ of } C = \frac{k_2}{k_1 + k_2} \times 100$$

$$(v) [A] = [A]_0 e^{-(k_1 + k_2)t}$$

➤ **Pseudo-order reaction :**

$$\text{Rate law} \rightarrow \text{rate} = k [A]^m [B]^n$$

Pseudo rate law :

$$\text{rate} = k_1 [A]^m$$

[B] assumed constant in two cases :

(i) B in large excess (ii) B → CATALYST

## RADIOACTIVITY

- **All nuclear reactions are first order :**

Two types of nuclear reaction :     **(a)** Artificial radioactivity **(b)** Radioactivity (spont.)

First order

$$\lambda t = 2.303 \log \frac{N_0}{N_t}$$

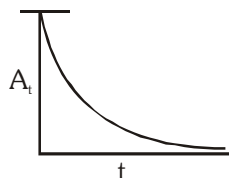
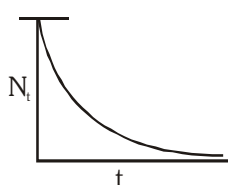
$\lambda \rightarrow$  Decay constant

$N_0 \rightarrow$  Initial nuclei

$N_t \rightarrow$  Nuclei at 't'

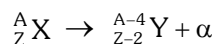
$$\text{Activity} = A_t = \frac{-dN_t}{dt} = \lambda N_t; \text{Nuclei/sec.}$$

$A_t =$  Rate of decay



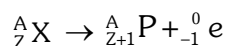
$$\Rightarrow t_{1/2} = \frac{\ln 2}{\lambda}$$

- **$\alpha$  decay** =  ${}^4_2\text{He}$  Particles at high velocity



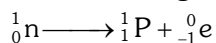
To  $\downarrow$  size of large nuclei

- **$\beta$  decay** =  ${}^0_{-1}e$  at high velocity



# To  $\downarrow$   $\frac{n}{p}$  ratio.

# Nuclear change in  $\beta$  decay



- **$\gamma$ -decay :**

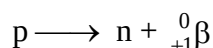
Photons from excited nuclei after  $\alpha$  – or  $\beta$  – decay

No effect on  $n/p$  ratio

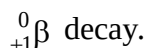
High energy e/m radiation.

$$\text{Mean life, } t_{\text{avg}} = \frac{1}{\lambda}$$

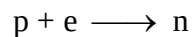
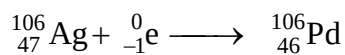
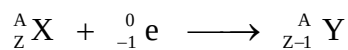
- ◆  **$({}^0_{+1}\beta)$  Positron decay**



\* Those nucleus which have low value of  $n/p$  ratio (lie below the stability belt) undergoes



## ◆ K electron capture



- \* Electron capture can occur when ever the mass of original neutral atom is larger than that of final atom.
- \* Those nucleus having low n/p ratio can capture K shell electron.
- \* X-rays are emitted during the process.

## ➤ Parallel decay :

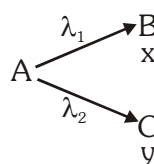
$$t = 0 \quad N_0$$

$$t = t \quad N_0 - x - y$$

$$\lambda_{\text{eff.}} = \lambda_1 + \lambda_2$$

$$\frac{1}{t_{\text{eff.}}} = \frac{1}{(t_{1/2})_1} + \frac{1}{(t_{1/2})_2}$$

$\lambda \rightarrow$  No dependence on temp.



# LIQUID SOLUTION

- **Vapour Pressure :** Pressure of any volatile substance at any given temperature.

$T \uparrow \Rightarrow V.P. \uparrow$

Attractive forces  $\uparrow \Rightarrow V.P. \downarrow$

- **Raoult's law :**

Non volatile solute and volatile solvent solution.

If  $\begin{cases} B = \text{Non volatile solid} \\ P_B = 0 \end{cases}$

$$P_A = P_A^\circ X_A$$

- **Colligative Properties :** Properties depends on no. of particles of Non volatile solute in solution.

No. of particle of Non volatile solute  $\uparrow \Rightarrow$  Colligative Properties  $\uparrow$

## (1) Relative lowering of V.P. :

$$\frac{P_A^\circ - P_A}{P_A^\circ} = i \frac{n_B}{n_A + n_B} \approx i \frac{n_B}{n_A}$$

Where  $n_B$  = mole of Non-volatile solute.

$i$  = Vant Hoff's factor.

## (2) Elevation in B.P. :

$$\Delta T_b = (T'_b - T_b) = i \cdot k_b \times m.$$

$$\text{where } K_b = \frac{RT_b^2}{1000 \times \ell_v}$$

where  $T_b$  = B.P. of pure solvent.

$\ell_v$  = Latent heat of vapourization per gm

$K_b$  = molal elevation constant

$M$  = molar mass

$$\text{where } \ell_v = \left( \frac{\Delta H_{\text{vap}}}{M} \right)$$

## (3) Depression in FP.

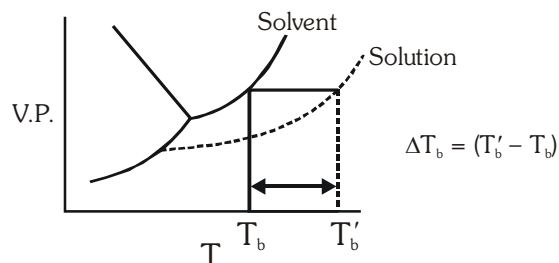
$$\Delta T_f = T_f - T'_f = i k_f \times m$$

$$\text{where } k_f = \frac{RT_f^2}{1000 \times \ell_f}$$

$T_f$  = f.p. of pure solvent

$k_f$  = molal depression constants

$\ell_f$  = latent heat of fusion per gm.



## (4) Osmotic pressure :

$$\pi \propto (P_A^\circ - P_A)$$

$$\pi = iC \cdot S.T.$$

where  $\pi$  = osmotic pressure

C = molarity (mole/lit)

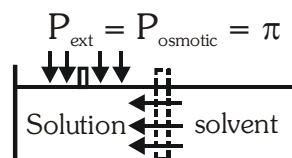
S = R = const. for solution.

Sol.(1)

Sol (2)

If  $\pi_1 = \pi_2$  Isotonic

If  $\pi_1 > \pi_2$   $\begin{cases} \text{sol}^n(1) \text{ hypertonic} \\ \text{sol}^n(2) \text{ hypotonic} \end{cases}$



## Van't Hoff factor for different Cases of solutes undergoing Ionisation and Association :

| Solute                              | Example                                         | Ionisation/association<br>(x degree)              | y*            | van'thoff<br>factor                                         | Abnormal<br>mol. wt. ( $m_1'$ )                              |
|-------------------------------------|-------------------------------------------------|---------------------------------------------------|---------------|-------------------------------------------------------------|--------------------------------------------------------------|
| Non-electrolyte                     | urea-glucose, sucrose etc.                      | none                                              | 1             | 1                                                           | normal mol.wt. ( $m_1$ )                                     |
| Binary electrolyte<br>$A^+B^-$      | NaCl, KCl, HCl<br>$CH_3COOH$ ,<br>$FeSO_4$ etc. | $AB \xrightleftharpoons[1-x]{x} A^+ + B^-$        | 2             | $(1+x)$                                                     | $\frac{m_1}{(1+x)}$                                          |
| Ternary electrolyte<br>$A_2B, AB_3$ | $K_2SO_4, BaCl_2$                               | $A_2B \xrightleftharpoons[1-x]{2x} 2A^+ + B^{2-}$ | 3             | $(1+2x)$                                                    | $\frac{m_1}{(1+2x)}$                                         |
|                                     | $K_3[Fe(CN)_6], FeCl_3$                         | $AB_3 \xrightleftharpoons[1-x]{3x} A^{3+} + 3B^-$ | 4             | $(1+3x)$                                                    | $\frac{m_1}{(1+3x)}$                                         |
| Associated Solute                   | benzoic acid in benzene forming dimer           | $2A \rightleftharpoons A_2$                       | $\frac{1}{2}$ | $\left(1 - \frac{x}{2}\right) = \left(\frac{2-x}{2}\right)$ | $\frac{2m_1}{(2-x)}$                                         |
|                                     | any solute forming polymer $A_n$                | $nA \rightleftharpoons A_n$                       | $\frac{1}{n}$ | $\left[1 + \left(\frac{1}{n} - 1\right)x\right]$            | $\left[\frac{m_1}{1 + \left(\frac{1}{n} - 1\right)x}\right]$ |
| General                             | one mole of solute giving y mol of products     | $A \rightleftharpoons yB$                         | y             | $[1+(y-1)x]$                                                | $\frac{m_1}{[1+(y-1)x]}$                                     |



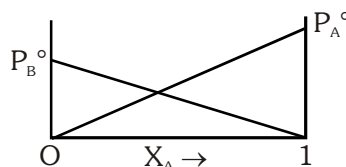
\* number of products from one mole solute

**Raoult's law :**

(1) Volatile binary liquid mix :

|                   |             |                                         |
|-------------------|-------------|-----------------------------------------|
| Volatile liq.     | A           | B                                       |
| Mole fraction     | $X_A/Y_A$   | $X_B/Y_A \Rightarrow \text{liq/vapour}$ |
| V.P. of pure liq. | $P_A^\circ$ | $P_B^\circ$                             |

Binary liquid solution :

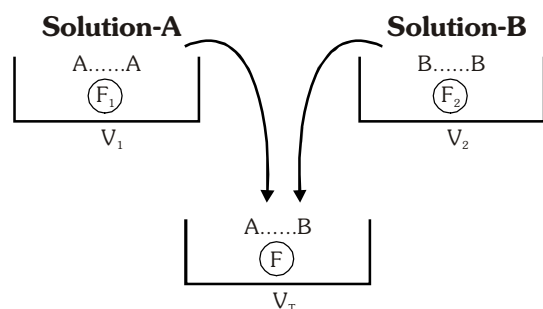


$$\text{By Raoult's law} \Rightarrow P_T = P_A^\circ X_A + P_B^\circ X_B = P_A + P_B \quad \text{.....(i)}$$

$$\text{By Dalton's law} \Rightarrow P_A = Y_A P_T \quad \text{.....(ii)}$$

$$P_B = Y_B P_T \quad \text{.....(iii)}$$

➤ **Ideal and Non-Ideal solution :**



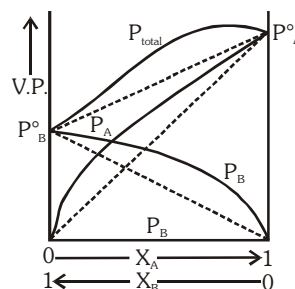
$$\text{Ideal solution : } \begin{cases} F_1 \approx F_2 \approx F \\ V_T = V_1 + V_2 \end{cases} \Rightarrow \Delta H_{\text{solution}} = 0$$

**Non-Ideal solution :**

(1) **Solution showing +ve deviation :**

$$F < F_1 \text{ \& } F_2$$

$$V_T > V_1 + V_2 \Rightarrow \Delta H_{\text{solution}} > 0$$

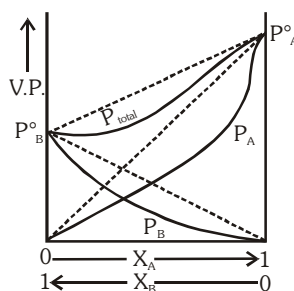


(2) **Solution showing -ve deviation :**

$$\Rightarrow F > F_1 \text{ \& } F_2$$

$$\Rightarrow V_T < (V_1 + V_2)$$

$$\Rightarrow \Delta H_{\text{solution}} < 0$$



## DEVIATION FROM RAOULT'S LAW

|       | Positive deviation<br>( $\Delta H = +ve$ )          | Negative deviation<br>( $\Delta H = -ve$ ) | Zero deviation<br>( $\Delta H = 0$ ) |
|-------|-----------------------------------------------------|--------------------------------------------|--------------------------------------|
| (i)   | ethanol + cyclohexane                               | acetone + chloroform                       | benzene + toluene                    |
| (ii)  | acetone + carbon disulphide<br>n-hexane + n-heptane | benzene + chloroform                       |                                      |
| (iii) | acetone + benzene                                   | nitric acid + chloroform                   | ethyl bromide + ethyl iodide         |
| (iv)  | ethanol + acetone                                   | acetone + aniline                          | chlorobenzene + bromo benzene        |
| (v)   | ethanol + water                                     | water + nitric acid                        |                                      |
| (vi)  | carbon tetrachloride<br>chloroform                  | diethyl ether +<br>chloroform              |                                      |

➤ **Azeotropic mixtures :**

Some liquids on mixing form azeotropes which are binary mixture having same composition in liquid and vapour phase and boil at a constant temperature. Azeotropic mixture cannot be separated by fractional distillation.

➤ **Types of Azeotropic mixtures****(i) Minimum boiling Azeotropic mixtures**

The mixture of two liquids whose boiling point is less than either of the two pure components. They are formed by non-ideal solutions showing positive deviation. For example (95.5%) + water (4.5%) + water boils at 351.15 K.

**(ii) Maximum boiling Azeotropic mixtures**

The mixture of two liquids whose boiling point are more than either of the two pure components. They are formed by non-ideal solutions showing negative deviation. For example  $\text{HNO}_3$  (68%) + water (32%) mixture boils at 393.5 K.

# SOLID STATE

## Distinction between Crystalline and Amorphous Solids

| Property                                      | Crystalline Solids                                                                                                 | Amorphous Solids                                                                   |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Shape                                         | Definite characteristic geometrical shape                                                                          | Irregular shape                                                                    |
| Melting point                                 | Melt at a sharp and characteristic temperature                                                                     | Gradually soften over a range of temperature                                       |
| Cleavage property                             | When cut with a sharp edged tool, they split into two pieces and the newly generated surfaces are plane and smooth | When cut with a sharp edged tool, they cut into two pieces with irregular surfaces |
| Heat of fusion                                | They have a definite and characteristic heat of fusion                                                             | They do not have definite heat of fusion                                           |
| Anisotropy                                    | Anisotropic in nature                                                                                              | Isotropic in nature                                                                |
| Nature                                        | True solids                                                                                                        | Pseudo solids or super cooled liquids                                              |
| Order in arrangement of constituent particles | Long range order                                                                                                   | Only short range order.                                                            |

## TYPES OF THE CRYSTALLINE SOLID

| Types of Solid                                                                   | Constituent Particles                           | Bonding/ Attractive forces  | Examples                                                                 | Physical Nature                | Electrical Conductivity                                                          | Melting Point |
|----------------------------------------------------------------------------------|-------------------------------------------------|-----------------------------|--------------------------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------|---------------|
| (1) Molecular Solids<br>(i) Non polar<br><br>(ii) Polar<br>(iii) Hydrogen bonded | Molecules                                       | Dispersion or London forces | Ar, CCl <sub>4</sub> , H <sub>2</sub> , I <sub>2</sub> , CO <sub>2</sub> | Soft                           | Insulator                                                                        | Very low      |
|                                                                                  |                                                 | Dipole-dipole               | HCl, SO <sub>2</sub>                                                     | Soft                           | Insulator                                                                        | Low           |
|                                                                                  |                                                 | Hydrogen bonding            | H <sub>2</sub> O (ice)                                                   | Hard                           | Insulator                                                                        | Low           |
| (2) Ionic Solids                                                                 | Ions                                            | Coulombic or electrostatic  | NaCl, MgO, ZnS, CaF <sub>2</sub>                                         | Hard but brittle               | Insulator in solid state but conductors in molten state and in aqueous solutions | High          |
| (3) Metallic Solids                                                              | Positive ions in a sea of delocalised electrons | Metallic bonding            | Fe, Cu, Ag, Mg                                                           | Hard but malleable and ductile | Conductors in solid state as well as in molten state                             | Fairly high   |
| (4) Covalent or network Solids                                                   | Atoms                                           | Covalent bonding            | SiO <sub>2</sub> (quartz), SiC, C (diamond), AlN,                        | Hard                           | Insulators                                                                       | Very high     |
|                                                                                  |                                                 |                             | C(graphite)                                                              | Soft                           | Conductor                                                                        |               |

## THE SEVEN CRYSTAL SYSTEMS

|    | Name of System           | Axes              | Angles                                                  | Bravais Lattices                                      |
|----|--------------------------|-------------------|---------------------------------------------------------|-------------------------------------------------------|
| 1. | Cubic                    | $a = b = c$       | $\alpha = \beta = \gamma = 90^\circ$                    | Primitive, Face-centred, Body centred = 3             |
| 2. | Tetragonal               | $a = b \neq c$    | $\alpha = \beta = \gamma = 90^\circ$                    | Primitive, Body centred = 2                           |
| 3. | Rhombohedral or Trigonal | $a = b = c$       | $\alpha = \beta = \gamma \neq 90^\circ$                 | Primitive = 1                                         |
| 4. | Orthorhombic or Rhombic  | $a \neq b \neq c$ | $\alpha = \beta = \gamma = 90^\circ$                    | Primitive, Face-centred, Body centred End centred = 3 |
| 5. | Monoclinic               | $a \neq b \neq c$ | $\alpha = \gamma = 90^\circ$ ;<br>$\beta \neq 90^\circ$ | Primitive, End - centred = 2                          |
| 6. | Triclinic                | $a \neq b \neq c$ | $\alpha \neq \beta \neq \gamma \neq 90^\circ$           | Primitive = 1                                         |
| 7. | Hexagonal                | $a = b \neq c$    | $\alpha = \beta = 90^\circ$<br>$\gamma = 120^\circ$     | Primitive = 1<br>Total = 14                           |

## CUBIC UNIT CELL

| Unit cell          | Relation between $r$ and $a$ | Packing fraction | Co-ordination number | Effective number of particle |
|--------------------|------------------------------|------------------|----------------------|------------------------------|
| Simple cubic       | $r = \frac{a}{2}$            | 52.4%            | 6                    | 1                            |
| Body centred cubic | $r = \frac{a\sqrt{3}}{4}$    | 68%              | 8                    | 2                            |
| Face centred cubic | $r = \frac{a\sqrt{2}}{4}$    | 74%              | 12                   | 4                            |

$$\text{Density : } d = \frac{ZM}{N_A \times a^3} \text{ gm/cm}^3$$

Where  $Z$  = effective number of particle

$M$  = molar mass

$N_A$  = Avogadro's number

$a$  = edge length (cm)

## Three dimensional close packing :

## ➤ Hexagonal close packing (HCP) :

Effective number of particle = 6

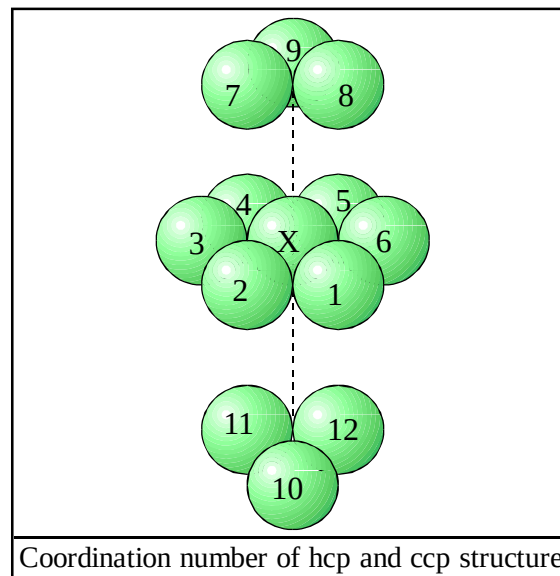
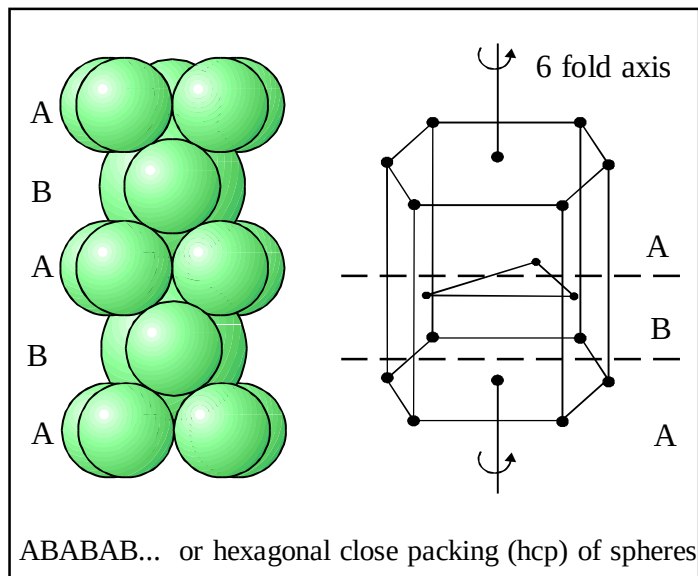
Effective number of octahedral void = 6

Effective number of tetrahedral void = 12

Packing fraction

= 74% ; co-ordination number = 12

$$a = \frac{r}{2} ; b = 4 \sqrt{\frac{2}{3}} r$$



➤ **Cubic close packing (CCP) :**

Effective number of particle = 4

Effective number of octahedral void = 4

Effective number of tetrahedral void = 8

Packing fraction = 74% ;

co-ordination number = 12

$$\frac{a\sqrt{2}}{4} = r$$

➤ **Different type of voids and their radius ratio :**

| Limiting radius ratio = $r/R$ | Limiting radius ratio for various types of sites |                                            |                        |
|-------------------------------|--------------------------------------------------|--------------------------------------------|------------------------|
|                               | Coordination Number of cation                    | Structural Arrangement (Geometry of voids) | Example                |
| 0.155 - 0.225                 | 3                                                | Plane Trigonal                             | Boron Oxide            |
| 0.225 - 0.414                 | 4                                                | Tetrahedral                                | ZnS, SiO <sub>2</sub>  |
| 0.414 - 0.732                 | 4                                                | Square planar                              | —                      |
| 0.414 - 0.732                 | 6                                                | Octahedral                                 | NaCl, MgO <sub>2</sub> |
| 0.732 - 1.000                 | 8                                                | Cubic                                      | CsCl                   |

## TYPES OF IONIC CRYSTAL

| Type of Ionic Crystal                                        | Geometry                                                                                                                                                                                                           | Co-ordination Number                               | No. of formula's per U.C.C.                                      | Examples                                                                                                                                      |  |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--|
| 1. NaCl (1 : 1)<br>(Rock Salt Type)                          | $\begin{array}{c} \text{Na}^+ \rightarrow \text{Every element of C.C.P.} \\ \text{C.C.P.} \swarrow \searrow \\ \text{Cl}^- \rightarrow \text{At every OHV} \end{array}$                                            | 6 : 6                                              | $4\text{Na}^+ + 4\text{Cl}^-$<br>$4\text{NaCl}$<br>(4)           | Halides of (Li, Na, K, Rb)<br>Oxides and sulphides of II-A (Some exception)<br>AgF, AgCl, AgBr, $\text{NH}_4\text{X}$                         |  |
| 2. CsCl Type (1 : 1)                                         | $\begin{array}{c} \text{Cs}^+ \rightarrow \text{at every corner} \\ \text{B.C.C.} \swarrow \searrow \\ \text{Cl}^- \rightarrow \text{at Body centre or at cubic void} \end{array}$                                 | 8 : 8                                              | $1\text{Cs}^+ + 1\text{Cl}^-$<br>$1\text{CsCl}$<br>(1)           | Halides of 'Cs'<br>TlCl, TlBr, CaS                                                                                                            |  |
| 3. ZnS Type (1 : 1)<br>(Zinc Blende Type)<br>(Sphalerite)    | $\begin{array}{c} \text{Zn}^{+2} \rightarrow \text{Every element of C.C.P.} \\ \text{C.C.P.} \swarrow \searrow \\ \text{S}^{2-} \rightarrow \text{At 50\% of T.H.V. or at alternate tetrahedral void} \end{array}$ | 4 : 4                                              | $4\text{Zn}^{+2} + 4\text{S}^{2-}$<br>$4\text{ZnS}$<br>(4)       | BeS,<br>BeO, CaO, AgI,<br>CuCl, CuBr, CuI                                                                                                     |  |
| 4. $\text{CaF}_2$ Type (1 : 2)<br>(Fluorite Type)            | $\begin{array}{c} \text{Ca}^{+2} \rightarrow \text{Every element of C.C.P.} \\ \text{C.C.P.} \swarrow \searrow \\ \text{F}^- \rightarrow \text{At every T.H.V.} \end{array}$                                       | $4\text{Ca}^{+2}, 8\text{F}^-$<br><del>8 : 4</del> | $4\text{Ca}^{+2} + 8\text{F}^-$<br>$4\text{CaF}_2$<br>(4)        | $\text{BaCl}_2$ , $\text{BaF}_2$<br>$\text{SrCl}_2$ , $\text{SrF}_2$<br>$\text{CaCl}_2$ , $\text{CaF}_2$                                      |  |
| 5. $\text{Na}_2\text{O}$ Type (2 : 1)<br>(Antifluorine)      | $\begin{array}{c} \text{Na}^+ \rightarrow \text{At every T.H.V.} \\ \text{C.C.P.} \swarrow \searrow \\ \text{O}^{2-} \rightarrow \text{Every element of C.C.P.} \end{array}$                                       | $8\text{Na}^+, 4\text{O}^{2-}$<br><del>4 : 8</del> | $8\text{Na}^+ + 4\text{O}^{2-}$<br>$4\text{Na}_2\text{O}$<br>(4) | $\text{Li}_2\text{O}$ , $\text{Li}_2\text{S}$<br>$\text{Na}_2\text{O}$ , $\text{Na}_2\text{S}$<br>$\text{K}_2\text{O}$ , $\text{K}_2\text{S}$ |  |
| 6. ZnS Type (1 : 1)<br>(Wurtzite)<br>another geometry of ZnS | $\begin{array}{c} \text{Zn}^{+2} \rightarrow \text{Every element of H.C.P.} \\ \text{H.C.P.} \swarrow \searrow \\ \text{S}^{2-} \rightarrow 50\% \text{ of T.H.V. or (at alternate T.H.V.)} \end{array}$           | 4 : 4                                              | $6\text{Zn}^{+2} + 6\text{S}^{2-}$<br>$6\text{ZnS}$<br>(6)       | Same as sphalerite                                                                                                                            |  |

**GASEOUS STATE**

| GAS LAW                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | GRAHAM'S DIFFUSION LAW                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>♦ <b>Boyle's law</b> : <math>V \propto \frac{1}{P}</math> (<math>n, T = \text{const}</math>) <math>P_1 V_1 = P_2 V_2</math></p> <p>♦ <b>Charle's law</b> : <math>V \propto T</math> (<math>n, P = \text{const}</math>) <math>\frac{V_2}{V_1} = \frac{T_2}{T_1}</math></p> <p>♦ <b>Gay lussac's law</b>: <math>P \propto T</math> (<math>n, V = \text{const}</math>) <math>\frac{P_2}{P_1} = \frac{T_2}{T_1}</math></p> <p>♦ <b>Avogadro's law</b> :<br/> <math>V \propto \text{moles} \propto \text{number of molecules}</math> (<math>P, T = \text{const}</math>)<br/>                     Ideal gas equation <math>PV = nRT</math><br/> <math>R = 0.0821 \text{ lit atm mol}^{-1} \text{ K}^{-1}</math><br/> <math>R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}</math> or <math>8.314 \text{ J mol}^{-1} \text{ K}^{-1}</math><br/> <math>R = 2 \text{ cal K}^{-1} \text{ mol}^{-1}</math>, <math>R = 8.314 \times 10^7 \text{ erg K}^{-1} \text{ mol}^{-1}</math></p> | <p>It is applicable for non reacting gases</p> <p><math>r \propto \frac{1}{\sqrt{d}}</math> ; <math>r \propto \frac{1}{\sqrt{VD}}</math> ; <math>r \propto \frac{1}{\sqrt{M_w}}</math> (<math>P, T = \text{constant}</math>)</p> <p><math>VD = \frac{d_{\text{gas}}}{dH_2} = \frac{M_w}{2}</math></p> <p>Rate of diffusion</p> <p><math>r = \frac{\ell_{\text{diffused gas}}}{t_{\text{time taken}}} ; r = \frac{V_{\text{diffused gas}}}{t_{\text{time taken}}} ; r = \frac{n_{\text{diffused gas}}}{t_{\text{time taken}}}</math></p> <p>(Where, <math>\ell</math> = distance travelled by diffused gas)</p> |

| KINETIC GAS EQUATION : $PV = \frac{1}{3} mN V_{\text{rms}}^2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DALTON'S LAW OF PARTIAL PRESSURE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | AVERAGE KINETIC ENERGY ( $KE_{\text{av}}$ )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p><math>P_{\text{mixture}} = \frac{P_1 + P_2 + P_3 + \dots}{\text{Partial pressure}} \dots (T \&amp; V \text{ const})</math></p> <p><math>P_{\text{moist gas}} = P_{\text{dry gas}} + P_{\text{water vapours}}</math></p> <p>It is applicable for non reacting gases.</p> <p>Methods of determination of partial pressure (<math>P_A</math> &amp; <math>P_B</math> are partial pressure)</p> <p>♦ from ideal gas equation<br/> <math>P_A V = n_A RT</math> &amp; <math>P_B V = n_B RT</math></p> <p>♦ In the form of mole fraction.<br/> <math>P_A = X_A P_T = \frac{n_A}{n_t} P_T</math> ; <math>P_B = X_B P_T = \frac{n_B}{n_t} P_T</math><br/> <math>[X_A + X_B = 1]</math><br/> <math>P_T</math> = sum of partial pressure of all gases</p> <p>♦ In the form of volume fraction<br/> <math>P_A = \frac{V_A}{V} P_T</math> &amp; <math>P_B = \frac{V_B}{V} P_T</math></p> <p>♦ If individual pressure and individual volume are given<br/> <math>P_A = \frac{V_A}{V} P_1</math> and <math>P_B = \frac{V_B}{V} P_2</math><br/> <math>P_1, P_2</math> = pressure of gases before mixing<br/> <math>P_A, P_B</math> = pressure of gases after mixing</p> | <p><math>KE_{\text{av}} = \frac{3}{2} nRT</math> (<math>n</math> moles)<br/> <math>KE_{\text{av}} = \frac{3}{2} RT</math> (1 mol or <math>N_A</math> molecules)<br/> <math>KE_{\text{av}} = \frac{3}{2} KT</math> (1 molecule)<br/> <math>K = 1.38 \times 10^{-23} \text{ JK}^{-1} \text{ molecule}^{-1}</math><br/> <math>K</math> is called Boltzman's constant</p> <p><math>v_{\text{rms}} = \sqrt{\frac{v_1^2 + v_2^2 + \dots + v_n^2}{N}}</math>      <math>v_{\text{av}} = \frac{v_1 + v_2 + v_3 + \dots + v_n}{N}</math></p> <p><math>v_{\text{rms}} = \sqrt{\frac{3RT}{M_w}}</math>      <math>v_{\text{av}} = \sqrt{\frac{8RT}{\pi M_w}}</math>      <math>v_{\text{mp}} = \sqrt{\frac{2RT}{M_w}}</math></p> <p><math>v_{\text{rms}} = \sqrt{\frac{3PV}{M_w}}</math>      <math>v_{\text{av}} = \sqrt{\frac{8PV}{\pi M_w}}</math>      <math>v_{\text{mp}} = \sqrt{\frac{2PV}{M_w}}</math></p> <p><math>v_{\text{rms}} = \sqrt{\frac{3P}{d}}</math>      <math>v_{\text{av}} = \sqrt{\frac{8P}{\pi d}}</math>      <math>v_{\text{mp}} = \sqrt{\frac{2P}{d}}</math></p> <p><math>v_{\text{rms}} : v_{\text{av}} : v_{\text{mp}} = \sqrt{3} : \sqrt{\frac{8}{\pi}} : \sqrt{2}</math><br/> <math>= 1 : 0.92 : 0.82</math></p> <p><math>v_{\text{mp}} : v_{\text{av}} : v_{\text{rms}} = \sqrt{2} : \sqrt{\frac{8}{\pi}} : \sqrt{3}</math><br/> <math>= 1 : 1.128 : 1.224</math></p> <p>Compressibility factor (<math>z</math>) = <math>\frac{(V_m)_{\text{obs}}}{V_i} = \frac{P(V_m)_{\text{obs}}}{RT}</math><br/>                     If <math>z = 1</math>, the gas show ideal gas behaviour.<br/>                     If <math>z &gt; 1</math>, the gas show positive deviation.<br/>                     If <math>z &lt; 1</math>, the gas show negative deviation.</p> |

## VANDERWAAL'S EQUATION

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

$$P_i = P_R + \frac{an^2}{V^2} \Rightarrow P_i > P_R$$

$a \uparrow$  force of attraction  $\uparrow$

liquification  $\uparrow$ ;

$b \uparrow$ , effective size of molecule  $\uparrow$ ,

incompressible vol  $\uparrow$ ,

compressible vol  $\downarrow$

- ♦ At high pressure, Vanderwaal's eq<sup>n</sup> is

$$PV_m - Pb = RT$$

- ♦ At low pres. or Moderate pressure Vanderwaal's eq<sup>n</sup> is

$$PV_m + \frac{a}{V_m} = RT$$

- ♦ At very low pressure, high temp. Vander waal's Equation is

$$VP = nRT$$

Ideal gas behavior.

- ♦ Gases having  $\uparrow$ value of  $a$ ; will have  $\uparrow T_c$ ;  
 $\uparrow$ rate of liquefaction.

**IMPORTANT NOTES**





## BOHR'S ATOMIC MODEL

Theory based on quantum theory of radiation and the classical laws of physics

- $\frac{K(Ze)(e)}{r^2} = \frac{mv^2}{r}$
- $mvr = \frac{nh}{2\pi}$  or  $mvr = n\hbar$
- Electron remains in stationary orbit where it does not radiate its energy.
- **Radius :**  $r = 0.529 \times \frac{n^2}{Z} \text{ \AA}$
- **Velocity :**  $v = 2.188 \times 10^6 \frac{Z}{n} \text{ ms}^{-1}$
- Energy(KE + PE)  
 $= \text{Total energy} = -13.6 \times \frac{Z^2}{n^2} \text{ eV/atom}$
- $TE = -\frac{KZe^2}{2r}$ ,  $PE = \frac{-KZe^2}{r}$ ,  $KE = \frac{KZe^2}{2r}$   
 $PE = -2KE$ ,  $KE = -TE$ ,  $PE = 2TE$
- Revolutions per sec =  $\frac{v}{2\pi r}$
- Time for one revolution =  $\frac{2\pi r}{v}$
- Energy difference between  $n_1$  and  $n_2$  energy level  
 $\Delta E = E_{n_2} - E_{n_1} = 13.6Z^2 \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \frac{\text{eV}}{\text{atom}} = IE \times \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$   
 where IE = ionization energy of single electron species.
- **Ionization energy**  $= E_\infty - E_{G.S.} = 0 - E_{G.S.}$   
 $E_{G.S.}$  = Energy of electron in ground state

## HYDROGEN SPECTRUM

## • Rydberg's Equation :

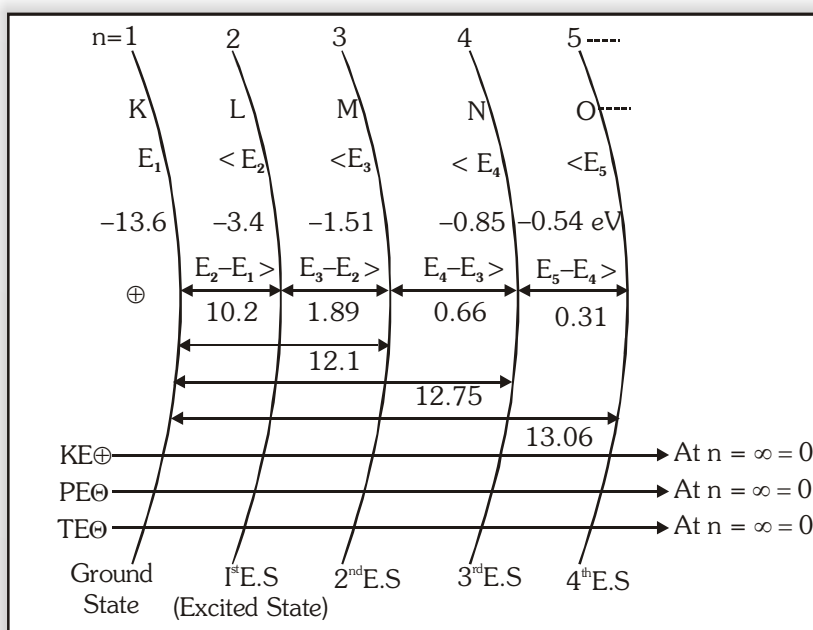
$$\frac{1}{\lambda} = \bar{\nu} = R_H \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \times Z^2$$

$$R_H \cong 109700 \text{ cm}^{-1} = \text{Rydberg constant}$$

- For first line of a series  $n_2 = n_1 + 1$
- Limiting spectral line (series limit) means  $n_2 = \infty$
- $H_\alpha$  line means  $n_2 = n_1 + 1$ ; also known as line of longest  $\lambda$ , shortest  $\nu$ , least E
- Similarly  $H_\beta$  line means  $n_2 = n_1 + 2$
- When electrons de-excite from higher energy level ( $n$ ) to ground state in atomic sample, then number of spectral lines observed in the spectrum =  $\frac{n(n-1)}{2}$
- When electrons de-excite from higher energy level ( $n_2$ ) to lower energy level ( $n_1$ ) in atomic sample, then number of spectral line observed in the spectrum

$$= \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

- No. of spectral lines in a particular series =  $n_2 - n_1$



### DE-BROGLIE HYPOTHESIS

- All material particles possess wave character as well as particle character.
- $\lambda = \frac{h}{mv} = \frac{h}{p}$
- The circumference of the  $n^{\text{th}}$  orbit is equal to  $n$  times of wavelength of electron i.e.,  $2\pi r_n = n\lambda$   
Number of waves =  $n$  = principal quantum number
- Wavelength of electron ( $\lambda$ )  $\approx \sqrt{\frac{150}{V(\text{volts})}} \text{ \AA}$
- $\lambda = \frac{h}{\sqrt{2mKE}}$

### HEISENBERG UNCERTAINTY

- According to this principle, "it is impossible to measure simultaneously the position and momentum of a microscopic particle with absolute accuracy"  
If one of them is measured with greater accuracy, the other becomes less accurate.
- $\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$  or  $(\Delta x)(\Delta v) \geq \frac{h}{4\pi m}$   
where  $\Delta x$  = Uncertainty in position  
 $\Delta p$  = Uncertainty in momentum  
 $\Delta v$  = Uncertainty in velocity  
 $m$  = mass of microscopic particle
- Heisenberg replaced the concept of orbit by that of orbital.

## QUANTUM NUMBER

- Principal Quantum number (By Bohr)**
  - Indicates = Size and energy of the orbit, distance of  $e^-$  from nucleus
  - Values  $n = 1, 2, 3, 4, 5, \dots$
  - Angular momentum =  $n \times \frac{h}{2\pi}$
  - Total number of  $e^-$ s in an orbit =  $2n^2$
  - Total number of orbitals in an orbit =  $n^2$
  - Total number of subshell in an orbit =  $n$
- Azimuthal/Secondary/Subsidiary/Angular momentum quantum number ( $\ell$ )**
  - Given by = Sommerfeld
  - Indicates = Sub shells/sub orbit/sub level
  - Values  $\Rightarrow 0, 1, \dots, (n-1)$
  - Indicates shape of orbital/Sub shell

| Value of $n$      | Values of $\ell$ [Shape]                                                                                                  | Initial from word                             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| eg.<br>If $n = 4$ | $\ell = 0$ (s) [Spherical]<br>$\ell = 1$ [p] [Dumb bell]<br>$\ell = 2$ [d] [Double dumb bell]<br>$\ell = 3$ [f] [Complex] | Sharp<br>Principal<br>Diffused<br>Fundamental |

- Total no. of  $e^-$ s in a suborbit =  $2(2\ell + 1)$
- Total no. of orbitals in a suborbit =  $(2\ell + 1)$
- Orbital angular momentum

$$= \sqrt{\ell(\ell+1)} \frac{h}{2\pi} = \sqrt{\ell(\ell+1)} \hbar$$

$h$  = Planck's constant

- For H & H like species all the subshells of a shell have same energy.  
i.e.  $2s = 2p$   $3s = 3p = 3d$

- Magnetic Quantum number ( $m$ )**

- Given by Linde
- Indicates orientation of orbitals i.e. direction of electron density.
- Value of  $m = -\ell, \dots, 0, \dots, +\ell$
- Maximum no. of  $e^-$ s in an orbital = 2 (with opposite spin)

$m$  for  $p$  sub shell =  $p_x \quad p_y \quad p_z$   
 $-1 \quad +1 \quad 0$

$m$  for  $d$  sub shell =

$d_{xy} \quad d_{yz} \quad d_{z^2} \quad d_{xz} \quad d_{x^2-y^2}$   
 $-2 \quad -1 \quad 0 \quad +1 \quad +2$

- Spin Quantum Number ( $m_s$  or  $s$ )**

Given by Uhlenback & Goudsmit

Values of  $s = \pm \frac{1}{2}$

Total value of spin in an atom =  $\pm \frac{1}{2} \times \text{number of unpaired electrons}$

$$\text{Spin Angular momentum} = \sqrt{s(s+1)} \frac{h}{2\pi}$$

### RULES FOR FILLING OF ORBITALS

- Aufbau principle** : The electrons are filled up in increasing order of the energy in subshells.  
 $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^6 6s^2 4f^{14} 5d^{10} 6p^6 7s^2 5f^{14} 6d^{10}$
- ( $n + \ell$ ) rule** : The subshell with lowest ( $n + \ell$ ) value is filled up first, but when two or more subshells have same ( $n + \ell$ ) value then the subshell with lowest value of  $n$  is filled up first.
- Pauli exclusion principle** : Pauli stated that no two electrons in an atom can have same values of all four quantum numbers.
- Hund's rule of maximum multiplicity** : Electrons are distributed among the orbitals of subshell in such a way as to give maximum number of unpaired electrons with parallel spin.

## SURFACE CHEMISTRY

## CLASSIFICATION BASED ON INTERACTION OF PHASES :-

LYOPHILIC  
AND  
LYOPHOBIC  
SOLS

Colloidal solutions in which the dispersed phase has considerable affinity for the dispersion medium, are called lyophilic sols (solvent - liking).

**For example** - dispersion of gelatin, starch, gum and proteins in water.

Colloidal solutions in which the dispersed phase has no affinity or attraction for the medium or for the solvent are called Lyophobic colloidal (Solvent hating) solutions.

## COMPARISON OF LYOPHOBIC AND LYOPHILIC SOLS

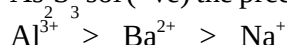
| Property                        | Lyophilic sol (Emulsoid)                                                                     | Lyophobic sol (suspensoid)                                                                    |
|---------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 1. Preparation                  | Can be easily prepared by shaking or warming the substance with solvent                      | Can not be prepared easily, special methods are required                                      |
| 2. Stability                    | are more stable                                                                              | are less stable                                                                               |
| 3. Reversibility                | are reversible                                                                               | are irreversible                                                                              |
| 4. viscosity                    | viscosity is much higher than that of solvent                                                | viscosity is nearly same as that of the solvent                                               |
| 5. Surface tension              | Surface tension is usually low                                                               | Surface tension is almost same as that of solvent                                             |
| 6. Hydration or solvation       | These are highly solvated as the particles have great affinity for solvent                   | These are less solvated as the particles have less affinity for the solvent                   |
| 7. Charge                       | The particles have little charge or no charge at all                                         | The particles carry a characteristic charge- either positive or negative                      |
| 8. Visibility                   | Particles can not be seen under microscope                                                   | Particles can be seen under microscope                                                        |
| 9. Coagulation or precipitation | Precipitated by high concentration of electrolytes                                           | Precipitated by low concentration of electrolytes                                             |
| 10. Tyndall effect              | Less Scattering                                                                              | More Scattering                                                                               |
| 11. Migration in electric field | may or may not migrate as they may or may not carry charge.                                  | migrate towards anode or cathode as these particles carry charge.                             |
| 12. General Ex.                 | <b>Mostly of organic nature</b><br>Ex. Gelatin, Starch,<br>Gum, Albumin & Cellulose Solution | <b>Mostly of Inorganic nature</b><br>Ex. Transition metal salt in water like<br>Gold, As etc. |

## PEPTIZATION

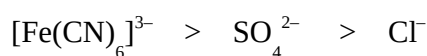
The dispersion of a freshly precipitated material into colloidal solution by the action of an electrolyte in solution is termed as peptization. The electrolyte used is called a **Peptizing agent**.

**Hardy Schulze Rule** - This rule states that the precipitating effect of an ion on dispersed phase of opposite charge increases with the valency of the ion.

The higher the valency of the flocculating ion, the greater is its precipitating power. Thus for the precipitation of  $\text{As}_2\text{S}_3$  sol (-ve) the precipitating power of  $\text{Al}^{3+}$ ,  $\text{Ba}^{2+}$ , and  $\text{Na}^+$  ions is in the order



Similarly for precipitating  $\text{Fe}(\text{OH})_3$  sol (positive) the precipitating power of  $[\text{Fe}(\text{CN})_6]^{3-}$ ,  $\text{SO}_4^{2-}$  and  $\text{Cl}^-$  ions is in the order



The minimum concentration of an electrolyte in milli moles required to cause precipitation of 1 litre sol in 2 hours is called FLOCCULATION VALUE. The smaller the flocculating value, the higher will be the coagulating power of

the ion.

$$\text{Flocculation value} \propto \frac{1}{\text{Flocculation power}}$$

**GOLD  
NUMBER**

The number of **milligrams** of a hydrophilic colloid that will just prevent the precipitation of 10 ml of standard gold sol on addition of 1 ml of 10% NaCl solution is known as **Gold number** of that protector (Lyophilic colloid).

The precipitation of the gold sol is indicated by a colour change from red to blue when the particle size just increases.

The **smaller the gold number** of a protective Lyophilic colloid, **greater is its protection power**.

**Note :** Gelatin and starch have the maximum & minimum protective powers.

$$\text{Protection Capacity} \propto \frac{1}{\text{Protection Number (Gold number)}}$$

## TYPES OF COLLOIDS ACCORDING TO THEIR SIZE

| Multi Molecular                                                                        | Macro Molecular                                                                              | Associated colloids                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Formation by aggregation of a large number of atoms or smaller molecules of substance. | Formation by aggregation of big size molecules. These are polymers with high molecular mass. | These are the substances which behave as normal electrolytes at low concentration but get associated at higher concentration and behave as colloidal solutions. These associated particles are also called micelles. |
| Ex. → <b>Gold Sol (Au)</b><br><b>Sulphur sol (S<sub>8</sub>)</b>                       | Ex. → <b>Starch, Cellulose, Protein etc.</b>                                                 | Ex. → <b>Soap &amp; Detergent</b>                                                                                                                                                                                    |

## COMPARISON OF PHYSI-SORPTION AND CHEMI-SORPTION

|    | Physical Adsorption                                        | Chemical Adsorption (Activated ad.)                            |
|----|------------------------------------------------------------|----------------------------------------------------------------|
| 1. | It is caused by intermolecular vander waal's forces.       | It is caused by chemical bond formation.                       |
| 2. | It is not specific.                                        | It is highly specific.                                         |
| 3. | It is reversible.                                          | It is irreversible.                                            |
| 4. | Heat of adsorption is low. (– 20 to –40 KJ/mol) (Enthalpy) | Heat of adsorption is high.<br>(–80 to –240 KJ/mol) (Enthalpy) |
| 5. | No appreciable activation is energy involved.              | High activation energy involved.                               |
| 6. | It forms multimolecular layers on adsorbent surface.       | It forms unimolecular layer under high pressure.               |

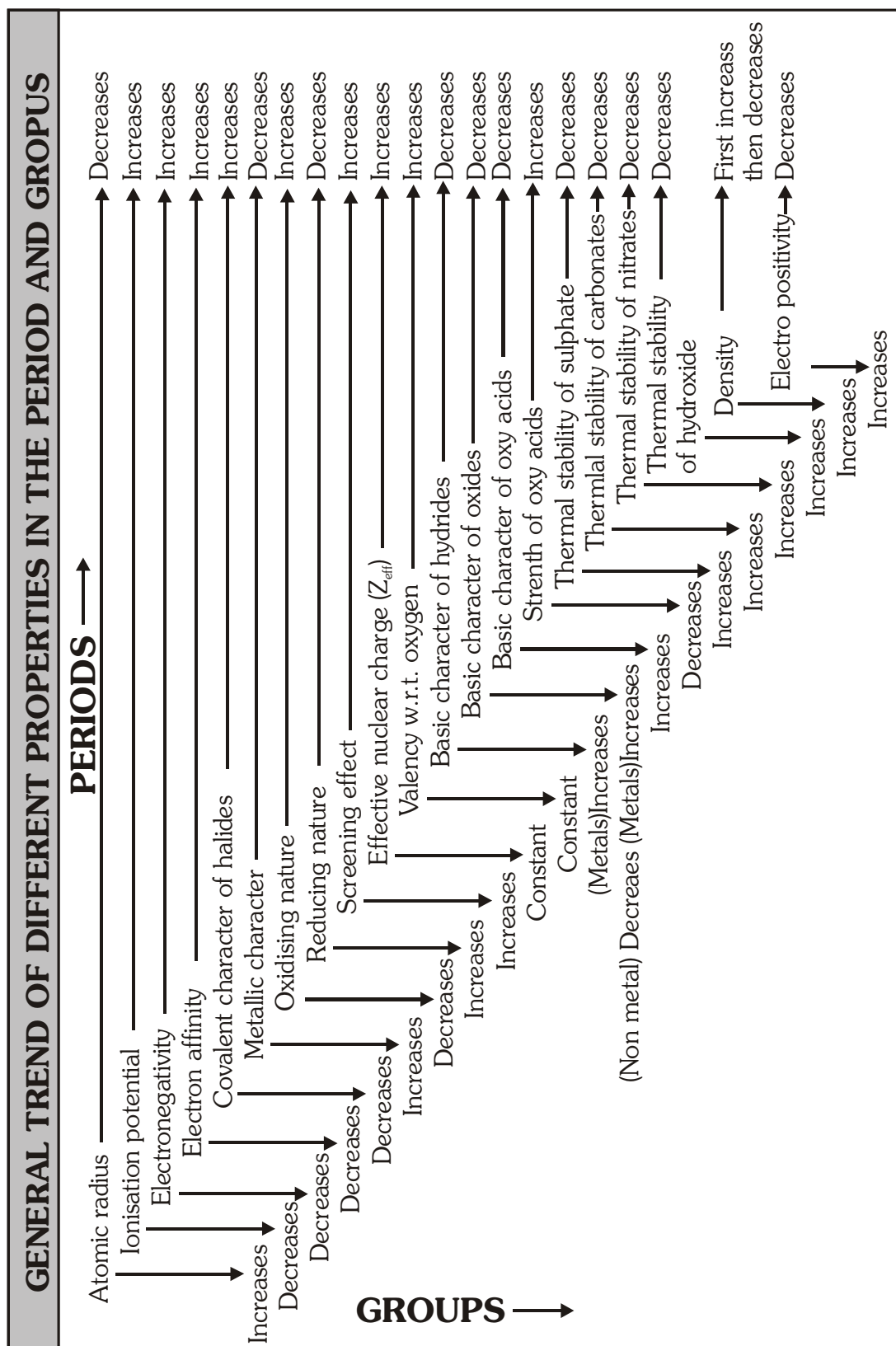
# GENERAL CHARACTERISTICS OF CATALYSTS

Critical temperature increases  
Ease of liquification increases  
Extent of adsorption increases  
(true for physisorption)

- |                                                                                                                                                                      |                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| (i) A catalyst remains unchanged in mass and chemical composition but can change their physical state.                                                               |                                                      |
| (ii) Only a very small amount of catalyst is sufficient to catalyse a reaction.                                                                                      | (vii) The catalyst are generally specific in nature. |
| (iii) A catalyst does not initiate a reaction.                                                                                                                       | (viii) Changes rate constant of reaction.            |
| (iv) Solid catalyst is more efficient when used in finely divided form.                                                                                              | (ix) Does not change free energy of reaction.        |
| (v) Generally catalyst does not change the nature of products.                                                                                                       | (x) Participates in mechanism of reaction.           |
| (vi) A catalyst does not change the equilibrium state of a reversible reaction but helps to achieve the equilibrium state or position of equilibrium in lesser time. |                                                      |

[illegible]

# Some Important Increasing Order



# PERIODIC PROPERTIES

## Periodicity :

Repetition of properties after regular interval is known as periodicity and these properties are known as periodic properties.

1. Effective Atomic Number
2. Atomic Radius
3. Ionisation Potential
4. Electron Affinity
5. Electro Negativity

### ATOMIC RADIUS

Distance between centre of nucleus to outermost electron.

Accurate value of atomic radius cannot be measured therefore operational definitions are used.

- (i) Covalent radius
- (ii) Metallic radius
- (iii) Vander Waal's radius

$$r_{\text{cov}} < r_{\text{metallic}} < r_{\text{vw}}$$

Vander wall radius mainly used for noble gases.

### Factors Affecting atomic radius :

- (1)  $AR \propto \text{Number of shells}$

$$(2) \quad AR \propto \frac{1}{Z_{\text{eff}}}$$

### Periodic trends :

- (1) Generally increases on moving down the group
- (2) Generally decrease when we are moving left to right in the period.

#### Note :

Atomic radius of  $Al > Ga$  : Due to poor shielding of 3d subshell electron

Atomic radius of 4d  $\approx$  5d series element due to lanthanide contraction

Lanthanide Contraction - decrease in atomic/ionic radius with increase in atomic number of lanthanide.

Actinide Contraction - decrease in atomic/ionic radius with increase in atomic number of actinide.

### Ionic Radius:



$$\begin{array}{ll} Z_{\text{eff}} & A^+ > A > A^- \\ \text{Ionic radius} & A^+ < A < A^- \end{array}$$

### Factors affecting ionic radius

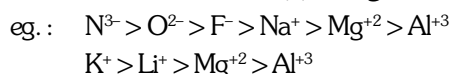
(+) charge  $\uparrow$  ionic radius  $\downarrow$

(-) charge  $\uparrow$  ionic radius  $\uparrow$

$Z_{\text{eff}}$   $\uparrow$  ionic radius  $\downarrow$

$n$   $\uparrow$  ionic radius  $\uparrow$

$\Rightarrow$  In isoelectronic species (+) charge  $\uparrow$  Ionic radius  $\downarrow$   
(-) charge  $\uparrow$  Ionic radius  $\uparrow$

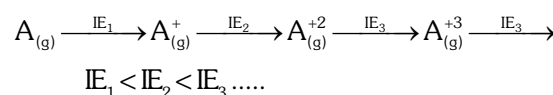


### IONIZATION ENERGY

Energy required to remove to loosely bonded  $e^-$  from isolated gaseous atom.

### Successive I.E. :

$I.E_1, I.E_2, \dots$  combinedly termed as successive I.E.



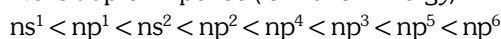
### Factors affecting :

- (1)  $IE \propto Z_{\text{eff}}$
- (2)  $IE \propto \frac{1}{\text{size}}$
- (3)  $IE \propto (+)\text{charge}$
- (4)  $IE \propto \frac{1}{(-)\text{charge}}$
- (5) half filled and full filled configuration (affected upto 4<sup>th</sup> period)
- (6) nature of subshell (affected upto 4<sup>th</sup> period)

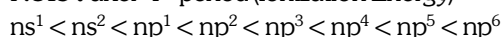
### Periodic trend (s & p-block) :

- (1) Generally decrease on moving down the group
- (2) Generally increase when we are moving left to right in the period.

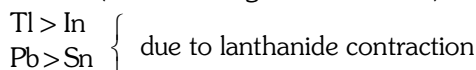
**Note :** upto 4<sup>th</sup> period (Ionization Energy)



**Note :** after 4<sup>th</sup> period (Ionization Energy)

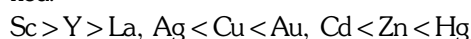


$Al < Ga$  (Poor shielding of 3d subshell  $e^-$ )



### Periodic trend (d-block) :

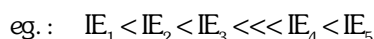
- (1) Generally increases on moving down the group
- (2) Generally increase when moving left to right in the period.



### Application of ionization energy :



**Note :** Number of valence  $e^-$  = highest jump in successive IE



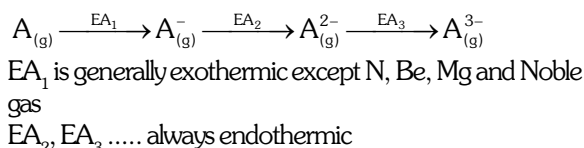
Highest jump between  $IE_3$  &  $IE_4$  therefore V  $e^-$  is 3



### Electron affinity & Electron gain enthalpy :

| Electron affinity                                                                                                   | Electron gain enthalpy                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| $A_{(g)} + e^- \rightarrow A_{(g)}^-$<br>amount of energy released when an $e^-$ is added to isolated gaseous atom. | $A_{(g)} + e^- \rightarrow A^-$<br>Change in enthalpy when an $e^-$ is added to isolated gaseous atom.<br>$\Delta H_{eg} = H_P - H_R$ |

### Successive electron affinity :



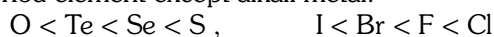
### Factors affecting :

- (1)  $EA \propto Z_{eff}$
- (2)  $EA \propto \frac{1}{size}$
- (3) Nature of subshell
- (4) half and full filled

### Periodic trend :

- (1) Generally decreases on moving down the group
- (2) Generally increase when moving left to right in the period.

**Note :** III<sup>rd</sup> period element having greater EA than II<sup>nd</sup> period element except alkali metal.



**Note :**  $|IE_A| = |EA_{A^+}|$  and  $|EA_A| = |IE_{A^-}|$

### Electronegativity :

Tendency to attract shared pair of  $e^-$  towards itself in covalent bond.

### Factors affecting :

- (1)  $EN \propto Z_{eff}$
- (2)  $EN \propto \frac{1}{size}$
- (3)  $EN \propto \frac{1}{(-)charge}$
- (4)  $EN \propto (+)charge$ ,
- (5)  $EN \propto \% s\text{-characters of hybrid orbital}$

### Periodic trend :

- (1) Generally decreases on moving down the group
- (2) Generally increase when moving left to right in the period.

### Application of electronegative

- (1) Polarity of bond  
 $\Delta EN = 0$  non-polar bond  
 $\Delta EN \neq 0$  polar bond
- (2) Bond parameter  
 $\Delta EN \uparrow$  Ionic character  $\uparrow$   
 $\Delta EN \uparrow$  bond length  $\downarrow$   
 $\Delta EN \uparrow$  bond strength  $\uparrow$
- (3) Nature of oxide & oxyacid :  
 E.N. of central atom increase acidic character of oxide and oxyacid increases.

## Some Important Increasing Order

### 1. Abundance of Elements

- Elements on earth crust - Fe, Al, Si, O
- Metals on earth crust - Ca, Fe, Al
- Non-metals - Si, O  
 In atmosphere - O, N  
 In universe - He, Si, H

### 2. Atomic / Ionic Size

- $Mg^{2+}, Na^+, F^-, O^{2-}, N^{3-}$   
 (Hint : Isoelectronic series)
- $Ca^{2+}, Ar, Cl^-, S^{2-}$
- O, C, S, Se
- B, Be, Li, Na
- $F, O, F^-, O^{2-}$

### 3. Ionization Energy

- Na, Al, Mg, Si
- Li, B, Be, C, O, N, F, Ne, He (I<sup>st</sup> I.P.)
- Be, C, B, N, F, O, Ne, He, Li (II<sup>nd</sup> I.P.)

### 4. Electron Affinity

- I, Br, F, Cl
- Cu, Ag, Au (EA, of Au is very high = 222 kJ mol<sup>-1</sup>)
- O, S, F, Cl
- N, P, O, S

### 5. Electronegativity

- As, P, S, Cl
- I, Br, Cl, F
- C, N, O, F

### 6. Hydration of Ions/Hydration Energy

- $Ba^{+2}, Sr^{+2}, Ca^{+2}, Mg^{+2}, Be^{+2}$
- $Cs^+, Rb^+, K^+, Na^+, Li^+$
- $Na^+, Mg^{+2}, Al^{+3}$

### 7. Ionic Radii in water

- $Cs^+, Rb^+, K^+, Na^+, Li^+$
- $Li^+, Be^{+2}$
- $Na^+, Mg^{+2}, Al^{+3}$

### 8. Molar Conductivity in Water

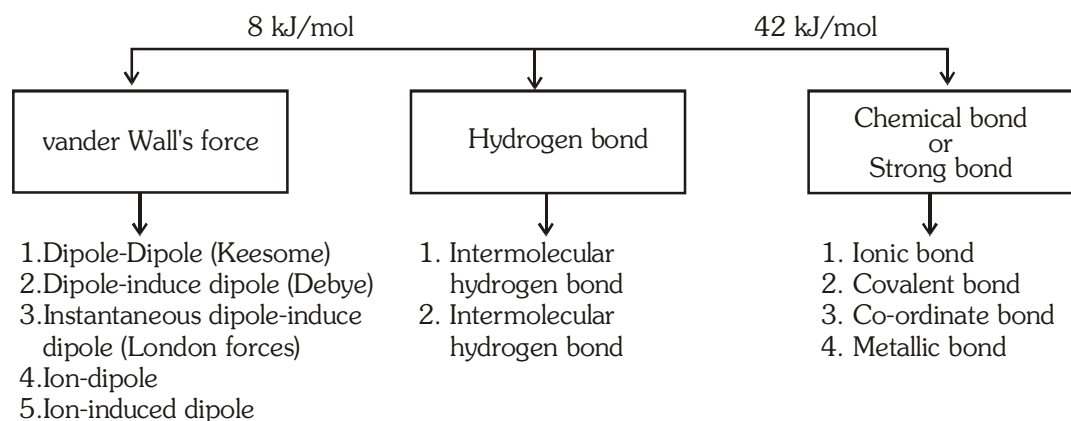
- $$Li^+, Na^+, K^+, Rb^+, Cs^+$$

**IMPORTANT NOTES**

# CHEMICAL BONDING

## CHEMICAL BOND :

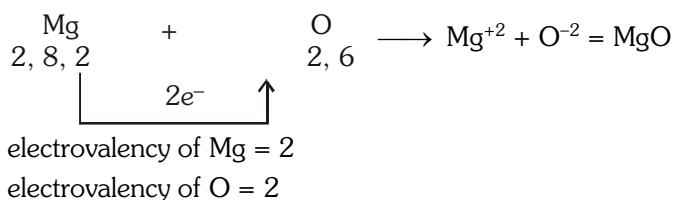
1. The force of attraction which hold together atom, molecule or ions with in chemical species.
2. It is always exothermic process.



## ❑ ELECTROVALENT OR IONIC BOND

The chemical bond formed between two or more atoms as a result of the complete transfer of one or more electrons from one atom to another is called as Ionic or electrovalent bond.

Ionic bonds are non-directional.



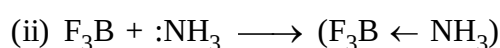
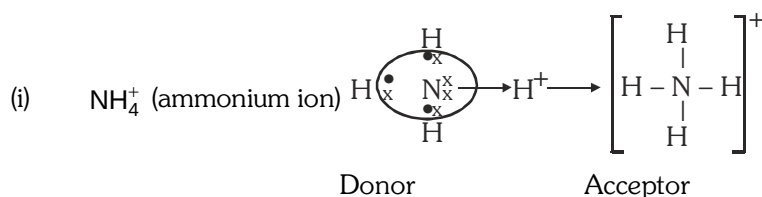
## ❑ COVALENT BOND

A covalent bond is formed by the mutual sharing of electrons between two atoms to complete their octet. (Except H which completes its duplet)



## ❑ COORDINATE BOND (DATIVE BOND)

The bond formed between two atom in which contribution of an electron pair is made by one of them while the sharing is done by both.



## EXCEPTION OF OCTET RULE

(a) electron deficient  
Central atom:  
No. of electron < 8  
 $\text{BeH}_2$   
 $\text{BF}_3$ ,  $\text{BCl}_3$ ,  $\text{BBr}_3$ ,  $\text{BI}_3$   
 $\text{AlCl}_3$ ,  $\text{AlBr}_3$ ,  $\text{AlI}_3$

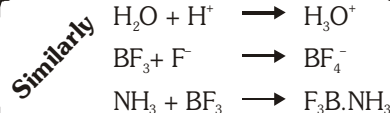
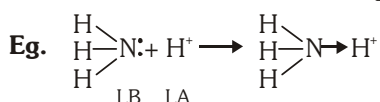
(b) electron rich  
Central atom:  
No. of electron > 8  
 $\text{PCl}_5$ ,  $\text{IF}_7$   
 $\text{SF}_6$ ,  $\text{XeF}_2$

(c) odd electron species  
Central atom :  
has odd electron  
 $\text{NO}$ ,  $\text{NO}_2$ ,  $\text{ClO}_2$   
 $\text{ClO}_3$

## CO-ORDINATE BOND (DATIVE BOND)

In this type of bond, shared pair of electron donates by one species but shared by both

- For this type of sharing
- One species - must have lone pair - act as donar known as Lewis base - acquire +ve charge.
- Another species - must have vacant orbital act as acceptor known as Lewis acid - acquire -ve charge.



- Donor atom follow octet rule

## MODERN APPROACH OF COVALENT BOND

Consider wave mechanical model of atom means electron has dual nature; wave nature as well as particle nature considered by these theories, there are two theories in this approach.

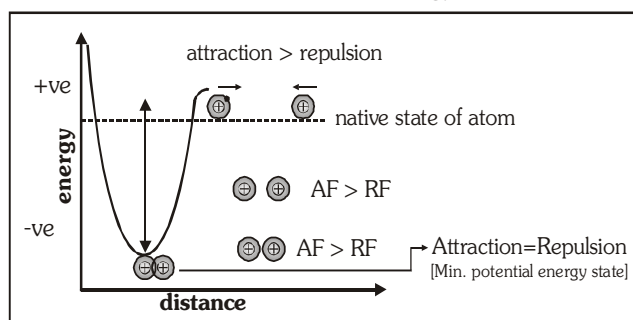
## 1. Valence Bond Theory

## 2. Molecular Orbital Theory

## VALENCE BOND THEORY

Proposed by Heitler & London as per VBT bonding takes place for attaining stability.

$$\text{Stability} \propto \frac{1}{\text{Potential energy}}$$



- Bond formation is an exothermic process.
- During this process some extent of electron cloud merge into each other; this part is known as overlapped region & this process is known as overlapping.

Atom  $\left\{ \begin{array}{l} \text{Nucleus} \\ \text{Shell - subshell - orbital - electron - cloud} \end{array} \right.$

- Only those orbitals of valence shell can exhibit overlapping which has

- Unpaired electron **For example**  
 $\text{H}-\text{Cl}$  bond form by overlapping of  $1s$  -  $3p$  orbitals.  
 $\text{H} \rightarrow 1s^1$   
 $\text{Cl} \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^5$

- Opposite spin

- **Strength of Covalent Bond**

Strength of covalent bond  $\propto$  extent of overlapping.

## 1. NATURE OF ORBITALS

- (a) No. of shell : lower the number of shell higher overlapping.

$$\text{Bond Strength} \propto \frac{1}{\text{No. of shell}} / \text{size of orbitals}$$

$$1-1 > 1-2 > 2-2 > 2-3$$

- **Exception** :  $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$  due to  
 $\text{O}-\text{O} < \text{S}-\text{S}$  lp-lp  
 $\text{N}-\text{N} < \text{P}-\text{P}$  repulsion

**(b) Type of Sub-shell**

Valence shell contain subshell s & p  
s-non-directional | Directional orbital has  
p-directional | higher extent of overlapping

**Possible combination & strength of overlapping**

$$s-s < s-p < p-p$$

\*\* This factor is applicable when number of shell is same otherwise shell factor prominent  
2s - 2s < 2s-2p < 2p-2p sub-shell factor  
1s - 1s > 1s-2s > 1s-3s shell factor

**2. PATTERN OF OVERLAPPING**

**(a) Axial overlapping :**

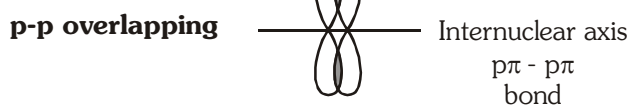
Along the internuclear axis; form sigma ( $\sigma$ ) bond, strong bond.



**(b) Co-lateral overlapping**

Side wise overlapping has less extent of overlapping form  $\pi$ - bond

Weak bond



- In case of multiple bond between two atom one bond is sigma and rest are pi-bonds.
- VBT was not able to define geometry of molecule therefore a new concept came into existence known as hybridisation.

**HYBRIDISATION**

- Intermixing of atomic orbitals and formation of new orbital, these orbitals are known as hybrid orbital and this concept is known as hybridisation.
- It is hypothetical concept.
- Only those orbitals can participate in hybridisation which has slight difference in energy.
- No. of hybrid orbitals : No. of atomic orbitals participate in intermixing
- Hybrid orbitals oriented at maximum possible distance three dimensionally.
- On the basis of type of orbitals participating in hybridisation, we can divide hybridisation into following categories.

| S.No. | Type of orbital           | No. of hybrid orbital             | 3D orientation         | Example                              |
|-------|---------------------------|-----------------------------------|------------------------|--------------------------------------|
| 1.    | one s + one p             | 2; sp                             | Linear                 | BeH <sub>2</sub> , BeCl <sub>2</sub> |
| 2.    | one s + two p             | 3; sp <sup>2</sup>                | Triangular             | BCl <sub>3</sub> , BF <sub>3</sub>   |
| 3.    | one s + three p           | 4; sp <sup>3</sup>                | Tetrahedral            | CH <sub>4</sub> , CCl <sub>4</sub>   |
| 4.    | one s + three p + one d   | 5; sp <sup>3</sup> d              | Triangular bipyramidal | PCl <sub>5</sub>                     |
| 5.    | one s + three p + two d   | 6; sp <sup>3</sup> d <sup>2</sup> | Octahedral             | SF <sub>6</sub>                      |
| 6.    | one s + three p + three d | 7; sp <sup>3</sup> d <sup>3</sup> | Pentagonal bipyramidal | IF <sub>7</sub>                      |

## VALENCE SHELL ELECTRON PAIR REPULSION THEORY

- Given by Nyholm & Gillespie to define shape of molecule.
- Shape of molecule define on the basis of electron pairs orientation present on central atom.
- Electron pairs present on central atom repel each other therefore these electron pair occupy such position on central atom; where they experience minimum repulsion at maximum possible distance three dimensionally.
- **Order of repulsion :**  $lp-lp > lp-bp > bp-bp$        $mb-mb > mb-sb > sb-sb$  (mb = multiple bond; sb = single bond)

## TYPE OF HYBRIDISATION &amp; POSSIBLE STRUCTURE

| Type of Hybridisation                                                                                                                                          | No. of B.P.      | No. of L.P.      | Shape                                                                                               | Examples                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. sp-hybridisation                                                                                                                                            | 2                | -                | Linear                                                                                              | BeF <sub>2</sub> , CO <sub>2</sub> , CS <sub>2</sub> , BeCl <sub>2</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2. (a) sp <sup>2</sup> -hybridisation<br>(b) sp <sup>2</sup> -hybridisation                                                                                    | 3<br>2           | -<br>1           | Trigonal planar<br>V-shape, Angular                                                                 | BF <sub>3</sub> , AlCl <sub>3</sub> , BeF <sub>3</sub> <sup>-</sup><br>NO <sub>2</sub> <sup>-</sup> , SO <sub>2</sub> , O <sub>3</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 3. (a) sp <sup>3</sup> -hybridisation<br>(b) sp <sup>3</sup> -hybridisation<br>(c) sp <sup>3</sup> -hybridisation                                              | 4<br>3<br>2      | 0<br>1<br>2      | Tetrahedral<br>Pyramidal<br>V-shape<br>Angular                                                      | CH <sub>4</sub> , CCl <sub>4</sub> , PCl <sub>4</sub> <sup>+</sup> , ClO <sub>4</sub> <sup>-</sup> , NH <sub>4</sub> <sup>+</sup> , BF <sub>4</sub> <sup>-</sup> , SO <sub>4</sub> <sup>2-</sup> , AlCl <sub>4</sub> <sup>-</sup><br>NH <sub>3</sub> , PF <sub>3</sub> , ClO <sub>3</sub> <sup>-</sup> , H <sub>3</sub> O <sup>+</sup> , PCl <sub>3</sub> , XeO <sub>3</sub> , N(CH <sub>3</sub> ) <sub>3</sub> , CH <sub>3</sub> <sup>-</sup><br>H <sub>2</sub> O, NH <sub>2</sub> <sup>-</sup><br>OF <sub>2</sub> , Cl <sub>2</sub> O, SF <sub>2</sub> , I <sub>3</sub> <sup>+</sup> |
| 4. (a) sp <sup>3</sup> d-hybridisation<br>(b) sp <sup>3</sup> d-hybridisation<br>(c) sp <sup>3</sup> d-hybridisation<br>(d) sp <sup>3</sup> d-hybridisation    | 5<br>4<br>3<br>2 | -<br>1<br>2<br>3 | Trigonal bipyramidal<br>See-Saw,<br>folded square distorted tetrahedral<br>almost T-shape<br>Linear | PCl <sub>5</sub> , SOF <sub>4</sub> , AsF <sub>5</sub><br>SF <sub>4</sub> , PF <sub>4</sub> <sup>-</sup> , AsF <sub>4</sub> <sup>-</sup><br>SbF <sub>4</sub> <sup>-</sup> , XeO <sub>2</sub> F <sub>2</sub><br>ClF <sub>3</sub> , ICl <sub>3</sub><br>I <sub>3</sub> <sup>-</sup> , Br <sub>3</sub> <sup>-</sup> , ICl <sub>2</sub> <sup>-</sup> , ClF <sub>2</sub> <sup>-</sup> , XeF <sub>2</sub>                                                                                                                                                                                    |
| 5. (a) sp <sup>3</sup> d <sup>2</sup> -hybridisation<br>(b) sp <sup>3</sup> d <sup>2</sup> -hybridisation<br>(c) sp <sup>3</sup> d <sup>2</sup> -hybridisation | 6<br>5<br>4      | -<br>1<br>2      | Square bipyramidal/octahedral<br>Square pyramidal/distorted octahedral<br>Square planar             | PCl <sub>6</sub> <sup>-</sup> , SF <sub>6</sub><br>XeOF <sub>4</sub> , ClF <sub>5</sub> , SF <sub>5</sub> <sup>-</sup> , XeF <sub>5</sub> <sup>+</sup><br>XeF <sub>4</sub>                                                                                                                                                                                                                                                                                                                                                                                                             |
| 6. (a) sp <sup>3</sup> d <sup>3</sup> -hybridisation<br>(b) sp <sup>3</sup> d <sup>3</sup> -hybridisation<br>(c) sp <sup>3</sup> d <sup>3</sup> -hybridisation | 7<br>6<br>5      | -<br>1<br>2      | Pentagonal bipyramidal<br>distorted octahedral / capped octahedral<br>Pentagonal planar             | IF <sub>7</sub><br>XeF <sub>6</sub><br>XeF <sub>5</sub> <sup>-</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## DIPOLE MOMENT

Measurement of Polarity in a molecule

$$\vec{\mu} = q \times d \quad \text{debye} = \text{esu-cm} \\ 1D = 10^{-18} \text{ esu.cm}$$

## (A) Identification of polar or Non-polar molecule.

Molecule : Symmetrical distribution of electron cloud- Non-polar.  
Molecule : Unsymmetrical distribution of electron cloud- Polar.

## Diatomic Molecule

(a) Homoatomic  $\Delta EN = 0 \rightarrow \vec{\mu} = 0 \rightarrow$  Non-polar  
H<sub>2</sub>, F<sub>2</sub>, Cl<sub>2</sub>, N<sub>2</sub> etc.

(b) Heteroatomic  $\Delta EN \neq 0 \rightarrow \vec{\mu}_{\text{net}} \neq 0 \rightarrow$  polar  
HF > HCl > HBr > HI

## Polyatomic molecule :

 $\mu_R \rightarrow$  Vector sum of bond moment

$$\mu_R \rightarrow \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2 \cos \theta}$$

## Important Order

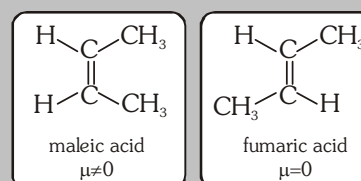
NH<sub>3</sub> > NI<sub>3</sub> > NBr<sub>3</sub> > NCl<sub>3</sub> > NF<sub>3</sub>NH<sub>3</sub> > SbH<sub>3</sub> > AsH<sub>3</sub> > PH<sub>3</sub>H<sub>2</sub>O > H<sub>2</sub>SCH<sub>3</sub>Cl > CH<sub>3</sub>F > CH<sub>3</sub>Br > CH<sub>3</sub>ICH<sub>3</sub>Cl > CH<sub>2</sub>Cl<sub>2</sub> > CHCl<sub>3</sub> > CCl<sub>4</sub>

## Applications

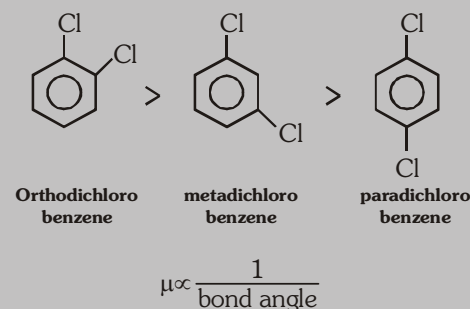
Predict shape and polarity of molecule

Symmetrical geometry  $\rightarrow \mu = 0 \rightarrow$  non-polarUnsymmetrical geometry  $\rightarrow \mu \neq 0 \rightarrow$  polar

Distinguish between cis &amp; trans form



## Dipole moment in Aromatic Compounds



## HYDROGEN BONDING

- ⇒ It is dipole-dipole type of interaction.
- ⇒ Electrostatic force of attraction between hydrogen (covalently bond with F/N/O) & highly electronegative atom.

- ⇒ Hydrogen bond
  - Intermolecular hydrogen bond → between the molecules
  - Intramolecular hydrogen bond → within the molecules

\* Intramolecular H-bonding takes place mainly in ortho derivatives of benzene.

### Note :

1. Boric acid solid at room temperature (with 2D sheet structure) due to intermolecular hydrogen bonding.
2. In vapour state or in non-polar solvent  $\text{CH}_3\text{COOH}$  as dimer due to intermolecular hydrogen bonding.
3. In vapour phase HF exist as dimer and  $(\text{HF})_6$ , due to intermolecular hydrogen bonding.
4. Due to intermolecular hydrogen bonding ice has 3D network structure with tetrahedral unit and having open cage structure.  
The density of ice is less than water.
5. DNA having hydrogen bonds.
6. In hydrated chloral intramolecular hydrogen bond is present.

| Strength                                                          |           |                   |
|-------------------------------------------------------------------|-----------|-------------------|
| Intermolecular H-bond > Intramolecular H-bond                     |           |                   |
| ● Intramolecular H-bonding takes place in ortho derivatives only. |           |                   |
| Applications of H-bonding                                         |           |                   |
| Physical State (dense nature)                                     | $\propto$ | H-bond            |
| Melting Point (mp)                                                | $\propto$ | H-bond            |
| Boiling Point (bp)                                                | $\propto$ | H-bond            |
| Viscosity                                                         | $\propto$ | H-bond            |
| Surface Tension                                                   | $\propto$ | H-bond            |
| Volatility                                                        | $\propto$ | $1/\text{H-bond}$ |
| Vapour Pressure                                                   | $\propto$ | $1/\text{H-bond}$ |

## MOLECULAR ORBITAL THEORY (MOT)

The molecular orbital theory was developed by F. Hund and R.S. Mulliken in 1932. The salient features are:

- (i) Just as electrons of any atom are present in various atomic orbitals, electrons of the molecule are present in various molecular orbitals.
- (ii) Molecular orbitals are formed by the combination of atomic orbitals of comparable energies and proper symmetry.
- (iii) An electron in an atomic orbital is influenced by one nucleus, while in a molecular orbital it is influenced by two or more nuclei depending upon the number of the atoms in the molecule. **Thus an atomic orbital is monocentric while a molecular orbital is polycentric.**
- (iv) The number of molecular orbitals formed is equal to the number of combining atomic orbitals. When two atomic orbitals combine, two molecular orbitals called **bonding molecular orbital** and **anti-bonding molecular orbital** are formed.
- (v) The bonding molecular orbital has lower energy and hence greater stability than the corresponding antibonding molecular orbital.

**Formation of Molecular Orbitals : Linear Combination of Atomic Orbitals(LCAO)**

**Case I :** When two waves are in same phase (constructive interference) the wave adds up and amplitude of new wave is the sum of wave functions of individual atomic orbitals.

$$\psi_{MO} = \psi_A + \psi_B \quad (\text{Bonding M.O.})$$

**Case II :** When two waves are out of phase, the waves are subtracted from each other so that the amplitude of new wave is :

$$\psi_{MO}^* = \psi_A - \psi_B \quad (\text{Antibonding M.O.})$$

**Condition for combination atomic orbitals :**

1. The combining atomic orbitals must have the same or nearly the same energy.
2. The combining atomic orbitals must have the same symmetry about the molecular axis.
3. The combining atomic orbitals must overlap to the maximum extent.

**Energy level diagram from MOs :****Molecular orbital energy diagram for up to  $N_2$  (molecule having  $\leq 14$  electrons)**

$$\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \pi_{2p_x} = \pi_{2p_y} < \sigma_{2p_z} < \pi_{2p_x}^* = \pi_{2p_y}^* < \sigma_{2p_z}^*$$

**Molecular orbital energy diagram for  $O_2$  and  $F_2$  (molecule having  $> 14$  electrons)**

$$\sigma_{1s} < \sigma_{1s}^* ; < \sigma_{2s} < \sigma_{2s}^* < \sigma_{2p_z} < \pi_{2p_x} = \pi_{2p_y} < \pi_{2p_x}^* = \pi_{2p_y}^* < \sigma_{2p_z}^*$$

$\sigma^*, \pi^* =$  antibonding molecular orbital

$\sigma, \pi =$  bonding molecular orbital

**Rules of Filling up of Molecular Orbital with Electrons :**

- (1) The molecular orbital with lower energy will be filled first. (Aufbau Principle)
- (2) The molecular orbital can accommodate maximum only two electrons. (Pauli's exclusion principle)
- (3) If the two MOs have same energy then molecular orbital will first get singly filled and after that pairing will start. (Hunds Rule)

**□ BOND ORDER**

**Bond order (B.O.)** is defined as follows  $\text{Bond order (B.O.)} = \frac{1}{2} (N_b - N_a)$

A positive bond order (i.e.,  $N_b > N_a$ ) means a stable molecule while a negative value (i.e.,  $N_b < N_a$ ) (i.e.,  $N_b = N_a$ ) bond order means an unstable molecule. If bond order zero then molecular does not exist.

**□ NATURE OF THE BOND**

Integral bond order values of 1, 2 or 3 correspond to single, double or triple bonds respectively.

**□ BOND-LENGTH**

The bond order between two atoms in a molecule may be taken as an approximate measure of the bond length. The bond length decreases as bond order increases.

**□ MAGNETIC NATURE**

If one or more molecular orbitals are singly occupied it is paramagnetic (attracted by magnetic field), e.g.,  $O_2$  molecule. Otherwise diamagnetic (eg :  $N_2$ )



**Fractional bond order it will be always paramagnetic.**

| Sr. No.                                                                                                                                                          | No. of electrons in molecules | N <sub>b</sub> | N <sub>a</sub> | B.O. | Paramagnetic / diamagnetic                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------|----------------|------|--------------------------------------------|
| 1                                                                                                                                                                | 1                             | 1              | 0              | ½    | paramagnetic                               |
| 2                                                                                                                                                                | 2                             | 2              | 0              | 1    | diamagnetic                                |
| 3                                                                                                                                                                | 3                             | 2              | 1              | 0.5  | paramagnetic                               |
| 4                                                                                                                                                                | 4                             | 2              | 2              | 0    | —                                          |
| 5                                                                                                                                                                | 5                             | 3              | 2              | ½    | paramagnetic                               |
| 6                                                                                                                                                                | 6                             | 4              | 2              | 1    | diamagnetic                                |
| 7                                                                                                                                                                | 7                             | 4              | 3              | ½    | paramagnetic                               |
| 8                                                                                                                                                                | 8                             | 4              | 4              | 0    | —                                          |
| 9                                                                                                                                                                | 9                             | 5              | 4              | ½    | paramagnetic                               |
| 10                                                                                                                                                               | 10                            | 6              | 4              | 1    | paramagnetic                               |
| 11                                                                                                                                                               | 11                            | 7              | 4              | 1.5  | paramagnetic                               |
| 12                                                                                                                                                               | 12                            | 8              | 4              | 2    | both bond are π<br>C <sub>2</sub> molecule |
| 13                                                                                                                                                               | 13                            | 9              | 4              | 2.5  | paramagnetic                               |
| 14                                                                                                                                                               | 14                            | 10             | 4              | 3    | diamagnetic                                |
| 15                                                                                                                                                               | 15                            | 10             | 5              | 2.5  | paramagnetic                               |
| 16                                                                                                                                                               | 16                            | 10             | 6              | 2    | paramagnetic                               |
| 17                                                                                                                                                               | 17                            | 10             | 7              | 1.5  | paramagnetic                               |
| 18                                                                                                                                                               | 18                            | 10             | 8              | 1    | diamagnetic                                |
| 19                                                                                                                                                               | 19                            | 10             | 9              | 0.5  | paramagnetic                               |
| 20                                                                                                                                                               | 20                            | 10             | 10             | 0    | —                                          |
| <p>In case of same bond order, stability depends upon<br/>No. of anti-bonding electrons</p> $\text{Stability} \propto \frac{1}{\text{No. of anti-bonding } e^-}$ |                               |                |                |      |                                            |

### BONDING PARAMETER

**1. Bond length** : Internuclear distance

### Factor affecting Bond length

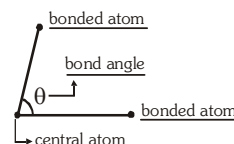
- (i) Atomic size : bond length  $\propto$  size [No. of shell]  
(ii)  $\Delta EN$ , Bond length  $\propto \frac{1}{\Delta EN}$

$$d_{A-B} = r_A + r_B - 0.09 \times \Delta EN \text{ \AA}$$

- (iii) Bond order : Bond length  $\propto \frac{1}{\text{B.O.}}$
  - (iv) Hybridisation : Bond length  $\propto \frac{1}{\% \text{age of s-character}}$

**Bond Angle :**

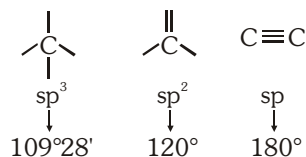
Angle between two adjacent bond is known as bond angle.



### FACTORS AFFECTING BOND ANGLE

### (i) Hybridisation

Bond angle  $\propto$  %age of s-character



## (ii) No. of lp/bp

[when hybridisation is same]

Bond angle  $\propto \frac{1}{lp}$  Eg. :  $\text{CH}_4 > \ddot{\text{N}}\text{H}_3 > \text{H}_2\ddot{\text{O}}:$

**(iii) Type of Central atom:**Applicable when :

- \* hybridisation same      \* No. of lp/bp same

Bond angle  $\propto$  EN of central atom

**Eg.**  $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$

**(iv) Type of bonded atom :** Applicable when

- \* hybridisation - same

- \* No. of lp/bp - same

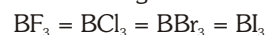
- \* Central atom - same

Bond angle  $\propto$  size of bonded species

Eg.  $\text{OF}_2 < \text{OCl}_2 < \text{OBr}_2 < \text{OI}_2$

**Note :**

Regular geometry / same hybridisation/  
bond angle same





### FACTORS AFFECTING POLARISATION

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(1) Polarisation <math>\propto</math> charge of cation or anion</p> <p><b>Eg.</b> (i) <math>\text{CrO}^{+2} &lt; \text{Cr}_2\text{O}_3^{+3} &lt; \text{CrO}_3^{+6}</math> Covalent character <math>\uparrow</math></p> <p>(ii) <math>\text{SF}_2^{+2} &lt; \text{SF}_4^{+4} &lt; \text{SF}_6^{+6}</math> Covalent character <math>\uparrow</math></p> <p>(iii) <math>\text{LiF}^{-1} &lt; \text{Li}_2\text{O}^{-2} &lt; \text{Li}_3\text{N}^{-3}</math> Covalent character <math>\uparrow</math> (anion charge <math>\uparrow</math>)</p> | <p>(2) Polarisation <math>\propto \frac{\text{size of anion}}{\text{size of cation}}</math></p> <p><b>Eg.</b></p> <p>(i) <math>\text{LiF} &lt; \text{LiCl} &lt; \text{LiBr} &lt; \text{LiI}</math> ————— anion size <math>\uparrow</math><br/>polarisability <math>\uparrow</math><br/>covalent character <math>\uparrow</math></p> <p>(ii) <math>\text{BeCl}_2 &gt; \text{MgCl}_2 &gt; \text{CaCl}_2 &gt; \text{SrCl}_2 &gt; \text{BaCl}_2</math> ————— cation size <math>\uparrow</math><br/>polarisation <math>\downarrow</math><br/>covalent character <math>\downarrow</math></p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### SOLUBILITY

**For s-block  
same group cation depends upon  
Lattice Energy and Hydration Energy**

#### Increasing order of solubility

- (i)  $\text{BaCO}_3, \text{SrCO}_3, \text{CaCO}_3, \text{MgCO}_3, \text{BeCO}_3$
- (ii)  $\text{Be(OH)}_2, \text{Sr(OH)}_2, \text{Mg(OH)}_2, \text{Ca(OH)}_2, \text{Ba(OH)}_2$
- (iii)  $\text{BaSO}_4, \text{SrSO}_4, \text{CaSO}_4, \text{MgSO}_4, \text{BeSO}_4$
- (iv)  $\text{Li}_2\text{CO}_3, \text{Na}_2\text{CO}_3, \text{K}_2\text{CO}_3, \text{Rb}_2\text{CO}_3, \text{CsCO}_3$
- (v)  $\text{LiOH}, \text{NaOH}, \text{KOH}, \text{RbOH}, \text{CsOH}$
- (vi)  $\text{LiF}, \text{LiCl}, \text{LiBr}, \text{LiI}$
- (vii)  $\text{LiF}, \text{NaF}, \text{KF}, \text{RbF}, \text{CsF}$
- (viii)  $\text{BaF}_2, \text{SrF}_2, \text{MgF}_2, \text{CaF}_2, \text{BeF}_2$
- (ix)  $\text{CaF}_2, \text{CaCl}_2, \text{CaBr}_2, \text{CaI}_2$

**For all**  $\text{solubility} \propto \frac{1}{\text{cov. char.}}$

**solubility in org. solvent  $\propto \text{cov. char} \propto \frac{1}{\text{ionic char}}$**   
[CCl<sub>4</sub>, benzene, ether, alcohol, acetone]

- Eg.** (i)  $\text{PbF}_2 > \text{PbCl}_2 > \text{PbBr}_2 > \text{PbI}_2$   
(Anion size  $\uparrow$ , cov. char.  $\uparrow$ , solubility  $\downarrow$ )
- (ii)  $\text{Fe}^{+2}(\text{OH})_2 > \text{Fe}^{+3}(\text{OH})_3$   
(+) charge  $\uparrow$ , PP  $\uparrow$ , CC  $\uparrow$ , solubility  $\downarrow$
- (iii)  $\text{ZnCl}_2 > \text{CdCl}_2 > \text{HgCl}_2$   
 $Z_{\text{eff}} \uparrow$ , PP  $\uparrow$ , CC  $\uparrow$ , solubility  $\downarrow$
- (iv)  $\text{Na}_2\text{SO}_4 > \text{MgSO}_4$   
(+) charge  $\uparrow$ , PP  $\uparrow$ , CC  $\uparrow$ , solubility  $\downarrow$
- (v)  $\text{ZnCl}_2 > \text{CdCl}_2 > \text{HgCl}_2$   
 $Z_{\text{eff}} \uparrow$ , PP  $\uparrow$ , CC  $\uparrow$ , solubility  $\downarrow$
- (vi)  $\text{NaCl} > \text{CuCl}$   
PP  $\uparrow$ , CC  $\uparrow$ , solubility  $\downarrow$
- (vii)  $\text{AgF} > \text{AgCl} > \text{AgBr} > \text{AgI}$   
Anionic Size  $\uparrow$ , PP  $\uparrow$ , CC  $\uparrow$ , solubility  $\downarrow$

### HYBRIDISATION OF FOLLOWING SPECIES IN SPECIFIED STATE

| Species                          | Cationic part                            | Anionic part                             |
|----------------------------------|------------------------------------------|------------------------------------------|
| $\text{PCl}_5$                   | $\text{PCl}_4^+ (\text{sp}^3)$           | $\text{PCl}_6^- (\text{sp}^3\text{d}^2)$ |
| $\text{PBr}_5$                   | $\text{PBr}_4^+ (\text{sp}^3)$           | $\text{Br}^-$                            |
| $\text{XeF}_6$                   | $\text{XeF}_5^+ (\text{sp}^3\text{d}^2)$ | $\text{F}^-$                             |
| $\text{N}_2\text{O}_5$           | $\text{NO}_2^+ (\text{sp})$              | $\text{NO}_3^- (\text{sp}^2)$            |
| $\text{I}_2\text{Cl}_6$ (liquid) | $\text{ICl}_2^+ (\text{sp}^3)$           | $\text{ICl}_4^- (\text{sp}^3\text{d}^2)$ |
| $\text{Cl}_2\text{O}_6$          | $\text{ClO}_2^+ (\text{sp}^2)$           | $\text{ClO}_4^- (\text{sp}^3)$           |
| $\text{I}_2$ (molten state)      | $\text{I}_3^+ (\text{sp}^3)$             | $\text{I}_3^- (\text{sp}^3\text{d})$     |

## SILICATES

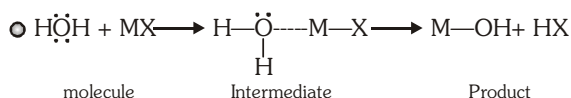
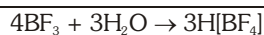
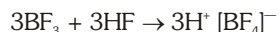
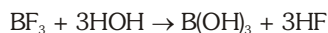
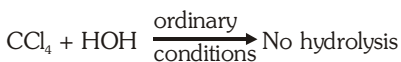
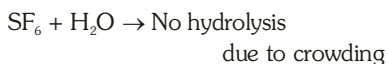
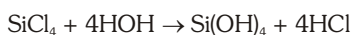
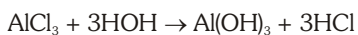
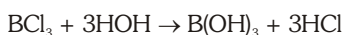
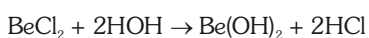
| Silicates                | Sharing of O-atom / Basic Tetrahedral unit | Contribution of O-atom/Basic Tetrahedral unit | General formula                      |
|--------------------------|--------------------------------------------|-----------------------------------------------|--------------------------------------|
| Ortho                    | 0                                          | 4                                             | $\text{SiO}_4^{4-}$                  |
| Pyro                     | 1                                          | 3.5                                           | $\text{Si}_2\text{O}_7^{6-}$         |
| Cyclic                   | 2                                          | 3                                             | $(\text{SiO}_3)_n^{2n-}$             |
| Single chain (pyroxene)  | 2                                          | 3                                             | $(\text{SiO}_3)_n^{2n-}$             |
| Double chain (Amphibole) | (3, 2) avg. = 2.5                          | $\frac{11}{4} = \left(\frac{5.5}{2}\right)$   | $(\text{Si}_4\text{O}_{11})_n^{6n-}$ |
| 2D or (Sheet)            | 3                                          | 2.5                                           | $(\text{Si}_2\text{O}_5)_n^{2n-}$    |
| 3D                       | 4                                          | 2                                             | $(\text{SiO}_2)_n$                   |

## HYDROLYSIS

Hydro - Water  
lysis - break down

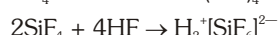
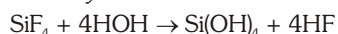
Break down of a molecule through water and formation of new product is known as hydrolysis.

- It is nucleophilic substitution reaction.

extent of hydrolysis  $\propto$  covalent character.

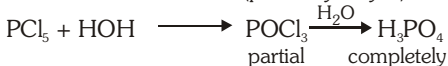
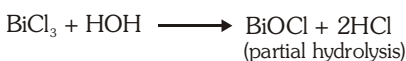
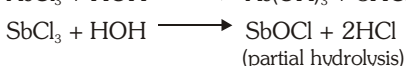
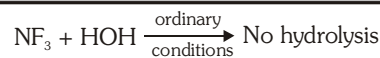
(partially hydrolysis)

Similarly



hydrolysis followed by  
Lewis acid-base reaction.

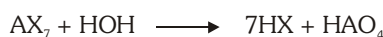
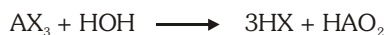
## 15th Group Halides



## Hydrolysis of higher covalent character containing salt

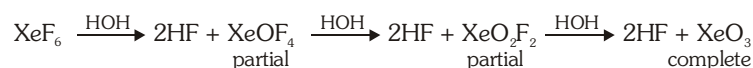
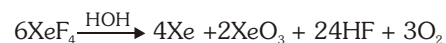
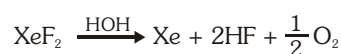


## Hydrolysis of Interhalogen Compounds



HX Hydrohalic acid  
HOA, HAO<sub>2</sub>, HAO<sub>3</sub>, HAO<sub>4</sub>  
oxyacid of halogen

## Some specific hydrolysis



### Back bonding :

It is type of  $\pi$ -interaction between lone pair & vacant orbital of adjacent atom in molecule.

**Condition :**

- (i) One atoms must have lone pair and another atoms must have vacant orbital.

**Type of back bond :**

- (i)  $(p\pi - p\pi)$  type of back bond  
eg.  $\text{BF}_3$ ,  $\text{BCl}_3$ ,  $\text{CF}_2$ ,  $\text{CCl}_2$
- (ii)  $(p\pi - d\pi)$  type of back bond  
eg.  $\text{CCl}_3^-$ ,  $\text{O}(\text{SiH}_3)_3$ ,  $\text{N}(\text{SiH}_3)_2$ ,  $\text{SiH}_5\text{O}^-$ , etc.

### Application of back bonding :

**Lewis acid character :**

$$\begin{array}{l} \text{BF}_3 < \text{BCl}_3 < \text{BrBr}_3 < \text{BI}_3 \\ \text{BeF}_2 < \text{BeCl}_2 < \text{BeBr}_2 < \text{BeI}_2 \\ \text{SiF}_4 > \text{SiCl}_4 > \text{SiBr}_4 > \text{SiI}_4 \end{array}$$

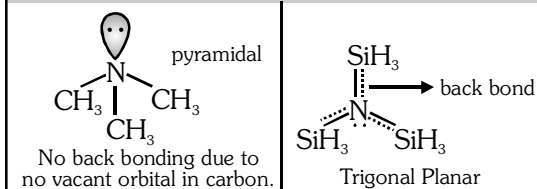
**Lewis basic character :**

$$\begin{array}{l} \text{N(CH}_3)_3 > \text{N(SiH}_3)_3 \\ \text{O(CH}_3)_3 > \text{O(SiH}_3)_3 \end{array}$$

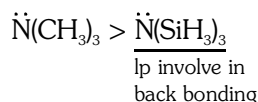
**Note :** Due to back bonding  $\underline{\text{B}}_3\text{N}_3\text{H}_6$ ,  $(\underline{\text{B}}\text{O}_2)_3^{3-}$ ,  $\underline{\text{N}}(\text{SiH}_3)_3$  is planar around under lined atom.

## Few more examples of back bonding

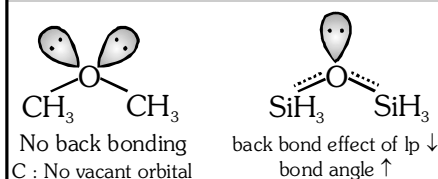
1. Shape of trimethyl amine pyramidal while shape of trisilyl amine is triangular planar.



## 2. Lewis base strength



3. Bond angle of  $(\text{CH}_3)_2\text{O}$  is very less than  $(\text{SiH}_3)_2\text{O}$



### Chemical Species having multicentered bond

(3c-2e) bond is also termed as banana bond.

Bridge bond is stronger than terminal bond.

Bridge bond is longer than terminal bond.

|                              | <b>Brigde<br/>bond</b> | <b>Hybridisation<br/>of central<br/>atom</b> |
|------------------------------|------------------------|----------------------------------------------|
| $\text{Be}_2\text{Cl}_4$     | (3c-4e)                | $\text{sp}^2$                                |
| $(\text{BeCl}_2)_n$          | (3c-4e)                | $\text{sp}^3$                                |
| $\text{Al}_2\text{Cl}_6$     | (3c-4e)                | $\text{sp}^3$                                |
| $\text{I}_2\text{Cl}_6$      | (3c-4e)                | $\text{sp}^3\text{d}^2$                      |
| $\text{B}_2\text{H}_6$       | (3c-2e)                | $\text{sp}^3$                                |
| $\text{Be}_2\text{H}_4$      | (3c-2e)                | $\text{sp}^2$                                |
| $(\text{BeH}_2)_n$           | (3c-2e)                | $\text{sp}^3$                                |
| $(\text{AlH}_3)_n$           | (3c-2e)                | $\text{sp}^3\text{d}^2$                      |
| $\text{Al}_2(\text{CH}_3)_6$ | (3c-2e)                | $\text{sp}^3$                                |

**Odd  $e^-$  species :** Total number of electron or valance electron in odd number.

|                                                                       | Hybridisation | Shape           | Magnetic behaviour |
|-----------------------------------------------------------------------|---------------|-----------------|--------------------|
| $\text{NO}_2$                                                         | $\text{sp}^2$ | V shape         | Para               |
| $\text{ClO}_2$                                                        | $\text{sp}^2$ | V shape         | Para               |
| $\text{ClO}_3$                                                        | $\text{sp}^3$ | Pyramidal       | Para               |
| $\bullet\text{CH}_3$                                                  | $\text{sp}^2$ | Trigonal planar | Para               |
| $\bullet\text{CH}_3 / \bullet\text{CHF}_2$<br>$/ \text{CH}_2\text{F}$ | $\text{sp}^3$ | Pyramidal       | Para               |

## OXY-ACIDS

- Mainly oxy-acids are hydroxide of Non-metal oxides.
- No. of  $H^+$  ion furnish by an oxyacid is known as their basicity.  
Oxyacid obtained by dissolving non-metal oxide in water.  
Eg.  $CO_2 + HOH \rightarrow H_2CO_3$  or  $OC(OH)_2$   
**Here :**  $CO_2 \rightarrow$  Non metal oxide - Anhydride of carbonic acid  
 $OC(OH)_2 \rightarrow$  Oxyacid
- $NO_2 \longrightarrow$  Mixed anhydride
- it gives  $\longrightarrow HNO_2$  &  $HNO_3$

## Oxide      Acid

- $N_2O_3 \longrightarrow HNO_2$  — Nitrous acid
- $N_2O_5 \longrightarrow HNO_3$  — Nitric acid
- $P_4O_{10} \longrightarrow H_3PO_4$  — Phosphoric acid
- $SO_2 \longrightarrow H_2SO_3$  — Sulphurous acid
- $SO_3 \longrightarrow H_2SO_4$  — Sulphuric acid
- $Cl_2O_7 \longrightarrow HClO_4$  — Perchloric acid
- Oxyacids of different elements

Order of acidic strength  
 $H_3PO_2 > H_3PO_3 > H_3PO_4$

Reducing nature  
 $H_3PO_2 > H_3PO_3 > H_3PO_4$

|   | Element    | Oxide    | Oxyacid                                                                                                                                                                                                                                                              | Basicity                                  |
|---|------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| 1 | Boron      | $B_2O_3$ | $B(OH)_3$<br>boric acid                                                                                                                                                                                                                                              | Not protonic acid<br>monobasic Lewis acid |
| 2 | Carbon     | $CO_2$   | $H_2CO_3$<br>carbonic acid                                                                                                                                                                                                                                           | Two                                       |
| 3 | Nitrogen   |          | $H_2N_2O_2$<br>Hyponitrous acid<br>$HNO_2$<br>Nitrous acid<br>$HNO_3$<br>Nitric acid<br>$HNO_4$<br>Pernitric acid                                                                                                                                                    |                                           |
| 4 | Phosphorus |          | $H_3PO_2$<br>Hypophosphorous acid<br>$H_3PO_3$<br>Ortho Phosphorous acid<br>$H_3PO_4$<br>Ortho phosphoric acid<br>$HPO_3$<br>Meta phosphoric acid<br>$H_4P_2O_5$<br>Pyrophosphorous acid<br>$H_4P_2O_7$<br>Pyrophosphoric acid<br>$H_4P_2O_6$<br>Hypophosphoric acid |                                           |

## OXYACIDS OF SULPHUR

- Sulphurous acid -  $H_2SO_3$
- Sulphuric acid -  $H_2SO_4$
- Thiosulphuric acid -  $H_2S_2O_3$
- Peroxymonosulphuric (Caro's acid) -  $H_2SO_5$  (Peroxide bond)
- Peroxydisulphuric acid (Marshall's acid) -  $H_2S_2O_8$  (Peroxide bond)
- Pyrosulphurous acid -  $H_2S_2O_5$  (S-S linkage)
- Pyrosulphuric acid -  $H_2S_2O_7$  (S-O-S linkage)
- Dithionus acid -  $H_2S_2O_4$
- Dithionic acid -  $H_2S_2O_6$
- Polythionic acid -  $H_2(S)_nO_6$  (S-S linkage)

## OXYACIDS OF HALOGEN (Cl)

- Hypochlorous acid -  $HClO$
- Chlorous acid -  $HClO_2$
- Chloric acid -  $HClO_3$
- Perchloric acid -  $HClO_4$

## Order of acidic strength

$HClO < HClO_2 < HClO_3 < HClO_4$

## Oxidising nature

$HClO > HClO_2 > HClO_3 > HClO_4$

## ALLOTROPY

- Those substance which are made up of same elements but having different bonding arrangement are known as allotropes & this phenomenon known as allotropy.
- Those elements which exhibit higher tendency of catenation exhibit higher tendency of allotropy.
- Therefore carbon, phosphorus & sulphur exhibit maximum allotropy.

## ALLOTROPE OF CARBON

## DIAMOND

$C-sp^3$ , tetrahedral structure  
C-C bond length 1.54 Å  
Compact 3 dimensional structure  
Hardest substance  
Very high mp (~ 3400°C)  
Very high density  
Non conductor  
Very high refractive index  
Exhibit total internal reflection  
Shines brightly in light

## GRAPHITE

## Hexagonal layer structure

All  $sp^2$  hybrid carbon  
Unhybrid orbital electron form  $\pi$ -bond. This  $\pi$ -bond exhibit resonance and due to resonance there is mobility of electrons and it becomes conductor of electricity.

## FULLERENE

## Latest discovered allotrope of carbon it is found in chimney sooty particle.

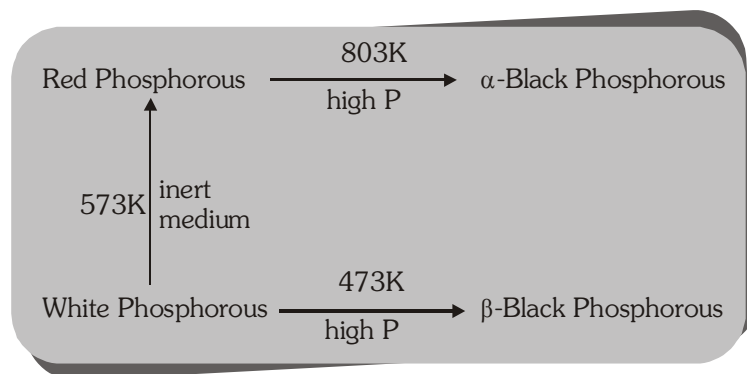
It contain  $C_{60}$ - $C_{320}$ , C :  $sp^2$  hybrid  
Contain pentagon & hexagonal structure  
 $C_{60}$ : Buckminster fullerene soccer ball (football) or bucky ball.  
 $C_{60}$ : 20 hexagon rings  
12 pentagon rings  
Purest form of carbon  
No dangling bond

### ALLOTROPES OF PHOSPHOROUS

(a) white phosphorous (b) Red phosphorous (c) Black phosphorous

| White phosphorous                                           | Red Phosphorous                                     |
|-------------------------------------------------------------|-----------------------------------------------------|
| Waxy solid                                                  | Brittle powder                                      |
| Poisonous                                                   | Non poisonous                                       |
| Soluble in CS <sub>2</sub> , Insoluble in water             | Insoluble in water & CS <sub>2</sub>                |
| Monomer of P <sub>4</sub>                                   | Polymer of P <sub>4</sub>                           |
| Highly reactive due to bond angle strain                    | More stable than white phosphorous                  |
| It glows in dark due to slow oxidation (phosphorescence)    | It does not glow in dark                            |
| It gives phosphine (PH <sub>3</sub> ) on reaction with NaOH | It gives hypo phosphoric acid on reaction with NaOH |

Order of stability or MP or density → white < red < black



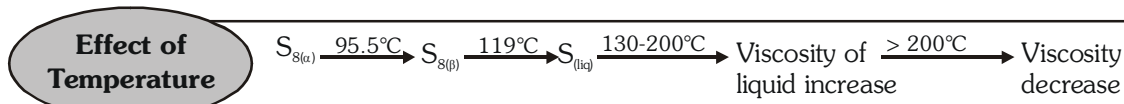
### ALLOTROPES OF SULPHUR

| S                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Crystalline                                                                                                                                                                                                                                                            | Amorphous                                                                                                                                                                                                                                                                                          |
| Rhombic sulphur (α-S)<br>most stable form<br><br>Monoclinic sulphur (β-S)<br>$\alpha\text{-S} \xrightleftharpoons[<95.6^\circ\text{C}]{>95.6^\circ\text{C}} \beta\text{-S}$ 95.6°C = transition Temp.<br>both are soluble in CS <sub>2</sub><br>but insoluble in water | Milk of sulphur<br>Plastic sulphur (γ-S)<br><br>Colloidal sulphur<br>$\text{H}_2\text{S} + 2\text{HNO}_3 \xrightarrow{\text{Redox}} \text{S} + 2\text{NO}_2 + 2\text{H}_2\text{O}$ <div style="display: flex; justify-content: space-around; width: 100%;"> <span>RA</span> <span>OA</span> </div> |

(a) density of αS > βS

(b) Both are puckered crown shape having S<sub>8</sub> units

(c) S<sub>2</sub> is paramagnetic sulphur which exist in vapour form at high temperature. (d) S<sub>6</sub> is chair form of S



- (i)  $\text{PH}_3, \text{AsH}_3, \text{NH}_3, \text{SbH}_3$
- (ii)  $\text{H}_2\text{S}, \text{H}_2\text{Se}, \text{H}_2\text{O}$
- (iii)  $\text{HCl}, \text{HBr}, \text{HI}, \text{HF}$
- (iv)  $\text{NH}_3, \text{HF}, \text{H}_2\text{O}$
- (v)  $\text{He}, \text{Ne}, \text{Ar}, \text{Kr}$
- (vi)  $\text{H}_2\text{O}, \text{D}_2\text{O}$
- (vii)  $\text{H}_2, \text{Cl}_2, \text{Br}_2$



### 10. Electrical Conductivity

Cr, Pt, Fe, Al, Au, Cu, Ag

### 11. Reactivity with water

- (i) Li, Na, K, Rb, Cs
- (ii) Be, Mg, Ca, Sr, Ba

### 12. Extent of Hydrolysis

- (i)  $\text{CCl}_4$ ,  $\text{MgCl}_2$ ,  $\text{AlCl}_3$ ,  $\text{SiCl}_4$ ,  $\text{PCl}_5$
- (ii)  $\text{BiCl}_3$ ,  $\text{SbCl}_3$ ,  $\text{AsCl}_3$ ,  $\text{PCl}_3$ ,  $\text{NCl}_3$

### 13. Bond Strength

- (i) HI, HBr, HCl, HF
- (ii)  $-\text{C}-\text{I}$ ,  $-\text{C}-\text{Br}$ ,  $-\text{C}-\text{Cl}$ ,  $-\text{C}-\text{F}$
- (iii)  $\text{N}-\text{N}$ ,  $\text{N}=\text{N}$ ,  $\text{N}\equiv\text{N}$
- (iv)  $\text{As}-\text{H}$ ,  $\text{Sb}-\text{H}$ ,  $\text{P}-\text{H}$ ,  $\text{N}-\text{H}$
- (v)  $\text{N}_2^{2-}$ ,  $\text{N}_2^-$ ,  $\text{N}_2^+$ ,  $\text{N}_2$
- (vi)  $\text{O}_2^{2-}$ ,  $\text{O}_2^-$ ,  $\text{O}_2$ ,  $\text{O}_2^+$ ,  $\text{O}_2^{2+}$   
 $\text{LiI}$ ,  $\text{LiBr}$ ,  $\text{LiCl}$ ,  $\text{LiF}$        $\text{NaI}$ ,  $\text{NaBr}$ ,  $\text{NaCl}$ ,  $\text{NaF}$   
 $\text{CsCl}$ ,  $\text{RbCl}$ ,  $\text{KCl}$ ,  $\text{NaCl}$        $\text{BaO}$ ,  $\text{SrO}$ ,  $\text{CaO}$ ,  $\text{MgO}$
- (vii)  $\text{F}_2$ ,  $\text{H}_2$ ,  $\text{O}_2$ ,  $\text{N}_2$
- (viii)  $\text{NO}^-$ ,  $\text{NO}$ ,  $\text{NO}^+$
- (ix)  $\text{I}_2$ ,  $\text{F}_2$ ,  $\text{Br}_2$ ,  $\text{Cl}_2$
- (x)  $\text{O}-\text{O}$ ,  $\text{S}-\text{S}$
- (xi)  $\text{F}-\text{F}$ ,  $\text{O}-\text{O}$ ,  $\text{N}-\text{N}$ ,  $\text{C}-\text{C}$ ,  $\text{H}-\text{H}$

### 14. Reducing Power

- (i)  $\text{PbCl}_2$ ,  $\text{SnCl}_2$ ,  $\text{GeCl}_2$
- (ii) HF, HCl, HBr, HI
- (iii) Ag, Cu, Pb, Fe, Zn
- (iv)  $\text{HNO}_3$ ,  $\text{H}_2\text{SO}_3$ ,  $\text{H}_2\text{S}$
- (v)  $\text{H}_3\text{PO}_4$ ,  $\text{H}_3\text{PO}_3$ ,  $\text{H}_3\text{PO}_2$

### 15. Covalent Character

- (i)  $\text{LiCl}$ ,  $\text{BeCl}_2$ ,  $\text{BCl}_3$ ,  $\text{CCl}_4$
- (ii)  $\text{SrCl}_2$ ,  $\text{CaCl}_2$ ,  $\text{MgCl}_2$
- (iii)  $\text{TiCl}_2$ ,  $\text{TiCl}_3$ ,  $\text{TiCl}_4$
- (iv)  $\text{LiCl}$ ,  $\text{LiBr}$ ,  $\text{LiI}$
- (v)  $\text{Na}_2\text{O}$ ,  $\text{Na}_2\text{S}$
- (vi)  $\text{AlF}_3$ ,  $\text{Al}_2\text{O}_3$ ,  $\text{AlN}$
- (vii) HF, HCl, HBr, HI

### 16. Strength of Hydrogen bonding ( $\text{X}\cdots\text{H}-\text{X}$ )

- (i) S, Cl, N, O, F
- (ii)  $\text{NH}_3$ ,  $\text{H}_2\text{O}$ , HF

### 17. Reactivity with Hydrogen

- (i) Cs, Rb, K, Na, Li
- (ii) Ba, Sr, Ca, Mg, Be

### 18. Reactivity Towards Air

Be, Mg, Cs, Sr, Ba

### 19. Bond Length

- (i)  $\text{N}_2$ ,  $\text{O}_2$ ,  $\text{F}_2$ ,  $\text{Cl}_2$
- (ii)  $\text{N}-\text{N}$ ,  $\text{C}-\text{N}$ ,  $\text{C}-\text{C}$
- (iii)  $\text{CO}$ ,  $\text{C}=\text{O}$ ,  $-\text{C}-\text{O}-$
- (iv)  $\text{NO}^+$ ,  $\text{NO}$ ,  $\text{NO}^-$
- (v)  $\text{O}_2$ ,  $\text{O}_3$ ,  $\text{H}_2\text{O}_2$  (O-O bond length)
- (vi)  $\text{CO}$ ,  $\text{CO}_2$ ,  $\text{CO}_3^{2-}$
- (vii)  $\text{N}_2$ ,  $\text{N}_2^-$ ,  $\text{N}_2^{2-}$
- (viii)  $\text{O}_2^{+2}$ ,  $\text{O}_2$ ,  $\text{O}_2^-$ ,  $\text{O}_2^{2-}$
- (ix) HF, HCl, HBr, HI

### 20. Dipole moments

- (i)  $\text{CCl}_4$ ,  $\text{CHCl}_3$ ,  $\text{CH}_2\text{Cl}_2$ ,  $\text{CH}_3\text{Cl}$
- (ii)  $\text{NF}_3$ ,  $\text{NH}_3$ ,  $\text{H}_2\text{O}$ , HF
- (iii) Cis-chloropropene, Trans-chloropropene
- (iv) p, m, o-dichlorobenzene
- (v)  $\text{CH}_3\text{I}$ ,  $\text{CH}_3\text{Br}$ ,  $\text{CH}_3\text{F}$ ,  $\text{CH}_3\text{Cl}$
- (vi)  $\text{NH}_3$ ,  $\text{SO}_2$ ,  $\text{H}_2\text{O}$ , HF
- (vii)  $\text{H}_2\text{S}$ ,  $\text{H}_2\text{O}$
- (viii) HI, HBr, HCl, HF
- (ix)  $\text{PH}_3$ ,  $\text{AsH}_3$ ,  $\text{SbH}_3$ ,  $\text{NH}_3$
- (x)  $\text{H}_2\text{O}$ ,  $\text{H}_2\text{O}_2$

| Group 15       | Bond angle      | Group 16              | Bond angle      |
|----------------|-----------------|-----------------------|-----------------|
| $\text{NH}_3$  | $107^\circ 48'$ | $\text{H}_2\text{O}$  | $104^\circ 28'$ |
| $\text{PH}_3$  | $93^\circ 36'$  | $\text{H}_2\text{S}$  | $92^\circ$      |
| $\text{AsH}_3$ | $91^\circ 48'$  | $\text{H}_2\text{Se}$ | $91^\circ$      |
| $\text{SbH}_3$ | $91^\circ 18'$  | $\text{H}_2\text{Te}$ | $90.5^\circ$    |

**INERT PAIR EFFECT**

➤ Inert pair effect the reluctance of ns electrons to take part in bond formation is called inert pair effect. This effect is more pronounced in heavier elements and that too for 13, 14 and 15 group. It results in the decrease in oxidation state by 2 units. For examples, Tl is more stable in oxidation state + 1 than the oxidation state +3.

In p-block elements the stability of the lower oxidation state increases on descending the group.

**Group 13**

B (+3)

Al (+3)

Ga (+3), (+1)

In (+3), (+1)

Tl (+3), (+1)

**Group 14**

C (+4)

Si (+4)

Ge (+4), (+2)

Sn (+4), (+2)

Pb (+4), (+2)

**Order of stability :**
 $\text{Tl}^{+1} > \text{In}^{+1} > \text{Ga}^{+1}$  (due to inert pair effect)
**Order of stability :**
 $\text{Pb}^{+2} > \text{Sn}^{+2} > \text{Ge}^{+2}$  (due to inert pair effect)
**Molecules that do not exist :**

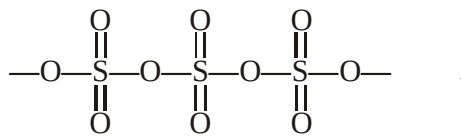
- (1)  $\text{SF}_4$ ,  $\text{SF}_6$  &  $\text{PF}_5$  exist while.  $\text{OF}_4$ ,  $\text{OF}_6$ ,  $\text{NF}_5$  do not exist
- (2) (a)  $\text{PI}_5(\text{vap})$  &  $\text{SCL}_6$  do not exist  
(b)  $\text{SCL}_6$  does not exist while  $\text{TeCl}_6$  exist  
(c)  $\text{PI}_5$  (Solid) exist
- (3)  $\text{SF}_6$ ,  $\text{PF}_5$ ,  $\text{XeF}_6$ ,  $\text{XeF}_4$  &  $\text{XeF}_2$  exist while  $\text{SH}_6$ ,  $\text{PH}_5$ ,  $\text{XeH}_6$ ,  $\text{XeH}_4$ ,  $\text{XeH}_2$  do not exist

**ALLOTROPIC FORM OF  $\text{SO}_3$** 

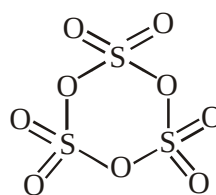
$\text{SO}_3$  have three allotropic forms  $\alpha\text{-SO}_3$ ,  $\beta\text{-SO}_3$  and

 $\gamma\text{-SO}_3$  $\alpha\text{-SO}_3$ 

Cross linked solid



Polymeric chain structure

 $\beta\text{-SO}_3$  $\gamma\text{-SO}_3$  $(\text{S}_3\text{O}_9)$ 

cyclic trimer

**IMPORTANT NOTES**

# s-BLOCK ELEMENTS

## ALKALI METALS

### Physical properties :

General electronic configuration  $ns^1$ .

General oxidation state +1.

Atomic/Ionic size  $Li < Na < K < Rb < Cs$ .

Density  $Li < K < Na < Rb < Cs$

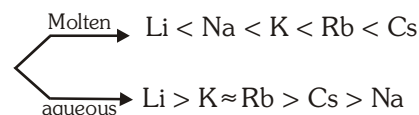
Ionisation energy  $Li > Na > K > Rb > Cs$

Flame colour

|    |                |
|----|----------------|
| Li | Crimsen red    |
| Na | Golden yellow  |
| K  | Pale violet    |
| Rb | Reddish violet |
| Cs | Blue           |

Basic nature increase down to the group

### Reducing property :

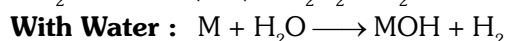
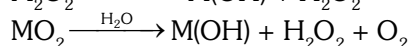
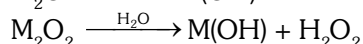
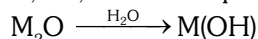


### Chemical properties :

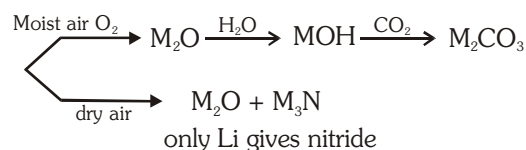
With  $O_2$  : Lithium forms normal oxide  $[Li_2O]$

Sodium forms peroxide  $(Na_2O_2)$

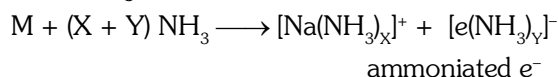
K, Rb, Cs forms superoxide  $KO_2, RbO_2, CsO_2$



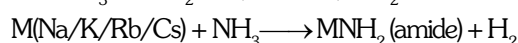
With Air :



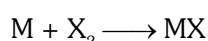
With  $NH_3$  :



Paramagnetic, blue colour. In excess of metal dimagnetic and copper bronze colour. on standing



With Halides :



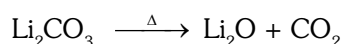
K, Rb, Cs forms poly halide due to large size

## SIMILARITIES BETWEEN LITHIUM AND MAGNESIUM

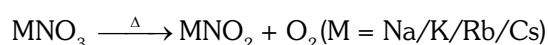
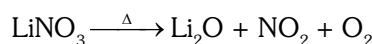
- Both lithium and magnesium are harder and lighter than other elements in the respective groups.
- Lithium and magnesium react slowly with cold water. Their oxides and hydroxides are much less soluble and their hydroxides decompose on heating. Both form a nitride by direct combination with nitrogen,  $Li_3N$  and  $Mg_3N_2$ .
- The oxides,  $Li_2O$  and  $MgO$  do not combine with excess oxygen to give a peroxide or a superoxide.
- The carbonates of lithium and magnesium decompose easily on heating to form the oxide and  $CO_2$ . Solid bicarbonates are not formed by lithium and magnesium.
- Both  $LiCl$  and  $MgCl_2$  are soluble in ethanol.
- Both  $LiCl$  and  $MgCl_2$  are deliquescent and crystallise from aqueous solution as hydrates,  $LiCl \cdot 2H_2O$  and  $MgCl_2 \cdot 8H_2O$ .

### Carbonates :

Only  $Li_2CO_3$  decomposes

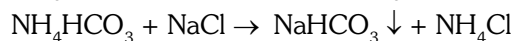


### Nitrates :

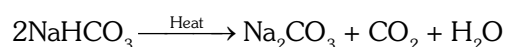


### $Na_2CO_3 \cdot 10H_2O$ (Washing Soda)

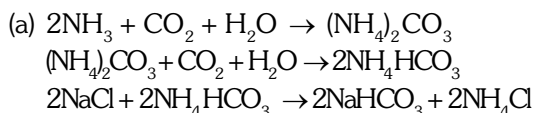
- In **Solvay's process**,  $CO_2$  gas is passed through saturated brine ( $NaCl$ ) solution when sparingly soluble  $NaHCO_3$  separates out.



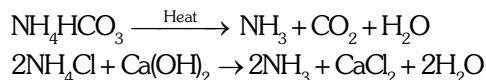
The  $NaHCO_3$  formed above is calcined to form  $Na_2CO_3$



- The reactions taking place at different stages during the manufacture of  $Na_2CO_3$  by Solvay process are given as under :



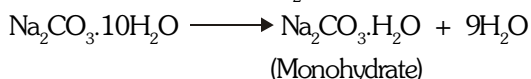
(b) Ammonia recovery tower



(c) Calcination of  $\text{NaHCO}_3$

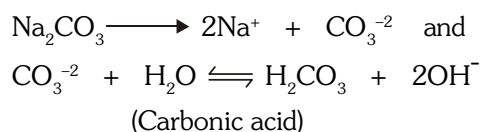


(i) **Efflorescence** :  $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$  when exposed to air it gives out nine out of ten  $\text{H}_2\text{O}$  molecules.



This process is called efflorescence. Hence washing soda losses weight on exposure to air.

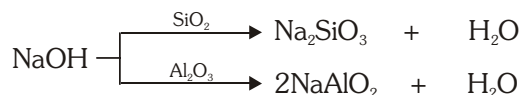
(ii) **Hydrolysis** : Aqueous solution of  $\text{Na}_2\text{CO}_3$  is alkaline in nature due to anionic hydrolysis.



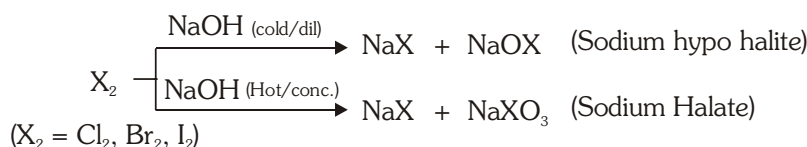
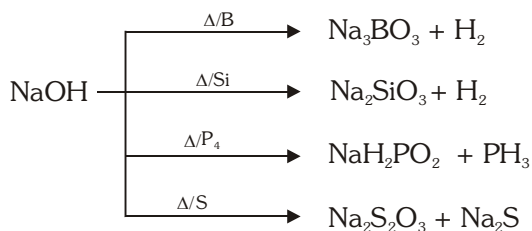
➤  $\text{Na}_2\text{CO}_3$  does not impart any colour to the flame but  $\text{NaCl}$  does because the thermal ionization of  $\text{Na}_2\text{CO}_3$  does not take place at the temperature of flame of the burner.

### □ Reaction of NaOH :

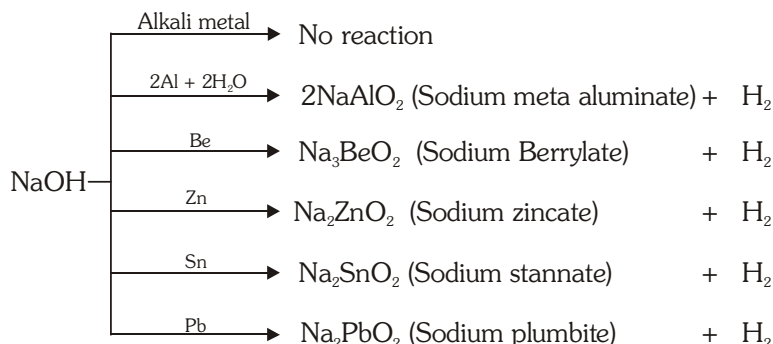
(i) NaOH is **strong base**



(ii) **Reaction with non metals** : no reaction with  $\text{H}_2$ ,  $\text{N}_2$  and C



(iii) **Reaction with Metal** :



## ALKALINE EARTH METALS

**Physical properties :** General electronic configuration  $ns^2$ .

General oxidation state +2.

Atomic/Ionic size  $\text{Be} < \text{Mg} < \text{Ca} < \text{Sr} < \text{Ba}$

Ionisation energy  $\text{Be} > \text{Mg} > \text{Ca} > \text{Sr} > \text{Ba}$

### Flame Test :

Among alkaline earth metals, Be and Mg do not impart any characteristic colour to the flame due to more ionization energies.

Ca - Brick red

Sr - Crimson red

Ba - Apple green

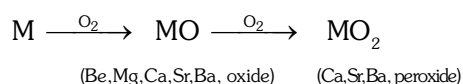
Basic nature of oxide

Reducing property : Order of reducing property in aqueous and gaseous medium is

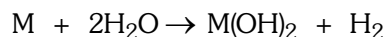
$\text{Be}^{+2} < \text{Mg}^{+2} < \text{Ca}^{+2} < \text{Sr}^{+2} < \text{Ba}^{+2}$

### Chemical properties :

#### With $\text{O}_2$ :



#### With Water :



#### With Air :

In moist air, except Be all the elements converts into carbonates.

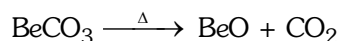
In dry air gives nitride and oxide both

#### With $\text{NH}_3$ :

Only Ca, Sr and Ba gives deep blue-black solution of ammoniated electron.

#### Carbonates :

- (i) Except  $\text{BeCO}_3$ , all the carbonates are stable towards heat



- (ii) Order of decreasing stability -



#### Nitrates :

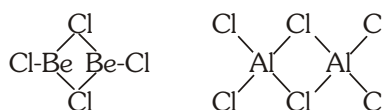
- (i) Alkaline earth metals forms  $\text{M}(\text{NO}_3)_2$  type nitrates. (M - Alkaline earth metal).
- (ii) All alkaline metals nitrates on heating gives oxides and  $\text{NO}_2 + \text{O}_2$



## DIAGONAL SIMILARITY BETWEEN BERYLLIUM AND ALUMINIUM

In many of its properties, beryllium resembles aluminium. Thus -

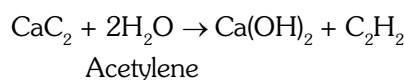
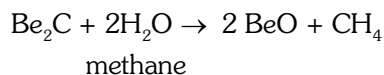
- (a) The two elements have same electronegativity and their charge/ radius ratios.
- (b) Both metals are fairly resistant to the action of acids due to a protective film of oxide on the surface. Both metals are acted upon by strong alkalis to form soluble complexes, beryllates  $[\text{Be}(\text{OH})_4]^{2-}$  and aluminates,  $[\text{Al}(\text{OH})_4]^-$ .
- (c) The chlorides of both beryllium and aluminium



have bridged chloride structures in vapour phase.

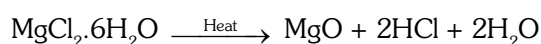
- (d) Salts of these metals form hydrated ions, Ex.  $[\text{Be}(\text{OH})_4]^{2-}$  and  $[\text{Al}(\text{OH})_6]^{3-}$  in aqueous solutions. Due to similar charge/ radius ratios of beryllium and aluminium ions have strong tendency to form complexes. For example beryllium forms tetrahedral complexes such as  $\text{BeF}_4^{2-}$  and  $[\text{Be}(\text{C}_2\text{O}_4)_2]^{2-}$  and aluminium forms octahedral complexes like  $\text{AlF}_6^{3-}$  and  $[\text{Al}(\text{C}_2\text{O}_4)_3]^{3-}$ .

- $\text{Be}_2\text{C}$  on treatment with  $\text{H}_2\text{O}$  forms  $\text{CH}_4$  while  $\text{CaC}_2$  forms  $\text{C}_2\text{H}_2$ .



### CHLORIDE OF ALKALINE EARTH METAL.

- $\text{BeCl}_2$  in the vapour phase above  $900^\circ\text{C}$  is monomeric ; below  $900^\circ\text{C}$  in the vapour exists as a mixture of monomer  $\text{BeCl}_2$  and dimer  $\text{Be}_2\text{Cl}_4$ , in the solid state, has a polymeric structure and when dissolved in a coordinating solvent it exists as a monomer.
- Anhydrous  $\text{MgCl}_2$  cannot be prepared by the simple heating of hydrated magnesium chloride  $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ , as it gets hydrolysed to magnesium oxide.



- Out of the oxides of group 2 elements only  $\text{BeO}$  is extremely hard, non volatile, has high melting point and it is amphoteric.

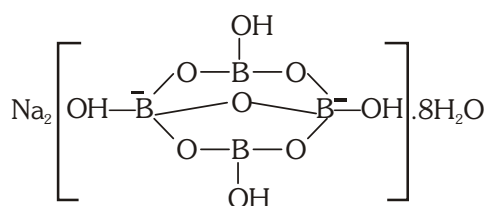
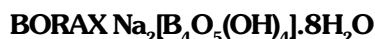


**IMPORTANT NOTES**

***p*-BLOCK**

## 13<sup>th</sup>-GROUP

- Some important ores of Boron are given as under.
- |                                         |                                                                          |                                                                                  |
|-----------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| (i) Boric acid, $\text{H}_3\text{BO}_3$ | (ii) Borax, $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$ | (iii) Colemanite, $\text{Ca}_2\text{B}_6\text{O}_{11} \cdot 5\text{H}_2\text{O}$ |
|-----------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------|
- Some important minerals of aluminium are given as under.
- |                                           |                                                                 |                                           |
|-------------------------------------------|-----------------------------------------------------------------|-------------------------------------------|
| (i) Corundum, $\text{Al}_2\text{O}_3$     | (ii) Bauxite, $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ | (iii) Cryolite, $\text{Na}_3\text{AlF}_6$ |
| (iv) Feldspar, $\text{KAlSi}_3\text{O}_8$ |                                                                 |                                           |



### Properties :

- (i) It is a white crystalline solid of formula  $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$ . In fact it contains the tetranuclear units  $[\text{B}_4\text{O}_5(\text{OH})_4]^{2-}$  and correct formula; therefore, is  $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$ .
- (ii) Hydrolysis  
$$\text{Na}_2\text{B}_4\text{O}_7 + 7\text{H}_2\text{O} \rightarrow 2\text{NaOH} + 4\text{H}_3\text{BO}_3$$
- (iii) Heating  
$$\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O} \xrightarrow{\Delta} \underset{\text{Swells up}}{\text{Na}_2\text{B}_4\text{O}_7} \xrightarrow{\Delta} \underset{\substack{\text{Sodium} \\ \text{metaborate}}}{2\text{NaBO}_2} + \underset{\substack{\text{Boric} \\ \text{anhydride}}}{\text{B}_2\text{O}_3}$$
- (iv) When borax is heated in a Bunsen burner flame with  $\text{CoO}$  on a loop of platinum wire, a blue coloured  $\text{Co}(\text{BO}_2)_2$  bead is formed.

## DIBORANE, B<sub>2</sub>H<sub>6</sub>

**Preparation :**

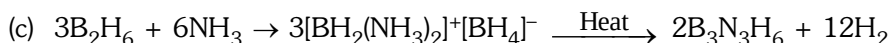
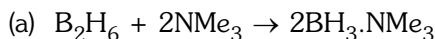
- (i)  $3\text{LiAlH}_4 + 4\text{BF}_3 \longrightarrow 3\text{LiF} + 3\text{AlF}_3 + 2\text{B}_2\text{H}_6$   
or  $\text{LiBH}_4$  or  $3(\text{BF}_3)$
- (ii) **Laboratory** method :  
 $2\text{NaBH}_4 + \text{I}_2 \rightarrow \text{B}_2\text{H}_6 + 2\text{NaI} + \text{H}_2$
- (iii) **Industrial scale** :  
 $2\text{BF}_3 + 6\text{NaH} \xrightarrow{450\text{K}} \text{B}_2\text{H}_6 + 6\text{NaF}$

### Properties :

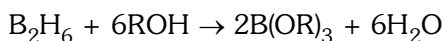
- (i) Diborane is a colourless, highly toxic gas with a b.p. of 180 K.
- (ii) Diborane catches fire spontaneously upon exposure to air.
- (iii) Reaction with Oxygen :  
$$\text{B}_2\text{H}_6 + 3\text{O}_2 \rightarrow \text{B}_2\text{O}_3 + 3\text{H}_2\text{O} ; \Delta_c H^\ominus = -1976 \text{ kJ mol}^{-1}$$
- (iv) Hydrolysis :  
$$\text{B}_2\text{H}_6(\text{g}) + 6\text{H}_2\text{O}(\ell) \text{ (Cold is enough)} \rightarrow 2\text{B}(\text{OH})_3(\text{aq}) + 6\text{H}_2(\text{g})$$
  
$$\text{B}_2\text{H}_6 + \text{HCl (dry)} \xrightarrow[\text{AlCl}_3]{\text{anh.}} \text{B}_2\text{H}_5\text{Cl} + \text{H}_2$$



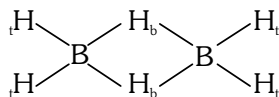
(v) Reaction with Lewis Bases



(vi) Reaction with ROH :



Diborane,  $B_2H_6$



dimer due to formation of  
3 centre-2e-bond

Inorganic benzene

**ORTHOBORIC ACID [ $H_3BO_3$  or  $B(OH)_3$ ]**

**From Borax :**  
 $Na_2B_4O_7 + H_2SO_4 + H_2O \rightarrow Na_2SO_4 + 4H_3BO_3$

**From Colemanite:**  
 $Ca_2B_6O_{11} + 4SO_2 + 11H_2O \rightarrow 2Ca(HSO_3)_2 + 6H_3BO_3$

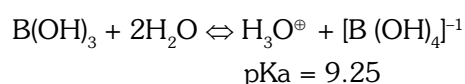
MOP Properties

**Monobasic Lewis acid**  
 $B(OH)_3 + HOH \rightarrow [B(OH)_4]^- + H^+$

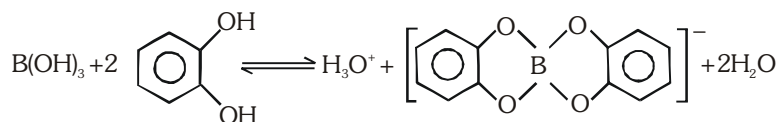
**Action of heat**  
 $H_3BO_3 \xrightarrow[ -H_2O ]{100^\circ C} HBO_2 \xrightarrow[ -H_2O ]{160^\circ C} H_2B_4O_7$   
metaboric      Tetraboric  
 $\downarrow \text{red hot}$   
 $B_2O_3$   
boron trioxide

**Reaction with alcohol  
(Test of boric acid)**  
 $H_3BO_3 + 3C_2H_5OH \rightarrow B(OC_2H_5)_3 + 3H_2O$   
Triethyl borate  
green edged flame

➤ Boric acid is a weak monobasic acid



➤ It is difficult to titrate boric acid against NaOH solutions and the end point cannot be located correctly. However, it can be successfully titrated in the presence of polyhydroxy alcohols (e.g. Glycerol, mannitol, catechol or sugar). The presence of these compounds greatly increase the acidity of boric acid.

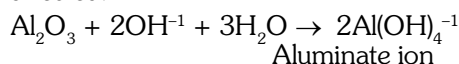


Boron is complexed by these compounds. These complex ions cannot interact with  $H^+$  ions as the boron atom has already acquired its maximum covalency of four. Consequently, boric acid in the presence of polyhydroxy alcohols can be titrated against NaOH to a definite end point.

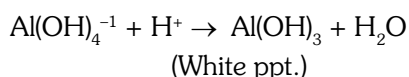
## EXTRACTION OF ALUMINIUM

Al is usually extracted from Bauxite,  $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ . First, bauxite is purified from the impurities (Red bauxite contains  $\text{Fe}_2\text{O}_3$  as impurity while white bauxite contains silica as impurity). Red bauxite can be purified by Baeyer's process or Hall's process while white bauxite is purified by Serpeck's process.

- **Baeyer's process**, involves the roasting of the ore to convert  $\text{FeO}$  to  $\text{Fe}_2\text{O}_3$  and then digested at 423K with conc.  $\text{NaOH}$  solution for a few hours when  $\text{Al}_2\text{O}_3$  gets dissolved to give a solution of  $[\text{Al}(\text{OH})_4]^{-1}$ . The basic oxide impurities such as  $\text{Fe}_2\text{O}_3$  are not affected.

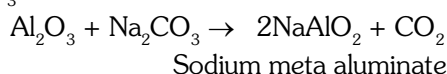


$\text{Fe}_2\text{O}_3$  left undissolved is filtered off. The treatment of  $[\text{Al}(\text{OH})_4]^{-1}$  solution with a weak acid precipitate pure  $\text{Al}(\text{OH})_3$ .

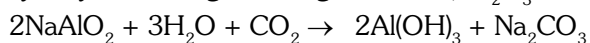


The  $\text{Al}(\text{OH})_3$  precipitate is removed by filtration and ignited to get alumina,  $\text{Al}_2\text{O}_3$ .

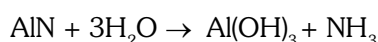
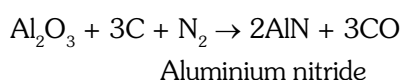
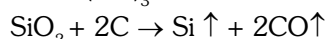
- **Hall's process**, Involves the fusion of the ore with  $\text{Na}_2\text{CO}_3$  when soluble sodium meta aluminate,  $\text{NaAlO}_2$  is produced. This is extracted with water when  $\text{Fe}_2\text{O}_3$  is left as a residue.



The water extract is heated upto 333K and  $\text{CO}_2$  is passed through it.  $\text{Al}(\text{OH})_3$  is precipitated due to hydrolysis and is ignited to get alumina,  $\text{Al}_2\text{O}_3$ .



- **Serpeck's process** involves the heating of bauxite with coke in a current of  $\text{N}_2$  at 2075K. The  $\text{SiO}_2$  present in the ore is reduced to silicon which volatilizes off and alumina gives aluminium nitride. This can be hydrolysed to  $\text{Al}(\text{OH})_3$  which on fusion gives alumina.



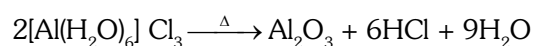
**Electrolysis of  $\text{Al}_2\text{O}_3$  to form aluminium.** Pure alumina is dissolved in fused cryolite,  $\text{Na}_3\text{AlF}_6$  at 1225 K by current of 100 amperes and 6–7 volts to get Al. The Al obtained is purified by Hoopes's process.

## ALUMINIUM TRIFLUORIDE

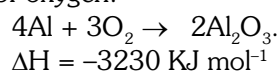
- **Aluminium trifluoride** ( $\text{AlF}_3$ ) is different from other trihalides of Al in being insoluble and nonvolatile. In  $\text{AlF}_3$  each Al is surrounded by a distorted octahedron of 6F atoms and 1 : 3 stoichiometry is achieved by the sharing of corner fluorine atoms between two octahedra.

## ALUMINIUM CHLORIDE

- Aluminium chloride ( $\text{AlCl}_3$ ) in the pure and anhydrous state is a white solid but commercial samples are yellowish due to the impurity of  $\text{FeCl}_3$ .
- Anhydrous  $\text{AlCl}_3$ , in the crystalline state possesses a closely packed layer structure with six coordinated aluminium octahedral arrangement.
- Anhydrous  $\text{AlCl}_3$  has a very high affinity for water. On treating  $\text{AlCl}_3$  with water the  $\text{Cl}^{-}$  ions go outside the coordination sphere to form  $[\text{Al}(\text{H}_2\text{O})_6]\text{Cl}_3$  with enthalpy of solution  $-330 \text{ KJ mol}^{-1}$ . Due to this strong Al–O linkage hydrate cannot be dehydrated on heating to form  $\text{AlCl}_3$ .



- Aluminium chloride, is a polymeric solid which exists as a dimer  $\text{Al}_2\text{Cl}_6$  between 200–400°C and then monomer up to 800°C.
- **Thermite welding**, Aluminium has got a very high affinity for oxygen.



It, therefore, displaces metals less electropositive than itself from oxides. The large amount of heat produced is used in welding rails or heavy machinery without removing them from their position. This is called thermite welding or Gold Schmidt's aluminothermic process. In this process, a mixture of Ferric oxide (3 parts) and aluminium powder (1 part) called thermite is used.

- **Boron nitride** (BN), has a structure similar to that of graphite.
- Green edged flame test for borate ( $\text{BO}_3^{3-}$ ) ion. A mixture of  $\text{C}_2\text{H}_5\text{OH}$  and  $\text{BO}_3^{3-}$  salt with conc.  $\text{H}_2\text{SO}_4$  burns with green edge flame due to the formation of ethyl borate.
- $$\text{H}_3\text{BO}_3 + 3\text{C}_2\text{H}_5\text{OH} \rightarrow \text{B}(\text{OC}_2\text{H}_5)_3 + 3\text{H}_2\text{O}$$
- Ethyl borate

- **Charcoal cavity test for aluminium.** On heating with  $\text{Na}_2\text{CO}_3$  and a drop of cobalt nitrate solution a blue coloured residue to **cobalt metaaluminate** (Thenard's blue) is obtained.

### ALUMS $[M_2SO_4.M_2'(SO_4)_3.24H_2O]$

**Alums are the double salts of Type**

$$[M_2SO_4 \cdot M_2'(SO_4)_3 \cdot 24H_2O]$$

**M : Monovalent Cation:**

 $\text{Na}^+, \text{K}^+, \text{Rb}^+, \text{Cs}^+, \text{NH}_4^+$ 

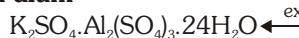
**M' : Trivalent cation :**

 $\text{Al}^{3+}, \text{Fe}^{3+} \text{ \& } \text{Cr}^{3+}$ 

## Soda alum



## Potash alum



## Ammonium alum



| example | Alum | properties |
|---------|------|------------|
|---------|------|------------|

properties

► aq. solution acidic

► Coagulant

► Mordant for dyeing

► Tanning of leather

**Pseudo alums** are double sulphates of a divalent and a trivalent metals which crystallize with twenty four water molecules of crystallization, for example.

(i)  $\text{MnSO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$       (ii)  $\text{FeSO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$

They are not isomorphous with true alums.

## IMPORTANT NOTES

**GROUP 14 ELEMENTS**

- The common oxidation states exhibited by these elements are +4 and +2. Carbon also exhibits negative oxidation states.
- Tin forms compounds in both oxidation states (Sn in +2 state is a reducing agent).
- Lead compounds in +2 state are stable and in +4 state are strong oxidising agents.
- SiO only exists at high temperature.
- The dioxides — CO<sub>2</sub>, SiO<sub>2</sub> and GeO<sub>2</sub> are acidic, whereas SnO<sub>2</sub> and PbO<sub>2</sub> are amphoteric in nature. Among monoxides, CO is neutral, GeO is distinctly acidic whereas SnO and PbO are amphoteric.
- Tin decomposes steam to form dioxide and dihydrogen gas.
- All halide of 14<sup>th</sup> group are covalent. Exceptions are SnF<sub>4</sub> and PbF<sub>4</sub>, which are ionic in nature.
- Stability of dihalides increases down the group.
- The order of catenation is C > Si > Ge ≈ Sn. Lead does not show catenation.

➤ **Catenation** is the unique tendency of 14 group elements to form long chains of different sizes and shapes. The tendency to show catenation is directly related to the strength of bond. The bond energies of 14 group elements decrease as under :

|                                      | C-C | Si-Si | Ge-Ge | Sn-Sn | Pb-Pb |
|--------------------------------------|-----|-------|-------|-------|-------|
| Bond energy (KJ mole <sup>-1</sup> ) | 348 | 222   | 167   | 155   | —     |

This is the reason why carbon forms many chains, Si, a few and Ge and Sn form practically no chains.

- CO<sub>2</sub> is a gas while SiO<sub>2</sub> is a solid at room temperature.
- SiO only exists at high temperature
- CO<sub>2</sub>, SiO<sub>2</sub>, GeO<sub>2</sub>, GeO are acidic, PbO, PbO<sub>2</sub> is SnO and SnO<sub>2</sub> are amphoteric and CO is neutral
- ⇒ Among 14<sup>th</sup> group element only Sn reacts with steam to produce H<sub>2</sub> gas.
- ⇒ Tetrahalide of 14<sup>th</sup> group element are covalent except SnF<sub>4</sub> and PbF<sub>4</sub>

Stability of oxidation state

- ⇒ C<sup>+4</sup> > ----- > Pb<sup>+4</sup> (Stability)
- C<sup>+2</sup> < ----- < Pb<sup>+2</sup> (Stability)
- ⇒ Pb<sup>+4</sup> compounds are strong oxidizing agent.
- ⇒ PbI<sub>4</sub> does not exist.

- Trimethylamine (CH<sub>3</sub>)<sub>3</sub>N is pyramidal while Trisilylamine N(SiH<sub>3</sub>)<sub>3</sub> is planar due to back bonding.

**SILICONES (ORGANO SILICONE POLYMER)**

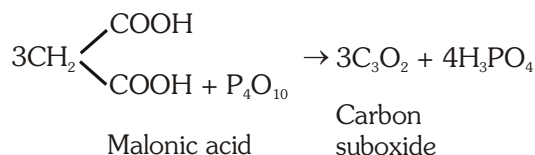
- Silicones are polymeric organosilicon compounds containing Si-O-Si linkage. They have high thermal stability of Si-O-Si chains and are also called high temperature polymers.

**General formula :** (R<sub>2</sub>SiO)<sub>n</sub>. Where R = -CH<sub>3</sub>, -C<sub>2</sub>H<sub>5</sub>, -C<sub>6</sub>H<sub>5</sub>

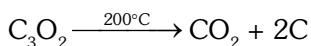
- (i)  $R_2SiCl_2 + 2H_2O \xrightarrow{-2HCl} R_2Si(OH)_2 \xrightarrow[-H_2O]{\Delta} \text{Linear Silicones/Cyclic Silicones}$
- (ii)  $R_3SiCl + H_2O \xrightarrow{HCl} R_3Si(OH) \xrightarrow[-H_2O]{\Delta} \text{Dimer Silicones}$
- (iii)  $RSiCl_3 + 3H_2O \xrightarrow{-3HCl} RSi(OH)_3 \xrightarrow[-H_2O]{\Delta} \text{Crossed Linked Silicones}$

## CARBON SUBOXIDE

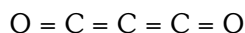
- Carbon suboxide ( $\text{C}_3\text{O}_2$ ) is an foul smelling gas which can be prepared by the dehydration of malonic acid with  $\text{P}_4\text{O}_{10}$ .



On heating upto  $200^{\circ}\text{C}$ , it decomposes into  $\text{CO}_2$  and carbon.



The molecule is thought to have a linear structure.



## OXIDE OF LEAD

- $\text{PbO} \xrightleftharpoons[\text{Litharge}]{\text{Cool}} \text{PbO}$   
massicot
- **Red lead ( $\text{Pb}_3\text{O}_4$ )** is considered to be mixture of lead monoxide and lead dioxide and it is written as  $(\text{PbO}_2 \cdot 2\text{PbO})$ .
- **Lead dioxide ( $\text{PbO}_2$ )**. It is a brown powder obtained by the treatment of red lead with  $\text{HNO}_3$ .  

$$\text{Pb}_3\text{O}_4 + 4\text{HNO}_3 \rightarrow 2\text{Pb}(\text{NO}_3)_2 + \text{PbO}_2 + 2\text{H}_2\text{O}$$

It is used as an active material of the positive plate in storage cells and finds use in match industry as an oxidizing agent.

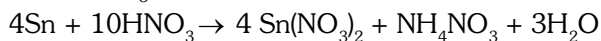
## REACTION OF LEAD OXIDES

| Oxide                          | ex NaOH                                                             | HCl                                 | H <sub>2</sub> SO <sub>4</sub>     | HNO <sub>3</sub>                                     |
|--------------------------------|---------------------------------------------------------------------|-------------------------------------|------------------------------------|------------------------------------------------------|
| PbO                            | Na <sub>2</sub> PbO <sub>2</sub>                                    | PbCl <sub>2</sub>                   | PbSO <sub>4</sub>                  | Pb(NO <sub>3</sub> ) <sub>2</sub>                    |
| PbO <sub>2</sub>               | Na <sub>2</sub> PbO <sub>3</sub>                                    | PbCl <sub>2</sub> + Cl <sub>2</sub> | PbSO <sub>4</sub> + O <sub>2</sub> | Pb(NO <sub>3</sub> ) <sub>2</sub>                    |
| Pb <sub>2</sub> O <sub>3</sub> | Na <sub>2</sub> PbO <sub>2</sub> + NaPbO <sub>3</sub>               | PbCl <sub>2</sub> + Cl <sub>2</sub> | PbSO <sub>4</sub> + O <sub>2</sub> | Pb(NO <sub>3</sub> ) <sub>2</sub> + PbO <sub>2</sub> |
| Pb <sub>3</sub> O <sub>4</sub> | Na <sub>2</sub> PbO <sub>2</sub> + Na <sub>2</sub> PbO <sub>3</sub> | PbCl <sub>2</sub> + Cl <sub>2</sub> | PbSO <sub>4</sub> + O <sub>2</sub> | Pb(NO <sub>3</sub> ) <sub>2</sub> + PbO <sub>2</sub> |

## TIN & ITS COMPOUNDS

- **Action of conc.  $\text{HNO}_3$  on tin**

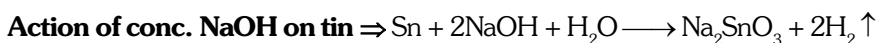
(a) Dilute  $\text{HNO}_3$



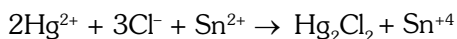
(b) Hot conc.  $\text{HNO}_3$



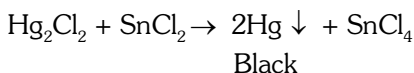
Metastannic acid



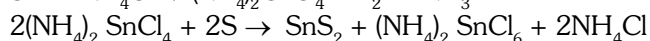
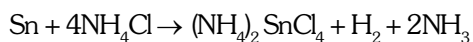
- $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$  on heating undergoes to form its basic chloride  $\text{Sn}(\text{OH})_2 \cdot \text{Cl}$ . The anhydrous salt, therefore, be obtained by heating the hydrated salt in the presence of  $\text{HCl}$  vapour.
- Stannous chloride reduces mercuric chloride ( $\text{HgCl}_2$ ) to a white precipitate of mercurous chloride ( $\text{Hg}_2\text{Cl}_2$ ) which finally turns to metallic mercury (dark grey or black).



Mercurous chloride



- **Mosaic gold** ( $\text{SnS}_2$ ). Stannic sulphide exists in yellow glistening scales which is used for decorative purposes under the name mosaic gold. It is prepared by heating mixture of tin fillings, sulphur and  $\text{NH}_4\text{Cl}$  in a retort.



- **Tin Cry**. Tin metal when bent produces a cracking noise due to rubbing of metal crystals over one another.

CARBON MONOXIDE [CO]

Colorless odourless, tasteless, neutral, poisonous gas

$\text{C} \equiv \text{O}$

CO

MOP

properties

**By dehydration of formic acid and oxalic acid**

$$\text{HCOOH} \xrightarrow[\text{H}_2\text{SO}_4]{\text{Conc.}} \text{CO} + \text{H}_2\text{O}$$

$$\begin{array}{c} \text{COOH} \\ | \\ \text{COOH} \end{array} \xrightarrow[\text{H}_2\text{SO}_4]{\text{Conc.}} \text{CO} + \text{CO}_2 + \text{H}_2\text{O}$$

**By incomplete combustion**

$$\text{C} + \frac{1}{2} \text{O}_2 \rightarrow \text{CO}$$

**By passing air on red hot coke**

dry air

$\rightarrow \text{CO} + \text{N}_2$

Producer gas

moist air

$\rightarrow \text{CO} + \text{N}_2 + \text{H}_2$

semi water gas

water vapour

$\rightarrow \text{CO} + \text{H}_2$

water gas or synthesis gas

**Complexing agent**

$$\text{Ni} + 4\text{CO} \rightarrow [\text{Ni}(\text{CO})_4]$$

**Reducing agent**

$$\text{FeO} + \text{CO} \rightarrow \text{Fe} + \text{CO}_2$$

$$\text{ZnO} + \text{CO} \rightarrow \text{Zn} + \text{CO}_2$$

**Combustion**

$$\text{CO} + \frac{1}{2} \text{O}_2 \rightarrow \text{CO}_2$$

blue flame

CARBIDES

Binary compounds of carbon with other elements (except hydrogen) are known as carbides

**CARBIDES**

**Ionic Carbide**

C + higher electropositive element  
(IA, IIA & Al)

---

Eg.  $\text{Be}_2\text{C}$ ,  $\text{Mg}_2\text{C}_3$ ,  $\text{Al}_4\text{C}_3$ ,  $\text{CaC}_2$

**Covalent Carbide**

C + Non-metals  
(like B & Si)

---

Eg.  $\text{B}_4\text{C}$ ,  $\text{SiC}$

**Interstitial Carbides**

C + transition metals

---

Eg. :  $\text{TiC}$ ,  $\text{WC}$

## GROUP 15 ELEMENTS

### NITROGEN FAMILY

1. Nitrogen and phosphorus are **non-metals**, arsenic and antimony **metalloids** and bismuth is a **typical metal**.
2. Except nitrogen all the elements show allotropy.
3. Bismuth hardly forms any compound in  $-3$  oxidation state.
4. The only well characterised Bi (V) compound is  $\text{BiF}_5$ .
5. In the case of nitrogen, all oxidation states from  $+1$  to  $+4$  tend to disproportionate in acid solution. For example,  
(i)  $3\text{HNO}_2 \rightarrow \text{HNO}_3 + \text{H}_2\text{O} + 2\text{NO}$
6. In case of phosphorus nearly all intermediate oxidation states disproportionate into  $+5$  and  $-3$  both in alkali and acid.  
exp:  $3\text{HNO}_2 \longrightarrow \text{HNO}_3 + 2\text{NO}$
7. The oxides of the type  $\text{E}_2\text{O}_3$  of nitrogen and phosphorus are purely acidic, that of arsenic and antimony amphoteric and those of bismuth predominantly basic.
8. In case of nitrogen, only  $\text{NF}_3$  is known to be stable.
9. Trihalides except  $\text{BiF}_3$  are predominantly covalent in nature.  
 $\Rightarrow$  The only well characterised Bi (V) compound in  $\text{BiF}_5$   
 $\Rightarrow$  All the oxidation states from  $+1$  to  $+4$  tend to disproportionate in acidic solution :  
 $\Rightarrow$  All the intermediate oxidation states disproportionate into  $+5$  and  $-3$   
 $4\text{H}_3\text{PO}_3 \xrightarrow{\Delta} 3\text{H}_3\text{PO}_4 + \text{PH}_3$   
 $\Rightarrow$  Hydride of 15<sup>th</sup> group elements :  
 $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3 > \text{BiH}_3$  Basic character  
 $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3 > \text{BiH}_3$  Stability  
 $\text{PH}_3 < \text{AsH}_3 < \text{NH}_3 < \text{SbH}_3 < \text{BiH}_3$  Boiling Point  
 $\text{PH}_3 < \text{AsH}_3 < \text{SbH}_3 < \text{NH}_3$  Melting Point  
 $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$  Bond Energy  
 $\Rightarrow$  Penta oxide of 15<sup>th</sup> group element is acidic.  $\text{N}_2\text{O}_3$ ,  $\text{P}_2\text{O}_3$  are acidic,  $\text{As}_2\text{O}_3$ ,  $\text{Sb}_2\text{O}_3$  are amphoteric with  $\text{Bi}_2\text{O}_3$  is basic.  
 $\Rightarrow$  Trihalide of 15<sup>th</sup> group elements are covalent except  $\text{BiF}_3$

### STRUCTURE OF OXIDES OF NITROGEN AND PHOSPHORUS

#### NITROGEN

| Oxide of Nitrogen                            | Oxid. state | Physical appearance | Structure                                       |
|----------------------------------------------|-------------|---------------------|-------------------------------------------------|
| $\text{N}_2\text{O}$ nitrous oxide           | +1          | Colourless gas      | $\text{N} \equiv \text{N} \rightarrow \text{O}$ |
| $\text{NO}$ Nitric oxide                     | +2          | Colourless gas      | $\text{N} = \text{O}$                           |
| $\text{N}_2\text{O}_3$ Dinitrogen trioxide   | +3          | Blue colour solid   | $\text{O}=\text{N}-\text{N}=\text{O}$           |
| $\text{N}_2\text{O}_4$ Dinitrogen tetraoxide | +4          | Colourless solid    | $\text{O}=\text{N}-\text{N}=\text{O}$           |
| $\text{NO}_2$ Nitrogen dioxide               | +4          | Brown gas           | $\text{O}=\text{N}-\text{O}$                    |
| $\text{N}_2\text{O}_5$ Dinitrogen pentaoxide | +5          | Colourless solid    | $\text{O}=\text{N}-\text{O}-\text{N}=\text{O}$  |

## Oxides of Nitrogen

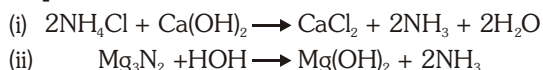
| Name                                          | Formula                       | Oxidation state of nitrogen | Common methods of preparation                                                                                                                          | Physical appearance and chemical nature   |
|-----------------------------------------------|-------------------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Dinitrogen oxide<br>[Nitrogen oxide]          | N <sub>2</sub> O              | +1                          | $\text{NH}_4\text{NO}_3 \xrightarrow{\text{Heat}} \text{N}_2\text{O} + 2\text{H}_2\text{O}$                                                            | Colourless gas, neutral                   |
| Nitrogen monoxide<br>[Nitrogen (II) oxide]    | NO                            | +2                          | $2\text{NaNO}_2 + 2\text{FeSO}_4 + 3\text{H}_2\text{SO}_4 \rightarrow \text{Fe}_2(\text{SO}_4)_3 + 2\text{NaHSO}_4 + 2\text{H}_2\text{O} + 2\text{NO}$ | Colourless gas, neutral                   |
| Dinitrogen trioxide<br>[Nitrogen (III) oxide] | N <sub>2</sub> O <sub>3</sub> | +3                          | $2\text{NO} + \text{N}_2\text{O}_4 \xrightarrow{250\text{K}} 2\text{N}_2\text{O}_3$                                                                    | Blue solid, acidic<br>blue liquid (–30°C) |
| Nitrogen dioxide<br>[Nitrogen (IV) oxide]     | NO <sub>2</sub>               | +4                          | $2\text{Pb}(\text{NO}_3)_2 \xrightarrow{673\text{K}} 4\text{NO}_2 + 2\text{PbO} + \text{O}_2$                                                          | brown gas, acidic                         |
| Dinitrogen tetroxide<br>[Nitrogen (IV) oxide] | N <sub>2</sub> O <sub>4</sub> | +4                          | $2\text{NO}_2 \xrightleftharpoons[\text{Heat}]{\text{Cool}} \text{N}_2\text{O}_4$                                                                      | Colourless solid/liquid, acidic           |
| Dinitrogen pentaoxide<br>[Nitrogen(V) oxide]  | N <sub>2</sub> O <sub>5</sub> | +5                          | $4\text{HNO}_3 + \text{P}_4\text{O}_{10} \rightarrow 4\text{HPO}_3 + 2\text{N}_2\text{O}_5$                                                            | colourless solid, acidic                  |

## COMPOUNDS OF NITROGEN FAMILY

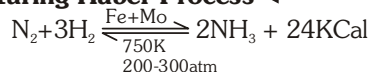
AMMONIA (NH<sub>3</sub>)

Colorless, Pungent Smell, basic in nature liquified easily, uses as coolant

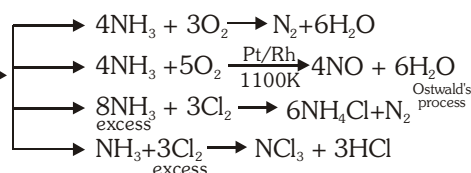
## Preparation



## Manufacturing Haber Process

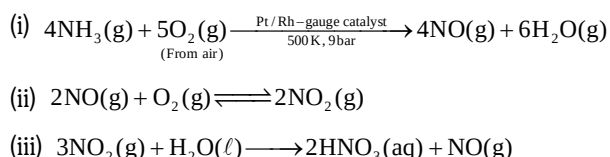


MOP

NITRIC ACID (HNO<sub>3</sub>)

## Preparation :

## Ostwald's process :

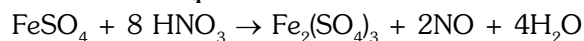


## Properties :

HNO<sub>3</sub>, nitric acid was earlier called as aqua fortis (meaning strong water).

It usually acquires yellow colour. due to its decomposition by sunlight into NO<sub>2</sub>.

It acts as a strong oxidising agent.

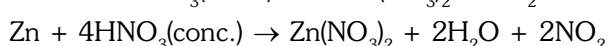
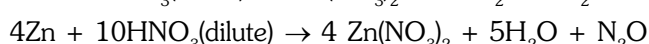
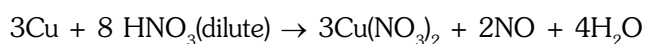
Reaction with FeSO<sub>4</sub> :

## Reaction with non-metals :

Non-metals converted into highest oxyacids by hot and conc.

HNO<sub>3</sub>. NO<sub>2</sub> gas is evolved (S to H<sub>2</sub>SO<sub>4</sub>; P to H<sub>3</sub>PO<sub>4</sub>; C to H<sub>2</sub>CO<sub>3</sub>; I<sub>2</sub> to HIO<sub>3</sub>; As to H<sub>3</sub>AsO<sub>4</sub>; Sb to H<sub>3</sub>SbO<sub>4</sub> and Sn to H<sub>2</sub>SnO<sub>3</sub>). Most of the metals except noble metals are attacked by HNO<sub>3</sub>. It plays double role in action on metals, i.e., it acts as an acids as well as an oxidising agent.

## Reaction with metals :





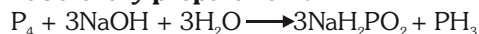
| Concentration of nitric acid | Metal              | Main products                                                  |
|------------------------------|--------------------|----------------------------------------------------------------|
| Very Dilute $\text{HNO}_3$   | Mg, Mn             | $\text{H}_2$ + metal nitrate                                   |
|                              | Fe, Zn, Sn         | $\text{NH}_4\text{NO}_3$ + metal nitrate                       |
|                              | Cu, Ag, Hg         | No reaction                                                    |
| Dilute $\text{HNO}_3$        | Fe, Zn             | $\text{N}_2\text{O}$ + metal nitrate                           |
|                              | Pb, Cu, Ag         | NO + metal nitrate                                             |
| Conc. $\text{HNO}_3$         | Sn                 | $\text{NO}_2$ + $\text{H}_2\text{SnO}_3$<br>(Metastannic acid) |
| Conc. $\text{HNO}_3$         | Fe, Co, Ni, Cr, Al | rendered passive                                               |

### PHOSPHINE ( $\text{PH}_3$ )

#### Preparation :



#### Laboratory preparation :



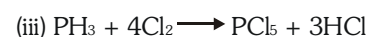
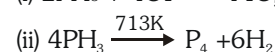
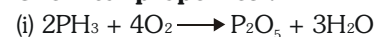
#### Uses :

As Holme's signals in deep seas and oceans.  
For the production of smoke screens.

#### Physical properties :

Colourless gas having smell of garlic or rotten fish, slightly soluble in water and slightly heavier than air.

#### Chemical properties :



### Oxoacids of Phosphorus

| Name                          | Formula                          | Oxidation state of Phosphorus | Characteristic bonds and their number          | Preparation                                              |
|-------------------------------|----------------------------------|-------------------------------|------------------------------------------------|----------------------------------------------------------|
| Hypophosphorus (Phosphinic)   | $\text{H}_3\text{PO}_2$          | + 1                           | One P — OH<br>Two P — H<br>One P = O           | white $\text{P}_4$ + alkali                              |
| Orthophosphorous (Phosphonic) | $\text{H}_3\text{PO}_3$          | + 3                           | Two P — OH<br>One P — H<br>One P = O           | $\text{P}_2\text{O}_3 + \text{H}_2\text{O}$              |
| Pyrophosphorous               | $\text{H}_4\text{P}_2\text{O}_5$ | + 3                           | Two P — OH<br>Two P — H<br>Two P = O           | $\text{PCl}_3 + \text{H}_3\text{PO}_3$                   |
| Hypophosphoric                | $\text{H}_4\text{P}_2\text{O}_6$ | + 4                           | Four P — OH<br>Two P = O<br>One P — P          | red $\text{P}_4$ + alkali                                |
| Orthophosphoric               | $\text{H}_3\text{PO}_4$          | + 5                           | Three P — OH<br>One P = O                      | $\text{P}_4\text{O}_{10} + \text{H}_2\text{O}$           |
| Pyrophosphoric                | $\text{H}_4\text{P}_2\text{O}_7$ | + 5                           | Four P — OH<br>Two P = O<br>One P — O — P      | heat phosphoric acid                                     |
| Metaphosphoric*               | $(\text{HPO}_3)_n$               | + 5                           | Three P — OH<br>Three P = O<br>Three P — O — P | phosphorous acid + $\text{Br}_2$ , heat in a sealed tube |



## GROUP 16 ELEMENTS

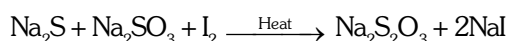
## OXYGEN FAMILY

- Oxygen is the most abundant of all the elements on earth crust.
  - Oxygen and sulphur are non-metals, selenium and tellurium metalloids, whereas polonium is a metal.
  - Polonium hardly shows  $-2$  oxidation state.
  - Reducing property of dioxide decreases from  $\text{SO}_2$  to  $\text{TeO}_2$ ;  $\text{SO}_2$  is reducing while  $\text{TeO}_2$  is an oxidising agent.
  - Sulphur hexafluoride,  $\text{SF}_6$  is exceptionally stable for steric reasons.
  - The well known monohalides are dimeric in nature. Examples are  $\text{S}_2\text{F}_2$ ,  $\text{S}_2\text{Cl}_2$ ,  $\text{S}_2\text{Br}_2$ ,  $\text{Se}_2\text{Cl}_2$  and  $\text{Se}_2\text{Br}_2$ . These dimeric halides undergo disproportionation as given below :  $2\text{Se}_2\text{Cl}_2 \rightarrow \text{SeCl}_4 + 3\text{Se}$
- ☐ The weakening of  $\text{M}-\text{H}$  bond with the increase in the size of  $\text{M}$  (not the electronegativity) explains the increasing acidic character of hydrides down the group.
- ☐ **Halides** : All these elements form a number of halides. The halides of oxygen are not very stable. Selenium does not form dihalides.
- e.g.  $\text{OF}_2$ ,  $\text{Cl}_2\text{O}$ ,  $\text{I}_2\text{O}_5$ , etc.

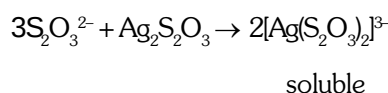
**SODIUM THIOSULPHATE  $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$**

**Preparation :**

- $\text{Na}_2\text{SO}_3 + \text{S} \xrightarrow{\text{Boil}} \text{Na}_2\text{S}_2\text{O}_3$
- $4\text{S} + 6\text{NaOH} \xrightarrow{\text{Boil}} \text{Na}_2\text{S}_2\text{O}_3 + 2\text{Na}_2\text{S} + 3\text{H}_2\text{O}$
- $2\text{Na}_2\text{S} + 3\text{SO}_2 \xrightarrow{\text{Boil}} 2\text{Na}_2\text{S}_2\text{O}_3 + \text{S}$
- Spring's reaction



**Concentrated  $\text{Na}_2\text{S}_2\text{O}_3$**  does not give a white ppt. with  $\text{AgNO}_3$ . This is because silver thiosulphate formed is readily soluble in excess of sodium thiosulphate forming soluble complex.

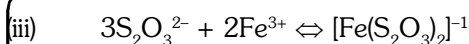


**Reaction with oxidizing agent :**

- (i)  $\text{Na}_2\text{S}_2\text{O}_3 + \text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{HCl} + \text{S}$
- (ii)  $2\text{Na}_2\text{S}_2\text{O}_3 + \text{I}_2 \rightarrow \text{Na}_2\text{S}_4\text{O}_6 + 2\text{NaI}$

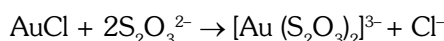
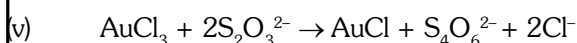
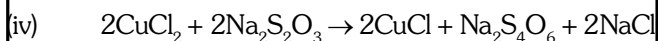
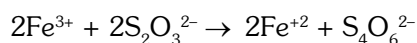
### Sodium tetrathionate

This reaction forms the basis of iodometric estimation of  $\text{Cr}_2\text{O}_7^{2-}$  and  $\text{Cu}^{2+}$  salts.



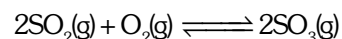
Violet

The violet colour disappears quickly due to the reduction of ferric chloride by  $\text{S}_2\text{O}_3^{2-}$  ions,

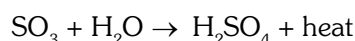


**Sulphur trioxides** ( $\text{SO}_3$ ) is a white crystalline solid with melting point 290K and boiling point 318K.

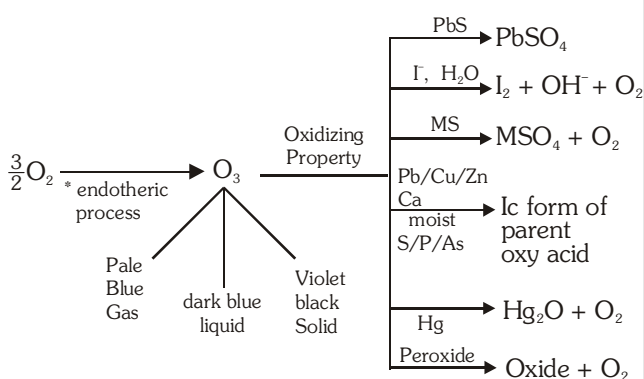
SO<sub>3</sub> is prepared by the direct oxidation of SO<sub>2</sub> with atmospheric oxygen in presence of finely divided Pt or V<sub>2</sub>O<sub>5</sub> at a pressure of 2 bar. and temperature 720K.



$\text{SO}_3$  reacts with water to produce  $\text{H}_2\text{SO}_4$  and large amount of heat is evolved.



## OZONE

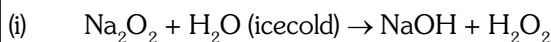


\* Fishy odour, Highly toxic.

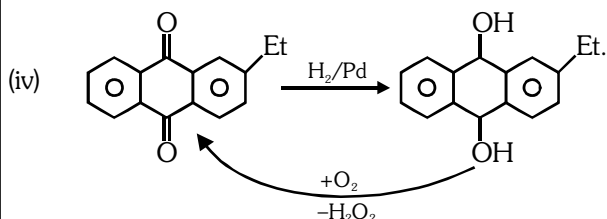
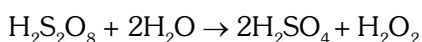
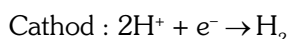
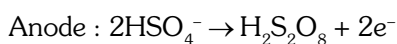
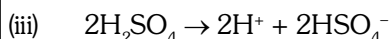
| <b>Gases</b>    | <b>Absorbent</b>                |
|-----------------|---------------------------------|
| O <sub>3</sub>  | Turpentine Oil/Oil of cinnamon. |
| NO              | FeSO <sub>4</sub>               |
| SO <sub>2</sub> | NaOH                            |
| CO              | Cu <sub>2</sub> Cl <sub>2</sub> |
| O <sub>2</sub>  | Pyrogallol                      |

## HYDROGEN PEROXIDE

### Preparation :

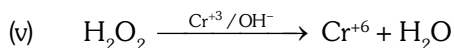
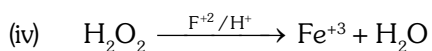
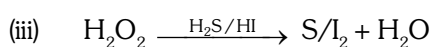
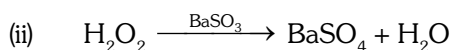
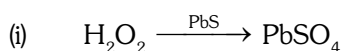


(Now a day conc.  $\text{H}_3\text{PO}_4$  is used)

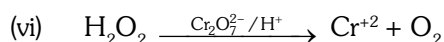
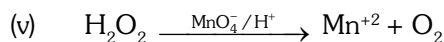
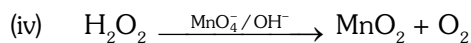
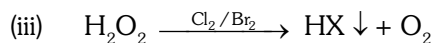
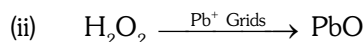
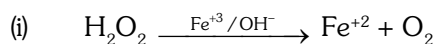


### Properties :

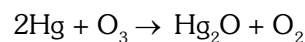
**Oxidizing property :**



### Reducing property :



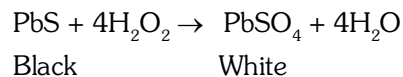
- Tailing of mercury is due to the formation of mercury oxide when Hg reacts with ozone.



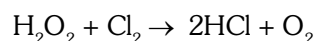
As a result of it, mercury loses its meniscus and starts sticking to the glass.

➤ The bleaching action of ozone and  $\text{H}_2\text{O}_2$  is permanent and is due to oxidation while that of  $\text{SO}_2$  is temporary and is due to reduction.

➤  $\text{H}_2\text{O}_2$  is used to restore the colour of old lead paintings which have been blackened due to the formation of  $\text{PbS}$  by the action of  $\text{H}_2\text{S}$  present in the air.

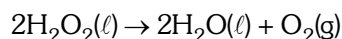


- Antichlor action of  $\text{H}_2\text{O}_2$  is due to its tendency to reduce the excess of chlorine from bleaching powder to  $\text{HCl}$ .


$$\Rightarrow \text{NH}_2.\text{NH}_2 + \text{H}_2\text{O}_2 \text{ as rocket propellant.}$$

## Storage

$\text{H}_2\text{O}_2$  decomposes slowly on exposure to light.



In the presence of metal surfaces or traces of alkali (present in glass containers), the above reaction is catalysed. It is, therefore, stored in wax-lined glass or plastic vessels in dark. Acetalanilide or Glycerol or Urea can be added as a stabiliser. It is kept away from dust because dust can induce explosive decomposition of the compound.

## SULPHURIC ACID ( $H_2SO_4$ )

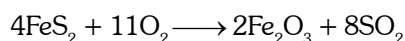
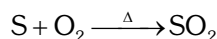
It is also known as oil of vitriol and king of chemicals.

### Manufacture of sulphuric acid :

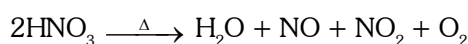
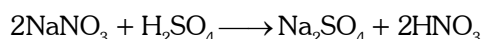
#### Lead chamber process :

The various steps involved are :

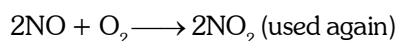
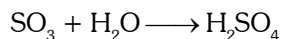
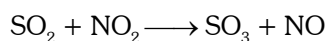
- (a) Production of  $SO_2$  : By burning S or iron pyrites.



- (b) Production of catalyst : Oxides of nitrogen.



- (c) Reaction in lead chamber

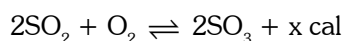


#### Contact process :

#### The steps involved are :

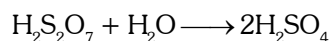
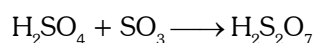
- (a) **Production of  $SO_2$**  : It is produced by burning sulphur or iron pyrites and purified by treating with steam to remove dust particles.

- (b) **Conversion of  $SO_2$  to  $SO_3$**  : It is done in container or catalyst chamber after being pre-heated to  $450^\circ C$ .



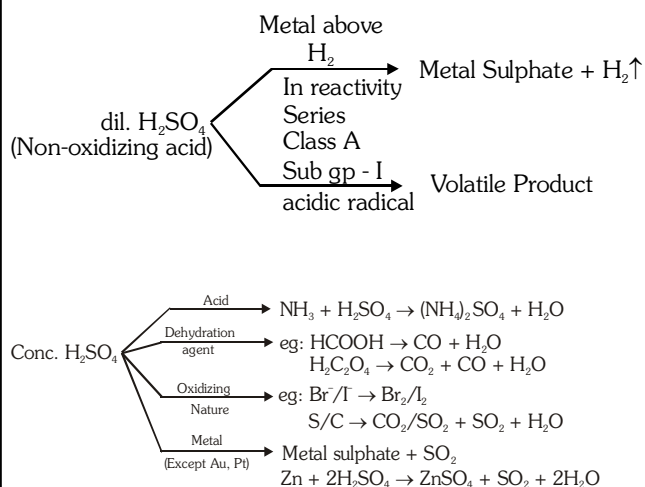
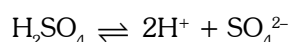
Catalyst : Formerly, platinised asbestos was used which is costly and easily poisoned. These days  $V_2O_5$  is used.

- (c)  $SO_3$  is absorbed by conc.  $H_2SO_4$  and then water is added to produce the acid of desired concentration.



**Properties** : Its specific gravity is 1.8 and it is 98% by weight.

It is strong dibasic acid.



#### Uses :

In lead storage batteries.

In manufacture of paints and pigments.

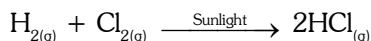
In metallurgy for electrolytic refining of metals.



## HYDROCHLORIC ACID, (HCl)

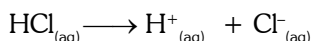
**Preparation :** By dissolving hydrogen chloride gas in water. Hydrogen chloride gas required in turn can be prepared by the following methods:

By the direct combination of hydrogen and chlorine.



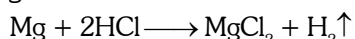
Hydrogen chloride gas can also be obtained by burning hydrogen in chlorine.

**Properties :** Hydrogen chloride is a covalent compound but when dissolved in water it ionizes to form hydrogen ions and chloride ions.

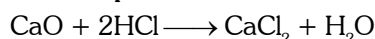


Thus anhydrous HCl does not show acidic properties. Only aqueous HCl or in presence of moisture, HCl behaves as an acid.

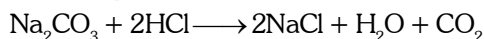
Metal + Hydrochloric acid  $\longrightarrow$  Metal chloride + Hydrogen



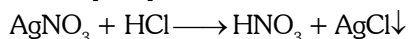
It reacts with bases and basic oxides or hydroxides to form their respective chlorides and water.



It reacts with metal carbonates, bicarbonates, sulphides, sulphites, thiosulphates and nitrites, etc, to form their respective chlorides.

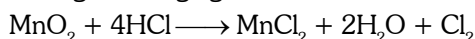


It reacts with silver nitrate and lead nitrate solution to form white precipitates.



### Reducing property :

HCl is a strong reducing agent.



### Uses :

In the production of dyes, paints, photographic chemicals, etc.

Used in the preparation of chlorides, chlorine, aqua-regia, etc.

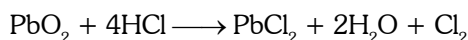
Used as a laboratory reagent.

## PSEUDO HALIDES

Pseudo halides are uninegative groups which show certain characteristics of halide ions. For example Cyanide (CN<sup>-</sup>), Cyanate (OCN<sup>-</sup>), Thiocyanate (SCN<sup>-</sup>), Selenocyanate (SeCN<sup>-</sup>), Azide (N<sub>3</sub><sup>-</sup>), Aziothio carbonate (SCSN<sub>3</sub><sup>-</sup>) and isocyanate (ONC<sup>-</sup>).

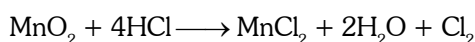
## CHLORINE (Cl<sub>2</sub>)

**Preparation :** By oxidation of conc. HCl.

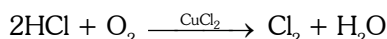


### Manufacture :

**Weldon's process :** By heating pyrolusite with conc. HCl.

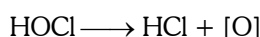
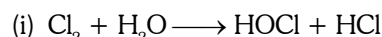


### Deacon process :



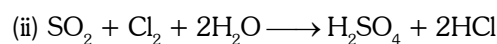
**Properties :** It is a yellowish green gas, poisonous in nature, soluble in water. Its aqueous solution is known as chlorine water which on careful cooling gives chlorine hydrate Cl<sub>2</sub>·8H<sub>2</sub>O.

Bleaching action and oxidising property

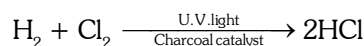


Coloured matter + nascent [O]  $\rightarrow$  Colourless matter

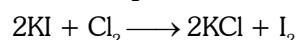
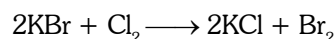
The bleaching action of chlorine is permanent and is due to its oxidising nature.



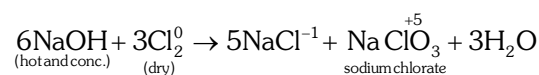
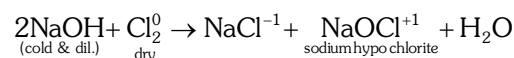
Action of hydrogen :



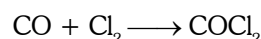
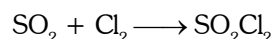
### Displacement reactions :



### Action of NaOH :



### Addition reactions :

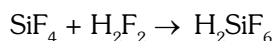
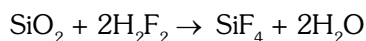


### Test for chlorine :

- It is a greenish yellow gas with irritating smell.
- It turns starch iodide paper blue.
- It bleaches litmus paper and indigo solution.

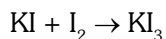
**NOTE**

- **Action of silica and glass** : Strong solution of HF attacks glass readily forming silicon fluoride which gives complex fluosilicic acid,  $\text{H}_2\text{SiF}_6$  with excess HF.



This is called etching of glass.

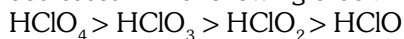
- Iodine is slightly soluble in water but much more soluble in KI due to the formation of  $\text{KI}_3$ .



The solution behaves as a simple mixture of KI and  $\text{I}_2$ .

**OXY-ACID**

The acidic strength of the oxoacids of halogen decreases in the following order.



The chlorine oxyanion in which the central atoms has highest oxidation number will have maximum number of oxygen atoms for participation in the extension of the  $\pi$ -bond. Thereby the charge on the ion is delocalized which greatly stabilizes the ion and thus decreases its tendency to accept a proton i.e., causes the ion to be very weak base with the result that the strength of the acid increases.

The acidity of oxoacids of different halogens having the same oxidation number decreases with the increases in atomic numbers of the halogen



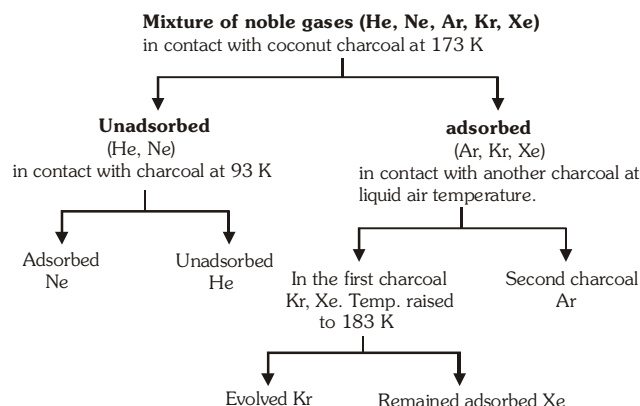


## GROUP 18 ELEMENTS

- Relative Abundance:  
Ar > Ne > Kr > He > Xe > Rn
- Melting and Boiling point:  
Rn > Xe > Kr > Ar > Ne > He
- He has the lowest boiling point among all element
- Noble gases are monoatomic, colourless, odourless, sparingly soluble in water and diffusing through most commonly used laboratory materials.
- O<sub>2</sub>[PtF<sub>6</sub>] is the first noble gas compound.
- Ar, Kr and Xe form clathrate compound when pass in ice with high pressure.  
A.6H<sub>2</sub>O (A = Ar/Kr/Xe)

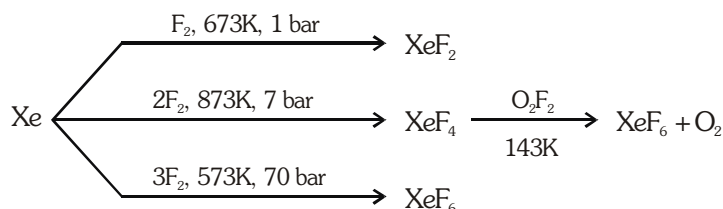
## NOBLE GAS

➤ In Dewar's method, the separation of noble gases is summarized below

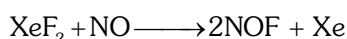
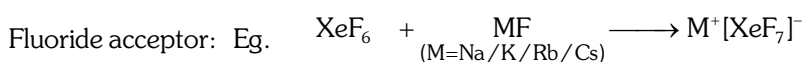
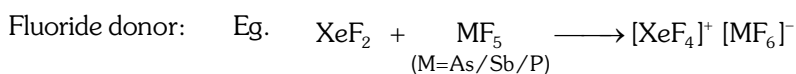


➤ The viscosity of He is extremely low, about 1/100<sup>th</sup> of hydrogen gas.

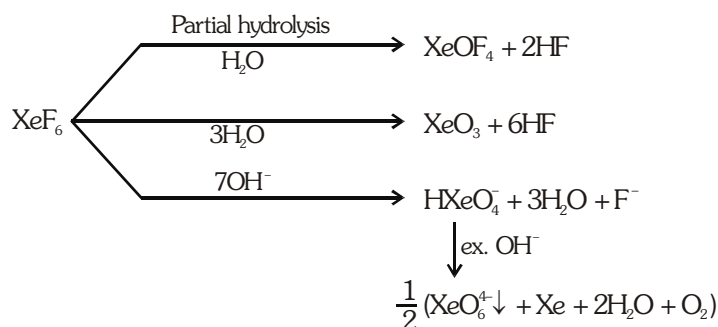
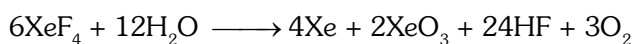
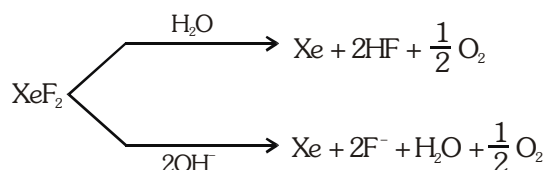
### Preparation of Xe-Fluoride:



### Properties:



### Hydrolysis:



# COORDINATION CHEMISTRY

## Addition compound

(two or more simple salt combining with fixed proportion of mass)

### Double salt

\* Loose their identity in aqueous solution

eg.  $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$  carnalite salt

$\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$

potash alum

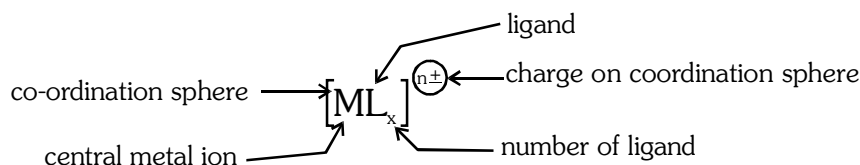
### C-ordination compound

\* Retain their identity in aqueous solution

eg.  $\text{K}_4[\text{Fe}(\text{CN})_6]$  Potassium

hexacyanidoferrate (II)

## REPRESENTATION OF COMPLEX COMPOUND



Co-ordination number = Number of electron pair accepted by central metal ion.

### LIGAND

Chemical species which can donate electron pair.

#### Classification on the basis of denticity

(Denticity : = Number of electron pair donated by central metal atom or ion)

**Monodentate** (denticity = 1) :

eg.  $\text{H}^-$ ,  $\text{X}^-$ ,  $\text{Py}$ ,  $\text{N}_2$ ,  $\text{N}_3^-$ ,  $\text{N}_2\text{H}_4$  etc.

**Bidentate** (denticity = 2) :

eg. en, pn, bn,  $\text{ox}^{2-}$ ,  $\text{acac}^-$ ,  $\text{gly}^-$ ,  $\text{dmg}^-$

**Polydentate** (denticity = >2) :

eg. dien, imda<sup>2-</sup>, trien, nta<sup>3-</sup>,  $\text{EDTA}^{4-}$

**Ambidentate** : Ligand which have more than one donor site but at the time of donation only atom can donate electron pair.

eg. :  $(\text{CN}^-, \text{NC}^-)$ ,  $(\text{NO}_2^-, \text{ONO}^-)$ ,  $(\text{SCN}^-, \text{NCS}^-)$ ,  $(\text{OCN}^-, \text{NCO}^-)$ ,  $(\text{S}_2\text{O}_3^{2-}, \text{SO}_2\text{S}^{2-})$

**Flexidentate ligand** : Show more than one type of denticity.

eg. :  $\text{CO}_3^{2-}$ ,  $\text{SO}_4^{2-}$ ,  $\text{CH}_3\text{COO}^-$

**Note** : Bidentate and Polydentate are also called chelating ligand.

### Classification on the basis of electron donating and accepting tendency

**Classical ligand** :  $\text{H}_2\text{O}$ ,  $\text{NH}_3$  etc.

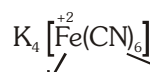
**Non-classical ligand** :  $\text{CN}^-$ ,  $\text{NO}^+$ ,  $\text{NO}$ ,  $\text{CO}$ ,  $\text{PPh}_3$  etc

$\Rightarrow \pi$ -donor ligand  $\Rightarrow \text{C}_2\text{H}_4$ ,  $\text{C}_2\text{H}_2$  etc.

### BONDING IN COORDINATION COMPOUND

#### Effective atomic number & Sidgwick rule :

Total number of electron present on central metal atom or ion after accepting the electron pair from ligand.



$$\text{EAN} = 26 - (+2) + 6(2) = 36$$

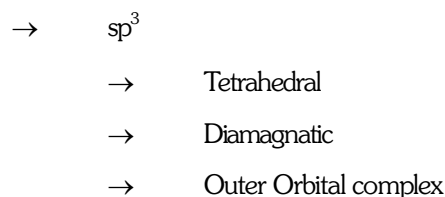
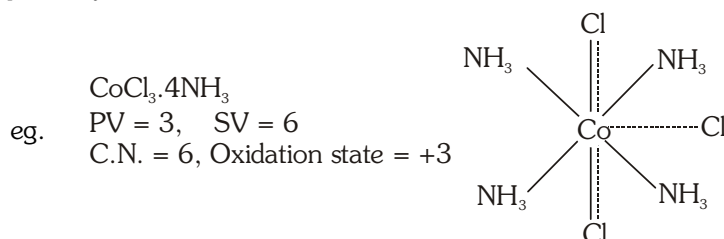
$\Rightarrow$  If EAN value is equal to atomic number of Noble gas then complex follow sidgwick rule of EAN.

$\Rightarrow$  In carbonyl complex if EAN value = Atomic number of Noble gas then carbonyl complex is more stable.

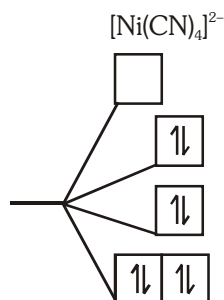
|     |                           |                            |                            |
|-----|---------------------------|----------------------------|----------------------------|
| eg. | $[\text{V}(\text{CO})_6]$ | $[\text{Cr}(\text{CO})_6]$ | $[\text{Mn}(\text{CO})_5]$ |
|     | act as oxidising agent    | stable complex             | act as reducing agent      |

|               |                               |    |
|---------------|-------------------------------|----|
| $\Rightarrow$ | Brown ring complex            | 37 |
|               | Sodium nitroprusside          | 36 |
|               | Zeise's Salt                  | 84 |
|               | $\text{Mn}_2(\text{CO})_{10}$ | 36 |
|               | $\text{Fe}_2(\text{CO})_9$    | 36 |
|               | $\text{Co}_2(\text{CO})_8$    | 36 |

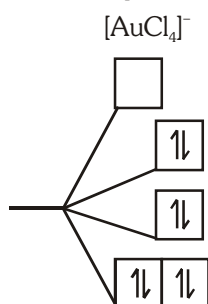
- ⇒ Metals possesses two types of valencies PV & SV.
- ⇒ PV is non-directional, represent by ..... (dotted line) is satisfied by negative charge species.
- ⇒ SV is directional, represent by \_\_\_\_\_ (solid line) and satisfied by negative or neutral species.
- ⇒ Now a days primary valency and secondary valency is consider as oxidation & co-ordination number respectively.







$\text{dsp}^2$ , sq. planar  
inner orbital complex,  
diamagnetic  
low spin complex



$\text{ds}^2$ , sq. planar  
inner orbital complex,  
diamagnetic  
low spin complex,

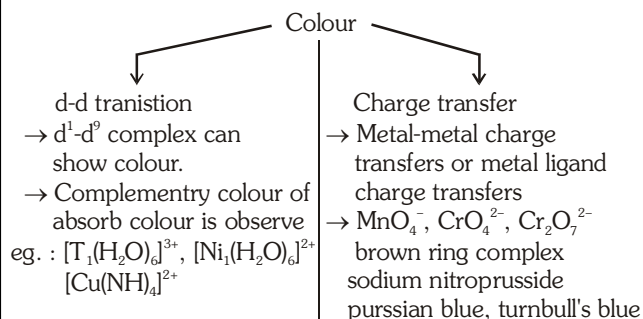
### Exception :

1.  $\text{d}^3\text{s}$  hybridisation, Td, diamagnetic, inner orbital complex  
eg.  
 $\text{MnO}_4^-$ ,  $\text{CrO}_4^{2-}$ ,  $\text{Cr}_2\text{O}_7^{2-}$ ,  $\text{CrO}_2\text{Cl}_2$ ,  $\text{CrO}_2\text{F}_2$ ,  $\text{VO}_4^{3-}$
2. Transference of electron  
eg.  $\text{Cu}^{+2}$  in  $\text{CN}^-$  with L  
(where  $\text{L} = \text{NO}_2^- / \text{CN}^- / \text{NH}_3$  etc.)

### Factor affecting splitting

- (i) Strength of ligand
- (ii) Oxidation state of central metal ion
- (iii) Transition series (d-series)
- (iv) Geometry (number of ligands).
- (v) Chelation

### Colour of complexes

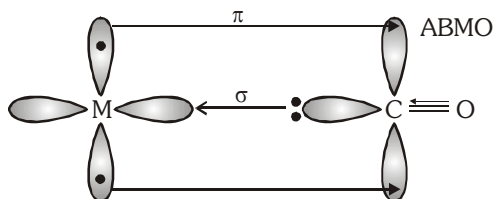


### ORGANOMETALLIC COMPOUNDS

Compounds in which the less E.N. (Ge, Sb, B, Si, P, As) central metal atoms are bonded directly to carbon atoms are called organometallic compounds.

- **$\sigma$ -bonded compounds** formed by nontransition elements.  
 $\text{R-Mg-X}$ ,  $(\text{CH}_3\text{-CH}_2)_2\text{Zn}$ , Ziegler natta catalyst, etc.
- **$\pi$ -bonded organometallic compounds** are generally formed by transition elements e.g. Zeise's salt, ferrocene, dibenzene chromium, etc.
- **$\sigma$ -and  $\pi$ -bonded organometallic compounds** : Metal carbonyls, compounds formed between metal and carbon monoxide belong to this class.  $\text{Ni}(\text{CO})_4$ ,  $\text{Fe}(\text{CO})_5$  etc.

### Synergic bonding



### IUPAC nomenclature of complex compounds :

- (A) For anionic complex (like  $\text{K}_4[\text{Fe}(\text{CN})_6]$ )  
Common name of normal cation (without numeral prefix) + name of ligands (with numeral prefix) + latin name of CMI along with suffix ate + Ox. St (in roman number).  
eg. : Potassium hexacyanoferrate (II)
- (B) For cationic complex like  $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$   
Name of ligands (with numeral prefix) + Common name of CMI + Ox. St (In roman number) + Name of anion (without numeral prefix)  
eg. : Tetraammine copper (II) sulphate.
- (C) For neutral complex (like  $[\text{Fe}(\text{CO})_5]$ )  
Name of ligands (with numeral prefix) + Common name of CMI + Ox. St. (In roman number)  
eg. : Pentacarbonyl iron (0)
- (D) Rule same just apply alphabetical order when write the name of ligands.  
e.g.  $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$   
Diamminedichloroplatinum (II)

**STRUCTURAL ISOMERISM**

- (i) **Ionisation isomerism** : Counter ion as a ligand & ligand act as counter ion.  
 $[\text{Co}(\text{H}_2\text{O})_5\text{Cl}]\text{SO}_4 \leftrightarrow [\text{Co}(\text{H}_2\text{O})_5\text{SO}_4]\text{Cl}$
- (ii) **Hydrate isomersim** : Number of water molecule inside & outside the co-ordinate sphere are different.  
 $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3 \leftrightarrow [\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O} \leftrightarrow [\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$
- (iv) **Linkage** : Exhibit when ambidentate ligand is present in co-ordination sphere.  
 $[\text{NC} \rightarrow \text{Ag} \leftarrow \text{CN}]^- \leftrightarrow [\text{NC} \rightarrow \text{Ag} \leftarrow \text{NC}]^- \leftrightarrow [\text{CN} \rightarrow \text{Ag} \leftarrow \text{NC}]^-$
- (iv) **Co-ordination isomerism** : Exhibit when both are cationic & anionic complex  
 $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_4] \leftrightarrow [\text{Pt}(\text{NH}_3)_3\text{Cl}][\text{Pt}(\text{NH}_3)_3\text{Cl}_3]$

**STEREO ISOMERISM****STEREO ISOMERISM IN CO-ORDINATION COMPOUND****CN-4**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>* Square planar complex does not show optical isomerism.</p> <p>* Square planar complex show optical activity if the co-ordinated ligand having chiral center.</p> <p>* Square planar complex<br/> <math>[\text{Ma}_2\text{b}_2]^{n+}</math>, <math>[\text{Ma}_2\text{bc}]^{n+}</math>, <math>[\text{Mabcd}]^{n+}</math>, <math>[\text{M}(\text{AB})\text{cd}]^{n+}</math>,<br/> <math>[\text{M}(\text{AB})(\text{CD})]^{n+}</math> show geometrical isomerism</p> <p>* <math>[\text{Mabcd}]^{n+}</math> form two cis and one trans.</p> | <p>* Tetrahedral complex <math>[\text{Mabcd}]^{n+}</math>, <math>[\text{M}(\text{AB})\text{cd}]^{n+}</math><br/> <math>[\text{M}(\text{AB})(\text{CD})]^{n+}</math> show optical isomerism</p> <p>* Tetrahedral complex does not show geometrical isomerism.</p> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**CN-6****NUMBER OF POSSIBLE ISOMERS FOR SPECIFIC COMPLEXES**

| Formula                                   | Number of stereoisomers | Pairs of Enantiomers | Number of G.I. |
|-------------------------------------------|-------------------------|----------------------|----------------|
| $\text{Ma}_4\text{b}_2$                   | 2                       | 0                    | 2              |
| $\text{Ma}_4\text{bc}$                    | 2                       | 0                    | 2              |
| $\text{Ma}_3\text{b}_3$                   | 2                       | 0                    | 2              |
| $\text{Ma}_3\text{b}_2\text{c}$           | 3                       | 0                    | 3              |
| $\text{Ma}_3\text{bcd}$                   | 5                       | 1                    | 4              |
| $\text{Ma}_2\text{b}_2\text{c}_2$         | 6                       | 1                    | 5              |
| $\text{Ma}_2\text{b}_2\text{cd}$          | 8                       | 2                    | 6              |
| $\text{Ma}_2\text{bcde}$                  | 15                      | 6                    | 9              |
| $\text{Mabcdef}$                          | 30                      | 15                   | 15             |
| $\text{M}(\text{AA})(\text{BC})\text{de}$ | 10                      | 5                    | 5              |
| $\text{M}(\text{AB})(\text{AB})\text{cd}$ | 11                      | 5                    | 6              |
| $\text{M}(\text{AB})(\text{CD})\text{ef}$ | 20                      | 10                   | 10             |
| $\text{M}(\text{AB})_3$                   | 4                       | 2                    | 2              |

**Note:** Uppercase letters represent chelating ligands and lowercase letters represent monodentate ligands.

[illegible]

# d-BLOCK (Transition Elements)

## DEFINITION

Transition elements are those d-block elements which has partially filled  $(n-1)d$  subshell either in their ground state or in their stable oxidation states. Therefore Zn, Cd & Hg – are d-block elements but not transition elements.

### GENERAL ELECTRONIC CONFIGURATION

$$ns^{0-2} (n-1)d^{1-10}$$

$$\text{Exceptions} \begin{cases} \text{Cr} = 4s^1 3d^5 \\ \text{Cu} = 4s^1 3d^{10}, \text{Pd} = 5s^0 4d^{10} \end{cases}$$

### TRANSITION SERIES

|                 |           |                                                            |            |
|-----------------|-----------|------------------------------------------------------------|------------|
| 1 <sup>st</sup> | 3d series | Sc <sub>21</sub> — Zn <sub>30</sub>                        | 9 + 1 = 10 |
| 2 <sup>nd</sup> | 4d series | Y <sub>39</sub> — Cd <sub>48</sub>                         | 9 + 1 = 10 |
| 3 <sup>rd</sup> | 5d series | La <sub>57</sub> , Hf <sub>72</sub> — Hg <sub>80</sub>     | 9 + 1 = 10 |
| 4 <sup>th</sup> | 6d series | Ac <sub>89</sub> , Unq <sub>104</sub> — Uub <sub>112</sub> | 9 + 1 = 10 |

### ATOMIC RADIUS

3d series Sc > Ti > V > Cr > Mn ≥ Fe ≈ Co ≈ Ni < Cu < Zn

**In a group** 3d to 4d series increases but 4d and 5d series nearly same due to poor shielding of f electron. (Lanthanide contraction)

$$3d < 4d \approx 5d$$

$$\text{e.g.: Ti} < \text{Zr} \approx \text{Hf} \begin{cases} \text{Smallest radius} - \text{Ni} \\ \text{Largest radius} - \text{La} \end{cases}$$

**Melting point** : s-block metals < d-block metals

In a series on increasing number of unpaired  $e^-$  m.p. increases and therefore m.p. increase upto Cr then decreases.

Sc < Ti < V < Cr > Mn < Fe > Co > Ni > Cu > Zn

↓  
Half filled  $d^5$   
∴ weak metallic bond

↓  
Fully filled  $d^{10}$   
∴ weak metallic bond

Melting point  $\begin{cases} \text{Zn} > \text{Cd} > \text{Hg} \\ \text{Cu} > \text{Ag} \leq \text{Au} \end{cases}$  (data based)

E.N. Exception Zn < Cd < Hg

**Density** : s-block metals < d-block metals.

### 3d series

Sc < Ti < V < Cr < Mn < Fe < Co ≤ Ni < Cu > Zn

**Density in a Group** 3d < 4d << 5d

**Metallic character** : They are solid, hard, ductile, malleable, good conductor of heat and electricity and exhibit metallic lusture, high tensile strength. Hg is liquid

Electrical conductor  $\text{Ag} > \text{Cu} > \text{Au} > \text{Al}$   
d-block p-block

### OXIDATION STATE

Transition elements exhibit variable oxidation state due to small energy difference of ns and  $(n-1)d$  electrons.

□ Sc(+3) and Zn(+2) exhibit only one oxidation state

□ Common oxidation state is +2 & +3

□ In 3d series highest oxidation state is +7 (Mn)

□ In d-block series highest oxidation state is +8 (Os, Ru)

□ In carbonyl compound the zero oxidation state of metals is stabilize by synergic effects.

□ Their higher oxidation states are more stable in fluoride and oxides.

□ Higher oxidation states in oxides are normally more stable than fluorides due to capability of oxygen to form multiple bonds.

eg. stable fluoride in higher oxidation state of Mn is  $\text{MnF}_4$  while oxide is  $\text{Mn}_2\text{O}_7$

⇒ Mn show maximum number of oxidation state (+2 to +7) among 3d series.

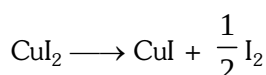
⇒ Beyond Mn trihalide are not observe except  $\text{FeX}_3$  (X = Cl/Br/I) &  $\text{CoF}_3$

⇒  $\text{V}_2\text{O}_3$  basic  $\text{V}_2\text{O}_4$  basic  $\text{V}_2\text{O}_5$  amphoteric

⇒ CrO basic  $\text{Cr}_2\text{O}_3$  amphoteric  $\text{CrO}_3$  acidic

⇒ MnO basic  $\text{MnO}_2$  amphoteric  $\text{Mn}_2\text{O}_7$  acidic

⇒  $\text{CuI}_2$  does not exist



⇒ In aqueous  $\text{Cu}^+$  disproportionated into Cu &  $\text{Cu}^{+2}$ .

In p-block lower oxidation states of heavier elements are more stable while in d-block heavier element, higher oxidation state are more stable.

eg. In VIB gp Mo(+6) & W(+6) are more stable than Cr(+6)



### MAGNETIC PROPERTY

All transition elements are paramagnetic due to presence of unpaired electrons. They attract when magnetic field is applied. Magnetic moment of unpaired electron is due to spin and orbital angular momentum.

"Spin only" magnetic moment can be calculated by using formula  $\mu = \sqrt{n(n+2)}$  Bohr magneton. (n is number of unpaired  $e^-$ .)

If      n is 1  $\mu = 1.73$  BM      n is 2  $\mu = 2.84$  BM      n is 3  $\mu = 3.87$  BM  
           n is 4  $\mu = 4.90$  BM      n is 5  $\mu = 5.92$  BM

Substances that are not attracted by applied magnetic field are diamagnetic. They have all the electrons paired. d-block element and ions having  $d^0$  and  $d^{10}$  configuration are diamagnetic.

### COLOUR

Colour in transition metal ions is associated with d-d transition of unpaired electron from  $t_{2g}$  to  $e_g$  set of energies. This is achieved by absorption of light in the visible spectrum, rest of the light is no longer white.

Colourless –  $Sc^{3+}$ ,  $Ti^{4+}$ ,  $Zn^{2+}$  etc

Coloured –  $Fe^{3+}$  yellow,  $Fe^{2+}$  green,  $Cu^{2+}$  blue,  $Co^{3+}$  blue etc

### ALLOYS

Solid mixture of metals in a definite ratio  
 (15% difference in metallic radius)

They are hard and having high melting point.

eg. Brass (Cu + Zn)

Bronze (Cu + Sn) etc.

Hg when mix with other metals form semisolid amalgam except Fe, Co, Ni, Pt.

### Interstitial compounds :

When less reactive nonmetals of small atomic size eg. H, B, N, C, Trapped in the interstitial space of transition metals, interstitial compounds are formed, like :-  $TiC$ ,  $Mn_4N$ ,  $Fe_3H$  etc.

They are nonstoichiometric compounds.

They have high melting point than metals.

They are chemically inert.

### CATALYTIC PROPERTIES

Most of the d-block compounds act as catalyst due to their variable oxidation state or complex formation tendency

or adsorption on their surface. Example :

Contact process =  $V_2O_5$

Ostwald process = Pt/Rh

Haber process =  $Fe_2O_3 + Al_2O_3 + K_2O$

Zeigter Natta =  $TiCl_4 + (C_2H_5)_3Al$

Phenton reagent =  $FeSO_4 + H_2O_2$

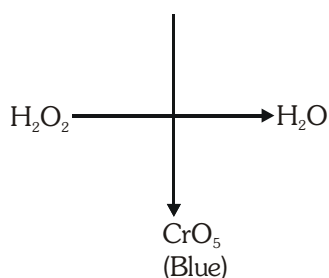
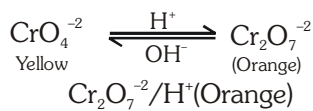
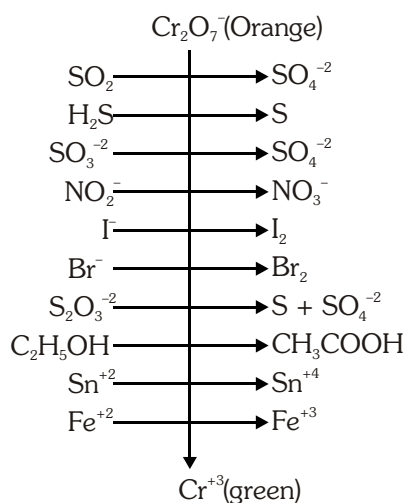
Hydrogenation of Alkene = Ni/Pd

Decomposition of  $KClO_3 = MnO_2$

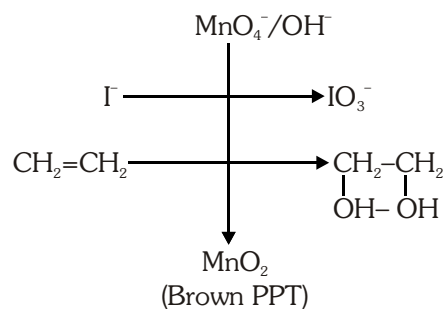
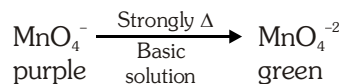
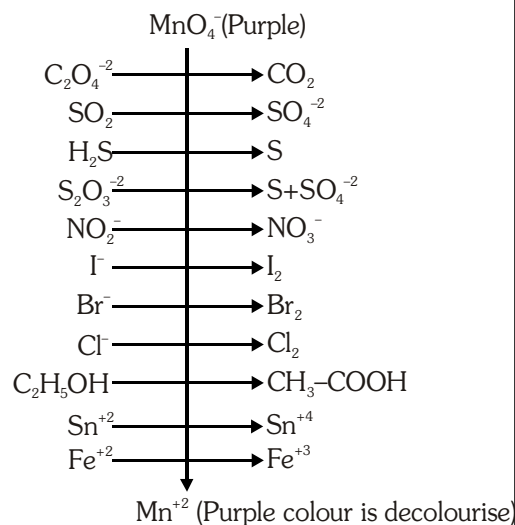
Wilkinson catalyst =  $RhCl + PPh_3$

## Important reactions of d-block elements

- (a)  $\text{Cu}^{2+} + 4\text{I}^- \longrightarrow \text{Cu}_2\text{I}_{2(s)} + \text{I}_2$
- (b)  $\text{CuSO}_4 + \text{KCN} \xrightarrow{\text{Excess}} \text{K}_2\text{SO}_4 + \text{Cu}(\text{CN})_2$   
 $2\text{Cu}(\text{CN})_2 \longrightarrow 2\text{CuCN} + (\text{CN})_2$   
 $\text{CuCN} + 3\text{KCN} \longrightarrow \text{K}_3[\text{Cu}(\text{CN})_4]$   
 (c)  $\text{Cu} \xrightarrow[\text{moist air}]{\text{H}_2\text{O} + \text{CO}_2} \text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$   
 (d)  $\text{AgNO}_3(s) \xrightarrow{\text{Heating}} \text{Ag} + \text{NO}_2 + \frac{1}{2}\text{O}_2$   
 $\text{AgCO}_3(s) \xrightarrow{\text{Heating}} \text{Ag} + \text{CO}_2 + \frac{1}{2}\text{O}_2$
- (e)  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \xrightarrow{100^\circ\text{C}} \text{CuSO}_4 \cdot \text{H}_2\text{O} \xrightarrow{230^\circ\text{C}} \text{CuSO}_4$   
 light greenish blue Colourless  
 $\xrightarrow{720^\circ\text{C}} \text{CuO} + \text{SO}_2 + \frac{1}{2}\text{O}_2$
- (f)  $\text{Hg}_2\text{Cl}_2 + \text{NH}_4\text{OH} \longrightarrow \text{Hg} \begin{matrix} \text{NH}_2 \\ \text{Cl} \end{matrix} + \text{Hg}$   
 Black
- (g)  $\text{NO}_3^- / \text{NO}_2^- \xrightarrow{\text{FeSO}_4 + \text{H}_2\text{SO}_4} [\text{Fe}(\text{H}_2\text{O})_5\text{NO}^+] \text{SO}_4$   
 Brown ring complex
- (h)  $\text{AgBr} + 2\text{Na}_2\text{S}_2\text{O}_3 \longrightarrow \text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2] + \text{NaBr}$   
 Photographic complex
- (i) **Chemical volcano :**  
 $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \xrightarrow{\Delta} \text{N}_2 + 4\text{H}_2\text{O} + \text{Cr}_2\text{O}_3$

Reactions of  $\text{Cr}_2\text{O}_7^{2-}$ :

(Stabilize in the presence of organic solvent such as pyridine etc.)

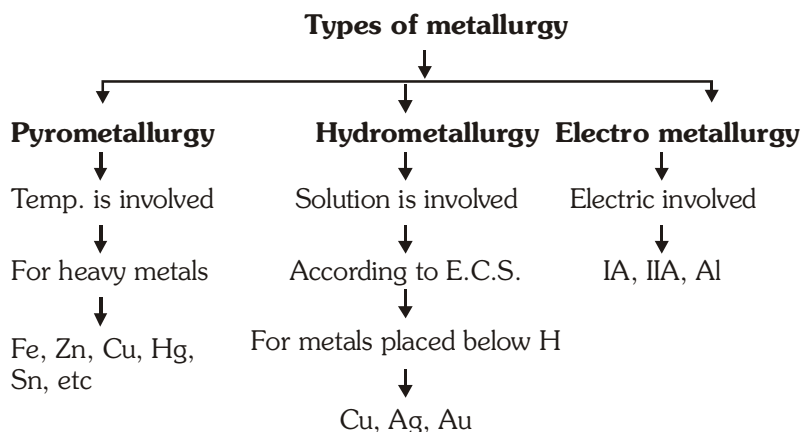
Reaction of  $\text{MnO}_4^-$ 

**IMPORTANT NOTES**

# METALLURGY

➤ Bunch of process to extract metal from their respective ore

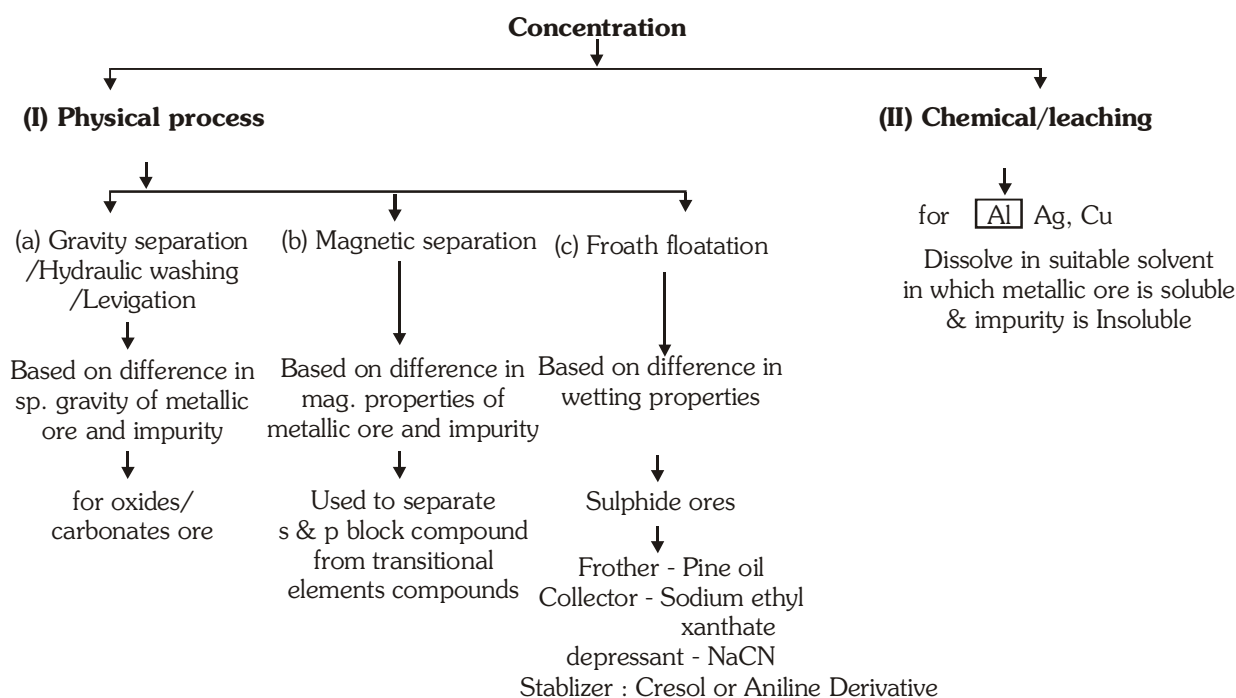
**Ore :** Minerals from which metal can be extracted economically :



**Metallurgical process :**

- 1. Mining :** Ore obtain in big lumps (less reactive)
- 2. Crushing/grinding/pulverization :** big lumps convert into powder (more reactive)
- 3. Concentration :** To remove matrix/gangue from ore

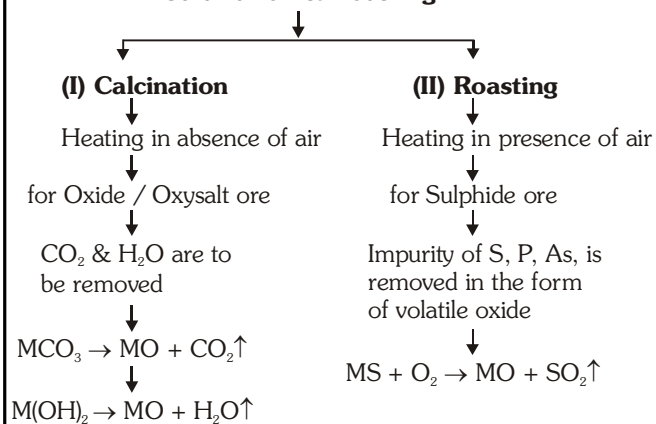
To increase the concentration of ore particle in ore sample.



Ag, Au, are concentrated by cyanide process.

4.

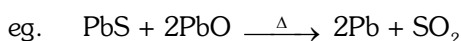
## Calcination & Roasting



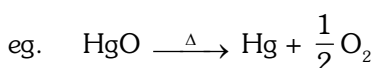
**5. Reduction :** To obtains metal (95 to 98%) from oxidized metallic ore.

### Self reduction :

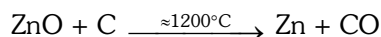
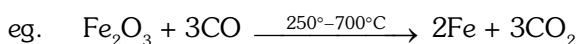
⇒ Sulphide ore itself act as reducing agent.



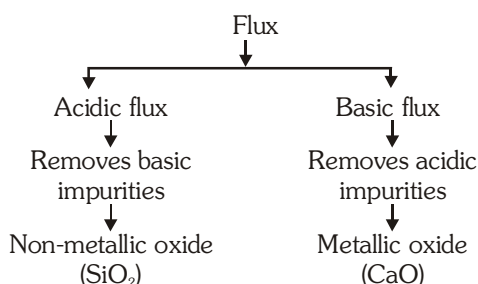
⇒ Thermal decomposition



**Carbon reduction (Smelting) :** Reducing agent C/CO



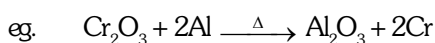
Flux - Substance to convert non-fusible impurities to fusible mass.



Imp. point – Above  $710^{\circ}\text{C}$  is reducing agent.  
– Below  $710^{\circ}\text{C}$  CO is better reducing agent.

**Aluminum reduction method :** (Thermite process)

$\Rightarrow$  Al act as reducing agent



Thermite mixture mass ratio :  $\text{Cr}_2\text{O}_3 : \text{Al} = 3 : 1$

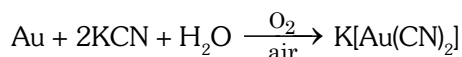
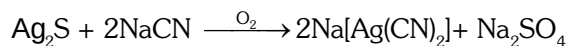
## METAL DISPLACEMENT REDUCTION

$\Rightarrow$  Metal placed below H. in E.C.S.

$$\Rightarrow \text{Ag, Au, Cu}$$

### Example of extraction of :

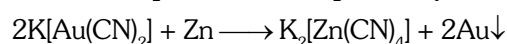
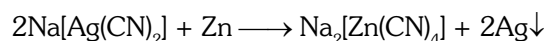
(i) Cynidation (Leaching Process)



NaCN / KCN → Leaching agent

$O_2 \rightarrow$  oxidizing agent

(ii) Recovery of Ag/Au (Metal displacement Reactions)



Zn → reducing agent

## ELECTROLYTIC REDUCTION

$\Rightarrow$  For IA, IIA, AI

⇒ Electrolysis of molten solution

### Example :-

(i) Extraction of Al(Hall-Herault)

- Al can be extracted from  $\text{Al}_2\text{O}_3$

–  $\text{Na}_3\text{AlF}_6$  &  $\text{CaF}_2$  act as auxiliary electrolyte increase the conductivity & to reduce the fusion temp. of mixture

(ii) At cathode impure Al is collected and at anode  $O_2, CO, CO_2$  is released.

(iii) Extraction of Na (Down cell process)

- Na can be extracted from NaCl

- Down Process Neutral flux ( $\text{CaCl}_2$ ) to be added to decrease the fusion temp. of NaCl

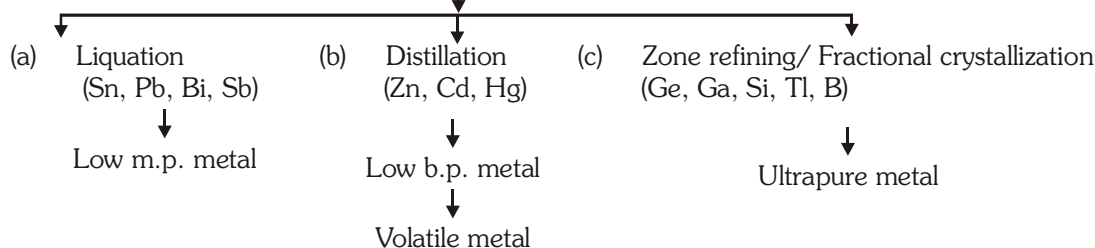
- Neutral flux - substance used to increase the conductivity of NaCl

Auxiliary Electrolyte – decrease the fusion temp. of ionic compounds of (IA, IIA, Al) which is more than the melting point of metal.

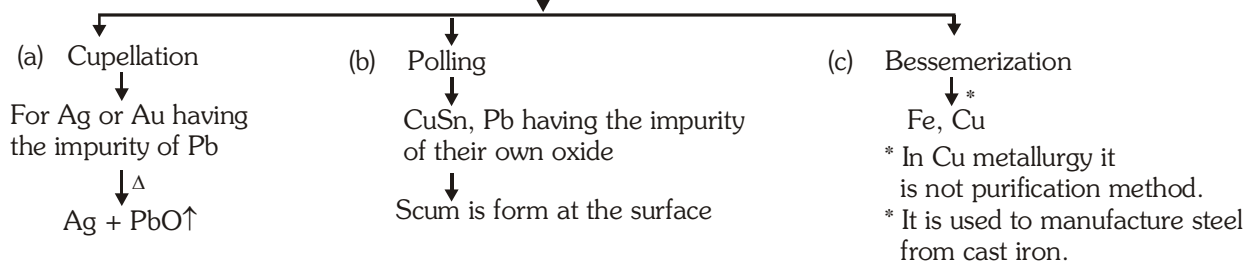
## REFINING

To obtain metal (99.98%)

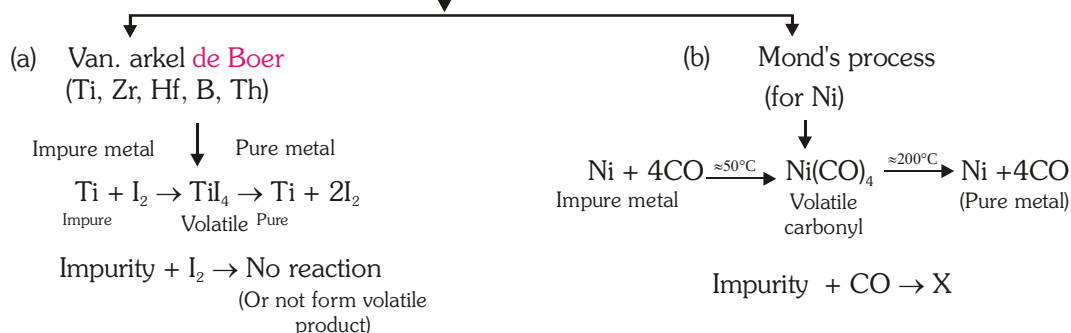
## Physical process



## Chemical Process



## Thermal decomposition (Vapour Phase Refining)



## (IV) Electrolytic refining

Anode – made up of impure metal

Cathode – pure metal

Impurity deposited below anode as anode mud or goes to electrolytic solution pure metal deposited at cathod.

## THERMODYNAMICS PRINCIPLE OF METALLURGY

- The graphical representation of Gibbs energy was first used by H.I.T. Ellingham. This provides a sound basis for considering the choice of reducing agent in the reduction of oxides. This is known as Ellingham diagram. Such diagrams help us in predicting the feasibility of thermal reduction of an ore.
- The criteria of feasibility is that at a given temperature, Gibbs energy of reaction must be negative.
- At high temperature 'C' is the best reducing agent.
- At low temperature 'CO' is the best reducing agent.
- In blast furnace reduction takes place at low temperature i.e. why CO is the reducing agent (For Fe).

A summary of the occurrence and Extraction of some Metals is presented in the following table :

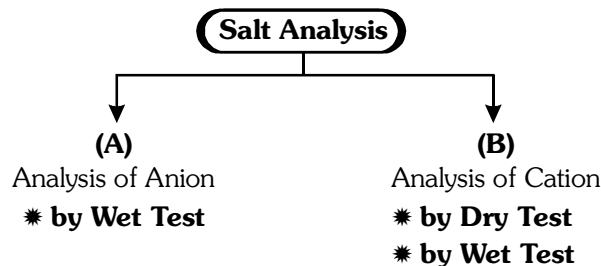
| Metal                          | Occurrence                                                                                                                                                        | Common method of extraction                                                                                                                           | Reffining                                                                 | Remarks                                                                                                                                                                |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aluminium<br>$E^\circ = -1.66$ | 1. Bauxite, $Al_2O_3 \cdot xH_2O$<br>2. Cryolite, $Na_3AlF_6$                                                                                                     | Electrolysis of $Al_2O_3$ dissolved in molten $Na_3AlF_6 + CaF_2$                                                                                     | Electrolytic refining by Hoop's cell                                      | For the extraction, a good source of electricity is required                                                                                                           |
| Iron<br>$E^\circ = -0.44$      | 1. Haematite, $Fe_2O_3$<br>2. Magnetite, $Fe_3O_4$<br>3. Limonite, $Fe_2O_3 \cdot 3H_2O$<br>4. Siderite, $FeCO_3$                                                 | Reduction of the oxide with CO and coke in Blast furnace                                                                                              | Bessemerization (impurities has more affinity for $O_2$ as compare to Fe) | Temperature approaching 2170K is required                                                                                                                              |
| Copper<br>$E^\circ = -0.34$    | 1. Copper pyrites, $CuFeS_2$<br>2. Copper glance, $Cu_2S$<br>3. Malachite, $CuCO_3 \cdot Cu(OH)_2$<br>4. Cuprite, $Cu_2O$<br>5. Azurite, $2CuCO_3 \cdot Cu(OH)_2$ | Roasting of sulphide partially and reduction                                                                                                          | (i) Polling<br>(ii) Electrolytic method.                                  | It is self reduction in a specially designed converted. The reduction takes place easily. Sulphuric acid leaching is also used in hydrometallurgy from low grade ores. |
| Zinc<br>$E^\circ = -0.76$      | 1. Zinc blende or Sphalerite, $ZnS$<br>2. Calamine, $ZnCO_3$<br>3. Zincite, $ZnO$                                                                                 | Roasting of sulphide ore or calcination of $ZnCO_3$ followed by reduction with coke                                                                   | The metal may be purified by fractional distillation                      | For $ZnO$ , carbon is better reducing agent then CO and Zn is obtain in vapours form<br>$ZnO + C \xrightarrow{1673K} Zn + CO$                                          |
| Mg<br>$E^\circ = -2.36$        | 1. Carnallite, $KCl \cdot MgCl_2 \cdot 6H_2O$<br>2. Magnesite, $MgCO_3$                                                                                           | Electrolysis of fused $MgCl_2$ with KCl                                                                                                               | -                                                                         | $MgCl_2 \cdot 6H_2O$ is heated in the excess current of dry HCl gas to produce anhydrous $MgCl_2$                                                                      |
| Sn<br>$E^\circ = -0.14$        | 1. Cassiterite, $SnO_2$ (Tin stone)                                                                                                                               | Reduction of the $SnO_2$ with carbon. $SnO_2 + 2C \rightarrow Sn + 2CO$                                                                               | Polling and Liquidation                                                   | Ore contain impurity of wolframite, $FeWO_4 + MnWO_4$ (magnetic separation)                                                                                            |
| Pb<br>$E^\circ = -0.13$        | 1. Galena, $PbS$                                                                                                                                                  | Roasting, then self reduction or Reduction of $PbO$ (Roasted ore) with carbon<br>$PbS + 2PbO \rightarrow 3Pb + SO_2$<br>$PbO + C \rightarrow Pb + CO$ | Liquidation & electrolytic method.                                        | -                                                                                                                                                                      |
| Ag<br>$E^\circ = 0.80$         | 1. Argentite- $Ag_2S$ , native Silver                                                                                                                             | Hydro metallurgy<br>$Ag_2S + 4NaCN \rightarrow 2NaAg(CN)_2 + N_2S$<br>$2NaAg(CN)_2 + Zn \rightarrow Na_2Zn(CN)_4 + 2Ag$                               | Cupelation & electrolytic method                                          | In hydro mettallurgy Ag obtain in the form of dark amorphous ppt.                                                                                                      |
| Au<br>$E^\circ = 1.40$         | Native ore                                                                                                                                                        | 1. Amalgamation.<br>2. Cynide process                                                                                                                 | Cupelation & electrolysis method.                                         | In hydro mettallurgy Au obtain in the form of dark amorphous ppt.                                                                                                      |

# SALT ANALYSIS

**Definition :**

The branch of chemical analysis which aims to find out the constituents of a mixture of compound is known as Qualitative Analysis.

The identification of a substances usually involves its conversion into a new substance possessing characteristic properties with the help of one or more substance of known composition. The substance which is used to bring about such change is called a Reagent.



## ANALYSIS OF ACIDIC RADICAL

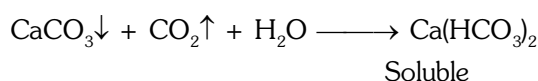
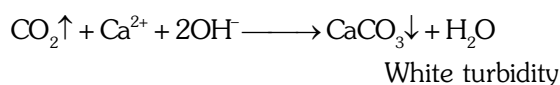
### Classification of acidic radical

| Class - A<br>(Form volatile product with acid)                                                                                                                                                                                                                 | Class - B<br>(Does not form volatile product with acid)                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sub group-I :</b><br>(Form volatile product with dil. HCl / dil. H <sub>2</sub> SO <sub>4</sub> )<br>$\text{CO}_3^{2-}, \text{HCO}_3^-, \text{SO}_3^{2-}, \text{HSO}_3^-, \text{S}_2\text{O}_3^{2-}, \text{S}^{2-}, \text{CH}_3\text{COO}^-, \text{NO}_2^-$ | <b>Sub group-I :</b><br>(Detected by precipitation reaction)<br>$\text{SO}_4^{2-}, \text{PO}_4^{3-}, \text{AsO}_3^{3-}, \text{AsO}_4^{3-}$ |
| <b>Sub group-II :</b><br>(Form volatile product with conc. H <sub>2</sub> SO <sub>4</sub> )<br>$\text{F}^-, \text{Cl}^-, \text{Br}^-, \text{I}^-, \text{NO}_3^-, \text{BO}_3^{3-}, \text{C}_2\text{O}_4^{2-}$ + sub group-I                                    | <b>Sub group-II :</b><br>(Detected by redox reaction)<br>$\text{CrO}_4^{2-}, \text{Cr}_2\text{O}_7^{2-}, \text{MnO}_4^-$                   |

- ⇒ CO<sub>2</sub> is colourless, odourless gas evolved with brisk effevesence, Detected by lime water test.
- ⇒ SO<sub>2</sub> is colourless, Suffocating gas with burning sulphur odour. Detected by lime water test or by passing in Cr<sub>2</sub>O<sub>7</sub><sup>2-</sup> / H<sup>+</sup> solution.
- ⇒ H<sub>2</sub>S is a colourless gas with rotten egg odour.
- ⇒ CH<sub>3</sub>COOH has colourless fumes having vinegar odour.
- ⇒ NO<sub>2</sub>/Br<sub>2</sub> is brown gas.
- ⇒ I<sub>2</sub> is violet gas.

### LIME WATER / BARYTA WATER TEST :

When CO<sub>2</sub> is passed in lime water or baryta water then white turbidity (milky appearance) is formed due to the formation of soluble carbonate but when excess of CO<sub>2</sub> is passed then white turbidity disappeared due to formation of soluble bicarbonate.



**Note :** SO<sub>2</sub> also gives similar observation.



Acidic Radical Table

|                             | dil. $\text{H}_2\text{SO}_4$                 | Conc. $\text{H}_2\text{SO}_4$                   | $\text{CaCl}_2$                              | $\text{BaCl}_2$                               | $\text{HgCl}_2$                                               | $\text{AgNO}_3$                                       | $\text{Pb(OAc)}_2$                                | $\text{MnO}_4^- / \text{H}^+$            |
|-----------------------------|----------------------------------------------|-------------------------------------------------|----------------------------------------------|-----------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------|------------------------------------------|
| $\text{CO}_3^{2-}$          | $\text{CO}_2 \uparrow$                       | $\text{CO}_2 \uparrow$                          | $\text{CaCO}_3 \downarrow$<br>white          | $\text{BaCO}_3 \downarrow$<br>white           | $\text{HgCO}_3 \cdot 3\text{HgO} \downarrow$<br>Reddish brown | $\text{Ag}_2\text{CO}_3 \downarrow$                   | $\text{PbCO}_3 \cdot 2\text{Pb(OH)}_2 \downarrow$ | —                                        |
| $\Delta + \text{HCO}_3^-$   | $\text{CO}_2 \uparrow$                       | $\text{CO}_2 \uparrow$                          | $\text{CaCO}_3 \downarrow$                   | $\text{BaCO}_3 \downarrow$                    | $\text{HgCO}_3 \cdot 3\text{HgO} \downarrow$<br>Reddish brown | $\text{Ag}_2\text{CO}_3 \downarrow$                   | $\text{PbCO}_3 \cdot 2\text{Pb(OH)}_2 \downarrow$ | —                                        |
| $\text{SO}_3^{2-}$          | $\text{SO}_2 \uparrow$                       | $\text{SO}_2 \uparrow$                          | $\text{CaSO}_3 \downarrow$<br>white          | $\text{BaSO}_3 \downarrow$<br>white           | —                                                             | $\text{Ag}_2\text{SO}_3 \downarrow$<br>white          | $\text{PbSO}_3 \downarrow$<br>white               | $\text{SO}_4^{2-}$                       |
| $\text{S}_2\text{O}_3^{2-}$ | $\text{SO}_2 \uparrow + \text{S} \downarrow$ | $\text{SO}_2 \uparrow$                          | —                                            | $\text{BaS}_2\text{O}_3 \downarrow$<br>white  | $\text{HgS}_2\text{O}_3 \downarrow$<br>white                  | $\text{Ag}_2\text{S}_2\text{O}_3 \downarrow$<br>white | $\text{PbS}_2\text{O}_3 \downarrow$<br>white      | $\text{SO}_4^{2-} + \text{S} \downarrow$ |
| $\text{S}^{2-}$             | $\text{H}_2\text{S} \uparrow$                | $\text{S} \downarrow + \text{SO}_2 \uparrow$    | —                                            | $\text{BaS} \downarrow$<br>black              | $\text{HgS} \downarrow$<br>black                              | $\text{Ag}_2\text{S} \downarrow$<br>black             | $\text{PbS} \downarrow$<br>black                  | $\text{S} \downarrow$                    |
| $\text{NO}_2^-$             | $\text{NO}_2 \uparrow$                       | $\text{NO}_2 \uparrow$                          | —                                            | —                                             | —                                                             | $\text{AgNO}_2 \downarrow$                            | —                                                 | $\text{NO}_3^-$                          |
| $\text{CH}_3\text{COO}^-$   | $\text{CH}_3\text{COOH} \uparrow$            | $\text{CH}_3\text{COOH} + \text{SO}_3 \uparrow$ | —                                            | —                                             | —                                                             | —                                                     | —                                                 | —                                        |
| $\text{I}^-$                | —                                            | $\text{I}_2 \uparrow$<br>violet                 | —                                            | —                                             | $\text{HgI}_2 \downarrow$<br>scarlet red                      | $\text{AgI} \downarrow$<br>yellow                     | $\text{PbI}_2 \downarrow$<br>dark yellow          | $\text{I}_2 \uparrow$                    |
| $\text{Cl}^-$               | —                                            | $\text{HCl} \uparrow$                           | —                                            | —                                             | —                                                             | $\text{AgCl} \downarrow$<br>white                     | $\text{PbCl}_2 \downarrow$<br>white               | $\text{Cl}_2 \uparrow$                   |
| $\text{Br}^-$               | —                                            | $\text{Br}_2 \uparrow$<br>brown                 | —                                            | —                                             | —                                                             | $\text{AgBr} \downarrow$<br>pale yellow               | $\text{PbBr}_2 \downarrow$<br>white               | $\text{Br}_2 \uparrow$                   |
| $\text{C}_2\text{O}_4^{2-}$ | —                                            | $\text{CO}_2 \uparrow + \text{CO} \uparrow$     | $\text{CaC}_2\text{O}_4 \downarrow$<br>white | $\text{BaC}_2\text{O}_4 \downarrow$<br>white  | —                                                             | $\text{Ag}_2\text{C}_2\text{O}_4 \downarrow$<br>white | —                                                 | $\text{CO}_2 \uparrow$                   |
| $\text{BO}_2^-$             | —                                            | $\text{H}_3\text{BO}_3 \uparrow$                | $\text{Ca(BO}_2)_2 \downarrow$<br>white      | $\text{Ba(BO}_2)_2 \downarrow$<br>white       | —                                                             | $\text{AgBO}_2 \downarrow$<br>white                   | —                                                 | —                                        |
| $\text{NO}_3^-$             | —                                            | $\text{NO}_2 \uparrow$                          | —                                            | —                                             | —                                                             | —                                                     | —                                                 | —                                        |
| $\text{SO}_4^{2-}$          | —                                            | —                                               | $\text{CaSO}_4 \downarrow$<br>white          | $\text{BaSO}_4 \downarrow$<br>curdy white ppt | $\text{HgSO}_4 \cdot 2\text{HgO} \downarrow$<br>yellow        | —                                                     | $\text{PbSO}_4 \downarrow$<br>white               | —                                        |
| $\text{PO}_4^{3-}$          | —                                            | —                                               | $\text{CaHPO}_4 \downarrow$<br>white         | $\text{BaHPO}_4 \downarrow$<br>white          | —                                                             | $\text{Ag}_3\text{PO}_4 \downarrow$<br>yellow         | —                                                 | —                                        |
| $\text{CrO}_4^{2-}$         | —                                            | —                                               | —                                            | $\text{BaCrO}_4 \downarrow$<br>yellow         | —                                                             | $\text{Ag}_2\text{CrO}_4 \downarrow$<br>brick red     | $\text{PbCrO}_4 \downarrow$<br>yellow             | —                                        |



# BASIC RADICAL ANALYSIS

## Dry test of cation :

**Flame test :** used for s-block cation (except  $\text{Be}^{+2}$ ,  $\text{Mg}^{+2}$ )

| Cation :                     | $\text{Li}^+$ | $\text{Na}^+$   | $\text{K}^+$ | $\text{Ca}^{+2}$ | $\text{Sr}^{+2}$ | $\text{Ba}^{+2}$ | $\text{Cu}^{+2}$ |
|------------------------------|---------------|-----------------|--------------|------------------|------------------|------------------|------------------|
| Observation :<br>(Naked eye) | Carmine red   | Golden yellow   | Lilac        | Brick Red        | Crimson Red      | Apple green      | Green flame      |
| Cobalt glass :               |               | flame disappear | crimson red  | Light Green      | Purple           | Bluish green     | Green flame      |

## Borax bead test :

- ⇒ Used for coloured cation (d-block cation)
- ⇒ Given salt heated on borax bead than metal metaborate are form.
- ⇒ Metal metaborate on heating show characteristic bead colour.

|    | Oxidizing flame |          | Reducing flame |            |
|----|-----------------|----------|----------------|------------|
|    | Hot             | Cold     | Hot            | Cold       |
| Mn | Violet          | Amethyst | Colourless     | Colourless |
| Cr | Yellow          | Green    | Green          | Green      |
| Fe | Yellowish brown | Yellow   | Green          | Green      |
| Co | Blue            | Blue     | Blue           | Blue       |
| Ni | Violet          | Brown    | Grey           | Grey       |
| Cu | Green           | Blue     | Colourless     | Opaque red |

## Charcoal cavity test :

Heat salt with  $\text{Na}_2\text{CO}_3$  in charcoal cavity  
 $\text{Zn}^{+2}$  In hot yellow and in cold white residue.  
 $\text{Pb}^{+2}$  Yellow residue in hot and grey metal in cold.  
 $\text{As}^{+3}$  White residue with garlic odour.  
 $\text{Cd}^{+2}$  Brown residue.

If white residue is obtain then add.  $\text{Co}(\text{NO}_3)_2$  and heat.

|                  |                               |                 |
|------------------|-------------------------------|-----------------|
| $\text{Zn}^{+2}$ | $\text{ZnO} \cdot \text{CoO}$ | Rinmann's Green |
| $\text{Al}^{+3}$ | $\text{Al}_2\text{O}_3$       | Thenard Blue    |
| $\text{Mg}^{+2}$ | $\text{MgO} \cdot \text{CoO}$ | Pink residue    |
| $\text{Sn}^{+2}$ | $\text{SnO} \cdot \text{CoO}$ | Bluish residue  |

## Classification of Basic Radical

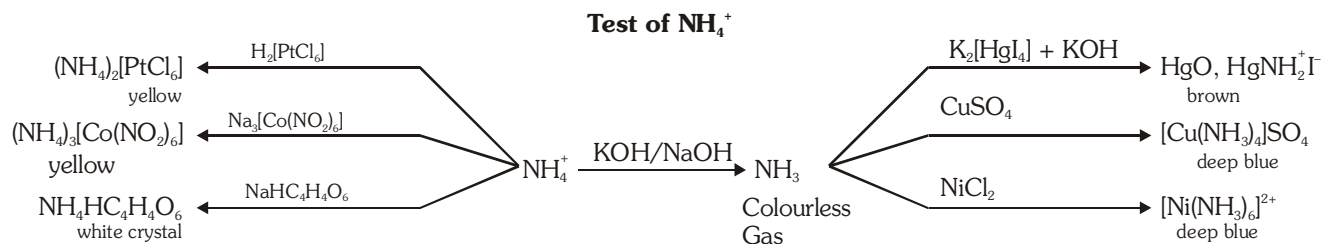
| Group     | Basic Radical                                                                                                                                                                                                   | Reagent used                                          | Precipitate form                                                                                                                                                                                                                                                                             |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Group-I   | $\text{Pb}^{+2}$ , $\text{Hg}_2^{+2}$                                                                                                                                                                           | dil HCl                                               | $\text{AgCl}$ $\text{PbCl}_2$ $\text{Hg}_2\text{Cl}_2$<br>white ppt.                                                                                                                                                                                                                         |
| Group-II  | $\text{Cu}^{+2}$ , $\text{Pb}^{+2}$ , $\text{Hg}^{+2}$ , $\text{Cd}^{+2}$ , $\text{Bi}^{+3}$<br>$\text{Sn}^{+2}$ , $\text{Sn}^{+4}$ , $\text{As}^{+3}$ , $\text{As}^{+5}$ , $\text{Sb}^{+3}$ , $\text{Sb}^{+5}$ | $\text{H}_2\text{S}$ + dil. HCl                       | $\text{Cu}_2\text{S}$ , $\text{PbS}$ , $\text{HgS}$ , $\text{CdS}$ , $\text{Bi}_2\text{S}_3$<br>Black<br>$\text{SnS}$ brown<br>$\text{SnS}_2$ yellow<br>$\text{As}_2\text{S}_3$ yellow<br>$\text{As}_2\text{S}_5$ orange<br>$\text{Sb}_2\text{S}_3$ orange<br>$\text{Sb}_2\text{S}_5$ orange |
| Group-III | $\text{Cr}^{+3}$ , $\text{Al}^{+3}$ , $\text{Fe}^{+3}$                                                                                                                                                          | $\text{NH}_4\text{OH}$ + $\text{NH}_4\text{Cl}$       | $\text{Al}(\text{OH})_3$ , $\text{Fe}(\text{OH})_3$ , $\text{Cr}(\text{OH})_3$<br>gelatinous reddish green<br>white brown                                                                                                                                                                    |
| Group-IV  | $\text{Zn}^{+2}$ , $\text{Mn}^{+2}$ , $\text{Ni}^{+2}$ , $\text{Co}^{+2}$                                                                                                                                       | $\text{H}_2\text{S}$ + $\text{NH}_4\text{OH}$         | $\text{ZnS}$ white<br>$\text{MnS}$ buff<br>$\text{NiS}$ black<br>$\text{CoS}$ black                                                                                                                                                                                                          |
| Group-V   | $\text{Ca}^{+2}$ , $\text{Sr}^{+2}$ , $\text{Ba}^{+2}$                                                                                                                                                          | $(\text{NH}_4)_2\text{CO}_3$ + $\text{NH}_4\text{OH}$ | $\text{CaCO}_3$ , $\text{SrCO}_3$ , $\text{BaCO}_3$<br>white                                                                                                                                                                                                                                 |

**Test of Basic Radical**

|                                                       | Ag <sup>+</sup>                                        | Hg <sub>2</sub> <sup>2+</sup>                                                        | Pb <sup>2+</sup>                                          | Cu <sup>2+</sup>                                                 | Hg <sup>2+</sup>                                             | Cd <sup>2+</sup>                                   | Al <sup>3+</sup>                            | Cr <sup>3+</sup>                                                  | Fe <sup>3+</sup>                                                       | Fe <sup>2+</sup>                                          | Zn <sup>2+</sup>                                               | Mn <sup>2+</sup>                                  | Ni <sup>2+</sup>                                                | Co <sup>2+</sup>                                             |
|-------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------|
| <b>KI</b>                                             | AgI<br>Yellow                                          | Hg <sub>2</sub> I <sub>2</sub><br>Green                                              | PbI <sub>2</sub><br>Yellow                                | CuI + I <sub>3</sub> <sup>-</sup><br>White solution              | HgI <sub>2</sub><br>Scarlet Red                              | —                                                  | —                                           | —                                                                 | FeI <sub>2</sub> + I <sub>3</sub> <sup>-</sup><br>yellowish brown sol. | —                                                         | —                                                              | —                                                 | —                                                               | —                                                            |
| <b>ex KI</b>                                          | —                                                      | Hg + [HgI <sub>4</sub> ] <sup>2-</sup><br>Black                                      | [PbI <sub>4</sub> ] <sup>2-</sup><br>soluble complex      | —                                                                | [HgI <sub>4</sub> ] <sup>2-</sup><br>Solution                | —                                                  | —                                           | —                                                                 | —                                                                      | —                                                         | —                                                              | —                                                 | —                                                               | —                                                            |
| <b>KCN</b>                                            | AgCN<br>White                                          | Hg + Hg(CN) <sub>2</sub><br>Black                                                    | Pb(CN) <sub>2</sub><br>White                              | CuCN + (CN) <sub>2</sub> ↑<br>White                              | —                                                            | Cd(CN) <sub>2</sub><br>White                       | —                                           | —                                                                 | Fe(CN) <sub>3</sub><br>Brown                                           | Fe(CN) <sub>2</sub><br>yellowish brown                    | Zn(CN) <sub>2</sub><br>White                                   | Mn(CN) <sub>2</sub> , Mn(OH) <sub>2</sub><br>Pink | Ni(CN) <sub>2</sub><br>green                                    | Co(CN) <sub>2</sub><br>Reddish Brown                         |
| <b>ex KCN</b>                                         | [Ag(CN) <sub>2</sub> ] <sup>-</sup>                    | —                                                                                    | —                                                         | K <sub>3</sub> [Cu(CN) <sub>4</sub> ]<br>soluble complex         | —                                                            | [Cd(CN) <sub>4</sub> ] <sup>2-</sup>               | —                                           | —                                                                 | K <sub>3</sub> [Fe(CN) <sub>6</sub> ]<br>Yellow                        | K <sub>4</sub> [Fe(CN) <sub>6</sub> ]<br>Pale yellow      | —                                                              | —                                                 | K <sub>2</sub> [Ni(CN) <sub>4</sub> ]<br>Soluble complex        | [Co(CN) <sub>6</sub> ] <sup>4-</sup><br>Brown Solution       |
| <b>NaOH</b>                                           | Ag <sub>2</sub> O<br>Brown                             | Hg <sub>2</sub> O<br>Black                                                           | Pb(OH) <sub>2</sub><br>White                              | Cu(OH) <sub>2</sub><br>Pale Blue                                 | HgO<br>yellow                                                | Cd(OH) <sub>2</sub><br>White                       | Al(OH) <sub>3</sub><br>Gelatinous white     | Cr(OH) <sub>3</sub><br>Green                                      | Fe(OH) <sub>3</sub><br>Reddish Brown ppt                               | Fe(OH) <sub>2</sub><br>Dirty Green                        | Zn(OH) <sub>2</sub><br>White                                   | Mn(OH) <sub>2</sub><br>Pink                       | Ni(OH) <sub>2</sub><br>Green                                    | Co(OH)Cl<br>Blue                                             |
| <b>ex NaOH</b>                                        | —                                                      | —                                                                                    | Na <sub>2</sub> [Pb(OH) <sub>4</sub> ]<br>soluble complex | —                                                                | —                                                            | —                                                  | Na[Al(OH) <sub>4</sub> ]<br>soluble complex | Na[Cr(OH) <sub>4</sub> ]<br>Yellow                                | —                                                                      | —                                                         | Na <sub>2</sub> [Zn(OH) <sub>4</sub> ]                         | —                                                 | —                                                               | Co(OH) <sub>2</sub><br>pink                                  |
| <b>NH<sub>4</sub>OH</b>                               | Ag <sub>2</sub> O<br>Brown                             | Hg + HgO.HgNH <sub>2</sub> <sup>+</sup> .NO <sub>3</sub> <sup>-</sup><br>black white | Pb(OH) <sub>2</sub><br>White                              | Cu(OH) <sub>2</sub><br>Pale Blue                                 | HgO.HgNH <sub>2</sub> <sup>+</sup> .Cl <sup>-</sup><br>White | Cd(OH) <sub>2</sub><br>White                       | Al(OH) <sub>3</sub><br>Gelatinous white     | Cr(OH) <sub>3</sub><br>Green                                      | Fe(OH) <sub>3</sub><br>Reddish Brown ppt                               | Fe(OH) <sub>2</sub><br>Dirty Green                        | Zn(OH) <sub>2</sub><br>White                                   | Mn(OH) <sub>2</sub>                               | Ni(OH) <sub>2</sub><br>Green                                    | Co(OH)Cl<br>Blue                                             |
| <b>ex NH<sub>4</sub>OH</b>                            | [Ag(NH <sub>3</sub> ) <sub>2</sub> ] <sup>+</sup>      | —                                                                                    | —                                                         | [Cu(NH <sub>3</sub> ) <sub>4</sub> ] <sup>2+</sup><br>Deep blue. | —                                                            | [Cd(NH <sub>3</sub> ) <sub>4</sub> ] <sup>2+</sup> | —                                           | [Cr(NH <sub>3</sub> ) <sub>3</sub> ] <sup>3+</sup><br>Pink/violet | —                                                                      | —                                                         | [Zn(NH <sub>3</sub> ) <sub>4</sub> ] <sup>2+</sup><br>White    | —                                                 | Ni[(NH <sub>3</sub> ) <sub>6</sub> ] <sup>2+</sup><br>Deep blue | [Co(NH <sub>3</sub> ) <sub>6</sub> ] <sup>2+</sup><br>Yellow |
| <b>H<sub>2</sub>S / (NH<sub>4</sub>)<sub>2</sub>S</b> | Ag <sub>2</sub> S<br>Black                             | Hg + HgS<br>black                                                                    | PbS<br>Black                                              | CuS<br>Black                                                     | HgS<br>Black                                                 | CdS<br>Yellow                                      | Al(OH) <sub>3</sub><br>Gelatinous white     | Cr(OH) <sub>3</sub><br>Green                                      | FeS + S<br>Black yellow                                                | FeS<br>Black                                              | ZnS<br>White                                                   | MnS<br>Pink                                       | NiS<br>Black                                                    | CoS<br>Black                                                 |
| <b>K<sub>2</sub>CrO<sub>4</sub></b>                   | Ag <sub>2</sub> CrO <sub>4</sub><br>Red                | Hg <sub>2</sub> CrO <sub>4</sub><br>Red                                              | PbCrO <sub>4</sub><br>Yellow                              | —                                                                | —                                                            | —                                                  | —                                           | —                                                                 | —                                                                      | —                                                         | —                                                              | —                                                 | —                                                               | —                                                            |
| <b>Na<sub>2</sub>S<sub>2</sub>O<sub>3</sub></b>       | Ag <sub>2</sub> S <sub>2</sub> O <sub>3</sub><br>White | —                                                                                    | PbS <sub>2</sub> O <sub>3</sub><br>White                  | Cu <sub>2</sub> S <sub>2</sub> O <sub>3</sub><br>White           | HgS <sub>2</sub> O <sub>3</sub><br>White                     | —                                                  | —                                           | —                                                                 | Fe <sup>2+</sup><br>Green solution                                     | —                                                         | —                                                              | —                                                 | —                                                               | —                                                            |
| <b>K<sub>4</sub>[Fe(CN)<sub>6</sub>]</b>              | Ag <sub>3</sub> [Fe(CN) <sub>6</sub> ]<br>White        | —                                                                                    | —                                                         | Cu <sub>3</sub> [Fe(CN) <sub>6</sub> ]<br>Brown                  | —                                                            | Cd <sub>3</sub> [Fe(CN) <sub>6</sub> ]             | —                                           | —                                                                 | Fe <sub>4</sub> [Fe(CN) <sub>6</sub> ]<br>Prussian blue                | K <sub>2</sub> Fe[Fe(CN) <sub>6</sub> ]<br>White          | K <sub>2</sub> Zn <sub>3</sub> [Fe(CN) <sub>6</sub> ]<br>White | Mn <sub>3</sub> [Fe(CN) <sub>6</sub> ]<br>White   | Ni <sub>2</sub> [Fe(CN) <sub>6</sub> ]<br>Light green           | Co <sub>2</sub> [Fe(CN) <sub>6</sub> ]<br>Green              |
| <b>K<sub>3</sub>[Fe(CN)<sub>6</sub>]</b>              | —                                                      | —                                                                                    | —                                                         | Cu <sub>3</sub> [Fe(CN) <sub>6</sub> ]<br>Green                  | —                                                            | —                                                  | —                                           | —                                                                 | Fe[Fe(CN) <sub>6</sub> ]<br>Brown                                      | Fe <sub>3</sub> [Fe(CN) <sub>6</sub> ]<br>Turnbull's blue | —                                                              | —                                                 | —                                                               | —                                                            |

Other important reaction of basic radical

| Basic radical    | Reagent                      | Product                                              | Observation           |
|------------------|------------------------------|------------------------------------------------------|-----------------------|
| $\text{Fe}^{+3}$ | $\text{SCN}^-$               | $[\text{Fe}(\text{H}_2\text{O})_5(\text{SCN})]^{2+}$ | Blood red colouration |
| $\text{Ni}^{+2}$ | $\text{dmg} / \text{NH}_4^+$ | $[\text{Ni}(\text{dmg})_2]$                          | Rosy red complex      |
| $\text{Co}^{+2}$ | $\text{KNO}_2$               | $\text{K}_3[\text{Co}(\text{NO}_2)_6]$               | Yellow ppt            |
| $\text{Hg}^{+2}$ | $\text{SnCl}_2(\text{ex})$   | $\text{Hg}$                                          | Grey / black          |
| $\text{K}^+$     | $\text{HClO}_4$              | $\text{KClO}_4$                                      | White ppt             |

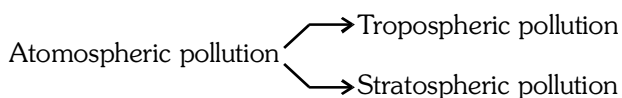


**IMPORTANT NOTES**

# ENVIRONMENT POLLUTION

## Pollutant :

Substance which cause pollution is known as pollutant.



⇒ Tropospheric pollution occurs due to presence of undesirable solid or gaseous particles in air.

## Gaseous pollutant :

### SO<sub>2</sub> :

- \* Cause respiratory diseases of asthma, bronchitis, emphysema etc & irritating to eyes.

### NO<sub>2</sub> :

- \* Form by fossil fuel burn, Damage lungs.
- \* Higher concentration of NO<sub>2</sub> damage the leaves of plant and retard rate of photosynthesis.

## Hydrocarbon :

- \* Form by incomplete combustion of fuel of automobile, Carcinogenic.

## Oxide of carbon :

### CO :

- \* Blocks the delivery of oxygen to organs and tissues.
- \* Carboxy hemoglobin is 300 times more stable than oxy hemoglobin about 3-4% of carboxy hemoglobin the oxygen carrying capacity is highly reduced.

### CO<sub>2</sub> :

- \* Main source is respiration, burning of fossil fuels, decomposition of lime stone in cement industry.
- \* Increase of CO<sub>2</sub> cause global warming.

## Global warming and Green House Effect :

Some of the gases such as CO<sub>2</sub>, CH<sub>4</sub>, O<sub>3</sub>, CFC(s) and water vapour trapped the heat and does not radiates back to the atmosphere. This cause global warming.

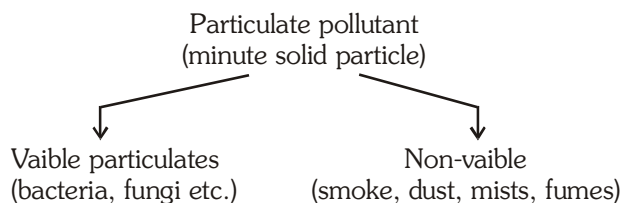
## Acid Rain :

- \* Normally the pH rain water is 5.6 due to the reaction between rain water and CO<sub>2</sub>.
- \* When pH less than 5.6 then it is called acid rain.
- \* Source : burning of fuel (contain N & S) form SO<sub>2</sub> & NO<sub>2</sub>.

- \* Harmful to agriculture, tree and plants.

- \* Taj Mahal affected by acid rain.

## Particulate pollutant :



## Smoke :

Solid/mixture of solid and liquid particles formed from burning of fossil fuel, oil smoke etc.

## Dust :

Fine solid particle over 1µm diameter, produced by crushing, grinding etc.

## Mist :

Mist are produced by particle of spray liquid condensation of vapours, eg. herbicides mist etc.

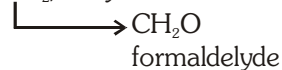
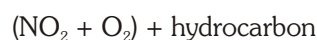
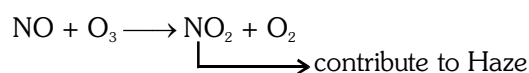
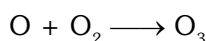
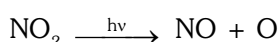
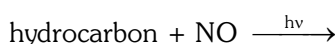
**Note :** Pb is major air pollutant.

## Smog (Smoke + fog) :

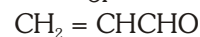
### Classical smog :

(Smoke + fog + SO<sub>2</sub>) also called reducing smog

### Photochemical smog :



or



A.croton

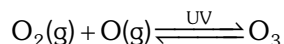
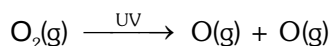
or



peroxy acetyl nitrate (PAN)

### Stratospheric pollution :

Formation & decomposition of oxone.

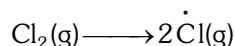
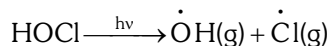
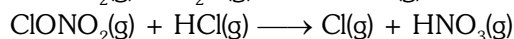
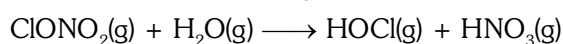
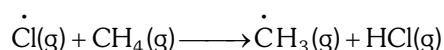
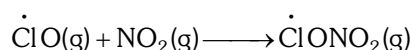
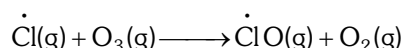
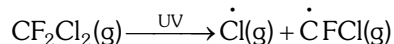


Ozone is thermodynamically unstable and thus dynamic equilibrium exist between production.

### Ozone Hole :

The main reason of ozone layer depletion the release of CFC(s) (also called as freous)

### Reaction of ozone depletion :



## WATER POLLUTION

### Cause of water pollution :

- (i) Pathogen
- (ii) Organic waste
- (iii) Organic waste

**BOD** : The amount of oxygen required by bacteria to break down the organic matter present in a certain volume of a sample of water, is called Biochemical **Oxygen Demand (BOD)**. Clean water would have BOD value of less than 5 ppm where as highly polluted water could have a BOD value of 17 ppm or more.

**Fluoride** : Soluble fluoride is often added to drinking water to bring its concentration upto 1 ppm or 1 mg dm<sup>-3</sup>.

However, F<sup>-</sup> ion concentration above 2 ppm causes brown mottling of teeth. At the same time, excess fluoride (over 10 ppm) causes harmful effect to bones and teeth.

**Lead** : The prescribed upper limit concentration of lead in drinking water is about 50 ppb. Lead can damage kidney, liver, reproductive system etc.

**Sulphate**: Excessive sulphate (>500 ppm) in drinking water causes laxative effect, otherwise at moderate levels it is harmless.

**Nitrate**: The maximum limit of nitrate in drinking water is 50 ppm. Excess nitrate in drinking water can cause disease such as methemoglobinemia ('blue baby' syndrome).

Table

### Maximum Prescribed Concentration of Some Metals in Drinking Water.

| Metal | Maximum concentration (ppm or mg dm <sup>-3</sup> ) |
|-------|-----------------------------------------------------|
| Fe    | 0.2                                                 |
| Mn    | 0.05                                                |
| Al    | 0.2                                                 |
| Cu    | 3.0                                                 |
| Zn    | 5.0                                                 |
| Cd    | 0.005                                               |

[illegible]



# TABLE FOR IUPAC NOMENCLATURE

The order of priority of functional groups used in IUPAC nomenclature of organic compounds.

| Functional group     | Structure                                                                                                                                                    | Prefix                            | Suffix         |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----------------|
| Carboxylic acid      | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{OH} \end{array}$                                                                                  | Carboxy                           | - oic acid     |
| Sulphonic acid       | $-\text{SO}_3\text{H}$                                                                                                                                       | Sulpho                            | sulphonic acid |
| Anhydride            | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C} \diagup \text{O} \\   \quad \diagdown \\ -\text{C} \quad \text{O} \\ \parallel \\ \text{O} \end{array}$ | ×                                 | oic-anhydride  |
| Ester                | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{OR} \end{array}$                                                                                  | Alkoxy carbonyl<br>or Carbalkoxy  | alkyl....oate  |
| Acid chloride        | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{Cl} \end{array}$                                                                                  | Chloroformyl or<br>Chlorocarbonyl | - oyl chloride |
| Acid amide           | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{NH}_2 \end{array}$                                                                                | Carbamoyl/<br>Amido               | - amide        |
| Carbonitrile/Cyanide | $-\text{C} \equiv \text{N}$                                                                                                                                  | Cyano                             | nitrile        |
| Aldehyde             | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{H} \end{array}$                                                                                   | Formyl or Oxo                     | - al           |
| Ketone               | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}- \end{array}$                                                                                           | Keto or oxo                       | - one          |
| Alcohol              | $-\text{OH}$                                                                                                                                                 | Hydroxy                           | - ol           |
| Thio alcohol         | $-\text{SH}$                                                                                                                                                 | Mercapto                          | thiol          |
| Amine                | $-\text{NH}_2$                                                                                                                                               | Amino                             | amine          |
| Ether                | $-\text{O}-\text{R}$                                                                                                                                         | Alkoxy                            | –              |
| Oxirane              | $\begin{array}{c} \text{C} - \text{C} \\ \diagdown \quad \diagup \\ \text{O} \end{array}$                                                                    | Epoxy                             | –              |
| Nitro derivative     | $-\text{NO}_2$                                                                                                                                               | Nitro                             | –              |
| Nitroso derivative   | $-\text{NO}$                                                                                                                                                 | Nitroso                           | –              |
| Halide               | $-\text{X}$                                                                                                                                                  | Halo                              | –              |
| Double bond          | $\text{C} = \text{C}$                                                                                                                                        | –                                 | ene            |
| Triple bond          | $\text{C} \equiv \text{C}$                                                                                                                                   | –                                 | yne            |

**IMPORTANT NOTES**



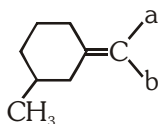
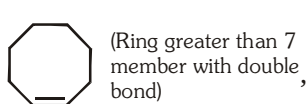
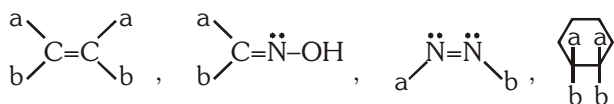
## (B) STEREOISOMERISM

Compounds with the same molecular formula and structural formula but having difference in the spatial arrangement of atoms or groups in 3D space are called stereoisomers and the phenomenon is called stereoisomerism.

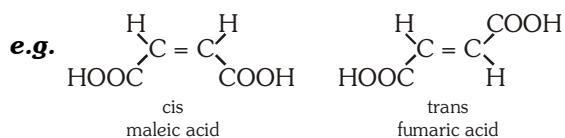
## TYPES OF STEREOISOMERISM

## GEOMETRICAL ISOMERISM

It is due to restricted rotation and is observed in following systems



- Cis-trans isomerism :** The cis compound is the one with the same groups on the same side of the bond, and the trans has the same groups on the opposite sides. Both isomers have different physical and chemical properties.



- General physical properties of geometrical isomer of but-2-ene

|                     |             |
|---------------------|-------------|
| (i) Stability       | trans > cis |
| (ii) Dipole moment  | cis > trans |
| (iii) Boiling point | cis > trans |
| (iv) Melting point  | trans > cis |

## Calculation of number of geometrical isomer

|               |                                   |
|---------------|-----------------------------------|
| Unsymmetrical | $2^n$                             |
| Symmetrical   | $2^{n-1} + 2^{m-1}$               |
|               | $m = \frac{n}{2}$ (If n is even)  |
|               | $m = \frac{n+1}{2}$ (If n is odd) |

\* Where n = number of sites where GI is possible.

## OPTICAL ISOMERISM

Compounds having similar molecular and structural formula but differing in the stereo chemical formula and behaviour towards plane polarised light are called optical isomers and this phenomenon is called optical isomerism.

## Types of optical isomers

- |                      |                        |
|----------------------|------------------------|
| (1) Optically active | (2) Optically inactive |
| • dextrorotatory (d) | • meso                 |
| • laevorotatory (l)  |                        |

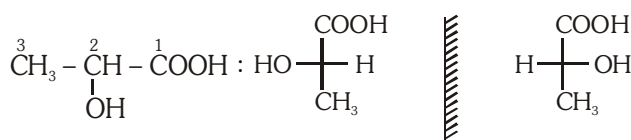
## Condition :

Molecule should be asymmetric or chiral i.e. symmetry element (POS & COS) should be absent.

- The carbon atom linked to four different groups is called **chiral carbon**.
- Fischer projection :** An optical isomer can be represented by Fischer projection which is planar representation of three dimensional structure.

Fischer projection representation of lactic acid

(2-hydroxypropanoic acid)



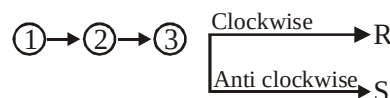
## Configuration of optical isomer :

- (a) Absolute configuration (R/S system)
- (b) Relative configuration (D/L system)

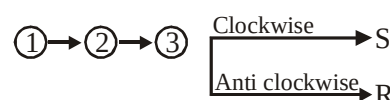
## Determination of R/S configuration :

**Rule-1** Assign the priority to the four groups attached to the chiral carbon according to priority rule.

**Rule-2** If lowest priority (4) is bonded to vertical line then moving



**Rule-3** If lowest priority (4) is bonded to horizontal line then moving

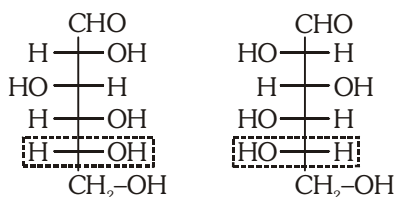


**DETERMINATION OF D/L SYSTEM :**

- Reference molecule glyceraldehyde
- It is used to assign configuration in carbohydrate, amino acid and similar compounds

**Rule:** Arrange parent carbon chain on the vertical line

- Placed most oxidised carbon on the top or nearest to top.
- On highest IUPAC numbered chiral carbon  
If OH group on RHS → D  
If OH group on LHS → L



D-Glucose

L-Glucose

- CIP SEQUENCE RULE :**

The following rules are followed for deciding the precedence order of the atoms or groups :-

- Highest priority is assigned to the atoms of higher atomic number attached to asymmetric carbon atom.
  - In case of isotopes, isotopes having higher atomic mass is given priority.
  - If the first atom of a group attached to asymmetric carbon atom is same then we consider the atomic number of 2<sup>nd</sup> atom or subsequent atoms in group.
  - If there is a double bond or triple bond, both atoms are considered to be duplicated or triplicated.
- Non-superimposable mirror images are called **enantiomers** which rotate the plane polarised light up to same extent but in opposite direction.

- Diastereomers** are stereoisomers which are not mirror images of each other. They have different physical and chemical properties.
- Meso compounds** are those compounds whose molecules are superimposable on their mirror images inspite of the presence of asymmetric carbon atom.
- An equimolar mixture of the enantiomers (d & l) is called **racemic mixture**. The process of converting d- or l- form of an optically active compound into racemic form is called **racemisation**.
- The process by which d/l mixture is separated into d and l forms with the help of chiral reagents or chiral catalyst is known as **resolution**.
- Compound containing chiral carbon may or may not be optically active but show optical isomerism.
- For optical isomer chiral carbon is not the necessary condition.

**Calculation of number of optical isomers**

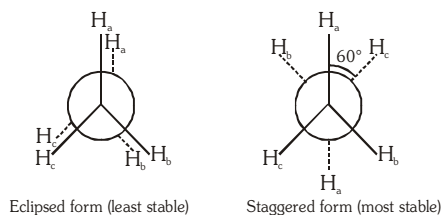
| The compound               | Optically active forms    | Optically inactive forms (meso) |
|----------------------------|---------------------------|---------------------------------|
| Unsymmetrical              | $2^n$                     | Zero                            |
| Symmetrical<br>If n = even | $2^{(n-1)}$               | $2^{\frac{n}{2}-1}$             |
| Symmetrical<br>If n = odd  | $2^{(n-1)} - 2^{(n-1)/2}$ | $2^{(n-1)/2}$                   |

\* Where n = no. of chiral carbon

The different arrangement of atoms in space that results from the carbon-carbon single bond free rotation by 0-360° are called conformations or conformational isomers or rotational isomers and this phenomenon is called conformational isomerism.

**CONFORMATIONAL ISOMERISM**

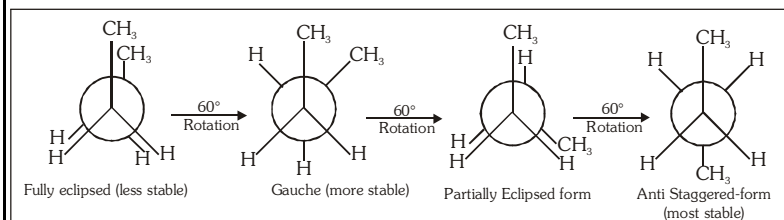
**Newmann projection :** Here two carbon atoms forming the  $\sigma$  bond are represented one by circle and other by centre of the circle. Circle represents rear side C and its centre represents front side carbon. The C-H bonds of front carbon are depicted from the centre of the circle while C-H bond of the back carbon are drawn from the circumference of the circle.



Eclipsed form (least stable)

Staggered form (most stable)

- Conformations of butane :**  ${}^4\text{CH}_3 - {}^3\text{CH}_2 - {}^2\text{CH}_2 - {}^1\text{CH}_3$



- The order of stability of conformations of n-butane.  
Anti staggered > Gauche > Partially eclipsed > Fully eclipsed.
- Relative stability of various conformation of cyclohexane is  
Chair > twist boat > boat > half chair

**IMPORTANT NOTES**

# REACTION MECHANISM

**Electrophiles** are electron deficient species.

eg.  $H^+$ ,  $R^+$ ,  $NO_2^+$ ,  $X^+$ ,  $PCl_3$ ,  $PCl_5$

( $NH_4^+$  and  $H_3O^+$  are not electrophile)

**Nucleophiles** are electron rich species.

e.g.  $Cl^-$ ,  $CH_3^-$ ,  $OH^-$ ,  $RO^-$ ,  $CN^-$ ,  $NH_3$ ,  $\ddot{O}H$ ,  $CH_2=CH_2$ ,  $CH\equiv CH$

**Relative electron withdrawing order (-I order)**

$-\overset{+}{N}F_3 > -\overset{+}{N}R_3 > -\overset{+}{N}H_3 > -NO_2 > -CN > -COOH > -X > -OR > -OH > -C\equiv CH > -NH_2 > -C_6H_5 > -CH=CH_2$

**Relative electron releasing order (+I order)**

$-\overset{-}{N}H > -\overset{-}{O} > -COO^- > 3^\circ \text{ alkyl} > 2^\circ \text{ alkyl} > 1^\circ \text{ alkyl} > -CH_3$

## RELATIVE STABILITY ORDER

### (A) Stability of carbocation

(i)  $\text{Cyclohexadienyl}^+ > (Ph)_3C^+ > (Ph)_2CH^+ > Ph-CH_2^+ > CH_2=CH-CH_2^+$

(ii)  $(CH_3)_3C^+ > (CH_3)_2CH^+ > CH_3CH_2^+ > CH_3^+ > CH_2=CH^+ > CH\equiv C^+$

### (B) Stability of free radical

$(Ph)_3\dot{C} > (Ph)_2\dot{C}H > Ph\dot{C}H_2 > CH_2=CH-\dot{C}H_2 >$

$(CH_3)_3\dot{C} > (CH_3)_2\dot{C}H > CH_3\dot{C}H_2 > \dot{C}H_3$

### (C) Stability of Carbanion

$(Ph)_3C^- > (Ph)_2CH^- > Ph-CH_2^- > CH_2=CH-CH_2^- >$

$CH_3^- > CH_3CH_2^- > (CH_3)_2CH^- > (CH_3)_3C^-$

**BASIC STRENGTH**  $\propto K_b \propto \frac{1}{pK_b}$

#### Basic strength of amine :-

**In aqueous medium**

$R \Rightarrow -CH_3 \quad 2^\circ > 1^\circ > 3^\circ > NH_3$

$R \Rightarrow -CH_2CH_3 \quad 2^\circ > 3^\circ > 1^\circ > NH_3$

**In gaseous medium**

$R \Rightarrow -CH_3 \quad 3^\circ > 2^\circ > 1^\circ > NH_3$

$R \Rightarrow -CH_2CH_3 \quad 3^\circ > 2^\circ > 1^\circ > NH_3$

#### Reactivity towards nucleophile (NAR)

(1)  $HCHO > CH_3CHO > (CH_3)_2CO$

(2)  $CCl_3CHO > CHCl_2CHO > CH_2ClCHO$

#### Reactivity order towards acyl nucleophilic substitution reaction

Acid chloride > anhydride > ester > amide

#### Order of electronic effect

Mesomeric > Hyperconjugation > Inductive effect

#### Stability of alkene $\propto$ no. of $\alpha$ -hydrogen

$R_2C=CR_2 > R_2C=CHR > R_2C=CH_2 > RCH=CHR$

trans form      cis form

$RCH=CH_2 > CH_2=CH_2$

**Heat of hydrogenation**  $\propto \frac{1}{\text{Stability of alkene}}$

## ACIDIC STRENGTH $\propto$ Stability of conjugate base

$$\propto K_a \propto \frac{1}{pK_a}$$

(i)  $H_2O > CH\equiv CH > NH_3$

(ii)  $CH\equiv CH > CH_2=CH_2 > CH_3-CH_3$

(iii)  $R-SO_3H > R-COOH > \text{Phenol} > R-OH$

(iv)  $HCOOH > CH_3COOH > CH_3CH_2COOH$

(v)  $\text{2,4,6-Trinitrophenol} > HCOOH > C_6H_5COOH > CH_3COOH$

(vi)  $CCl_3COOH > CHCl_2COOH > CH_2ClCOOH$

(vii)  $CH_3-CH_2-CH(F)-COOH > CH_3-CH(F)-CH_2COOH > CH_2F-CH_2CH_2COOH$

(viii)  $\text{Phenol} > m > p > o$

(ix)  $p > o > m > \text{Phenol}$

(x)  $\text{2,4,6-Trinitrophenol} > \text{2,4-Dinitrophenol} > \text{3-Nitrophenol} > \text{Phenol}$

(xi)  $o > p > m > \text{benzoic acid}$

(xii)  $o > \text{benzoic acid} > m > p$

(xiii)  $o > m > \text{benzoic acid} > p$

(xiv)  $o > m > p > \text{benzoic acid}$

[illegible]



# PRACTICAL ORGANIC CHEMISTRY

## PURIFICATION METHODS DISTILLATION TECHNIQUES

Type :

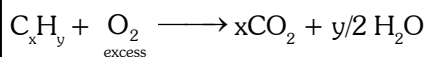
|                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>(A) SIMPLE DISTILLATION</b><br/><b>Conditions</b><br/>(i) When liquid sample has non volatile impurities<br/>(ii) When boiling point difference is 30 K or more.<br/><b>Examples</b><br/>(i) Mixture of chloroform (BP = 334K) and Aniline (BP = 457K)<br/>(ii) Mixture of Ether (b.p. = 308K) &amp; Toluene (b.p. = 384K)<br/>(iii) Hexane (342K) and Toulene(384K)</p> | <p><b>(B) FRACTIONAL DISTILLATION</b><br/>When b.p. difference is 10K<br/><b>Examples</b><br/>(i) Crude oil in petroleum industry<br/>(ii) Acetone (329 K) and Methyl alcohol (338K)</p>     |
| <p><b>(C) DISTILLATION UNDER REDUCED PRESSURE (Vacuum distillation)</b><br/>When liquid boils at higher temperature and it may decompose before b.p. is attained.<br/><b>Example</b> (i) Concentration of sugar juice<br/>(ii) Recovery of glycerol from spent lye.<br/>(iii) Glycerol</p>                                                                                     |                                                                                                                                                                                              |
| <p><b>(D) STEAM DISTILLATION</b><br/>When the substance is immiscible with water and steam volatile.<br/><b>Example :</b><br/>(i) Aniline is separated from water<br/>(ii) Turpentine oil<br/>(iv) Bromo Benzene<br/>(vi) o-Nitrophenol</p>                                                                                                                                    | <p><math display="block">P = P_1 + P_2</math><br/>Vapour pressure of organic liquid = Vapour pressure of water + Vapour pressure of water</p> <p>(iii) Nitro Benzene<br/>(v) Naphthalene</p> |

## LASSAIGNE'S METHOD (detection of elements)

| Element                              | Sodium extract                                                                       | Confirmed test                                                                                                                                                                                            |
|--------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Nitrogen</b>                      | $Na + C + N \xrightarrow{\Delta} NaCN$                                               | $(NaCN + FeSO_4 + NaOH) \xrightarrow{\text{boil and cool}} Fe_4[Fe(CN)_6]_3 + FeCl_3 + conc. HCl \rightarrow \text{Prussian blue colour}$                                                                 |
| <b>Sulphur</b>                       | $2Na + S \xrightarrow{\Delta} Na_2S$                                                 | (i) $Na_2S + Na_2[Fe(CN)_5NO] \xrightarrow{\Delta} Na_4[Fe(CN)_5NOS] \rightarrow \text{a deep violet colour}$<br>(ii) $Na_2S + CH_3COOH + (CH_3COO)_2Pb \rightarrow \text{A black ppt. (PbS)} \downarrow$ |
| <b>Halogen</b>                       | $Na + X \xrightarrow{\Delta} NaX$                                                    | $NaX + HNO_3 + AgNO_3$<br>(i) White ppt. soluble in aq. $NH_3$ confirms Cl.<br>(ii) Yellow ppt. partially soluble in aq. $NH_3$ confirms Br.<br>(iii) Yellow ppt. insoluble in aq. $NH_3$ confirms I.     |
| <b>Nitrogen and sulphur together</b> | $Na + C + N + S \xrightarrow{\Delta} NaCNS$<br>Sodium thiocyanate (Blood red colour) | As in test for nitrogen; instead of green or blue colour, blood red colouration confirms presence of N and S both                                                                                         |

## QUANTITATIVE ANALYSIS OF ORGANIC COMPOUNDS

### Estimation of carbon and hydrogen - Leebig's method



$$\% \text{ of C} = \frac{12}{44} \times \frac{\text{wt. of } CO_2}{\text{wt. of org. compd}} \times 100$$

$$\% \text{ of H} = \frac{2}{18} \times \frac{\text{wt. of } H_2O}{\text{wt. of org compd}} \times 100$$

**Note :** This method is suitable for estimation if organic compound contains C and H only. In case if other elements e.g., N, S, halogens are also present the organic compound will also give their oxides which is being absorbed in KOH and will increase the percentage of carbon and therefore following modification should be made.

### ESTIMATION OF NITROGEN

#### Duma's method :

The nitrogen containing organic compound yields nitrogen gas on heating it with copper (II) oxide in the presence of  $CO_2$  gas. The mixture of gases is collected over potassium hydroxide solution in which  $CO_2$  is absorbed and volume of nitrogen gas is determined.

$$\% \text{ of N} = \frac{28}{22400} \times \left( \frac{\text{Vol. of } N_2 \text{ collected at N.T.P.}}{\text{Wt. of organic compound}} \right) \times 100$$

**Note :** This method can be used to estimate nitrogen in all types of organic compounds

#### Kjeldahl's method :

In this method nitrogen containing compound is heated with conc.  $H_2SO_4$  in presence of copper sulphate to convert nitrogen into ammonium sulphate which is decomposed with excess of alkali to liberate ammonia. The ammonia evolved is

$$\% \text{ of N} = \frac{1.4 \times \text{volume of acid (ml)} \times \text{normality of acid}}{\text{wt of organic compound}}$$

**Note :** This method is simpler and more convenient and is mainly used for finding out the percentage of nitrogen in food stuffs, soil, fertilizers and various agricultural products. This method cannot be used for compound having nitro groups, azo group ( $-N=N-$ ) and nitrogen in the ring (pyridine, quinole etc.) Since nitrogen in these compounds is not quantitatively converted in to ammonium sulphate.

[illegible]







**Ethanal and methanal (Iodoform test)**

- $\text{CH}_3\text{CHO} + \text{I}_2 + \text{NaOH} \xrightarrow{\text{Iodoform test}} \text{CHI}_3 + \text{HCOONa}$   
Ethanal Iodoform (Yellow ppt.)
- $\text{HCHO} + \text{I}_2 + \text{NaOH} \xrightarrow{\text{Iodoform test}} \text{No yellow ppt.}$   
Methanal

**Acetophenone and benzophenone (Iodoform test)**

- $\text{C}_6\text{H}_5\text{COCH}_3 + \text{I}_2 + \text{NaOH} \xrightarrow{\text{Iodoform test}} \text{CHI}_3 + \text{C}_6\text{H}_5\text{COONa}$   
(Acetophenone) (Yellow ppt.)
- $\text{C}_6\text{H}_5\text{COC}_6\text{H}_5 + \text{I}_2 + \text{NaOH} \xrightarrow{\text{Iodoform test}} \text{No ppt.}$   
(Benzophenone)

**Benzoic acid and ethylbenzoate**

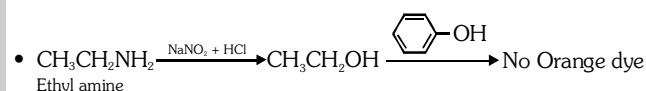
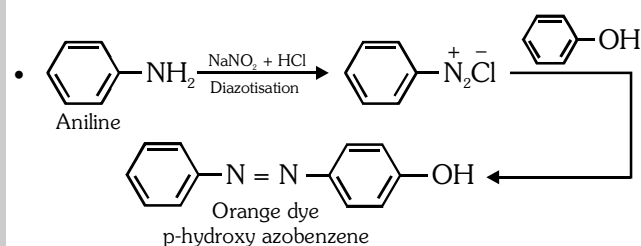
- $\text{C}_6\text{H}_5\text{COOH} + \text{NaHCO}_3 \rightarrow \text{C}_6\text{H}_5\text{COONa} + \text{CO}_2 \uparrow + \text{H}_2\text{O}$   
effervescence
- Ethyl benzoate + Sodium bicarbonate  $\rightarrow$  No effervescence

**Benzaldehyde and acetophenone (Tollen's test)**

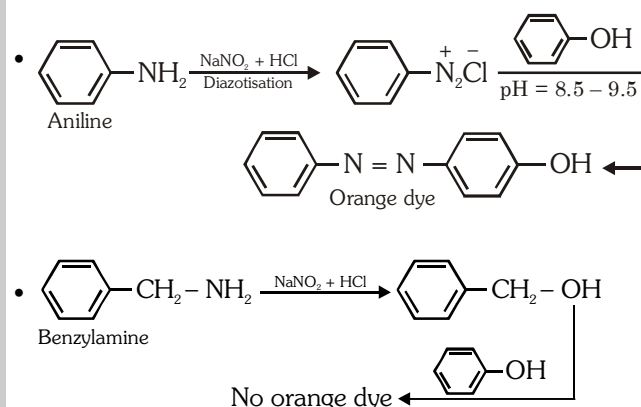
- Benzaldehyde + Tollen's reagent  $\rightarrow$  Silver mirror
- $\text{C}_6\text{H}_5\text{CHO} + 3\text{OH}^- + 2[\text{Ag}(\text{NH}_3)_2]^+ \rightarrow \text{C}_6\text{H}_5\text{COO}^- + 2\text{H}_2\text{O} + 4\text{NH}_3 + 2\text{Ag} \downarrow$   
(Tollen's reagent)
- Acetophenone + Tollen's reagent  $\rightarrow$  No silver mirror

**Methyl amine and dimethyl amine (Isocyanide test)**

- $\text{CH}_3\text{NH}_2 + \text{CHCl}_3 + 3\text{KOH} \xrightarrow{\text{alc.}} \text{CH}_3\text{NC} + 3\text{KCl} + 3\text{H}_2\text{O}$   
Methyl amine Methyl isocyanide (Offensive smell)
- $\text{H}_3\text{C}-\text{NH}-\text{CH}_3 + \text{CHCl}_3 + 3\text{KOH} \xrightarrow{\text{alc.}} \text{No offensive smell}$   
Di-methyl amine

**Aniline and ethyl amine (Diazotisation)****Aniline and N-methylaniline (Isocyanide test)**

- $\text{C}_6\text{H}_5\text{NH}_2 + \text{CHCl}_3 + 3\text{KOH} \xrightarrow{\text{alc.}} \text{C}_6\text{H}_5\text{NC} + 3\text{KCl} + 3\text{H}_2\text{O}$   
Aniline Phenyl isocyanide (Offensive smell)
- $\text{C}_6\text{H}_5\text{NHCH}_3 + \text{CHCl}_3 + 3\text{KOH} \xrightarrow{\text{alc.}} \text{No offensive smell}$   
N-Methylaniline

**Aniline and Benzylamine (Diazotisation + phenol)****Glucose and fructose**

- Glucose +  $\text{Br}_2 + \text{H}_2\text{O} \rightarrow \text{Gluconic acid} + 2\text{HBr}$   
(Brown colour) (Colourless)
- Fructose +  $\text{Br}_2 + \text{H}_2\text{O} \rightarrow \text{Brown colour}$   
(Brown colour) (no change in colour)

**Glucose and sucrose**

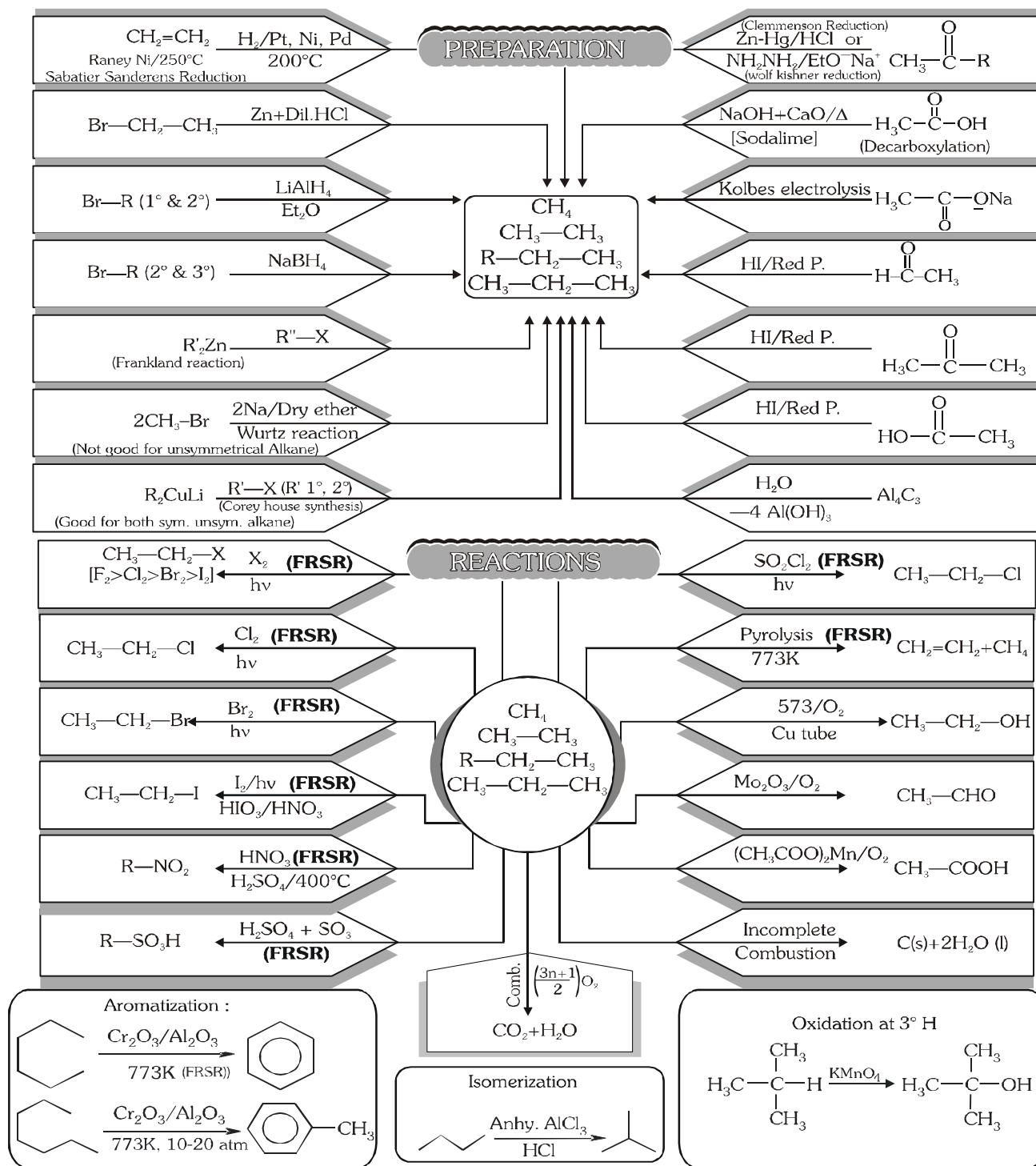
- Glucose + Tollen's reagent  $\rightarrow$  Silver mirror
- Sucrose + Tollen's reagent  $\rightarrow$  No silver mirror

**Glucose and starch**

- Glucose + Fehling's solution  $\rightarrow$  Red ppt.
  - Starch + Fehling's solution  $\rightarrow$  No red ppt.
- OR**
- Glucose +  $\text{I}_2$  solution  $\rightarrow$  No blue colour
  - Starch +  $\text{I}_2$  solution  $\rightarrow$  Blue colour

[illegible]

## ***HYDROCARBON - ALKANE***

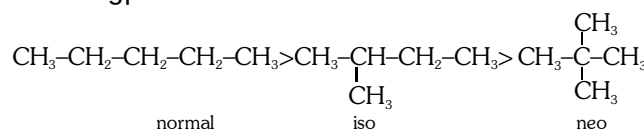


- Reactivity of alkane towards free radical halogenation is  $\propto$  stability of free radical  
 $\text{C}_6\text{H}_5-\text{CH}_3 > \text{CH}_2=\text{CH}-\text{CH}_3 > (\text{CH}_3)_3\text{CH}$   
 $> \text{CH}_3-\text{CH}_2-\text{CH}_3 > \text{CH}_3-\text{CH}_3 > \text{CH}_4$
- Reactivity of halogen towards free radical substitution  
 $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$
- Knocking tendency of petroleum as fuel decrease with increase in side chain. Straight chain  $>$  Branched chain

Knocking tendency is in the order

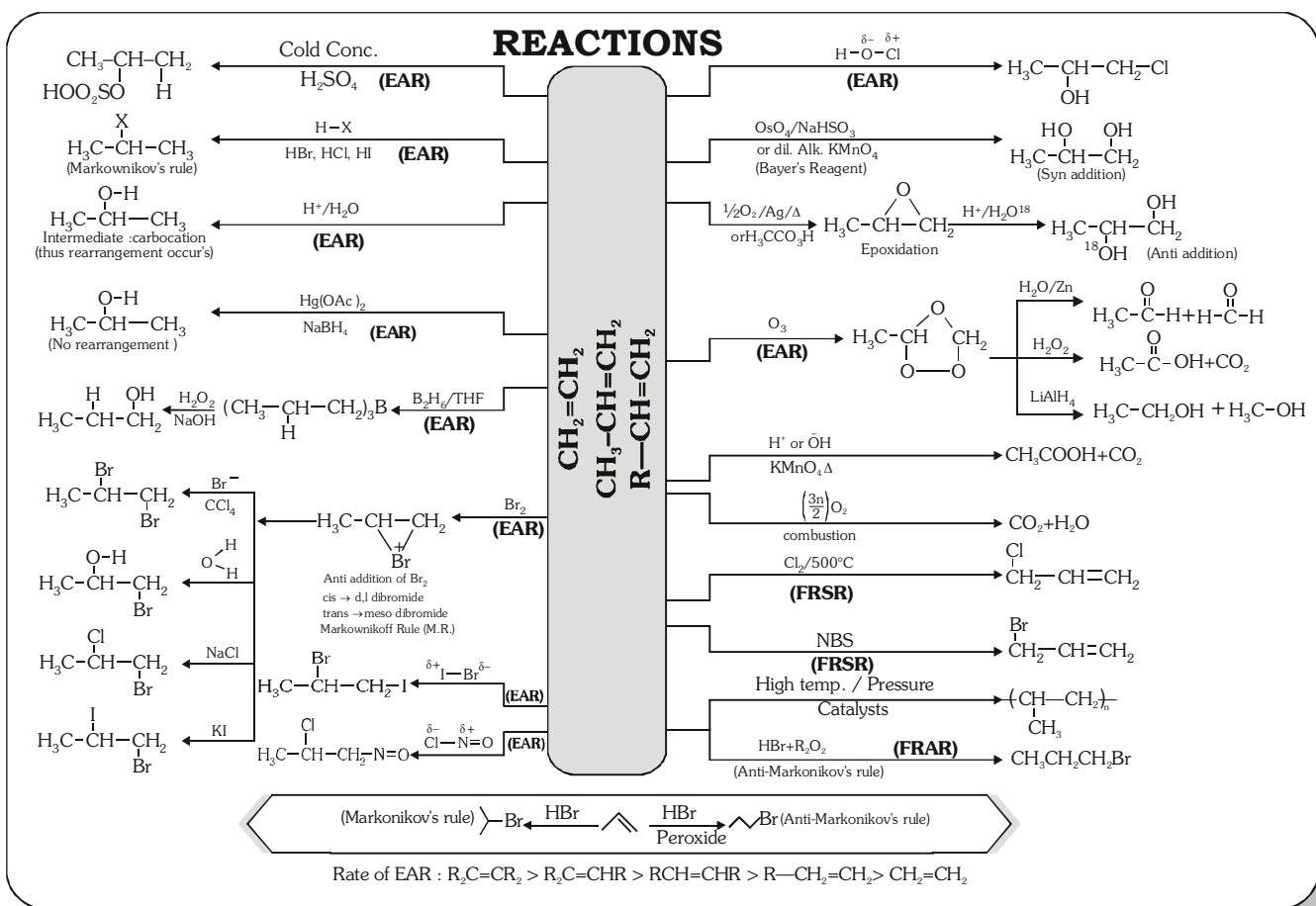
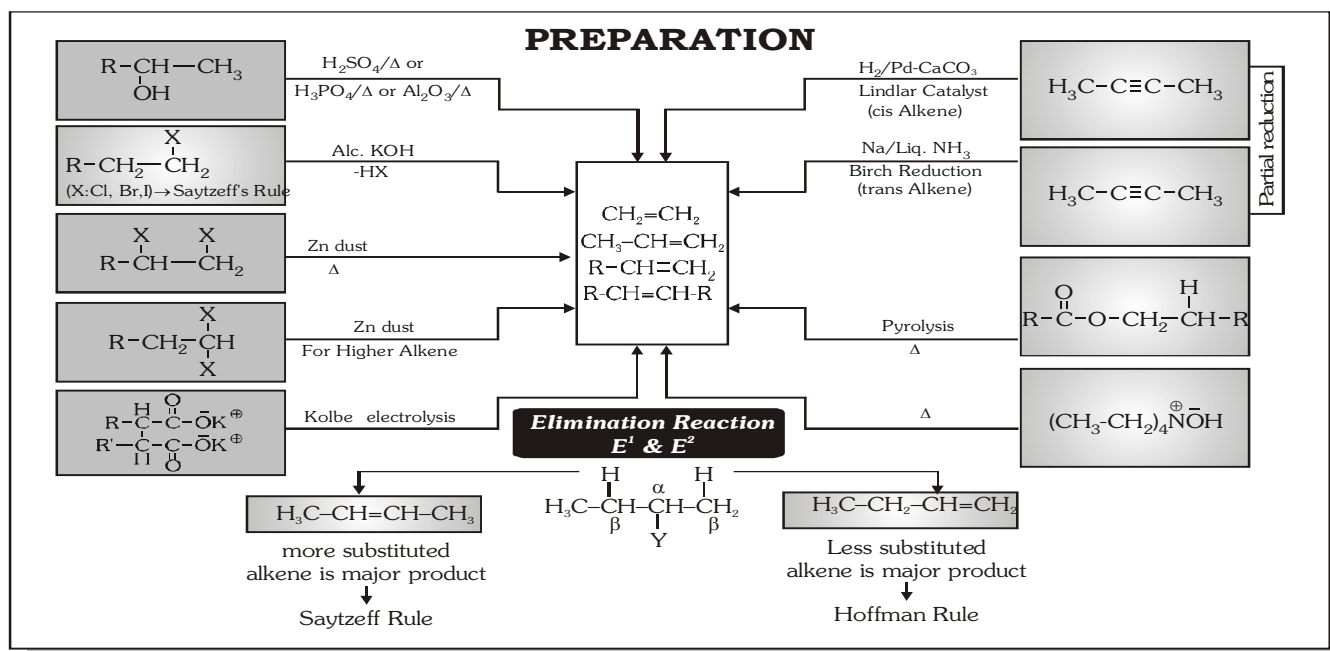
Olefin > cycloalkane > aromatic

- Boiling point decrease with increase in number of side chain.



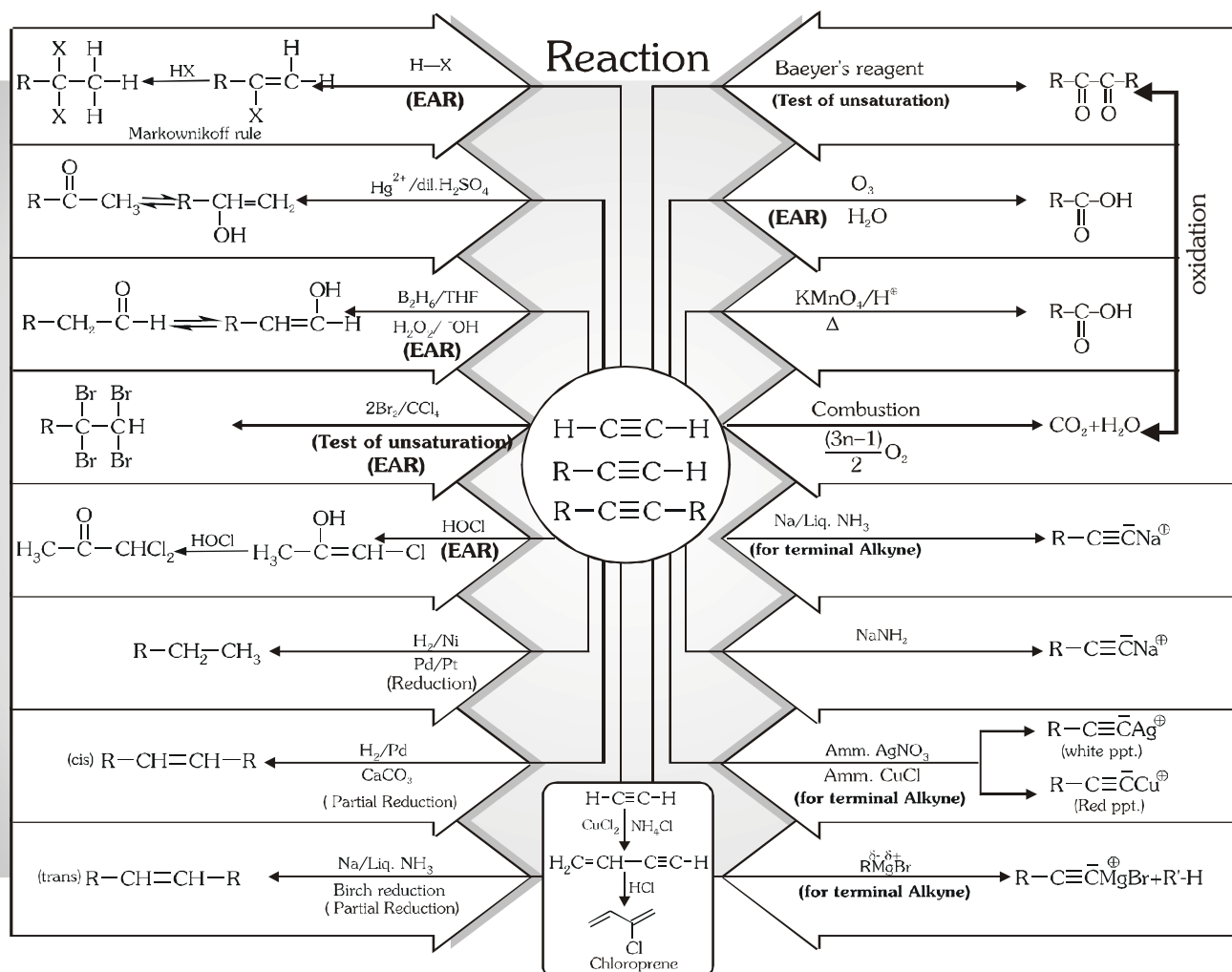
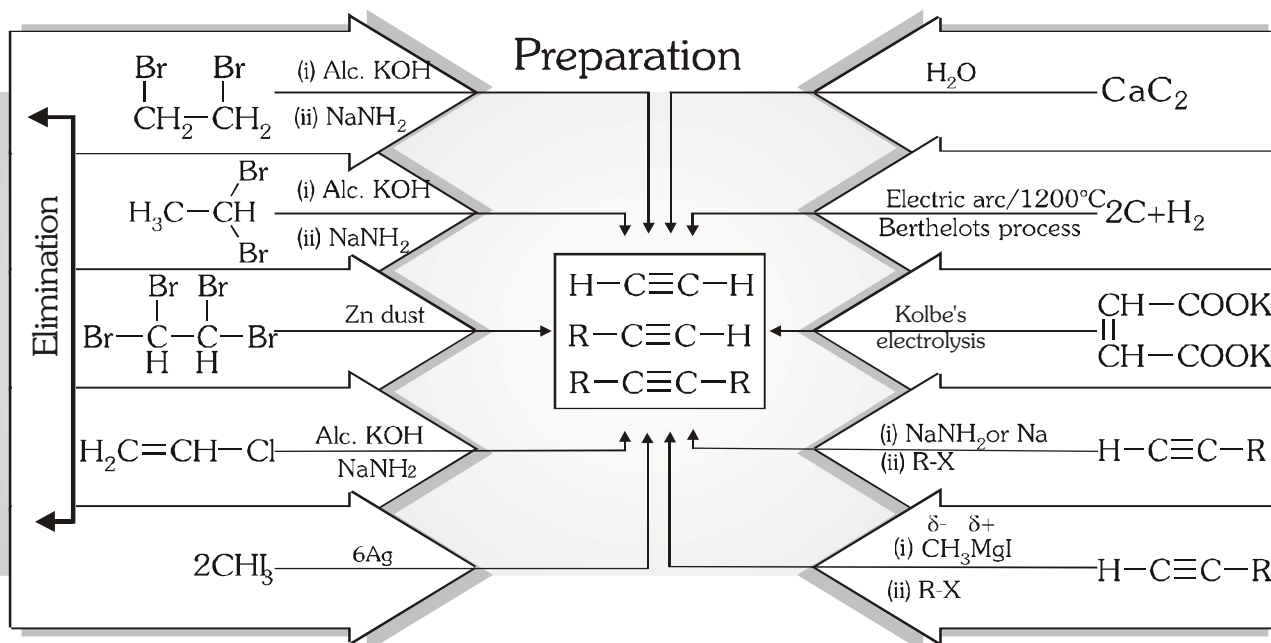


# HYDROCARBON - ALKENE



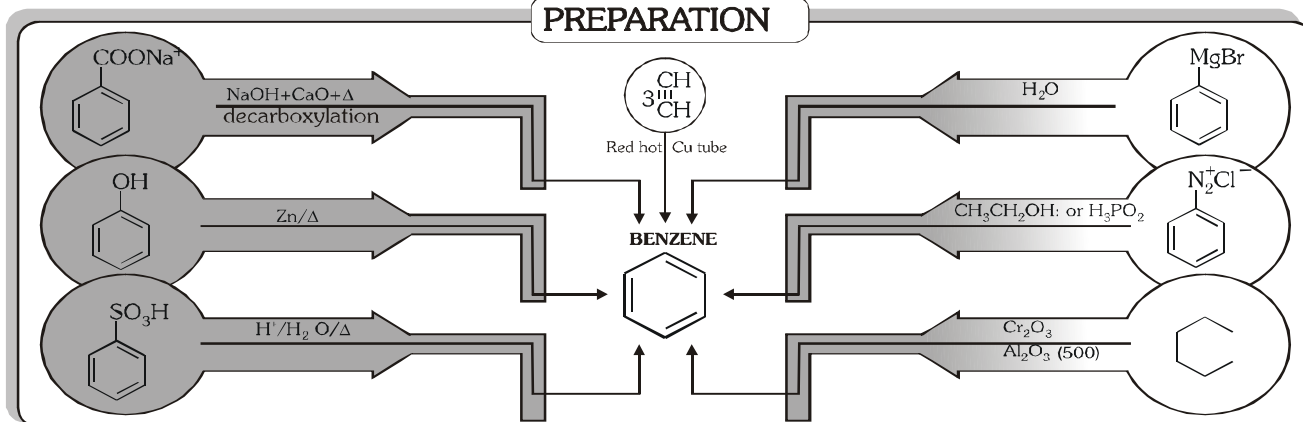
- Order of reactivity of olefins for hydrogenation  $\text{CH}_2=\text{CH}_2 > \text{R}-\text{CH}=\text{CH}_2$  (Reverse of stability)
- Order of reactivity of alkene towards hydration  $\text{CH}_3-\text{C}(\text{CH}_3)=\text{CH}_2 > \text{CH}_3-\text{CH}=\text{CH}_2 > \text{CH}_2=\text{CH}_2$

## HYDROCARBON - ALKYNE

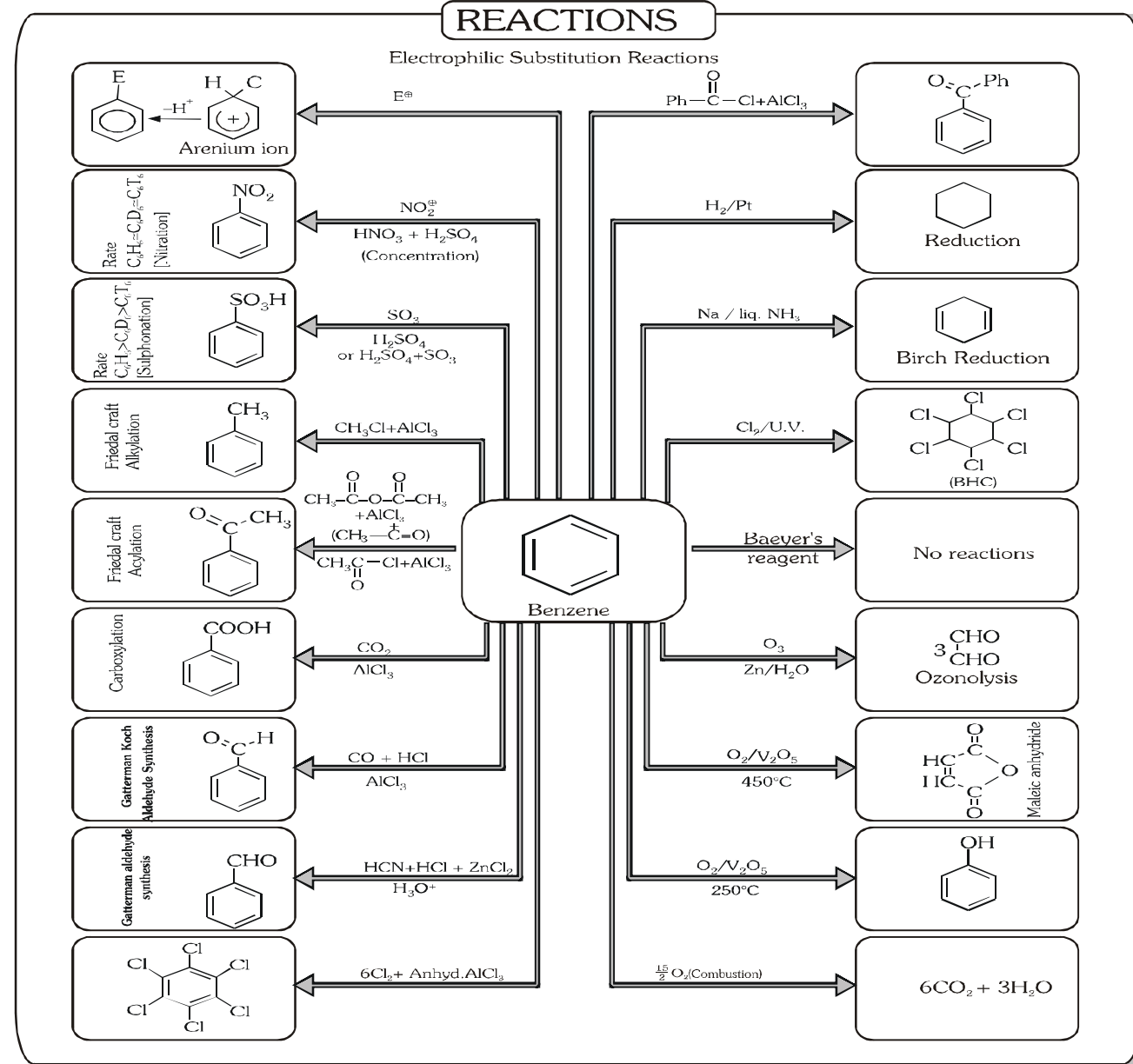


# HYDROCARBON - BENZENE

## PREPARATION



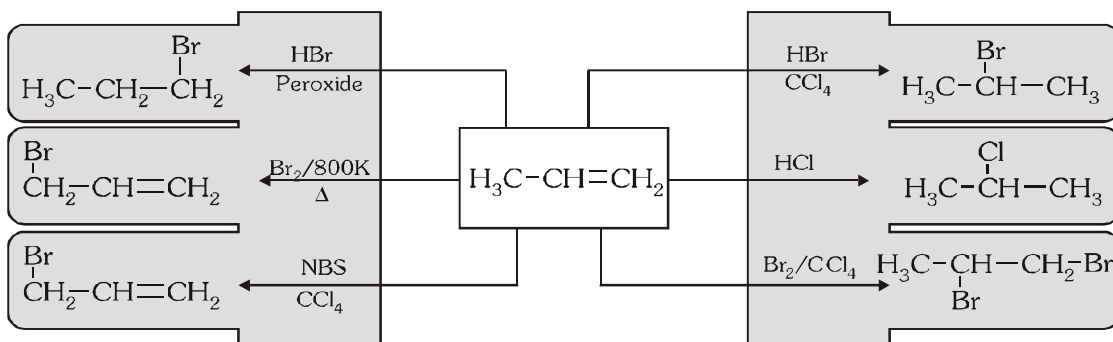
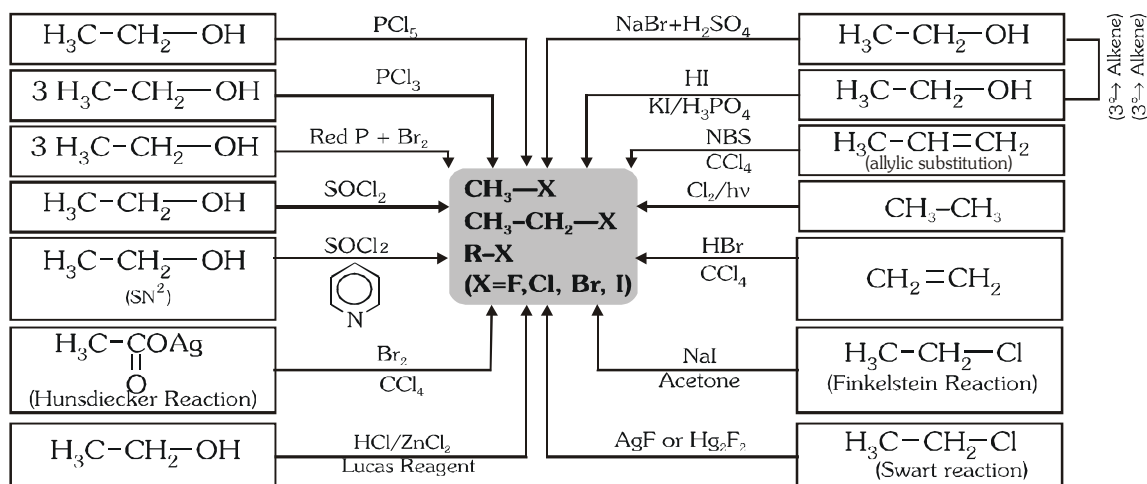
## REACTIONS



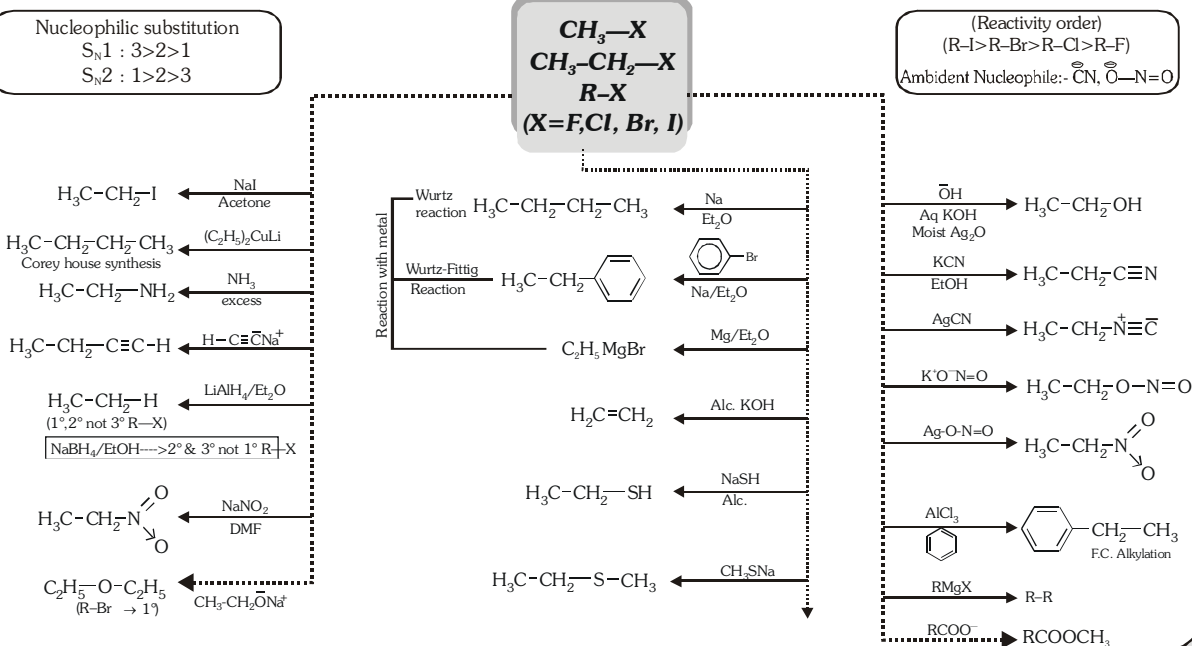
[illegible]

# HALOALKANE

## PREPARATION



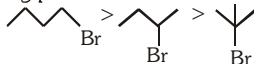
## REACTIONS



## PHYSICAL PROPERTIES

- (1) Dipole moment:  $\text{CH}_3\text{-Cl} > \text{CH}_3\text{-F} > \text{CH}_3\text{-Br} > \text{CH}_3\text{-I}$
- (2) Bond enthalpies:  $\text{CH}_3\text{-F} > \text{CH}_3\text{-Cl} > \text{CH}_3\text{-Br} > \text{CH}_3\text{-I}$

- (3) Boiling point:  $\text{R-I} > \text{R-Br} > \text{R-Cl} > \text{R-F}$

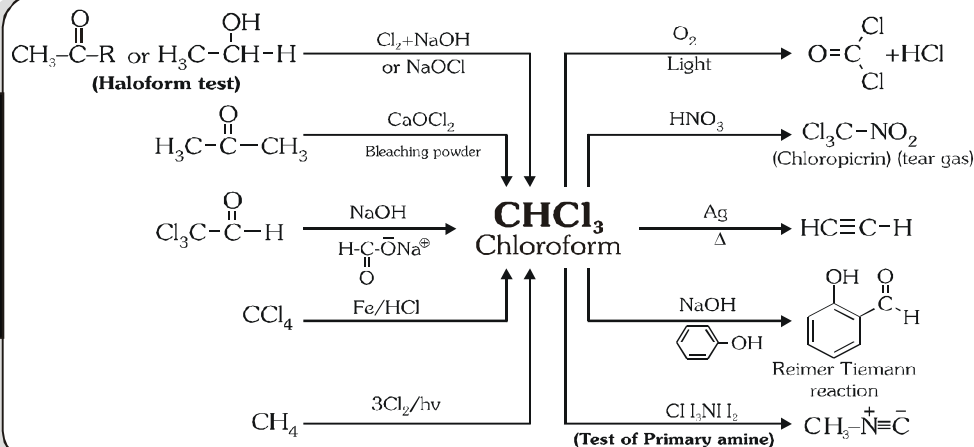


- (4) Density:  $n\text{-C}_3\text{H}_7\text{Cl} < n\text{-C}_3\text{H}_7\text{Br} < n\text{-C}_3\text{H}_7\text{I}$

- (5) Solubility  $\rightarrow$  slightly soluble in water

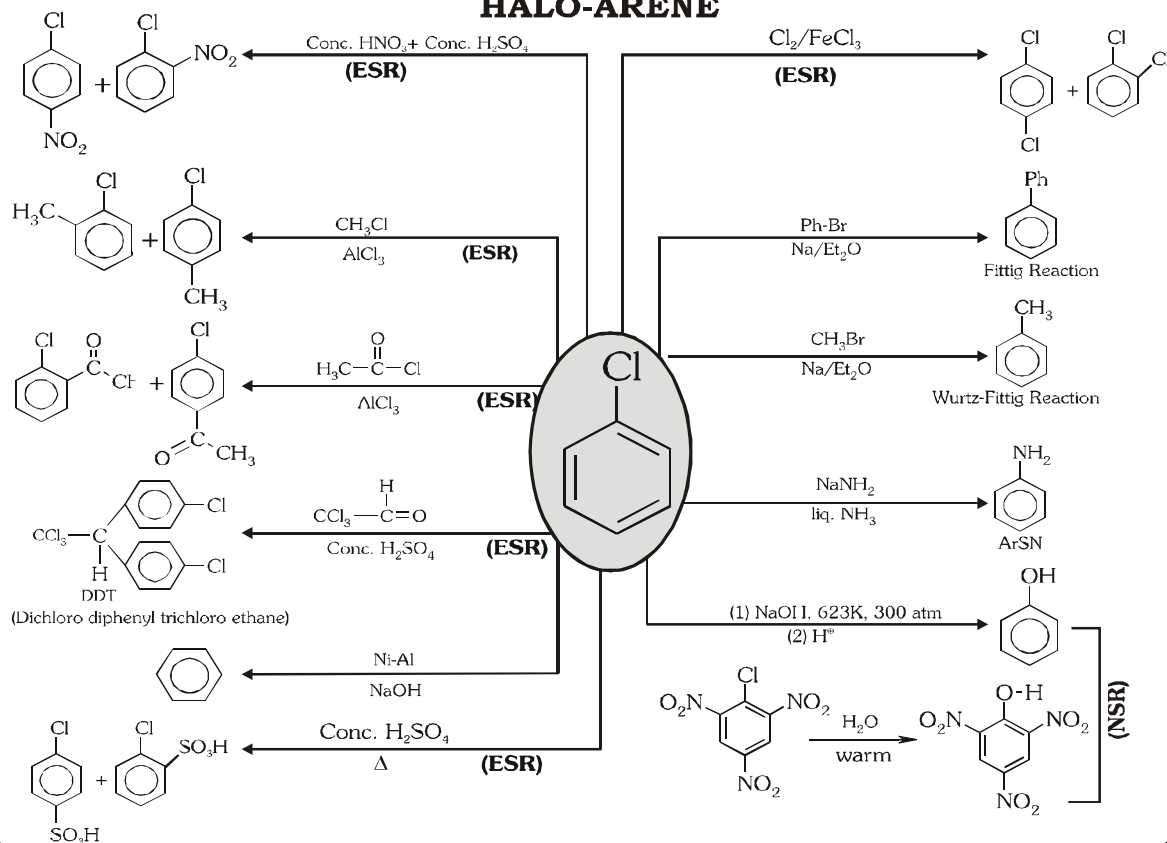
# TRI-HALO ALKANE

Preparation

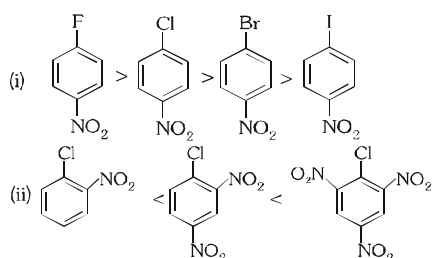


Reaction

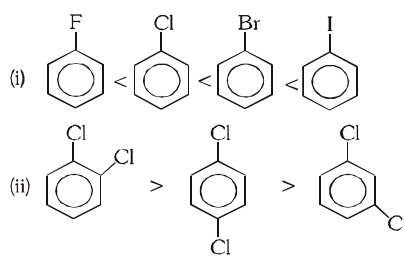
## HALO-ARENE



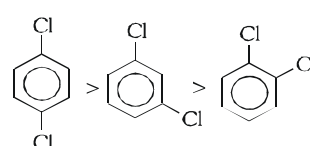
## (A) [Reactivity toward Nucleophile]



## (B) Boiling point



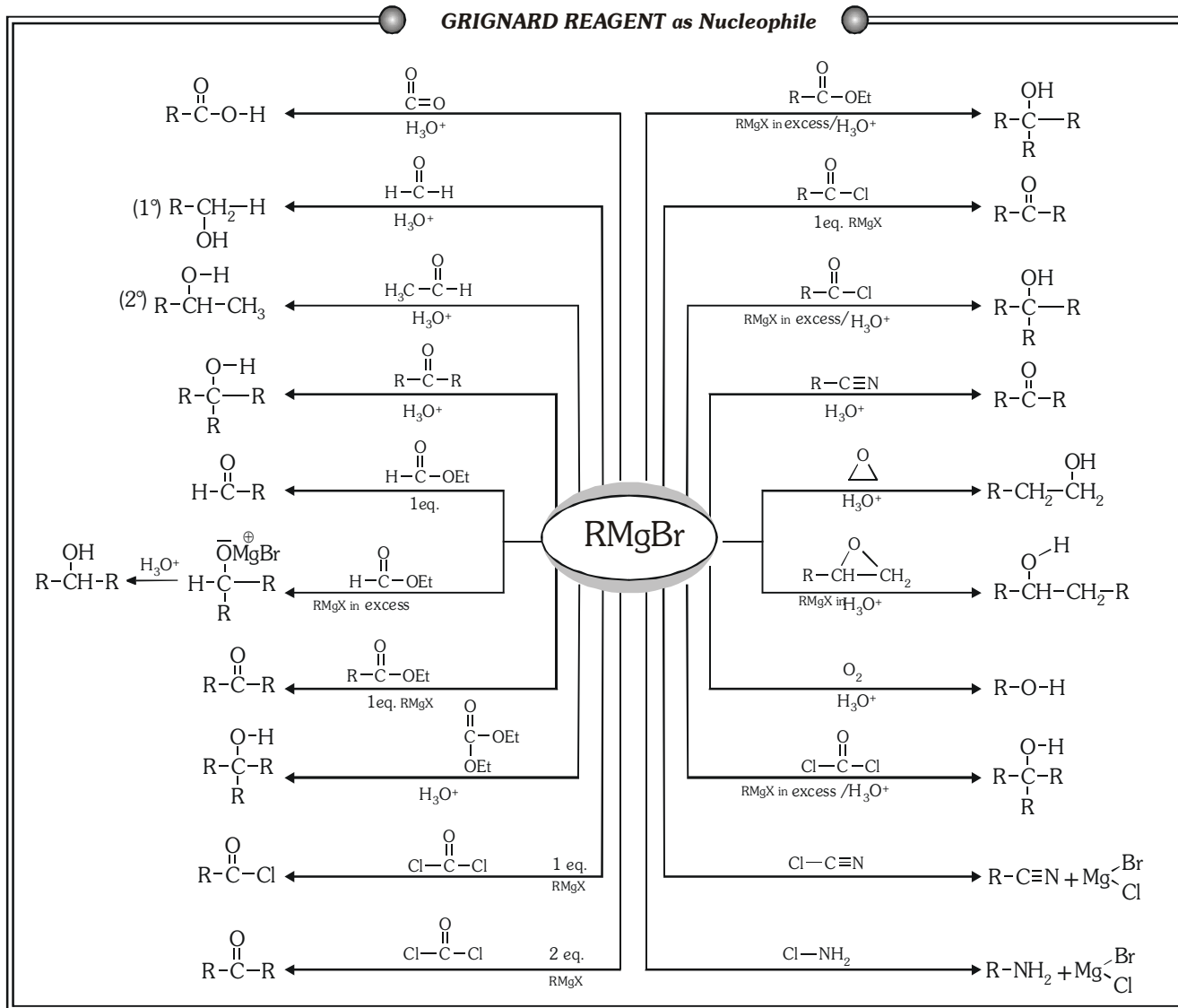
## (C) Melting point



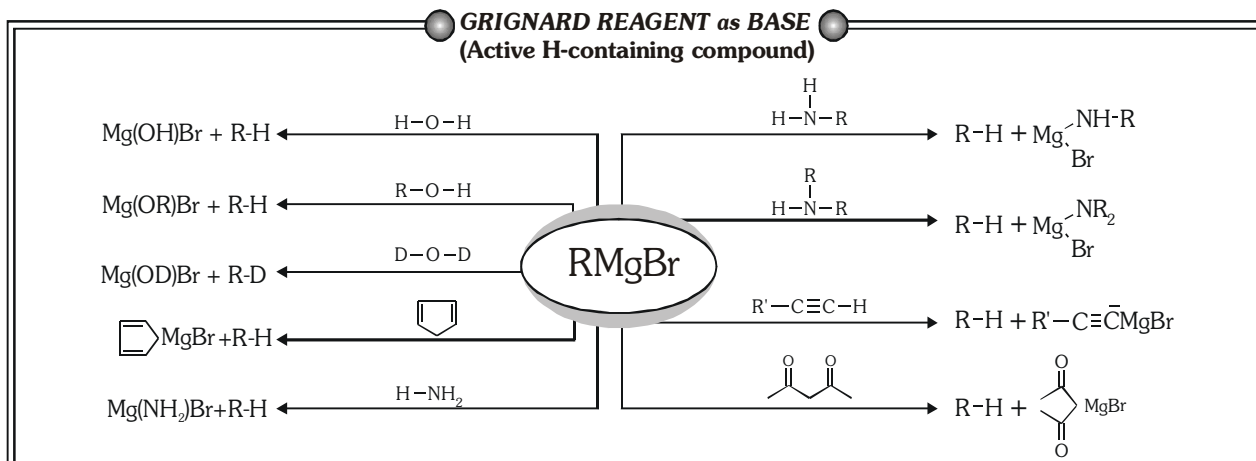
# GRIGNARD REAGENT

## REACTION

### GRIGNARD REAGENT as Nucleophile



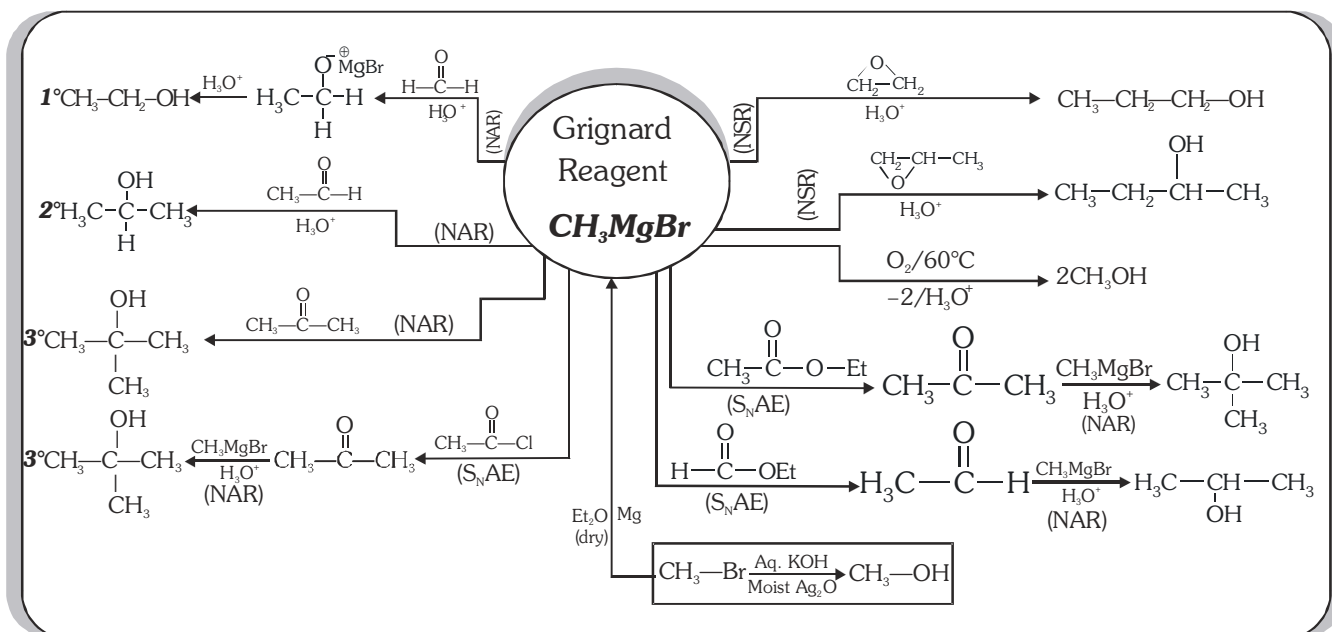
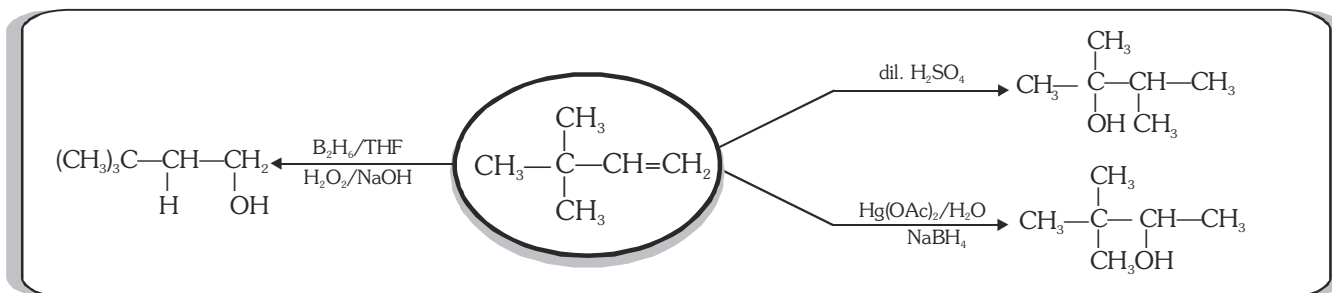
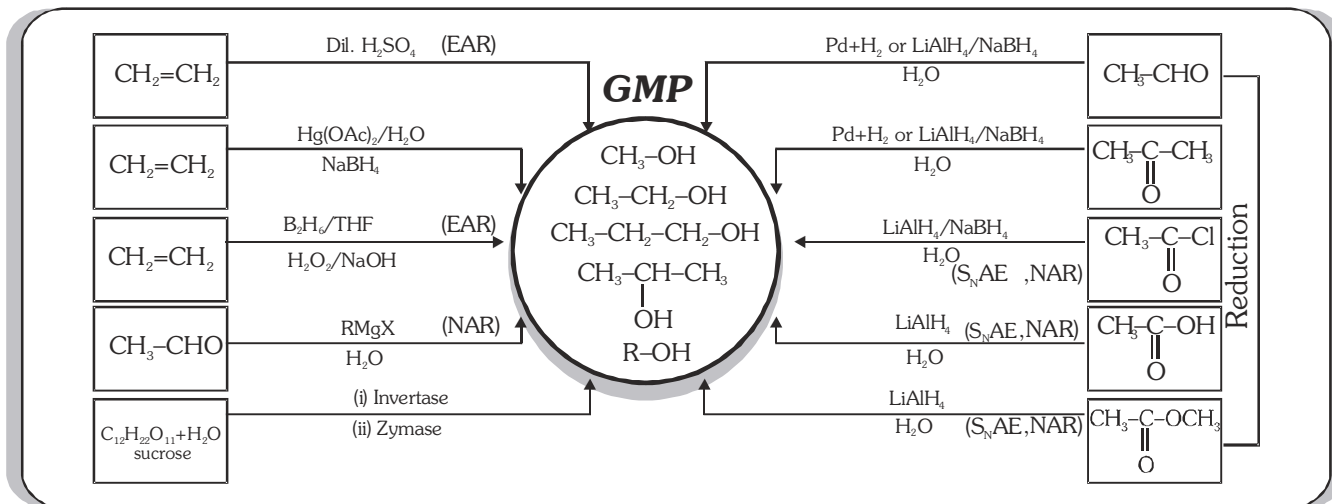
### GRIGNARD REAGENT as BASE (Active H-containing compound)



**IMPORTANT NOTES**

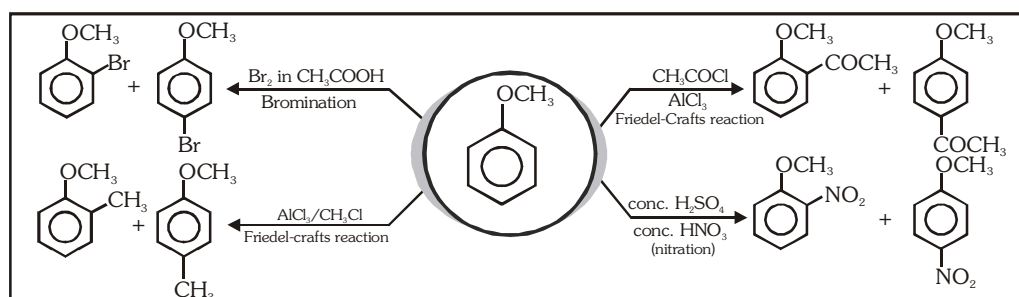
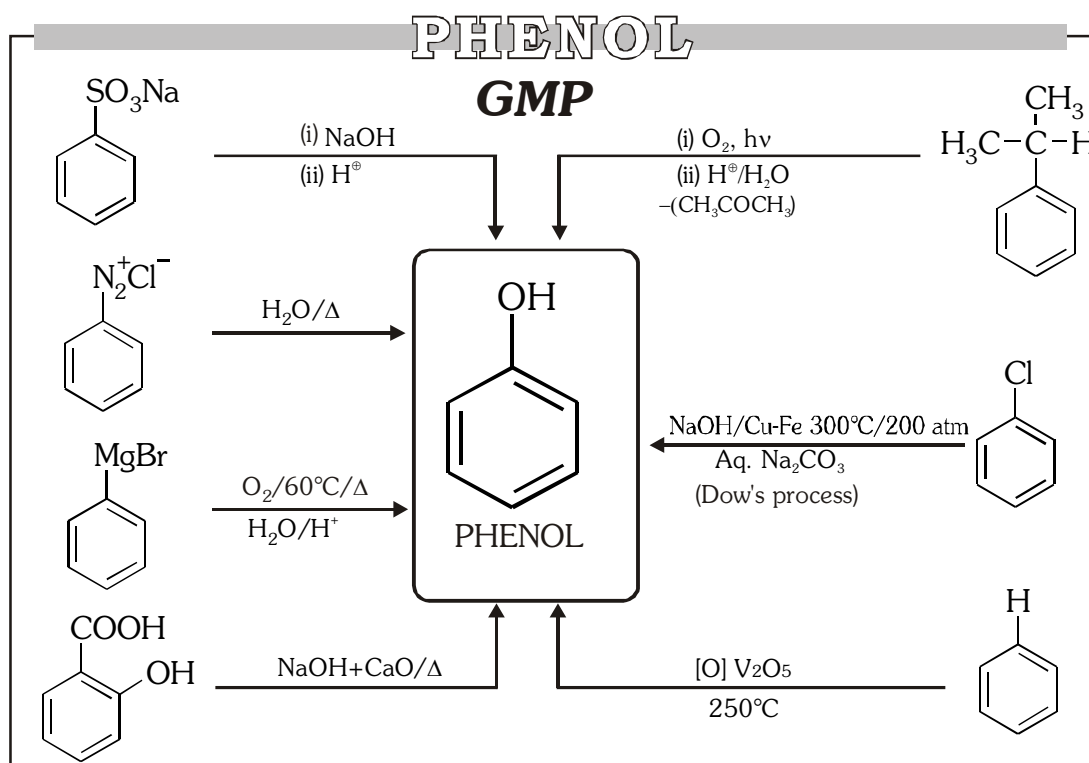
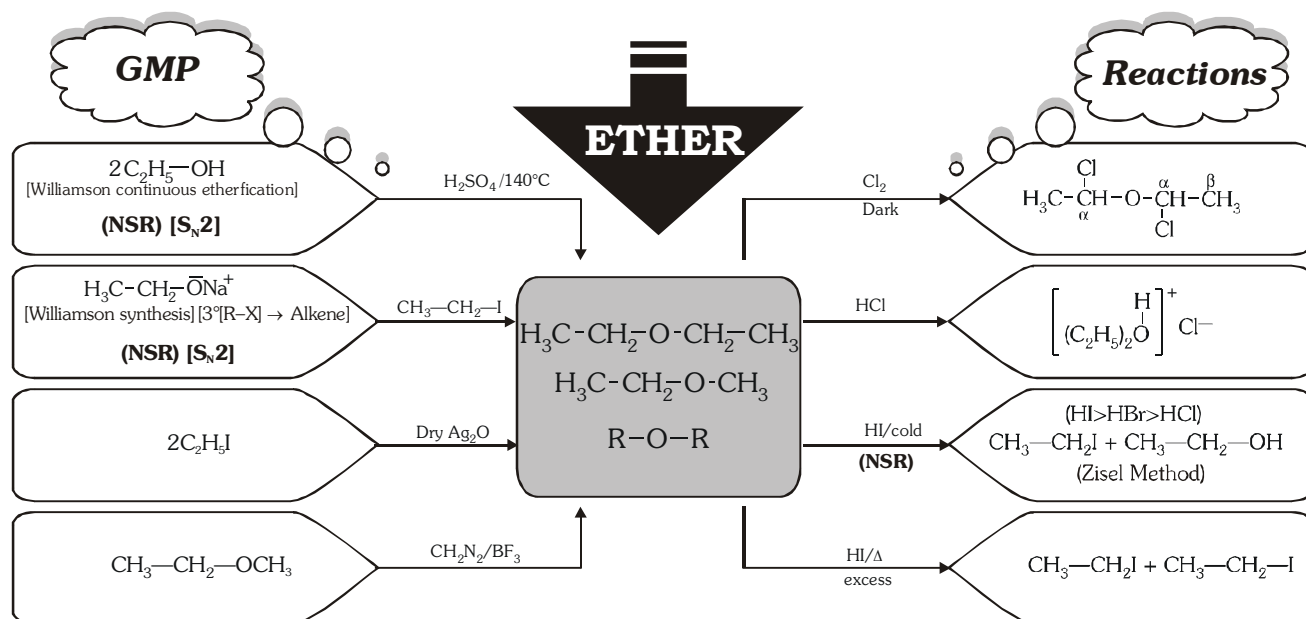


# ALCOHOL

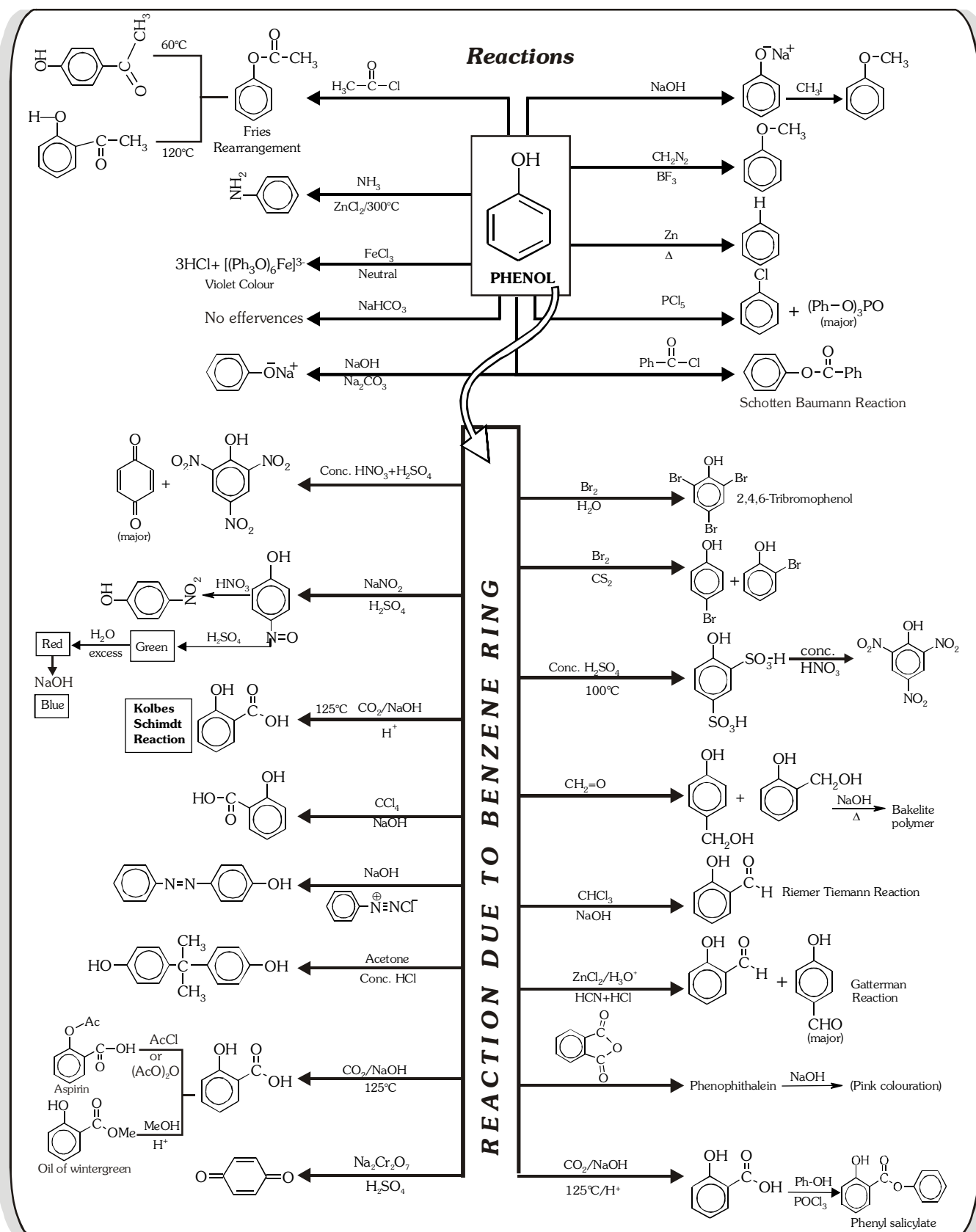


- Solubility of alcohol increase with increase in branching  $n < \text{iso} < \text{neo}$  (isomeric)
- Relative order of reactivity
  - (i)  $1^\circ > 2^\circ > 3^\circ$  (O-H bond fission)
  - (ii)  $3^\circ > 2^\circ > 1^\circ$  (C-O bond fission)
  - (iii)  $3^\circ > 2^\circ > 1^\circ$  (Dehydration)





# PHENOL



### Comparison of $S_N1$ and $S_N2$

| REACTIONS                | $S_N1$                              | $S_N2$                              |
|--------------------------|-------------------------------------|-------------------------------------|
| A Kinetics               | 1 <sup>st</sup> order               | 2 <sup>nd</sup> order               |
| B Rate                   | $k[RX]$                             | $k[RX][Nu:]$                        |
| C Stereochemistry        | Racemisation                        | Inversion                           |
| D Substrate (reactivity) | $3^\circ > 2^\circ > 1^\circ > MeX$ | $MeX > 1^\circ > 2^\circ > 3^\circ$ |
| E Nucleophile            | Rate Independent                    | Needs Strong Nu                     |
| F Solvent                | Good ionizing                       | Faster in aprotic                   |
| G Leaving Group          | Needs Good LG                       | Needs Good LG                       |
| H Rearrangement          | Possible                            | Not Possible                        |

### Comparison of $E_1$ and $E_2$

| REACTIONS         | $E_1$                         | $E_2$                         |
|-------------------|-------------------------------|-------------------------------|
| A Kinetics        | 1 <sup>st</sup> order         | 2 <sup>nd</sup> order         |
| B Rate            | $k[RX]$                       | $k[RX][B:]$                   |
| C Stereochemistry | No special geometry           | Anti-periplanar               |
| D Substrate       | $3^\circ > 2^\circ > 1^\circ$ | $3^\circ > 2^\circ > 1^\circ$ |
| E Base Strength   | Rate Independent              | Needs Strong bases            |
| F Solvent         | Good ionizing                 | Polarity not import           |
| G Leaving Group   | Needs Good LG                 | Needs Good LG                 |
| H Rearrangement   | Possible                      | Not Possible                  |

### Summary of $S_N1$ , $S_N2$ , $E_1$ and $E_2$ Reactions

| RX | Mechanism | $Nu/B$                                | Solvent       | Temp.  |
|----|-----------|---------------------------------------|---------------|--------|
| 1° | $S_N2$    | Better<br>$OH^-, C_2H_5O^-$           | Polar aprotic | Low    |
|    | $E_2$     | Strong & bulky base<br>$(CH_3)_3CO^-$ |               | High   |
| 2° | $S_N2$    | $HO^-; C_2H_5O^-$                     | Polar aprotic | Low    |
|    | $E_2$     | $(CH_3)_3CO^-$                        |               | High   |
|    | $(S_N1)$  | (Solvent)                             | Polar aprotic | (Low)  |
|    | $(E_1)$   | (Solvent)                             |               | (High) |
| 3° | $S_N1$    | Solvent                               | Protic        | Low    |
|    | $E_1$     | Solvent                               | Protic        | High   |

|                              | Primary (1°)   | Secondary (2°)                                 | Tertiary (3°)                                 |
|------------------------------|----------------|------------------------------------------------|-----------------------------------------------|
| Strong nucleophile           | $S_N2 \gg E_2$ | $S_N2 + E_2$<br>(if weak base, $S_N2$ favored) | $E_2$                                         |
| Weak nucleophile weak base   | Mostly $S_N2$  | Mostly $S_N2/S_N1$                             | Mostly $S_N1$ at low T mostly $E_1$ at high T |
| Weak nucleophile strong base | Mostly $E_2$   | Mostly $E_2$                                   | $E_2$                                         |

#### Order of reactivity of Alkyl Halide towards

$S_N1 \propto \text{Benzylic} > \text{Allylic} > 3^\circ > 2^\circ > 1^\circ$

$S_N1 \propto \text{Stability of carbocation}$

$S_N2 \propto 1^\circ > 2^\circ > 3^\circ$

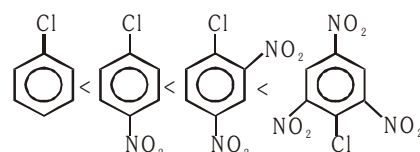
$S_N2 \propto \frac{1}{\text{Steric hindrance}}$

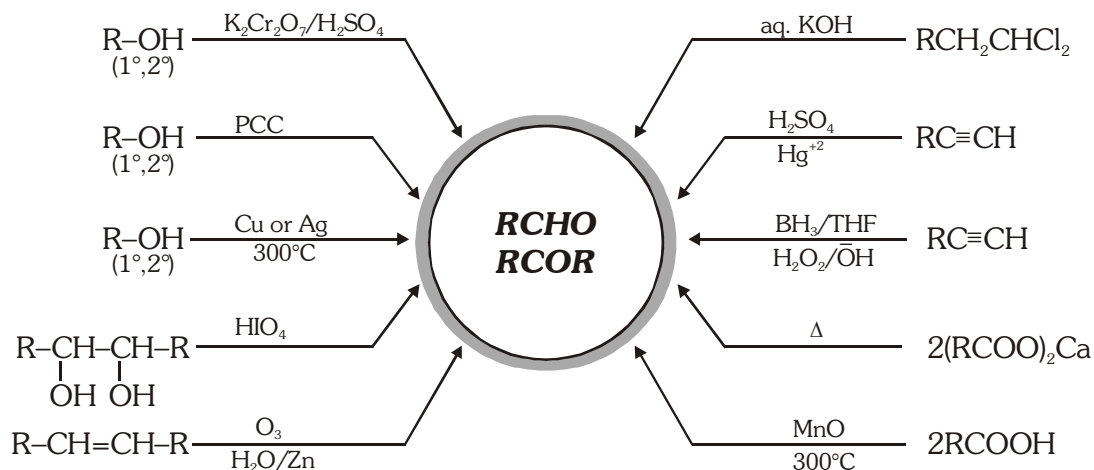
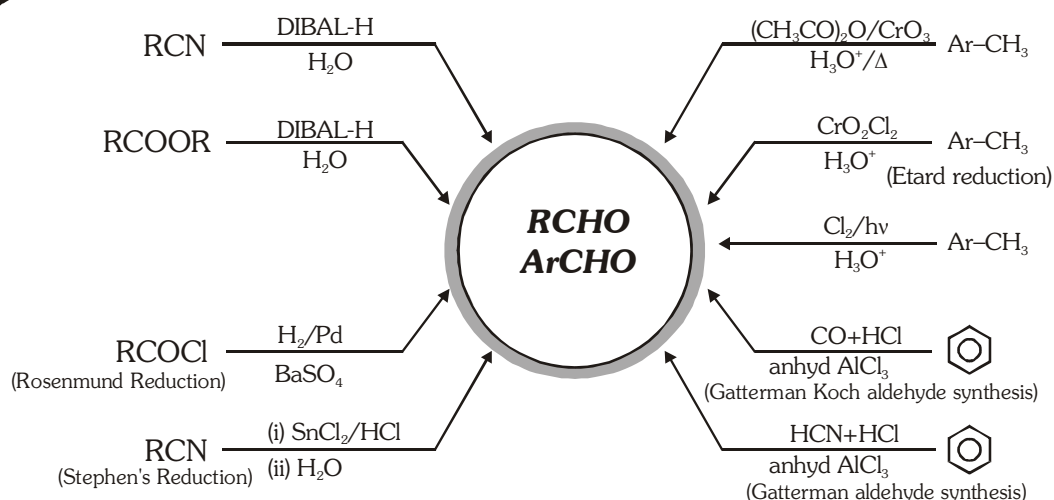
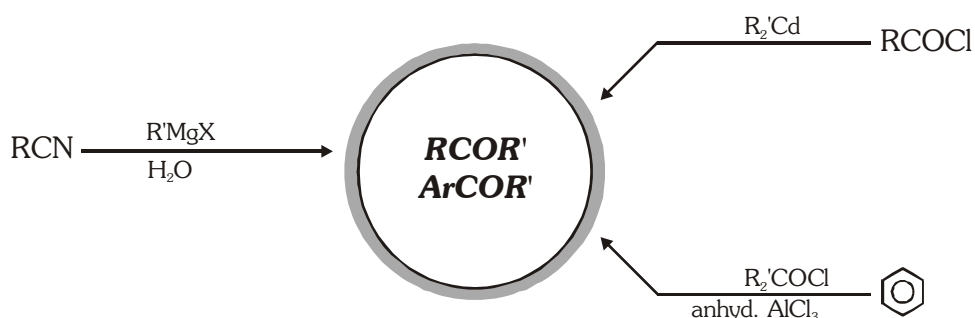
#### Reactivity order towards

$S_N1$  or  $S_N2$  and  $E_1$  or  $E_2$

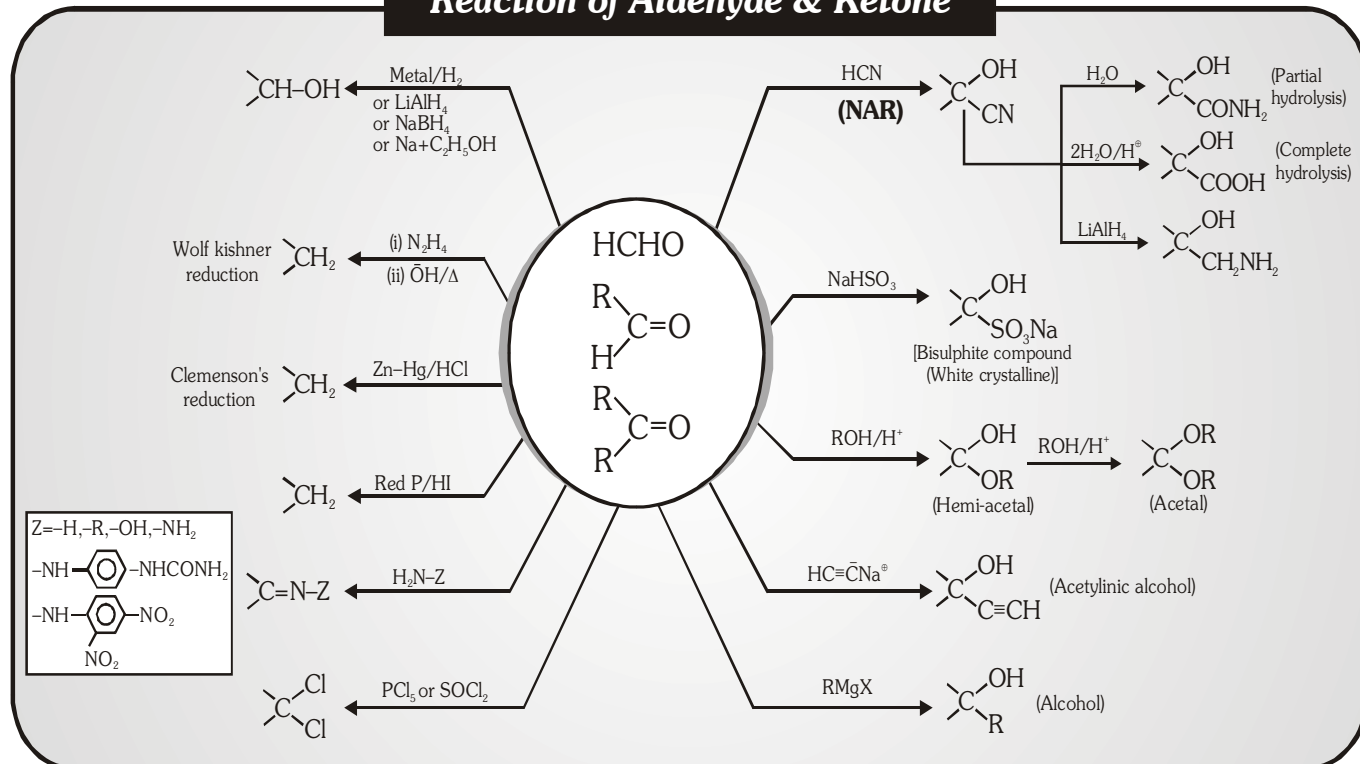
$R-I > R-Br > R-Cl > R-F$

With increase in number of strong electron withdrawing group at ortho and para position, reactivity of X towards aromatic nucleophilic substitution increases.

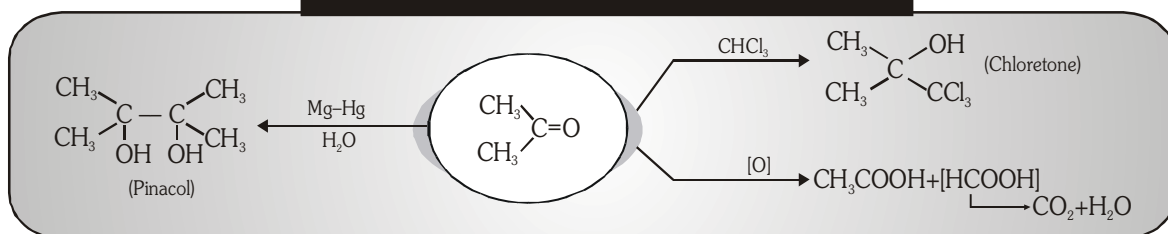


**Preparation of Aldehyde & Ketone both****Preparation of Aldehyde only****Preparation of Ketone only**

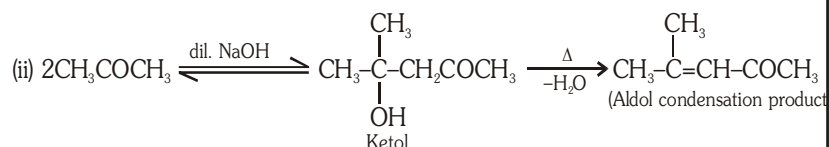
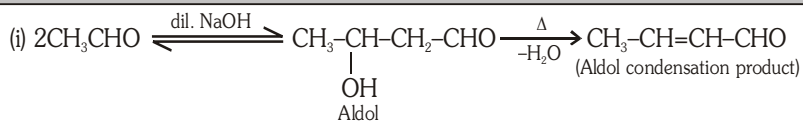
### Reaction of Aldehyde & Ketone



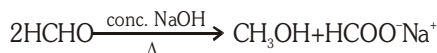
### Reaction of only Ketone



#### Aldol Reaction (Aldehyde or ketone with αH)

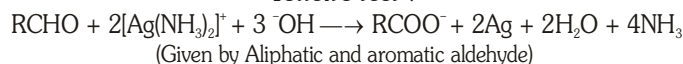


#### Cannizzaro reaction (Aldehyde with no α H)

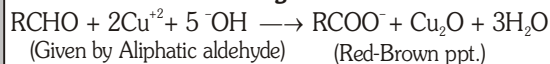


#### TESTS

##### Tollen's test :-



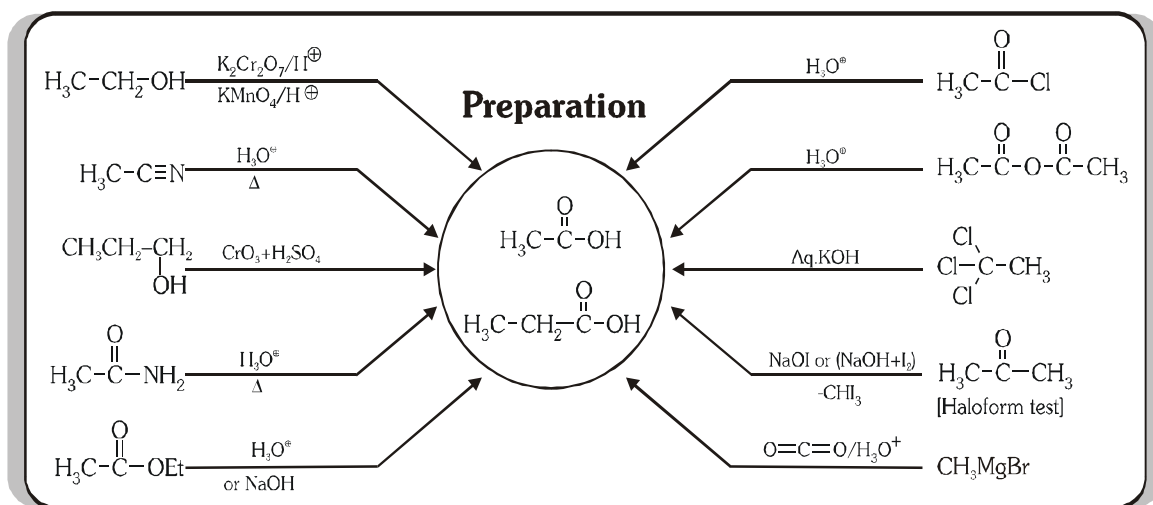
##### Fehling's test



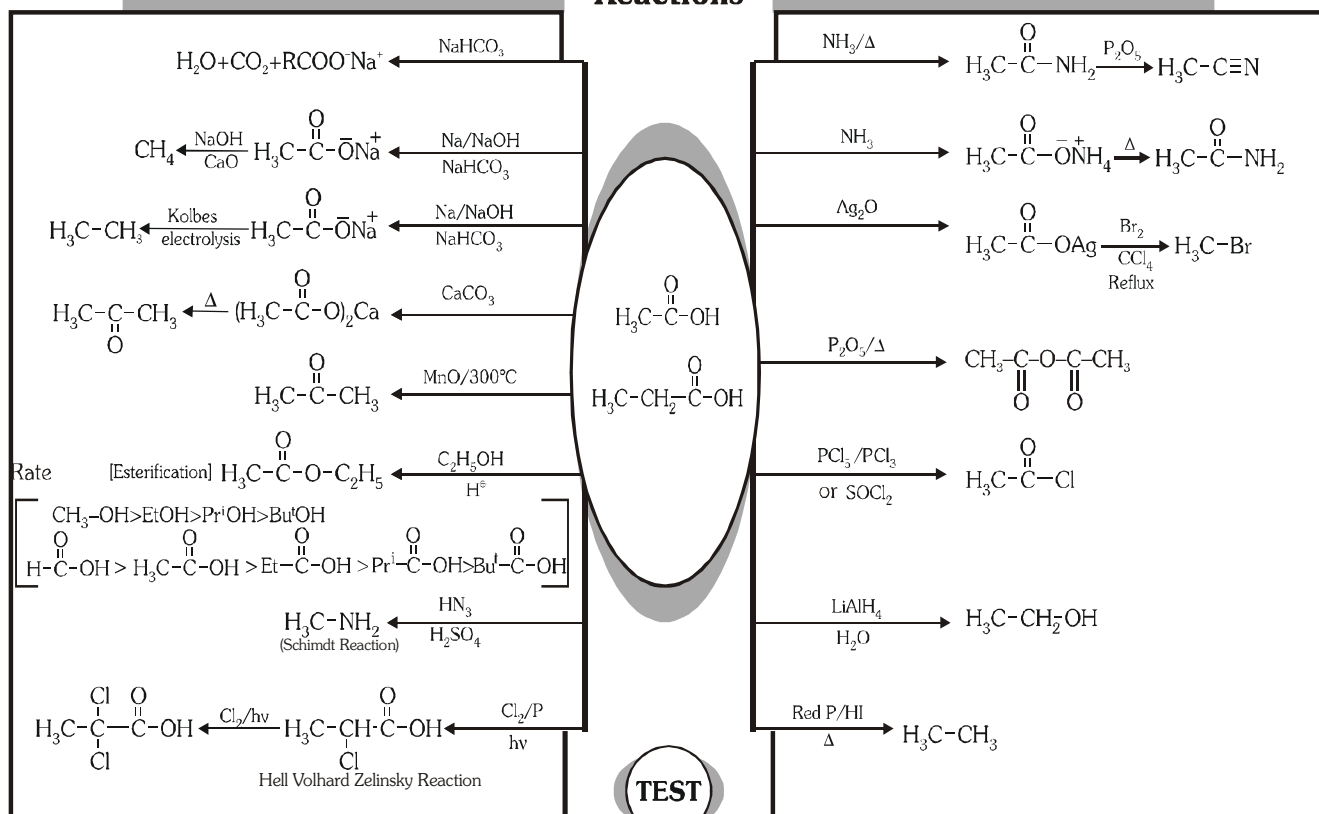
**IMPORTANT NOTES**



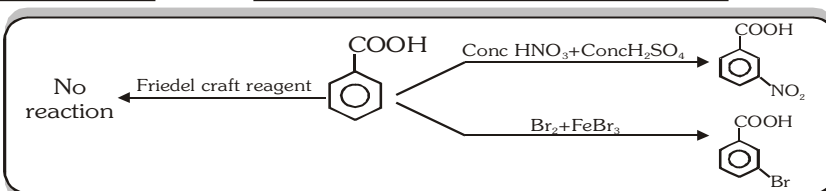
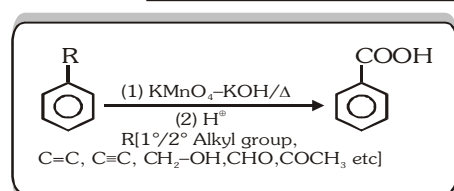
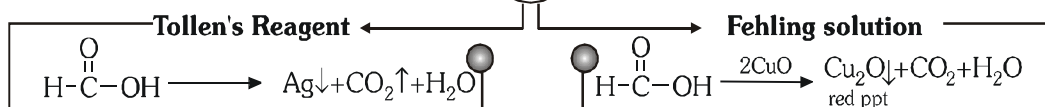
## CARBOXYLIC ACID

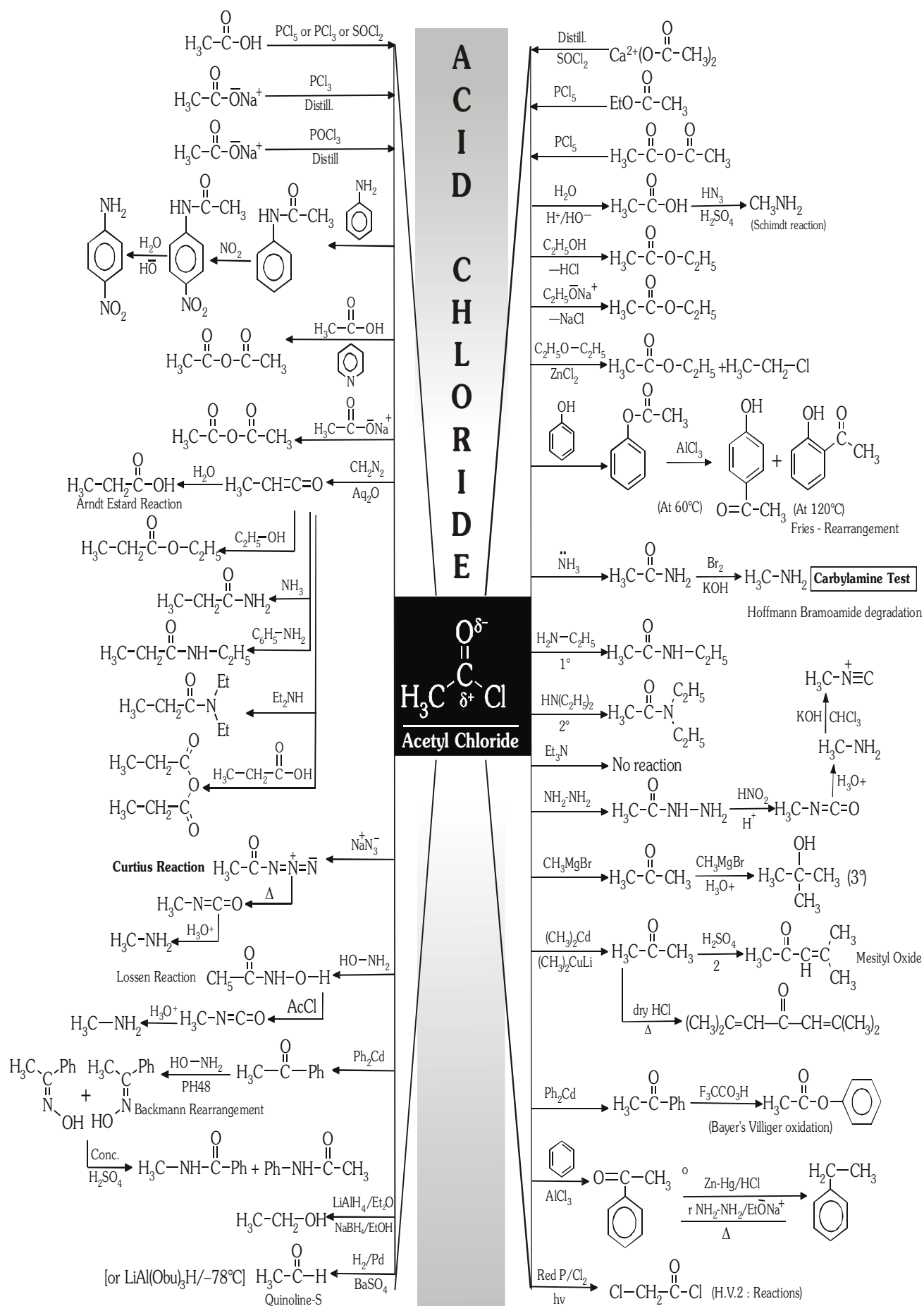


## Reactions

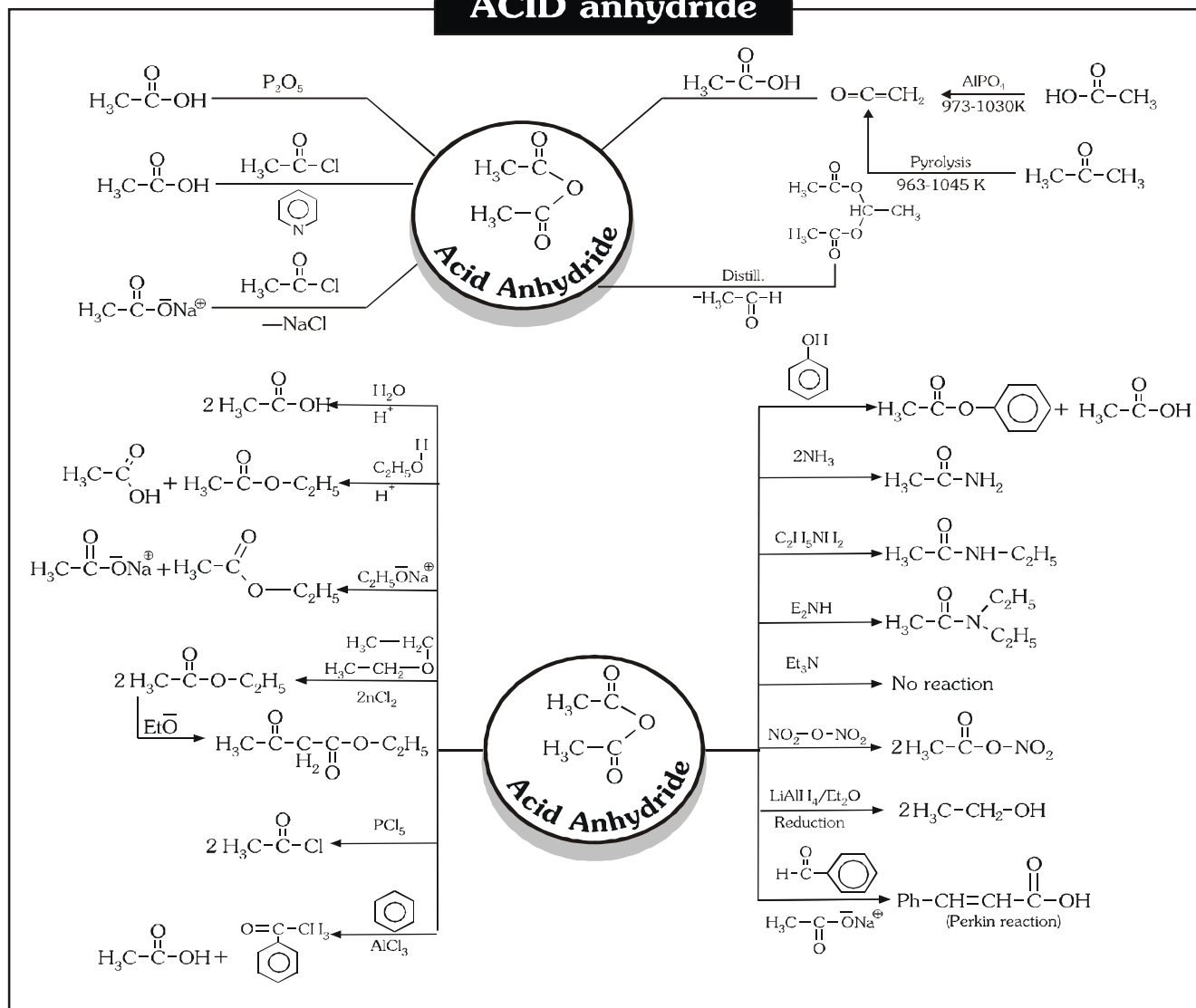


## TEST

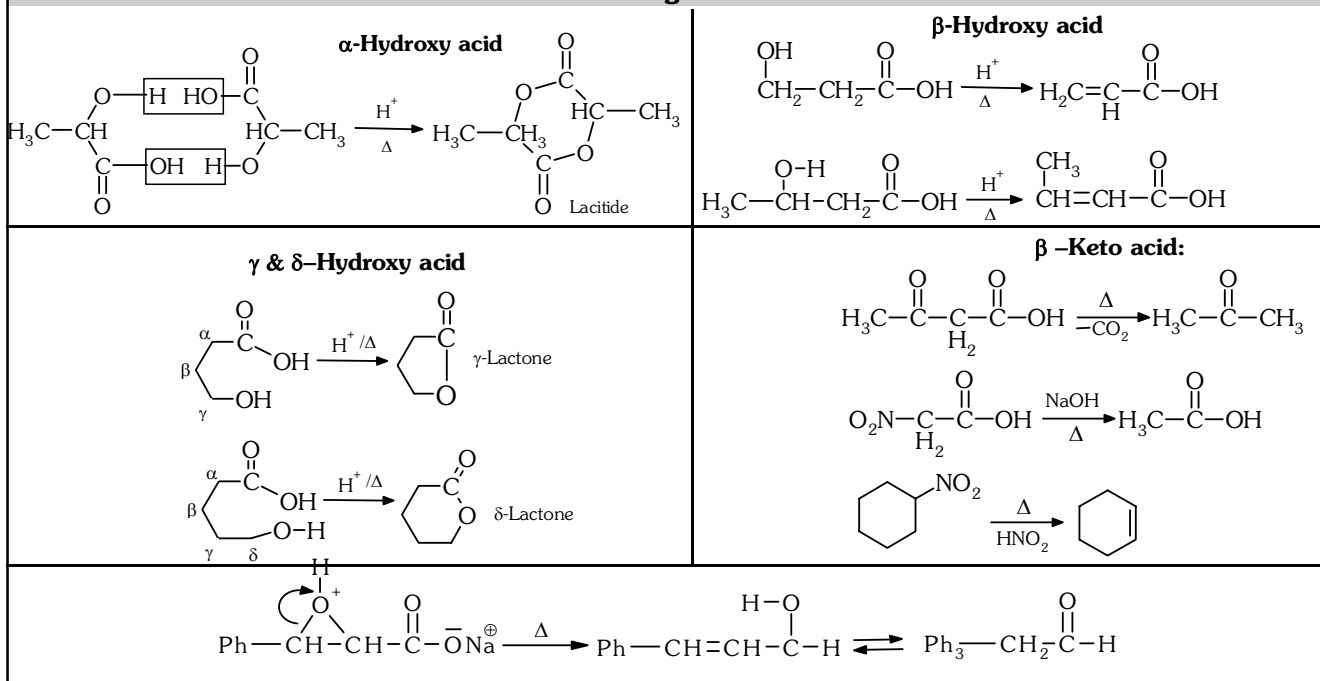




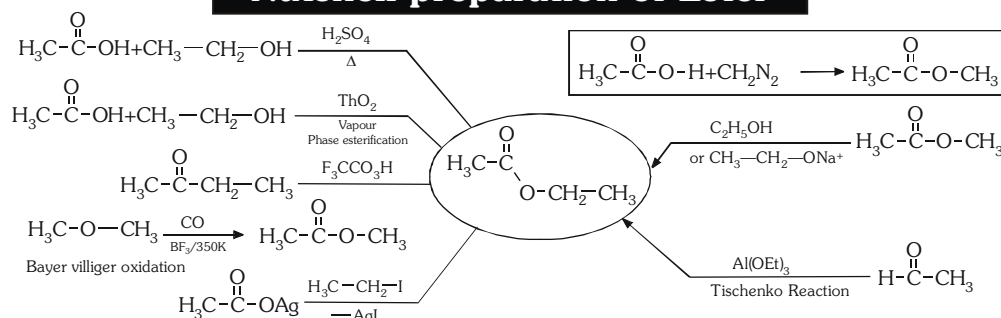
# ACID anhydride



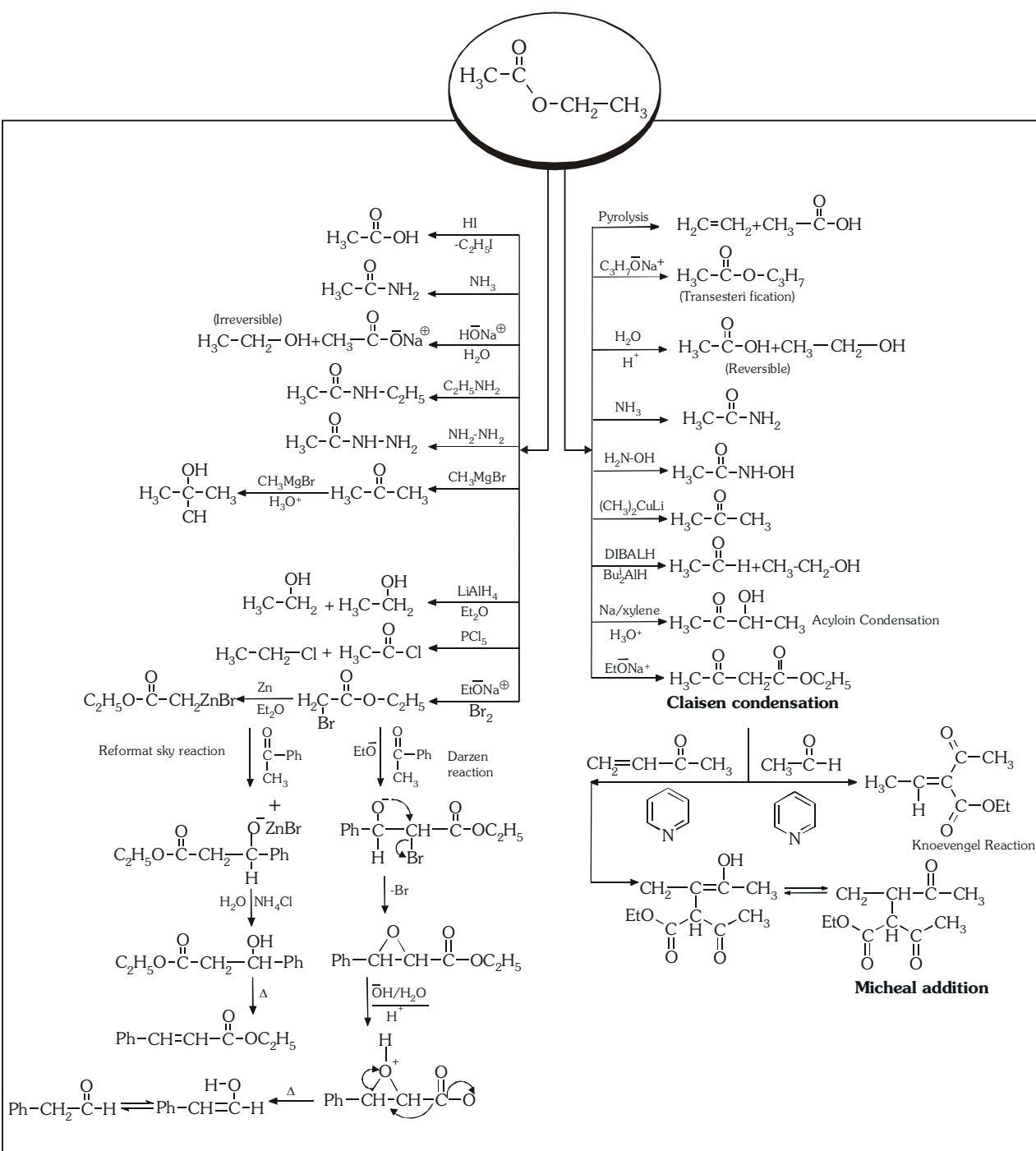
## Heating effect



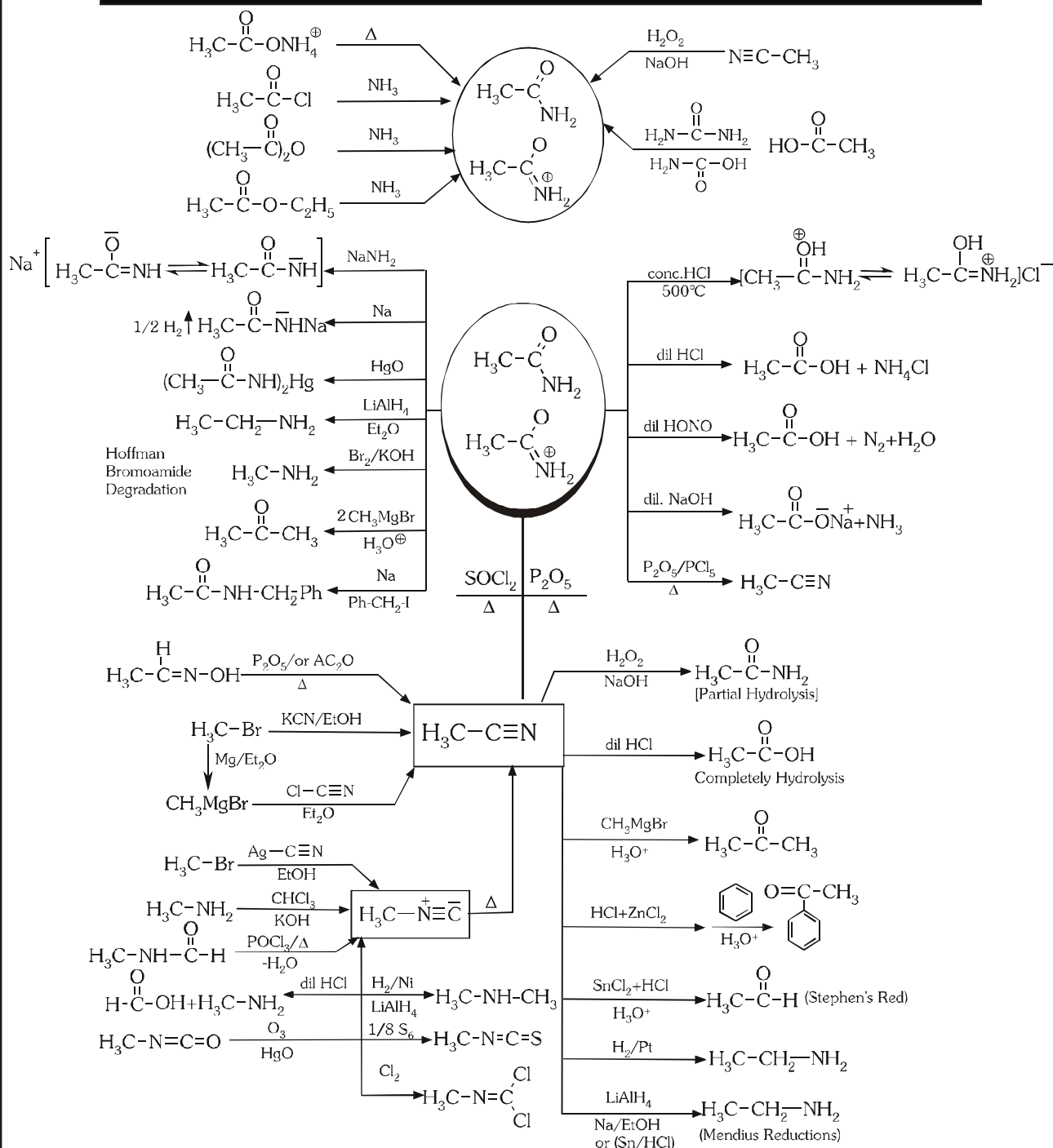
## Nutshell preparation of Ester



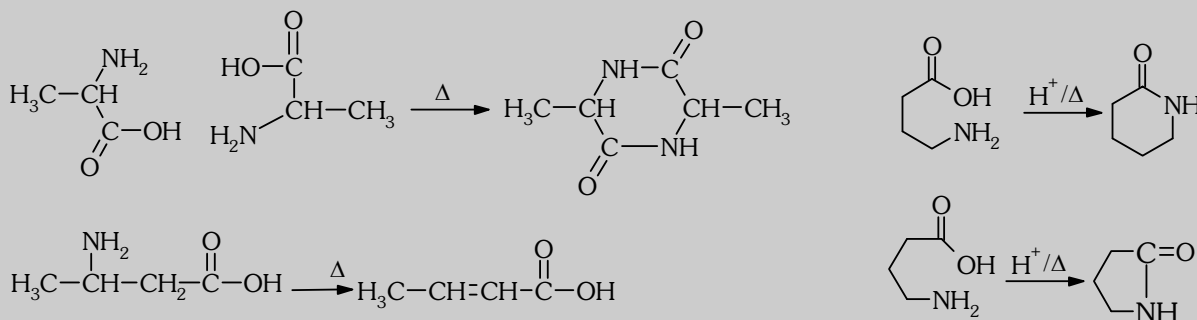
## Nutshell chemical properties of Ester



## Nutshell preview and review of Amide, Cyanide and Isocyanide



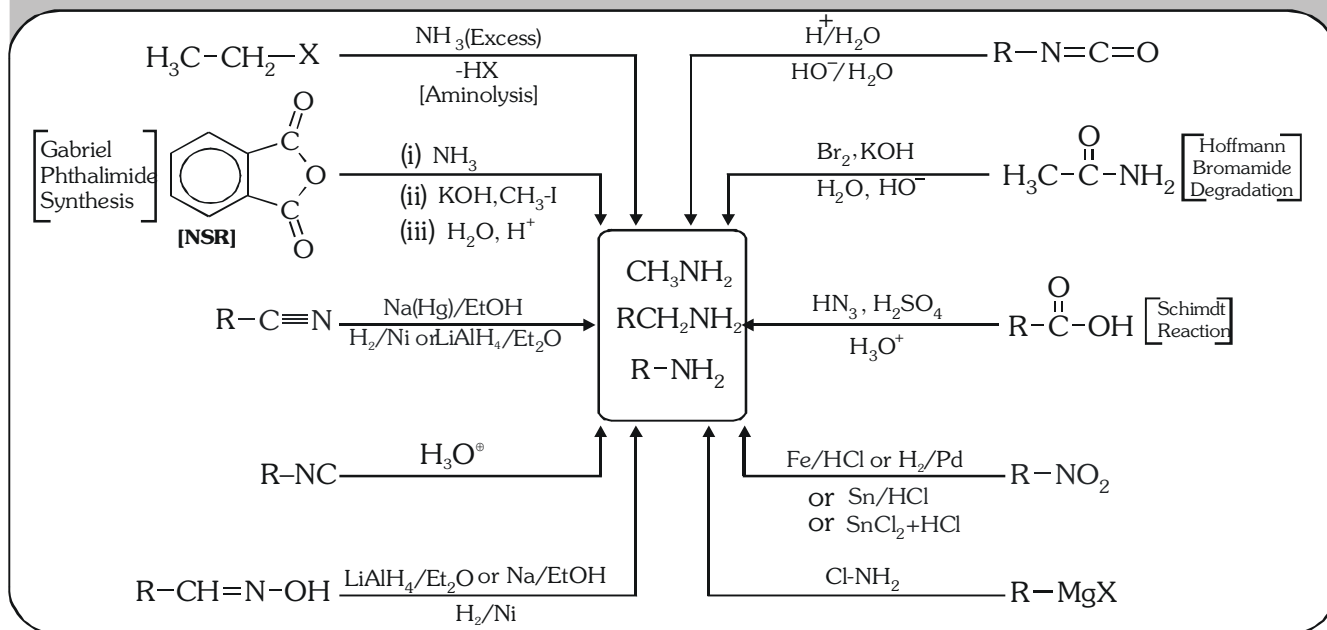
**Amino Acid :**



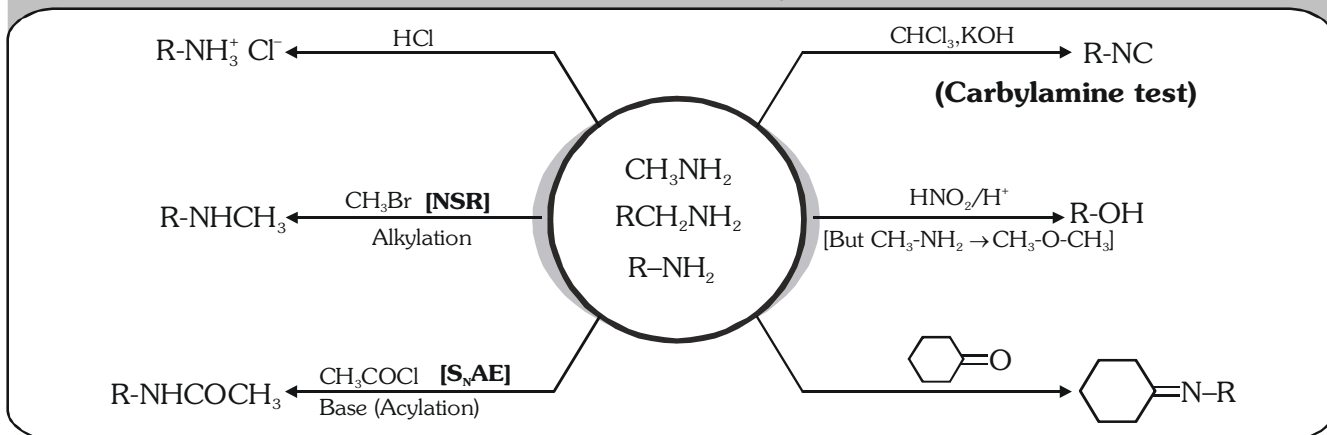
[illegible]

# AMINES

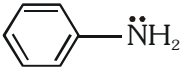
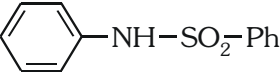
## PREPARATION



## REACTION



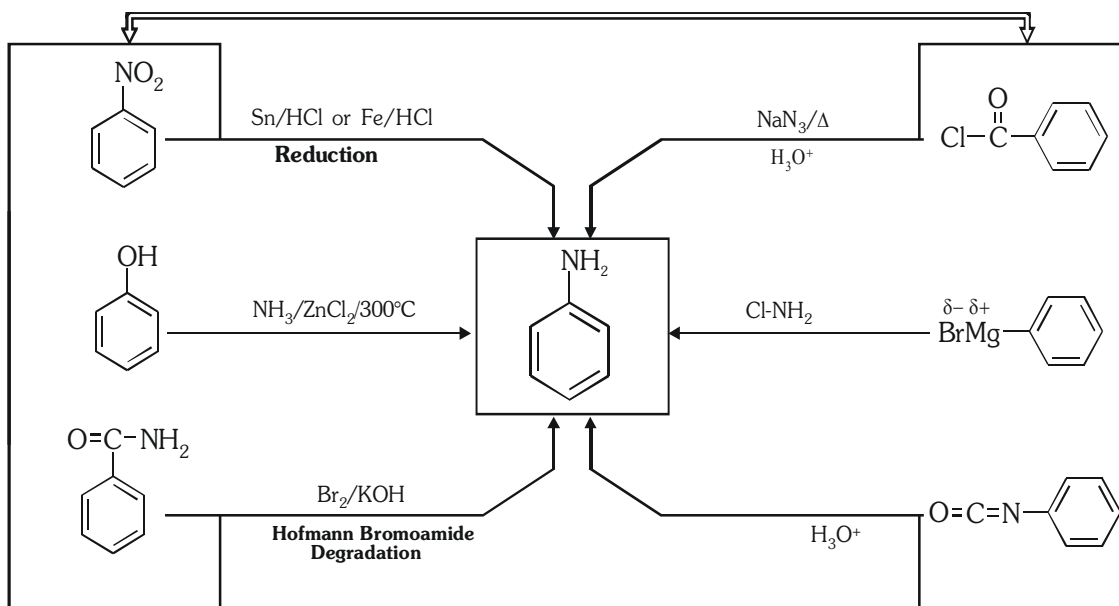
## TEST

| Reagent                                                                                                                                    | $R-\ddot{N}H_2(1^\circ)$                                                                                                                                                | $R_2\ddot{N}H(2^\circ)$                                                                                                                                    | $R_3\ddot{N}(3^\circ)$ |  |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------------------------------------------------------------------------------------|
| Ph-SO <sub>2</sub> Cl<br>(Hinsberg reagent)                                                                                                | $  \begin{array}{c}  R-\text{N}-\text{SO}_2-\text{Ph} \\    \\  \text{H} \\  \downarrow \text{NaOH} \\  [R-\text{N}^--\text{SO}_2-\text{Ph}]\text{Na}^+  \end{array}  $ | $  \begin{array}{c}  \text{O} \\     \\  R_2\text{N}-\text{S}-\text{Ph} \\    \\  \text{O} \\  \downarrow \text{NaOH} \\  \text{Insoluble}  \end{array}  $ | No reaction            |  |
| $  \begin{array}{c}  \text{S} \\     \\  \text{C}=\text{S} \\  \Delta/\text{HgCl}_2 \\  \text{Mustard} \\  \text{oil test}  \end{array}  $ | $  \begin{array}{c}  \text{S} \\     \\  R-\text{NH}-\text{C}-\text{SH} \\  \text{HgCl}_2 \downarrow \Delta \\  R-\text{N}=\text{C}=\text{S}+\text{HgS}  \end{array}  $ | $  \begin{array}{c}  \text{S} \\     \\  R_2\text{N}-\text{C}-\text{SH} \\  \text{HgCl}_2 \downarrow \Delta \\  \text{No reaction}  \end{array}  $         | No reaction            | $\xrightarrow{\text{KOH}} \text{Ph}-\text{N}=\text{C}=\text{S}+\text{HgS}$            |

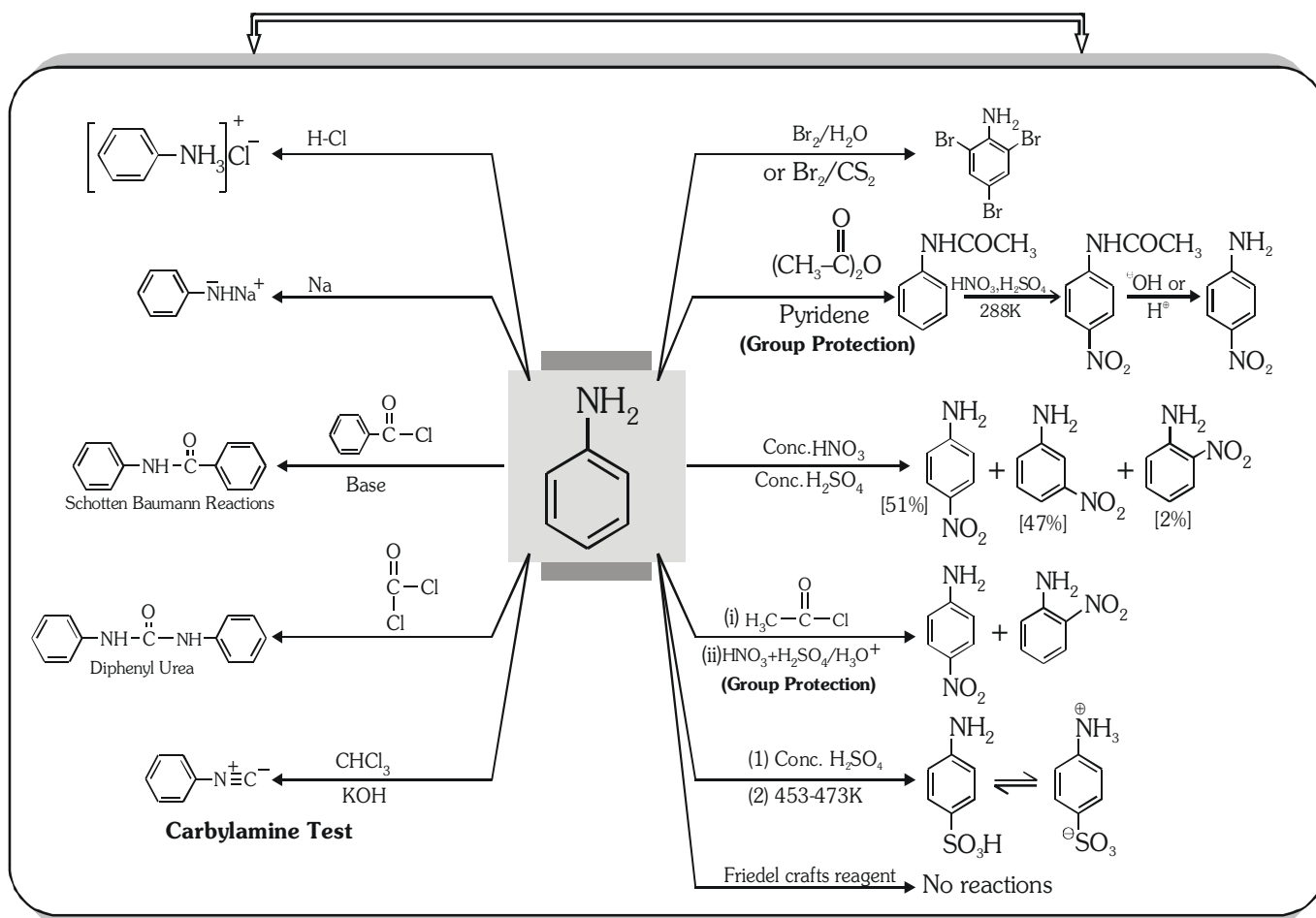


# ANILINE

## PREPARATION



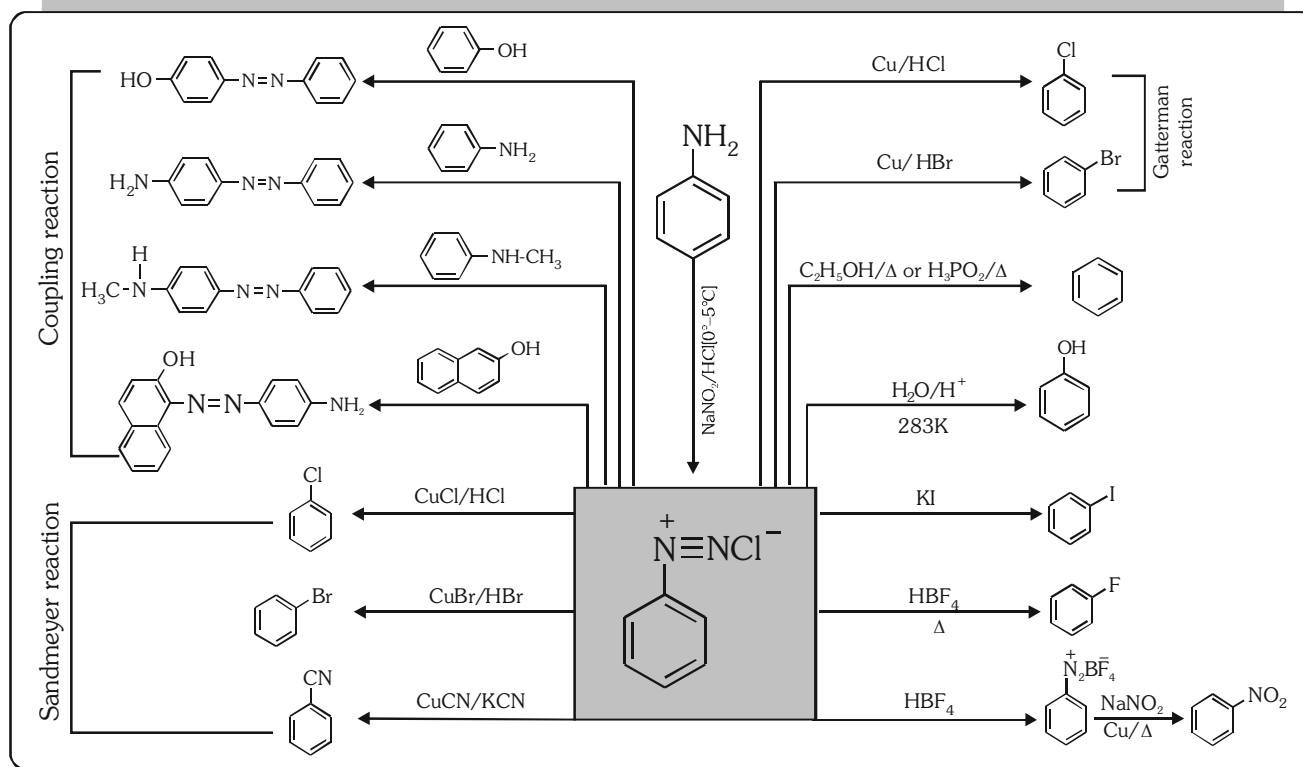
## REACTION





# ***BENZENE DIAZONIUM CHLORIDE***

## REACTION



|                   |                                                  |
|-------------------|--------------------------------------------------|
| FrSR              | Free Radical Substitution reaction               |
| ESR               | Electrophilic Substitution reaction              |
| NSR               | Nucleophilic Substitution reaction               |
| S <sub>N</sub> AE | Substitution Nucleophilic (addition elimination) |
| FrAR              | Free radical addition reaction                   |
| NAR               | Nucleophilic addition reaction                   |
| EAR               | Electrophilic addition reaction                  |
| FrER              | Free Radical Elimination reaction                |

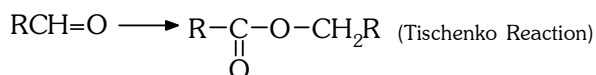
[illegible]

## Nutshell review & preview of ORGANIC REAGENTS

### 1. Alcoholic KOH

$R-X \rightarrow$  Alkene ; Elimination

### 2. Aluminium Ethoxide



(Aldehyde)

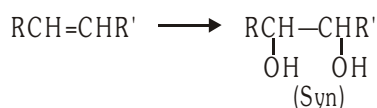
(Ester)

### 3. Aqueous KOH/NaOH

$R-X \rightarrow ROH$

Nucleophilic substitution reaction also used for Cannizzaro reaction with aldehyde.

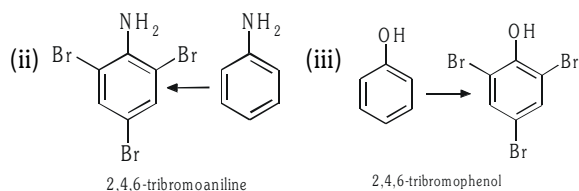
### 4. Baeyer's Reagent (Alkaline cold dilute $KMnO_4$ )



alkene  $\longrightarrow$  1, 2 diol  
(used to detect unsaturation)

### 5. Bromine water

(i) used to detect unsaturation;



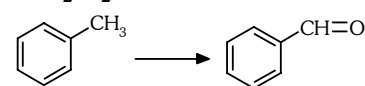
### 6. Benedict's solution

Used to detect aldehyde group  $RCHO \rightarrow RCO_2^-$   
[ketone gives -ve test]

### 7. $Cu_2Cl_2 + NH_4OH$

Used to Detect Terminal Alkyne  
Red Precipitate observed

### 8. $CrO_2Cl_2$

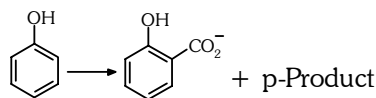


Etard reaction

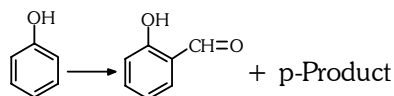
### 9. $CrO_3$

- (i)  $RCH_2OH \rightarrow RCHO$ ,
- (ii)  $R_2CHOH \rightarrow R_2C=O$
- (iii)  $R_3COH \rightarrow$  no reaction

### 10. $CCl_4 + OH^-$ (Reimer Tiemann)

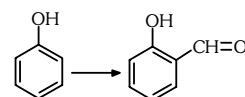


### 11. $CO + HCl + AlCl_3$



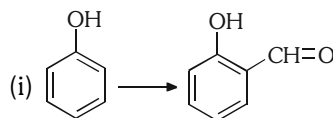
Gatterman koch reaction

### 12. $HCN + HCl + AlCl_3$



Gatterman Aldehyde Synthesis

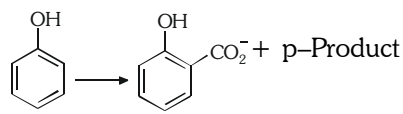
### 13. $CHCl_3 + KOH$



Reimer Tiemann reaction

(ii)  $RNH_2 \rightarrow RNC$  (**Carbyl amine reaction**)  
(used to detect  $1^\circ$  Amine) (Isocyanide test)

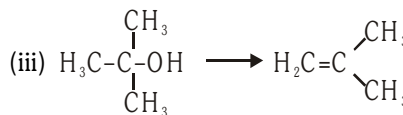
### 14. $CO_2 + OH^-$ (high temp. + Pressure)



Kolbe's reaction

### 15. $Cu/\Delta$

- (i)  $RCH_2OH \rightarrow RCHO$ ,
- (ii)  $R_2CHOH \rightarrow R_2C=O$



### 16. 2,4 - D.N.P.

used to detect carbonyl group (orange ppt observed)

### 17. DMSO

Polar aprotic solvent: favour  $S_N2$  mechanism



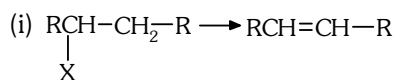
### 37. Ninhydrin

Detection of amino acid

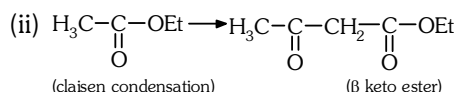
Observation : Purple coloured ion

### 38. NaOR

**Strong base :**



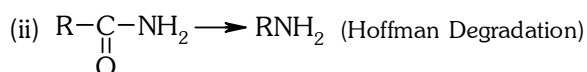
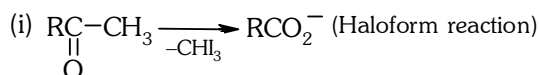
(Saytzeff Product : E<sub>2</sub> elimination)



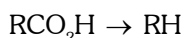
(claisen condensation)

(β keto ester)

### 39. NaOH + X<sub>2</sub> or NaOX



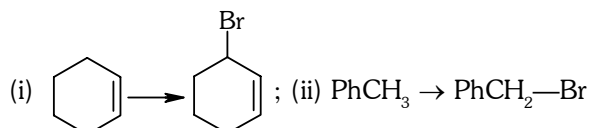
### 40. NaOH + CaO



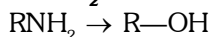
### 41. MnO / 300°C

used for -CO<sub>2</sub> & -H<sub>2</sub>O in carboxylic acid.

### 42. NBS



### 43. NaNO<sub>2</sub> + HCl



### 44. NaNH<sub>2</sub> in paraffin

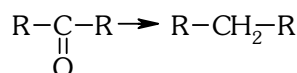
Non-terminal Alkyne → Terminal Alkyne

(2-Butyne → 1-butyne)

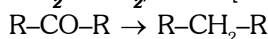
### 45. Na/EtOH

Reduce all except c/c double & triple bond

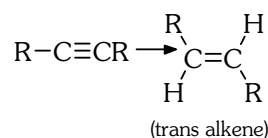
### 46. Zn(Hg) + HCl [Clemmensen's reduction]



### 47. NH<sub>2</sub>-NH<sub>2</sub>/OH<sup>-</sup> [Wolf Kishner reduction]

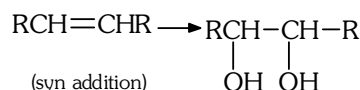


### 48. Na in Liq. NH<sub>3</sub> [Birch reduction]

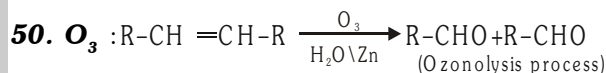


(trans alkene)

### 49. OsO<sub>4</sub> + H<sub>2</sub>O



(syn addition)

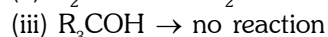
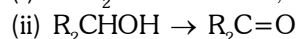
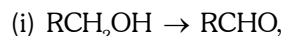


(Ozonolysis process)

### 51. Oxirane followed by H<sup>+</sup>

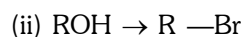
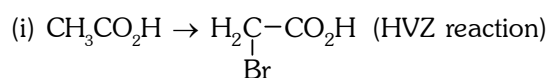


### 52. PCC

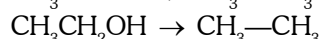
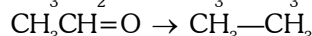
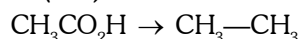


(Mild oxidizing reagent)

### 53. P(red) + Br<sub>2</sub>

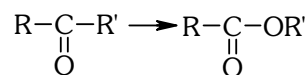


### 54. P (red) + HI



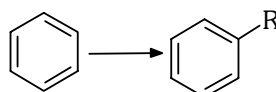
(strong reducing agent can reduce any oxygen or halogen containing compound to alkane)

### 55. Perbenzoic acid [Baeyer Villiger Oxidation]

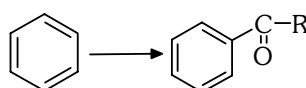


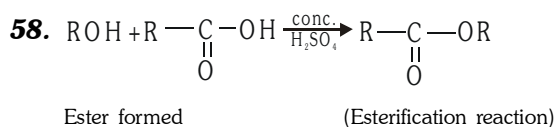
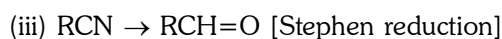
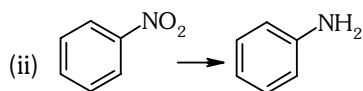
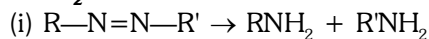
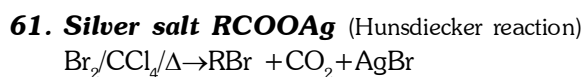
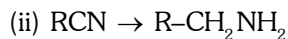
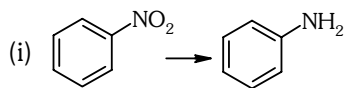
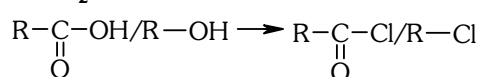
R' having more migrating tendency than R

### 56. RCl + AlCl<sub>3</sub> [Friedel craft alkylation]



### 57. RCOCl + AlCl<sub>3</sub> [Friedel craft acylation]



59.  $\text{SnCl}_2 + \text{HCl}$ 60.  $\text{Sn} + \text{HCl}$ 63.  $\text{SOCl}_2$ 64. **Tollens Reagent Test**

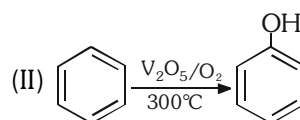
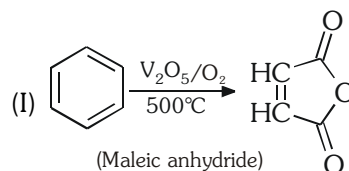
- (i) Terminal alkyne gives
- (ii) Aldehyde Group gives
- (iii) Ketone gives -ve test
- (iv)  $\alpha$ -hydroxy ketone gives
- (v)  $\text{HCOOH}$  gives
- (vi) Hemi acetal gives
- (vii)  $\text{PhNH}-\text{OH}$  gives

65. **Benzene sulphonyl chloride**

It is used to distinguish and separate (Hinsberg reagent)  
 $1^\circ$ ,  $2^\circ$  and  $3^\circ$  amines.

66. **Tetra ethyl lead (TEL)**

Used as antiknock compound

67.  $\text{V}_2\text{O}_5$ 

**IMPORTANT NOTES**

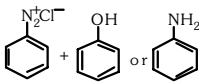
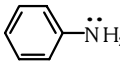
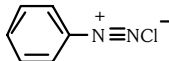
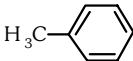
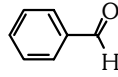
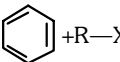
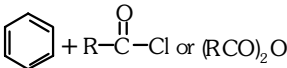
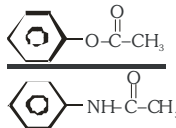
## Nutshell review & preview of ORGANIC NAME REACTIONS

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| • <b>Aldol Condensation</b>             | $\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\text{H} \xrightarrow{\text{OH}^-} \text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 \xrightarrow{\text{H}^+} \text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\overset{\text{OH}}{\text{CH}}-\text{CH}_3 \xrightarrow{\text{OH}^-/\Delta} \text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}=\text{CH}-\text{CH}_3$                                                                                              |
| • <b>Claisen Condensation</b>           | $\text{H}_3\text{C}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OEt} \xrightarrow{\text{EtO}^-} \text{EtO}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 \xrightarrow{\text{H}^+} \text{EtO}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 \xrightarrow{-\text{OEt}} \text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{OEt}$                                                     |
| • <b>Perkin Condensation</b>            | $\text{C}_6\text{H}_5-\overset{\text{H}}{\text{C}}=\overset{\text{O}}{\parallel}{\text{C}}-\text{H} + \text{H}_3\text{C}-\overset{\text{H}}{\text{C}}-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 \xrightarrow{\text{H}_3\text{C}-\overset{\text{O}}{\parallel}{\text{C}}-\text{ONa}^+} \text{C}_6\text{H}_5-\overset{\text{H}}{\text{C}}=\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}(\text{H})-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ <p style="text-align: center;">Cinnamic acid</p> |
| • <b>Benzoin Condensation</b>           | $2 \text{C}_6\text{H}_5-\overset{\text{O}}{\parallel}{\text{C}}-\text{H} \xrightarrow[\text{EtOH}]{\text{KCN}} \text{C}_6\text{H}_5-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}(\text{OH})-\text{C}_6\text{H}_5 \text{ Benzoin}$                                                                                                                                                                                                                                                                                                                             |
| • <b>Haloform Reaction</b>              | $\text{H}_3\text{C}-\overset{\text{OH}}{\text{CH}}-\text{R}^1 \xrightarrow{\text{NaOI}} \text{H}_3\text{C}-\overset{\text{O}}{\parallel}{\text{C}}-\text{R}^1 \xrightarrow[\text{I}_2]{\text{NaOH}} \text{CHI}_3 + \text{R}^1-\overset{\text{O}}{\parallel}{\text{C}}-\text{ONa}^+$                                                                                                                                                                                                                                                                                |
| • <b>Carbylamine Test</b>               | $\text{R}-\text{N}(\text{H})_2 + \text{CHCl}_3 \xrightarrow{\text{KOH}} \text{R}-\text{N}^+(\text{H})_2-\text{C}(\text{Cl})_3 \xrightarrow{\text{OH}^-} \text{R}-\text{N}^+(\text{H})_2-\text{C}(\text{Cl})_2\text{OH} \xrightarrow{-\text{HCl}} \text{R}-\text{N}^+(\text{H})_2-\text{C}(\text{Cl})\text{OH} \xrightarrow{-\text{HCl}} \text{R}-\text{N}^+(\text{H})_2-\text{OH} \xrightarrow{-\text{HCl}} \text{R}-\text{N}^+(\text{H})-\text{OH} \xrightarrow{-\text{HCl}} \text{R}-\text{N}^+=\text{CH} \text{ Isocyanide}$                                    |
| • <b>Reimer Tiemann Reaction</b>        | $\text{C}_6\text{H}_5-\text{OH} \xrightarrow[\text{KOH}]{\text{CCl}_4} \text{C}_6\text{H}_5-\text{O}^- \xrightarrow{\text{CHCl}_3/\text{KOH}} \text{C}_6\text{H}_5-\text{OH}-\text{C}(=\text{O})-\text{H} \text{ (Salicylaldehyde)}$                                                                                                                                                                                                                                                                                                                               |
| • <b>Kolbe's Schimdt Reaction</b>       | $\text{C}_6\text{H}_5-\text{OH} \xrightarrow[\text{CO}_2/\text{H}^+]{\text{NaOH}, 125^\circ\text{C}} \text{C}_6\text{H}_5-\text{OH}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH} \text{ (Salicylic acid) major}$                                                                                                                                                                                                                                                                                                                                              |
| • <b>Hoffmann Bromamide Degradation</b> | $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2 \xrightarrow[\text{KOH}]{\text{Br}_2} \text{R}-\text{NH}_2 + \text{K}_2\text{CO}_3$                                                                                                                                                                                                                                                                                                                                                                                                                  |
| • <b>Curtius Reaction</b>               | $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl} \xrightarrow[\Delta, \text{H}_3\text{O}^+]{\text{NaN}_3} \text{R}-\text{NH}_2$                                                                                                                                                                                                                                                                                                                                                                                                                         |
| • <b>Schmidt Reaction</b>               | $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH} \xrightarrow[\text{H}_2\text{SO}_4]{\text{HN}_3} \text{R}-\text{N}=\text{C}=\text{O} \xrightarrow{\text{H}_3\text{O}^+} \text{R}-\text{NH}_2$                                                                                                                                                                                                                                                                                                                                                          |
| • <b>Cannizzaro reaction</b>            | $\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H} \xrightarrow[50\%]{\text{NaOH}} \text{H}_3\text{C}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H} \xrightarrow{\text{slow RDS}} \text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH} + \text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H} \rightarrow \text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{ONa}^+ + \text{CH}_3-\text{OH}$                                                                                                                                             |



|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>• <b>Bayer villiger oxidation</b></p>        | $\text{CH}_3\text{C}(=\text{O})\text{C}_6\text{H}_5 \xrightarrow[\text{(ii) H}^+]{\text{(i) F}_3\text{C}-\text{C}(=\text{O})-\text{OO}-\text{H (peracid)}} \text{H}_3\text{C}-\text{C}(=\text{O})-\text{OC}_6\text{H}_5$                                                                                                                                                                                                                                                                                                                                                    |
| <p>• <b>Cumene</b></p>                          | $\text{C}_6\text{H}_5\text{CH}(\text{CH}_3)_2 \xrightarrow[130^\circ\text{C}]{\text{O}_2} \text{C}_6\text{H}_5\text{C}(\text{CH}_3)_2\text{OOH} \xrightarrow{\text{HCl}} \text{C}_6\text{H}_5\text{OH} + \text{CH}_3\text{C}(=\text{O})\text{CH}_3$ <p style="text-align: center;">Cumene hydroperoxide</p>                                                                                                                                                                                                                                                                 |
| <p>• <b>Pinacol-Pincolone rearrangement</b></p> | $\text{C}_6\text{H}_5\text{C}(\text{OH})(\text{C}_6\text{H}_5)\text{C}(\text{OH})(\text{C}_6\text{H}_5)_2 \xrightarrow[\text{-H}_2\text{O}]{\text{H}^+} \text{C}_6\text{H}_5\text{C}(\text{OH})(\text{C}_6\text{H}_5)\text{C}^+(\text{C}_6\text{H}_5)_2 \xrightarrow{-\text{H}^+} \text{C}_6\text{H}_5\text{C}(=\text{O})\text{C}(\text{Ph})_2$                                                                                                                                                                                                                             |
| <p>• <b>Birch Reduction</b></p>                 | $\text{C}_6\text{H}_6 \xrightarrow[\text{liq. NH}_3]{\text{Na}} \text{C}_6\text{H}_8$ $\text{C}_6\text{H}_5\text{OR} \xrightarrow[\text{liq. NH}_3]{\text{Na}} \text{C}_6\text{H}_7\text{OR} \xrightarrow[\text{liq. NH}_3]{\text{Na}} \text{C}_6\text{H}_7\text{COOH}$ $\text{R}-\text{C}\equiv\text{C}-\text{R}' \xrightarrow[\text{liq. NH}_3]{\text{Na}} \text{trans-R}-\text{C}=\text{C}-\text{H}$ $\text{R}-\text{C}\equiv\text{C}-\text{H} \xrightarrow[\text{liq. NH}_3]{\text{Na}} \text{R}-\text{C}\equiv\text{C}^-\text{Na}^+ \quad (\text{Acid-base reaction})$ |
| <p>• <b>Gabriel Synthesis</b></p>               | $\text{C}_6\text{H}_4(\text{CO})_2\text{O} \xrightarrow{\text{NH}_3} \text{C}_6\text{H}_4(\text{CO})_2\text{N}-\text{H} \xrightarrow{\text{KOH}} \text{C}_6\text{H}_4(\text{CO})_2\text{N}^-\text{K}^+ \xrightarrow{\text{CH}_3\text{I}} \text{C}_6\text{H}_4(\text{CO})_2\text{N}-\text{CH}_3 \xrightarrow{\text{H}_3\text{O}^+} \text{C}_6\text{H}_4(\text{CO})_2\text{OH} + \text{CH}_3\text{NH}_2$                                                                                                                                                                      |

## NAME REACTIONS

| Name                                  | Reactant                                                                          | Reagent                                                       | Product                                                                                                      |
|---------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Clemmensen Reduction</b>           | Aldehyde & Ketone                                                                 | Zn-Hg/conc. HCl                                               | Alkane                                                                                                       |
| <b>Coupling Reaction</b>              |  | NaOH (phenol)<br>HCl (Aniline)                                | Azo Dye<br>(Detection of OH or NH <sub>2</sub> gr)                                                           |
| <b>Diazotization</b>                  |  | NaNO <sub>2</sub> + HCl<br>0° – 5°C                           |                           |
| <b>Etard reaction</b>                 |  | CrO <sub>2</sub> Cl <sub>2</sub> /CS <sub>2</sub>             | <br>(Benzaldehyde)        |
| <b>Fittig Reaction</b>                | Halo benzene                                                                      | Na/Dry ether                                                  | Diphenyl                                                                                                     |
| <b>Friedel Craft alkylation</b>       |  | Anhydrous AlCl <sub>3</sub>                                   | Alkyl Benzene                                                                                                |
| <b>Friedel Craft acylation</b>        |  | Anhydrous AlCl <sub>3</sub>                                   | Acyl Benzene                                                                                                 |
| <b>Gattermann aldehyde synthesis</b>  | C <sub>6</sub> H <sub>6</sub>                                                     | HCN + HCl / ZnCl <sub>2</sub> / H <sub>3</sub> O <sup>+</sup> | Benzaldehyde                                                                                                 |
| <b>Gattermann-Koch reaction</b>       | C <sub>6</sub> H <sub>6</sub> (CO + HCl)                                          | anhy AlCl <sub>3</sub>                                        | Benzaldehyde                                                                                                 |
| <b>Hell-Volhard-Zelinsky reaction</b> | carboxylic acid having α-hydrogen atom                                            | Br <sub>2</sub> / red P                                       | α-halogenated carboxylic acid                                                                                |
| <b>Hoffmann mustard oil reaction</b>  | primary aliphatic amine + CS <sub>2</sub>                                         | HgCl <sub>2</sub> /Δ                                          | CH <sub>3</sub> CH <sub>2</sub> -N=C=S + HgS (black)                                                         |
| <b>Hunsdiecker reaction</b>           | Ag salt of carboxylic acid                                                        | Br <sub>2</sub> /CCl <sub>4</sub> , 80°C                      | alkyl or aryl bromide                                                                                        |
| <b>Kolbe electrolytic reaction</b>    | alkali metal salt of carboxylic acid                                              | electrolysis                                                  | alkane, alkene and alkyne                                                                                    |
| <b>Mendius reaction</b>               | alkyl or aryl cyanide                                                             | Na/C <sub>2</sub> H <sub>5</sub> OH                           | primary amine                                                                                                |
| <b>Rosenmund reduction</b>            | acid chloride                                                                     | H <sub>2</sub> , Pd/BaSO <sub>4</sub><br>boiling xylene       | aldehyde                                                                                                     |
| <b>Sabatier-Senderens reaction</b>    | Unsaturated hydrocarbon                                                           | Raney Ni/H <sub>2</sub> ,<br>200—300°C                        | Alkane                                                                                                       |
| <b>Sandmeyer reaction</b>             | C <sub>6</sub> H <sub>5</sub> N <sub>2</sub> <sup>+</sup> Cl <sup>-</sup>         | CuCl/HCl or CuBr/HBr or<br>CuCN/KCN, heat                     | Halo or cyanobenzene                                                                                         |
| <b>Gattermann Reaction</b>            | C <sub>6</sub> H <sub>5</sub> N <sub>2</sub> <sup>+</sup> Cl <sup>-</sup>         | Cu/HX (HBr/HCl)                                               | Halobenzene                                                                                                  |
| <b>Schotten-Baumann reaction</b>      | (phenol or aniline or alcohol)                                                    | NaOH + C <sub>6</sub> H <sub>5</sub> COCl                     | benzoylated product<br> |
| <b>Stephen reaction</b>               | alkyl cyanide                                                                     | (i) SnCl <sub>2</sub> /HCl (ii) H <sub>2</sub> O              | Aldehyde                                                                                                     |
| <b>Williamson synthesis</b>           | alkyl halide                                                                      | sodium alkoxide or<br>sodium phenoxide                        | Ether                                                                                                        |
| <b>Wurtz-Fittig reaction</b>          | alkyl halide + aryl halide                                                        | Na/dry ether                                                  | alkyl benzene                                                                                                |

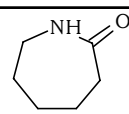
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# POLYMERS

## ADDITION POLYMERS

| S.No                      | Name of Polymer                                                      | Starting Materials                                                                                                                                    | Nature of Polymer                               |
|---------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>I. Polyolefins</b>     |                                                                      |                                                                                                                                                       |                                                 |
| 1.                        | Polyethylene or Polyethene                                           | $\text{CH}_2=\text{CH}_2$                                                                                                                             | Low density homopolymer (branched chain growth) |
| 2.                        | Polypropylene or Polypropene                                         | $\text{CH}_3\text{CH}=\text{CH}_2$                                                                                                                    | Homopolymer, linear, chain growth               |
| 3.                        | Polystyrene                                                          | $\text{C}_6\text{H}_5\text{CH}=\text{CH}_2$                                                                                                           | Homopolymer, linear, chain growth               |
| <b>II. Polydienes</b>     |                                                                      |                                                                                                                                                       |                                                 |
| 1.                        | Neoprene                                                             | $\begin{array}{c} \text{Cl} \\   \\ \text{CH}_2=\text{CH}-\text{C}=\text{CH}_2 \\ \text{Chloroprene or} \\ \text{2-Chloro-1,3-butadiene} \end{array}$ | Homopolymer, chain growth                       |
| 2                         | Buna S (Styrene-Butadiene, Rubber) SBR or GRS                        | $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ (and) $\text{C}_6\text{H}_5\text{CH}=\text{CH}_2$<br>1,3-butadiene Styrene                              | Copolymer, chain growth                         |
| <b>III. Polyacrylates</b> |                                                                      |                                                                                                                                                       |                                                 |
| 1.                        | Polymethylmethacrylate (Flexiglass Lucite, Acrylite or Perspex PMMA) | $\begin{array}{c} \text{CH}_3 \\   \\ \text{H}_2\text{C}=\text{C}-\text{COOCH}_3 \end{array}$                                                         | Homopolymer                                     |
| 2.                        | Polyethylacrylate                                                    | $\text{H}_2\text{C}=\text{CH}-\text{COOC}_2\text{H}_5$                                                                                                | Homopolymer                                     |
| 3.                        | Polyacrylonitrile or Orlon PAN                                       | $\text{CH}_2=\text{CH}-\text{CN}$                                                                                                                     | Homopolymer                                     |
| <b>IV. Polyhalofins</b>   |                                                                      |                                                                                                                                                       |                                                 |
| 1.                        | Polyvinyl chloride PVC                                               | $\text{CH}_2=\text{CH}-\text{Cl}$                                                                                                                     | Homopolymer, chain growth                       |
| 2.                        | Polytetrafluoroethylene or Teflon PTFE                               | $\text{F}_2\text{C}=\text{CF}_2$                                                                                                                      | Homopolymer                                     |
| 3.                        | Polymonochlorotrifluoro-ethylene PCTFE                               | $\text{ClFC}=\text{CF}_2$                                                                                                                             | Homopolymer                                     |

## CONDENSATION POLYMERS

| S.No                       | Name of Polymer                      | Starting Materials                                                                                                                                                                                                                         | Nature of Polymer              |
|----------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>I. Polyesters</b>       |                                      |                                                                                                                                                                                                                                            |                                |
| 1.                         | Terylene or Dacron                   | $\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}$<br>Ethylene glycol or Ethane-1,2-diol           And $\text{HO}-\text{C}(=\text{O})-\text{C}_6\text{H}_4-\text{C}(=\text{O})-\text{OH}$<br>Terephthalic acid or Benzene-1,4-dicarboxylic acid | Copolymer, step growth, linear |
| 2.                         | Glyptal or Alkyl resin               | $\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}$ (and) $\text{HOOC}-\text{C}_6\text{H}_4-\text{COOH}$<br>Ethylene glycol Phthalic acid or Benzene-1,2-dicarboxylic acid                                                                       | Copolymer, linear step growth  |
| <b>II. Polyamides</b>      |                                      |                                                                                                                                                                                                                                            |                                |
| 1.                         | Nylon-6,6                            | $\text{HO}-\text{C}(=\text{O})-(\text{CH}_2)_4-\text{C}(=\text{O})-\text{OH}$ (and) $\text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2$<br>Adipic acid Hexamethylene diamine                                                                  | Copolymer, linear, step growth |
| 2                          | Nylon-6,10                           | $\text{H}_2\text{N}(\text{CH}_2)_6\text{NH}_2$ (and) $\text{HOOC}(\text{CH}_2)_8\text{COOH}$<br>Hexamethylene diamine Sebacic acid                                                                                                         | Copolymer, linear, step growth |
| 3.                         | Nylon-6                              | <br>Caprolactam                                                                                                                                         | Homopolymer, linear            |
| <b>Formaldehyde resins</b> |                                      |                                                                                                                                                                                                                                            |                                |
| 1.                         | Phenolformaldehyde resin or Bakelite | Phenol and formaldehyde                                                                                                                                                                                                                    | Copolymer, step growth         |
| 2.                         | Melamine formaldehyde resin          | Melamine and formaldehyde                                                                                                                                                                                                                  | Copolymer, step growth         |

**IMPORTANT NOTES**

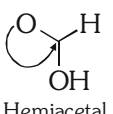
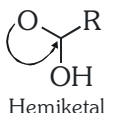
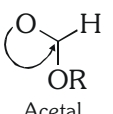
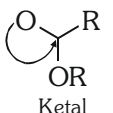
## CARBOHYDRATES

- Carbohydrates are defined as optically active polyhydroxy aldehydes or ketones or the compound which produce such units on hydrolysis.
- Monosaccharide** ( $C_nH_{2n}O_n$ ) : single unit, can't be hydrolysed : Glucose and fructose.
- Oligosaccharides** gives two to ten monosaccharides on hydrolysis.
- Disaccharides** (by glycosidic linkage)
 

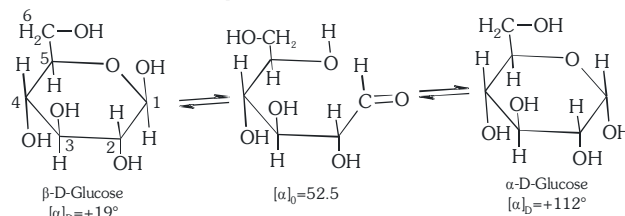
Sucrose  $\xrightarrow{H_3O^+}$   $\alpha$ -D. Glucose +  $\beta$ -D. Fructose;

Maltose  $\xrightarrow{H_3O^+}$  2  $\alpha$ -D. Glucose unit

Lactose  $\xrightarrow{H_3O^+}$   $\beta$ -D. Glucose +  $\beta$ -D. Galactose
- Polysaccharide** : Contain more than ten monosaccharide units  
 $(C_6H_{10}O_5)_n$  : Starch & cellulose.

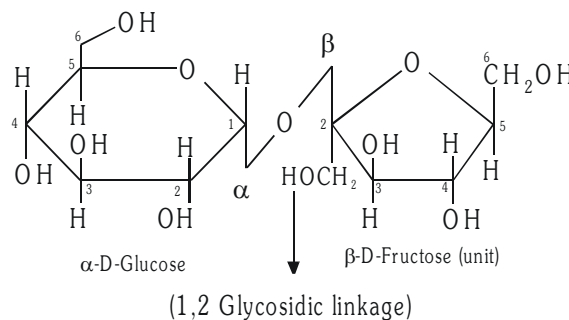
| TYPE OF SUGAR    |                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                         |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Give Test        | Reducing                                                                                                                                                                                                                                                                                   | Non Reducing                                                                                                                                                                            |
| Tollen's Reagent | +ve test                                                                                                                                                                                                                                                                                   | -ve test                                                                                                                                                                                |
| Fehling Reagent  | +ve test                                                                                                                                                                                                                                                                                   | -ve test                                                                                                                                                                                |
| Benedict Test    | +ve test                                                                                                                                                                                                                                                                                   | -ve test                                                                                                                                                                                |
| Mutarotation     | Yes                                                                                                                                                                                                                                                                                        | No                                                                                                                                                                                      |
| Functional Unit  | $-\overset{\alpha}{C}-C=O / -\overset{\alpha}{C}-\overset{O}{\underset{OH}{\parallel}}-C-$  Hemiacetal<br> Hemiketal |  Acetal<br> Ketal |
| Example          | All monosaccharides<br>Glucose<br>Fructose<br>Mannose<br>Galactose<br>Disaccharide<br>Maltose<br>lactose                                                                                                                                                                                   | Disaccharide<br>Sucrose<br>Polysaccharide<br>Starch<br>cellulose                                                                                                                        |

- Mutarotation**: When either form of D-glucose is placed in aq. solution it slowly form the other via open chain aldehyde and gradual change in specific rotation until specific rotation ( $\pm 52.5^\circ$ ) is reached.

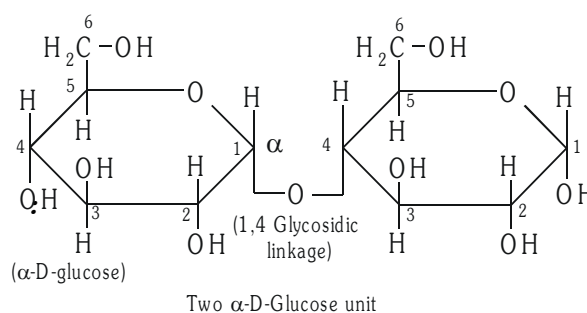


- Anomer's** : Differ in configuration at 1st carbon due to hemi (acetal or ketal) ring formation. The new-asymmetric carbon is referred to as Anomeric carbon.
- Epimer's** : Diastereomer's which differ in conformation at any one chiral carbon  
eg. D-Glucose & D-mannose  
D-Glucose & D-Galactose

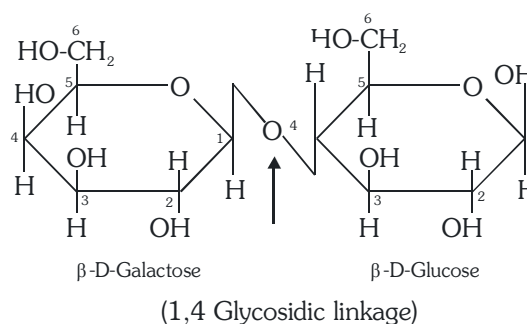
- Sucrose** :



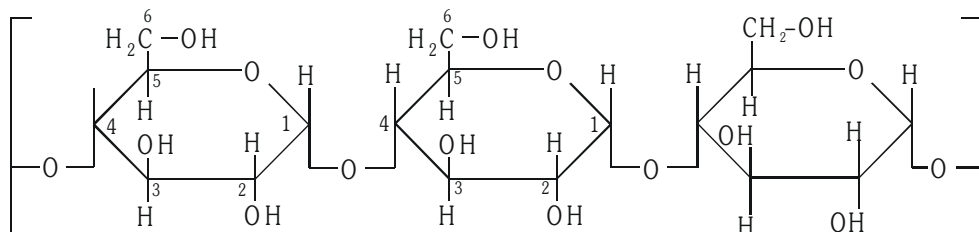
- Maltose**



- Lactose** :



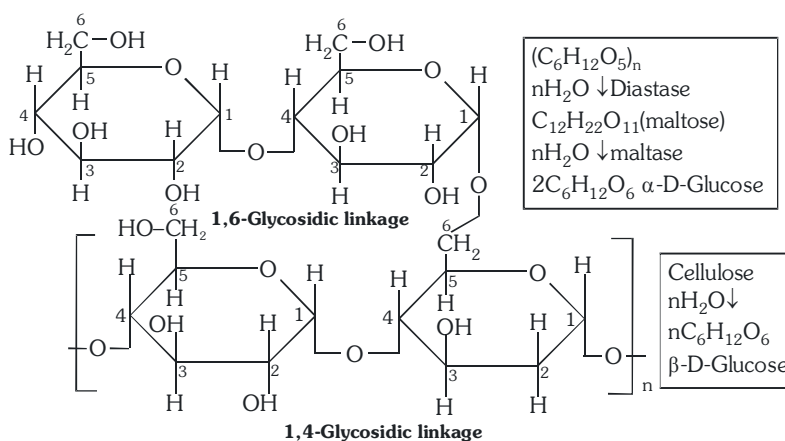
- **Starch** : (Amylose & Amylopectin)
- **Amylose : (Straight Chain) :**



( $\alpha$ -1,4 Glycosidic linkage)

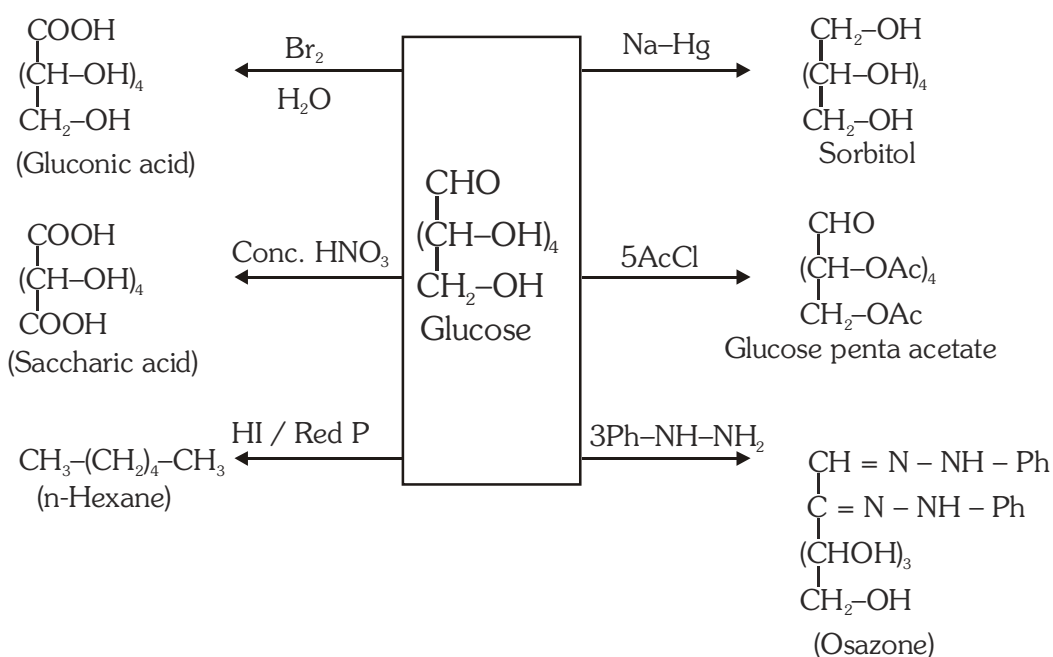
(i) Soluble in  $H_2O$  & gives blue colour with  $I_2$

- **Amylopectin (Branch chain) : ( $C_6H_{12}O_5$ )<sub>n</sub>**



### REACTION OF GLUCOSE (OPEN CHAIN STRUCTURE)

\* Fructose doesn't react with  $Br_2/H_2O$



**IMPORTANT NOTES**